The Institute of Chartered Accountants of Bangladesh (ICAB)

Business Strategy and Technology

Volume II



Workbook
For CA Professional Level Exams

CA BANGLADESH

THE INSTITUTE OF CHARTERED ACCOUNTANTS OF BANGLADESH

www.icab.org.bd

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Questions within the Workbook should be treated as preparation questions, providing you with a firm foundation before you attempt the exam-standard questions.

**** Italic denotes changes***

Chapter 11
Methods of development

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Introduction

Learning outcomes

- Select and justify a strategy or combination of strategies which will best achieve the organisation's objectives taking account of known constraints and capabilities, including stakeholder risk preferences and the potential for developing technologies to increase or mitigate risk.
- Select and evaluate methods of further developing an organisation which adjust existing strategies or implement new strategies to take account of changing position and risk.

Syllabus links

The concept of different forms of business alliance will be familiar to you from your Business, Technology and Finance studies. The discussion of acquisition and mergers here complements the studies you will be taking in Financial Management concerning business valuations.

Examination context

Having decided on a strategy for growth, an organisation must consider how that growth is to be achieved. This chapter considers the options available to a business that wishes to expand. In the exam you will be expected to apply the knowledge covered in this chapter to the scenario, to advise an organisation on the most appropriate method of expansion. This is a popular exam topic.

Earlier in the Workbook, we mentioned that some exam questions may specify the models candidates are expected to use when attempting particular questions. It is also important to understand that some questions may not specify which models to use at all, instead asking for 'appropriate' or 'relevant' models to be used. This illustrates the importance of being able to identify how the information in a scenario lends itself to a particular model or theory.

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this chapter.

Topic	Practical significance	Study approach	Exam approach	Interactive questions
1	**Methods of growth** The management of firms are under pressure to grow their businesses. Share prices, and management careers, depend on the ability to deliver better earnings next year than last year. Once a firm has committed to growing the	**Approach** Read quickly through this short section paying attention to Lynch's expansion method matrix. **Stop and think** Following a period of global expansion into a range of different business areas, Tesco experienced turbulent times. Accounting scandals and declining	In the exam you will be expected to apply the knowledge covered in this chapter to the scenario, in order to advise an organisation on the most appropriate method of expansion. This is a popular exam topic.	

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	business it needs to carefully identify the most appropriate method of growth.	profitability led to Tesco selling off some of its international operations, including its Homeplus retail operations in South Korea (in 2015), so it could concentrate on its core UK business. Did Tesco's expansions actually benefit its shareholders?		
2	**Organic growth** Organic growth, often called internal development, involves the expansion of an organisation using its own resources and capabilities, without taking over other firms.	**Approach** Understand the benefits and drawbacks of organic growth. **Stop and think** Apple's growth has been predominantly organic since its incorporation in 1977. What has been the key to their success?	In the exam avoid simply listing benefits and drawbacks of organic growth. Strong candidates will apply their answers to the scenario and discuss the implications of organic growth for the organisation in the question.	
3	**International expansion** International expansion can be a source of great opportunity and also risk for an organisation. Firms must be sure they have the resources to manage successful international expansion.	**Approach** Review and understand the reasons in favour of and against international expansion. **Stop and think** Consider an organisation you are familiar with that has expanded overseas. What led to the expansion? Has it always been successful?	Implementing a strategy by external or international growth will feature in many exam questions. Be able to apply the content of the topic to organisations in different circumstances.	
4	**Mergers and acquisitions** A merger is the integration of two or more businesses whereas an acquisition occurs where one business purchases another.	**Approach** Read through the topic paying particular attention to the worked examples, which explain real-life merger/acquisition scenarios. **Stop and think** Less than half of all	Carefully assess whether you think the mergers and acquisitions are appropriate, taking into consideration the specific circumstances in the exam scenario.	

Topic	Practical significance	Study approach	Exam approach	Interactive questions
		acquisitions are successful in terms of satisfying shareholders. Using the internet, quickly research a failed acquisition. What were the reasons behind the failure?		
5	**Joint ventures, alliances and franchising** These business arrangements allow organisations to pool resources and work together to gain competitive advantage.	**Approach** Read through the descriptions, advantages and disadvantages of each type of business structure. **Stop and think** Franchising is common in the fast-food industry. Consider a fast-food company you know. What are the benefits and problems they face as a result of being part of a franchise?	As with all topics in this chapter, ensure you are comfortable with the advantages and disadvantages of each approach. When recommending a suitable growth strategy apply the advantages and disadvantages to the scenario in the exam.	**IQ1: Ponda** This is a typical exam style question asking you to make recommendations by applying technical knowledge to the question scenario. Ensure you write the answer in the prescribed format.

Once you have worked through this guidance, you are ready to attempt the further question practice included at the end of this chapter.

1 Methods of growth

Section overview

- Many organisations pursue **growth**, such as increases in profits or market share. Growth may be achieved organically, or through a link to another firm. Once a firm has made its choice of strategies, it needs to choose an appropriate mechanism:
  - Develop the business from scratch.
  - Acquire or merge with an already existing businesses.
  - Co-operate in some way with another firm.

1.1 What is growth?

An organisation's growth may be expressed in a number of ways, for example:

- Sales revenue (a growth in the number of markets served)
- Profitability (in absolute terms, and as a return on capital)
- Number of goods/services sold; Number of employees
- Number of outlets/sites; Number of countries

The chapter, 'Strategic options,' looked at Ansoff's four directions (or vectors) of growth available to the business. Growth may be achieved by a number of mechanisms:

- Develop the business from scratch
- Acquire or merge with an already existing business
- Co-operate in some way with another firm

The main issues involved in choosing a method of growth are these.

- A firm may not be able to go it alone, or it may have plenty of **resources** to invest.
- Two different businesses might have **complementary skills.**
- Does a firm need to **move fast**?
- A firm might wish to **retain control** of a product or process.
- Is there an acquisition or joint venture partner with compatible **people/organisation culture**?
- **Risk**: a firm may either increase or reduce the level of risk to which it is subject.

1.2 Lynch

Lynch summarised possible expansion methods in the following matrix:

Figure 11.1: Lynch's expansion method matrix

		Company	
		Internal development	**External development**
New location	**Home country**	Internal Domestic Development	Joint venture Merge Acquisition Alliance Franchise/licence
	Abroad	Exporting Overseas office Overseas manufacture Multinational operation Global operation	Joint venture Merge Acquisition Alliance Franchise/licence

Professional skills focus: Structuring problems and solutions

Although a question may specify a particular model to use, there are instances when questions ask you to refer to relevant models. Be sure to know which models are appropriate in which instances. For example, Lynch's matrix is useful for analysing the method of growth, compared to Ansoff's matrix which can determine the direction of growth.

2 Organic growth

Section overview

- Organic growth is expansion by use of internal resources (internal development).
- Its advantages are the maintenance of overall control, and the fact that managers can concentrate on product-market issues, rather than concerns of organisation structure.
- Its drawbacks are that it can be slow and there may be barriers to entry preventing organic growth.

Definition

Organic growth: Involves the expansion of a firm's size, profits, and activities through the use of its own resources and capabilities without taking over other firms.

2.1 Benefits of organic growth

Firms pursue organic growth for a number of reasons.

- The **process of developing** a new product gives the firm the best understanding of the market and the product.

- It might be the only sensible way to pursue **genuine technological innovations.**
- There is **no suitable target for acquisition.**
- It can be planned and financed easily from the company's **current resources** and the costs are spread over time.
- The same **style of management** and corporate culture can be maintained so there is less disruption.
- **Hidden or unforeseen losses**, common in acquisitions, are less likely with organic growth.
- It provides **career development** opportunities for managers otherwise plateaued in their present roles.
- It could be **cheaper** because assets are being acquired without additional payments for goodwill (eg, future earnings foregone by the persons selling their business or shares).
- It is **less risky**. In acquisitions the purchaser may also take on liability for the effects of decisions made by the previous owners (eg, underpaid tax, liability to employees for health and safety breaches and so on).

2.2 Drawbacks of organic growth

- It may **intensify competition** in a given market compared to buying an existing player.
- It is too **slow** if the market is developing very quickly.
- The firm does **not gain access to the knowledge and systems** of an established operator so it can be more risky.
- It will initially lack economies of scale/experience effects.
- There may be prohibitive barriers to entry in new markets.

3 International expansion



Section overview

- International expansion is a big undertaking and firms must understand their reasons for it, and be sure that they have the resources to manage it, both strategically and operationally. The decision about which international markets to enter should be based upon assessment of market attractiveness, competitive advantage and risk.

3.1 Reasons for international expansion

Some of the reasons for international expansion cited by management are as follows:

- **Chance**: A company executive may recognise an opportunity while on a foreign trip or the firm may receive chance orders or requests for information from potential foreign customers.
- **Life cycle:** Home sales may be in the mature or declining stages of the product life cycle. International expansion may allow sales growth since products are often in different stages of the product life cycle in different countries.
- **Competition**: Intense competition in an overcrowded domestic market sometimes induces firms to seek markets internationally, where rivalry is less keen.
- **Reduce dependence**: Many companies wish to diversify away from an over-dependence on a single domestic market. Increased geographic diversification can help to spread risk.
- **Economies of scale**: Technological factors may be such that a large volume is needed either to cover the high costs of plant, equipment, R&D and personnel or to exploit a large

potential for economies of scale and experience. For these reasons firms in the aviation, ethical drugs, computer and automobile industries are often obliged to enter multiple countries.

- **Variable quality**: International expansion can facilitate the disposal of discontinued products and seconds since these can be sold abroad without spoiling the home market.
- **Finance**: Many firms are attracted by favourable opportunities such as the following:
  - The development of lucrative emerging markets (such as China and India)
  - Depreciation in their domestic currency values
  - Corporate tax benefits offered by particular countries
  - Lowering of import barriers abroad
- **Familial**: Many countries and companies trade because of family or cultural connections.
- **Aid agencies**: Countries that benefit from bilateral or unilateral aid often purchase goods which normally they would not have the money for.

3.2 International expansion

- **Reasons supporting international expansion**
  - **Profit margins** may be higher abroad.
  - Increase in **sales volume** from foreign sales may allow large reductions in unit costs.
  - The **product life cycle** may be extended if the product is at an earlier stage in the life cycle in other countries.
  - **Seasonal fluctuations** may be levelled out (peak periods in some countries coinciding with troughs in others).
  - It offers an opportunity of disposing of excess production in times of low domestic demand.
  - International activities spread the risk which exists in any single market (eg, political and economic changes).
  - **Obsolescent products** can be sold off to international customers without damage to the domestic market.
  - The firm's prestige may be enhanced by portraying a **global image**.
- **Reasons for avoiding international expansion**
  - Profits may be unduly affected by factors outside the firm's **control** (eg, due to fluctuation of exchange rates and foreign government actions).
  - The **adaptations** to the product (or other marketing mix elements) needed for international success will diminish the effects of economies of scale.
  - Extending the product life cycle is not always **cost effective**. It may be better to develop new products for the domestic market.
  - **Opportunity costs** of investing abroad – funds and resources may be better utilised at home.
  - In the case of marginal cost pricing, **anti-dumping duties** are more quickly imposed now than in the past.



Professional skills focus: Assimilating and using information

Although you may be able to recall many advantages and disadvantages of the various methods of expansion, this is unlikely to generate many marks in the exam. If citing a particular advantage, it is important you apply your answer to the scenario company, in order to explain how and why the organisation could benefit.

3.3 Issues for management to consider

3.3.1 Strategic issues

- Is the venture likely to yield an acceptable financial **return**?
- Does it fit with the company's overall **mission** and objectives?
- Does the organisation have (or can it raise) the **resources** necessary to exploit effectively international opportunities?
- What is the impact on the firm's **risk** profile?
- What **method of entry** is most suitable?

3.3.2 Tactical issues

- How can the company get to **understand customers' needs** and preferences in foreign markets?
- Does the company know how to **conduct business**s abroad, and deal effectively with potential partners there?
- Are there **foreign regulations** and associated hidden costs?
- Does the company have the necessary **management skills** and experience?

Such tactical issues may mean international expansion is more easily achieved through some form of acquisition or shared arrangement, rather than by organic growth.

4 Mergers and acquisitions

Section overview

- A merger is the integration of two or more businesses. An acquisition is where one business purchases another. This offers speedy access to new technologies and markets, but there are risks: only about half of acquisitions succeed.
- The mechanics of financing and undertaking a merger or acquisition are essential to its success and will be covered in the Financial Management syllabus.

4.1 The motives for acquiring companies

Definitions

Merger: Concerns the joining of two separate companies to form a single company.

Acquisition: Is the purchase of a controlling interest in another company.

Some acquisitions are dressed up to look like mergers (ie, a combination of equals) because it suggests agreement and may ease the integration of cultures.

In financial reporting, IFRS Accounting Standards do not permit the concept of a merger. Nevertheless, strategically and in terms of financial arrangements, a merger may best describe a business combination, rather than being forced to identify an acquirer.

The classic reasons for mergers/acquisitions as a part of strategy are as follows:

Reason	Effect on operations
Marketing advantages	• New product range • Market presence • Rationalise distribution and advertising • Eliminate competition
Production advantages	• Economies of scale • Technology and skills • Greater production capacity • Safeguard future supplies • Bulk purchase opportunities
Finance and management	• Management team • Cash resources • Gain assets • Tax advantages (eg, losses bought)
Risk-spreading	• Diversification
Retain independence	• Avoid being taken over by acquiring predator by becoming too big to buy
Overcome barriers to entry	• Acquired firm may have licences or patents
Outplay rivals	• Stop rival getting the target

4.2 Diversification

Diversification means a change of both products and markets from the company's present base. Because of the extent of the change there is clearly more risk involved than in expansion, so we must consider the reasons why firms nevertheless diversify:

- **Objectives can no longer be met without diversification**. This would be identified by the gap analysis. The reason for the dissatisfaction with the present industry might be due to poor return caused by product decline, with little chance of technological innovation in the same field, or due to a lack of flexibility, eg, unavoidable dependence on a single customer or a single product line.
- **The firm has more cash than it needs for expansion**. Whether it prefers to invest this outside the business or to seek opportunities for diversification will depend on the relative rates of return obtainable (in general the return from operations exceeds the return from outside investments, but of course more risk is involved) and management preference (management have to balance the internal flexibility achieved by keeping reserves in liquid form with external flexibility offered by diversification).
- **Firms may diversify even if their objectives are being or could be met within their industry**, if diversification promises to be more profitable than expansion.

  Related and conglomerate diversification were covered in detail in the chapter Strategic options.

4.3 Porter's attractiveness tests

An acquisition at a bargain price is unlikely to make up for a long-run lack of profits due to a flawed industry structure. For reasons of cost, the ideal acquisition is in an industry not yet attractive but capable of being made attractive.

Porter proposes two tests:

(a) **The 'cost of entry' test**

Unfortunately, attractive industries tend to have high costs of entry. Premiums likely to be paid for the acquisition of companies are an important consideration.

(b) **The 'better off' test**

The acquisition must do something for shareholders that they cannot do for themselves. Diversification for its own sake will not increase shareholder wealth. Asset stripping brings only one-off benefits and is not a sound basis for long-run investment.

4.4 Acquisitions and risk

Acquisitions provide a means of entering a market, or building up a market share, more quickly and/or at a lower cost than would be incurred if the company tried to develop its own resources.

Corporate planners must however consider the level of **risk** involved. Acquiring companies internationally is riskier, for a number of reasons.

The acquirer should attempt an evaluation of the following:

- The prospects of technological change in the industry
- The size and strength of competitors
- The reaction of competitors to an acquisition
- The likelihood of government intervention and legislation
- The state of the industry and its long-term prospects
- The amount of synergy obtainable from the merger or acquisition

4.5 Synergy as a motive for acquisitions



Definition

Synergy: The benefits gained from two or more businesses combining that would not have been available to each independently. Sometimes expressed as the 2 + 2 = 5 effect. Synergy arises because of complementary resources which are compatible with the products or markets being developed and is often realised by transferring skills or sharing activities.



Context example: Marriott group

The US-based Marriott group provides a good example of skill transfers. The group initially began in the restaurant business. One of its major skills was the use of standardised menus and hospitality routines. Much of its initial business was in the sale of takeaways to customers on the way to the airport. Accordingly, it diversified in turn into airline catering, in-house catering, family restaurants, gourmet restaurants, hotels, cruise ships, travel agents and theme parks. Interestingly some areas, such as gourmet restaurants and travel agents, where skills were not easily transferred, were subsequently divested.

Synergies arise from four sources:

(a) **Marketing and sales synergies**

(1) Benefits of conferring one firm's brands on products of another

(2) Use of common sales team and advertising

(3) Ability to offer wider product range to the client

(b) **Operating synergies**

(1) Economies of scale – in purchasing of inputs, capital equipment etc

(2) Economies of scope – including use of distribution channels and warehousing

(3) Rationalisation of common capacity (eg, logistics, stores, factories)

(4) Capacity smoothing (eg, one firm's peak demand coincides with the other's slack time)

(c) **Financial synergies**

(1) Risk spreading allows cheaper capital to be obtained

(2) Reduction in market competition if firms in similar industry

(3) Shared benefits from same R&D

(4) Possibly more stable cash flows

(5) Sale of surplus assets

(d) **Management synergies**

(1) Highly paid managers used to fix ailing firm rather than administer successful one

(2) Transfer of learning across businesses

(3) Increased opportunity for managerial specialisation in a larger firm

4.6 The mechanics of acquiring companies

Management will wish to assess the value of an acquisition since the price to be paid is critical in determining whether the acquisition is worthwhile. A number of share valuation techniques are available and there are different ways of paying for an acquisition (cash, shares, loan stock). These topics will be covered in your Financial Management syllabus.

Broadly speaking, there are two types of acquisition approaches:

(a) **Agreed bids:** Here, there have been discussions beforehand between the boards of the two companies and their significant investors and a price and management structure agreed. The bid is announced and the board of the target recommends acceptance to the shareholders.

(b) **Hostile (contested) bids**: the predator has either been turned down by the board in discussions or did not approach them to begin with, preferring to build a shareholding over the months beforehand at lower prices before announcing a bid to the market through the financial media.

4.7 Reasons for failure of acquisitions

Takeovers benefit the shareholders of the acquired company often more than the acquirer. According to the Economist Intelligence Unit, there is a consensus that fewer than half of all acquisitions are successful, if seen from the point of view of the buyer's shareholders.

The reasons for the poor performance of acquisitions include:

- Acquirers conduct financial audits but, according to research by London Business School, only 37% conducted anything approaching a management audit.
- Some major problems of implementation relate to human resources and personnel issues such as morale, performance assessment and culture. If key managers or personnel leave, the business will suffer.
- A further explanation may be that excessive prices are paid for acquisitions, resulting in shareholders in the target company being rewarded for expected synergy gains.
- Lack of actual strategic fit between the businesses
- Failure of new management to retain key staff and clients
- Failures by management to exert corporate governance and control over larger business

4.8 Reasons acquisitions still occur

- Many acquisitions and mergers are successful.
- Evidence of a loss of value resulting from a merger doesn't consider whether a worse outcome might have occurred if the firms had not combined.

- Vested interests of corporate financial advisors in pressing for the acquisition, ie, commissions and fees.
- Weak corporate governance allows domineering CEOs or boards to pursue personal agendas with shareholder funds (eg, 'empire building').
- Short-term need for boards to give impression of strategic action to convince investors that business is growing.

Professional skills focus: Applying judgement

Awareness of how business is conducted in the 'real-world' will help you to understand the reasons behind business success and failure. Ensure you regularly read a good quality business news source so that you have some real life examples of acquisitions, to help you with idea generation in the exam.

5 Joint ventures, alliances and franchising

Section overview

- These are other types of arrangement whereby businesses pool resources.

5.1 Choosing business partners

The following factors should be considered in choosing business partners for joint ventures, alliances and franchising:

Drivers	What benefits are offered by collaboration?
Partners	Which partners should be chosen?
Facilitators	Does the external environment favour a partnership?
Components	Activities and processes in the network.
Effectiveness	Does the previous history of alliances generate good results? Is the alliance just a temporary blip? For example, in the airline industry, there are many strategic alliances, but these arise in part because there are legal barriers to cross-border ownership.
Market-orientation	Alliance partners are harder to control and may not have the same commitment to the end-user.

5.2 Joint ventures

Definitions

Consortia: Organizations that co-operate on specific business prospects.

Joint venture: Is a contractual arrangement whereby two or more parties undertake an economic activity which is subject to joint control.

Joint ventures are especially attractive to **smaller or risk-averse firms**, or where very expensive new technologies are being researched and developed, since they allow risks and costs to be shared amongst the venture partners.

Other advantages of joint ventures are:

- They permit coverage of a **larger number of countries** since each one requires less investment.
- They can reduce the risk of **government intervention**.
- They can provide close **control** over operations.
- A joint venture with an indigenous firm provides **local knowledge**.
- They can also be a **learning exercise**.
- They provide funds for expensive **technology and research** projects.
- They are often an alternative to seeking to buy or build a wholly owned manufacturing operation abroad.
- Core competences, which are not available in one entity can be accessed from another venturer.

The major disadvantages of joint ventures are:

- Major **conflicts of interest** over profit shares, amounts invested, the management of the joint venture, and the marketing strategy.
- Problems in each party protecting **intellectual property** such as proprietary product designs, process methods.
- Danger that a **partner may seek to leave joint venture** if its priorities change (eg, shortage of funds) or it is acquired by another firm.
- **Lack of management interest**: the JV will be seen as a secondment outside of the main career hierarchy of the parent firms.
- Exit routes may be unclear, including sharing of the assets generated in the venture.
- Contractual rights may be difficult to enforce across geographical borders or regulatory boundaries.

5.3 Alliances

Some firms enter long-term **strategic alliances** with others for a variety of reasons:

- They share development costs of a particular technology.
- The regulatory environment prohibits take-overs (eg, most major airlines are in strategic alliances because in most countries there are limits to the level of control an 'outsider' can have over an airline).
- Complementary markets or technology.

Such alliances tend to be a looser contractual arrangement than a joint venture and no separate company is formed. There may be less commitment than to a joint venture so the benefits whilst similar, may not be so great. Strategic alliances only go so far, as there may be disputes over control of strategic assets.

Alliances have some **limitations**:

- **Core competence**: Each organisation should be able to focus on its core competence. Alliances may not enable it to create new competences.
- **Strategic priorities**: If a key aspect of strategic delivery is handed over to a partner, the firm loses flexibility. A core competence may not be enough to provide a comprehensive customer benefit.

5.3.1 Information systems based alliances

The cost of major Information system (IS) based methods of working, combined with their

inherent communications capability have made alliances based on IS a natural development. There are four common types:

- **Single industry partnerships**: For example, UK insurance brokers can use a common system called IVANS to research the products offered by all of the major insurance companies.
- **Multi-industry joint marketing partnerships:** Some industries are so closely linked with others that it makes sense to establish IS linking their offerings. A well-known example is holiday bookings, where a flight reservation over the internet is likely to lead to a seamless offer of hotel reservations and car hire.
- **Supply chain partnerships:** Greater and closer co-operation along the supply chain has led to the need for better and faster information flows. Electronic data interchange between customers and suppliers is one aspect of this improvement, perhaps seen most clearly in the car industry, where the big-name manufacturers effectively control the flow of inputs from their suppliers. However, as we discussed in the chapter Strategic capability, linking the information systems of customers and suppliers presents new challenges especially around data protection.
- **IT supplier partnerships**: A slightly different kind of partnership is not uncommon in the IT industry itself, where physical products have their own major software content. The development of these products requires close co-operation between the hardware and software companies concerned.

5.4 Licensing agreements

Definition

Licensing agreement: Permission from one party to another to manufacture or sell a product, or to use a brand name.

A licence grants a third-party organisation the rights to exploit an asset belonging to the licensor. Licences can be granted over:

- The use of brands and recipes (eg, Nestle selling Starbucks coffee 'under licence')
- A patent or technology (eg, an IT firm may be licensed to install and maintain some given software application)
- A particular asset (eg, a drinks firm may buy a license to exploit a mineral water source or a media firm will license the rights to produce and sell a particular film, book or recording in its country)

Licensees will pay an agreed proportion of the sales revenue to the licensor for the right to exploit the license in a given geographical area or for a given range of products.

Licence agreements will vary considerably in the constraints the place on the licensee. Some will dictate branding, pricing and marketing issues. Others will leave their decisions to the licensee.

A very common form of licensing is the franchise.

5.5 Franchising

Definition

Franchising: A method of expanding the business on less capital than would otherwise be possible. Franchisers include IKEA, McDonald's, Starbucks and Pizza Hut.

Franchising can be a method of financing rapid growth without having to raise as much capital as conventional business structures. The International Franchising Association provides prospective franchisees with a directory of current global franchise opportunities. Companies including American car repair business AAMCO offer franchises in Africa, while Dairy Queen are looking for investors in their domestic US market.

The mechanism

- The franchiser grants a licence to the franchisee allowing the franchisee to use the franchiser's name, goodwill, systems.
- The franchisee pays the franchiser for these rights and also for subsequent support services which the franchiser may supply.
- The franchisee is responsible for the day-to-day running of the franchise. The franchiser may impose quality control measures on the franchisee to ensure that the goodwill of the franchiser is not damaged.
- Capital for setting up the franchise is normally supplied by both parties. The franchiser may help the franchisee in presenting proposals to suppliers of capital; presenting a business plan based on a successful trading formula will make it easier to obtain finance. Thus far, the franchiser needs less equity capital than other business structures.
- The franchiser will typically provide support services, including national advertising, market research, research and development, technical expertise, and management support.

Advantages for the franchiser

- Rapid expansion and increasing market share with relatively little equity capital, since franchisees will put in some capital. Strategically, the franchiser will be able to pursue an aggressive expansion policy to cover all geographical areas and establish brand dominance which otherwise could not be afforded.
- The franchisee provides local knowledge and unit supervision. The franchiser specialises in providing a central marketing and control function, limiting the range of management skills needed.
- The franchiser has limited capital in any one unit and therefore has low financial risk.
- Economies of scale are quickly available to the franchiser as the network increases. Hence supply of branded goods, extensive advertising spend are justifiable.
- Franchisee has strong incentives.

The advantages for the franchisee are mainly in the set-up stages, where many new businesses often fail. The franchisee will adopt a brand name, trading format and product specification that have been tested and practised. The learning curve and attendant risks are minimised. The franchisee usually undertakes training, organised by the franchiser, which should provide a running start, thus further reducing risk.

Disadvantages

- A franchisee is largely independent and makes personal decisions about how to run his operation. In addition, the quality of product, customer satisfaction and goodwill is under his control. The franchiser will seek to maintain some control or influence over quality and service from the centre, but this will be difficult if the local unit sees opportunities to increase profit by deviating from the standards which the franchiser has established.
- There can be a clash between local needs or opportunities and the strategy of the franchiser. For example, in the late 1980s McDonald's pursued a strategy of widening the usage of outlets, in particular by encouraging breakfast trade and family groups in the evenings. As part of this it was determined that ideal locations would be in pedestrian areas, near to schools, hospitals, office centres. A franchisee in an industrial town may disagree with this strategy and want to locate on a busy approach road to an industrial estate.
- The franchiser may seek to update/amend the products/services on offer, while some franchisees may be slow to accept change or may find it necessary to write off existing inventory holdings.

- The most successful franchisees may break away and set up as independents, thereby becoming competitors.

5.6 Agency arrangements

These can be used as the distribution channel where local knowledge and contacts are important, eg, exporting. The agreements may be restricted to marketing and product support. Other situations where agents are used include:

- Sales of cosmetics
- Holidays
- Financial services, eg, insurance

The main problem for the company is that it is cut off from direct contact with the customer.



Interactive question 1: Ponda

Ponda plc is the second largest vehicle manufacturing firm in Europe. The board was recently concerned about the lack of 'infant' products in the firm's portfolio, and so decided to launch an electric car – the Greencar. Early research has shown that the demand is expected to be much higher in the US market initially than in the European market.

For the initial trial launch Ponda has decided to distribute the car using the American dealer Envirofriend Inc. The firm was keen to participate in the venture to bolster its environmental credentials in the market. If the launch proves to be successful, the arrangement will be reviewed. If the new product proves to be as popular as the board of Ponda hopes, it may be necessary to enlist additional distribution networks. However, the managing director has assured Envirofriend that the networks will include Envirofriend (so long as the Greencar continues to be sold in American markets).

Ponda had originally wanted to invoice Envirofriend in sterling, but reluctantly agreed to do so in dollars, on the understanding that all the cars would be paid for, whether sold or not. Envirofriend accepted this risk, as the car's remarkable technical properties were expected to result in near certain sales (according to research).

If the car is popular in the initial market, Ponda has decided to penetrate the mass market as quickly and efficiently as possible by setting up a manufacturing subsidiary in the US. It has never been the intention of the firm to distribute its own products; Ponda will continue to do so via third parties.

Franchising is currently being considered as part of the distribution mix.

Ponda's strategic planning department has investigated the possible effect of US Government policy on the Greencar venture. You have been given a rough draft of the research.

(1) It is likely that indigenous firms will be awarded preferential taxation treatment, such as 100% capital allowances, on research-related expenditure.

(2) Government grants for the purchase of the robotic machinery required for the manufacture of the electric cars are to be awarded to firms buying from US companies.

(3) Legislation is currently being planned to increase redundancy payments required in the hi-tech manufacturing industries. It is felt that this may lead to firms investigating ways in which to utilise their spare capacity.

(4) Unfortunately for Ponda the potential competitors are all government suppliers.

(5) The regulatory bodies in the US are likely to bring severe pressure to bear on Ponda – particularly with regard to price and environmental performance indicators.

(6) The Government has already decided to buy a large number of electric motorbikes for the state education sector, to provide free transport to children from low income families (as an alternative to public transport). The scheme has proved extremely popular in rural areas. The US supplier has profited considerably from the deal, and is considering diversification into related areas.

(7) Import tariffs are to be levied on all 'environmental' goods manufactured by firms which are not resident in the US.
(8) A network of 'recharging points' is to be set up by the US Government, designed for use by all types of 'electric vehicle.' The government is keen to set up the system as quickly as possible, and is therefore planning to develop a system ensuring compatibility with the first firm offering marketable vehicles.
(9) Output tax is to be levied on petrol, but not on electricity obtained from the 'recharging network.'

Apart from political research Ponda has also commissioned research into the environmental movement in the US. There is a board consensus in favour of electrically-powered vehicles, particularly in urban areas. The main lobby against such products argues that the switch from petrol engines to electrical engines merely transfers the pollution problem from the car to the power station. Ponda is finding it difficult to judge which lobby will win the greater support in the long term.

The technology associated with the Greencar is far superior to that developed by other manufacturers to date: the electric car is no longer the poor relation of the petrol car.

In fact, the Greencar has consistently outperformed its 'petrol using' rivals in respect of:

- Miles per $ of 'fuel'
- Top speeds
- Quietness inside the car
- Reliability of the engine
- Servicing frequency required
- Environmental pollution

Ponda is concerned about the expected speed of competitive reaction if the car causes the revolution in the industry which its performance specification would merit. Unfortunately, a patent could not be obtained for the design, so Ponda is expecting its rivals to introduce 'copycat' versions a year or so after the initial launch. One strategy, currently under serious consideration, would be to sell the manufacturing technology to allow production of the Greencar under licence. In any case the directors are sure of the need to establish a strong brand image as quickly as possible, so that the generic product – the electric car – becomes strongly identified in the mind of the consumer with the brand name Greencar, rather as the generic product 'vacuum cleaner' is strongly associated with the brand 'Hoover.'

Requirement

Write a memorandum to the managing director of Ponda concerning the strategic aspects of the Greencar project.

Your memorandum should include the following:

- US Government policy
- A discussion on whether the manufacturing plant should be constructed by Ponda or acquired or whether manufacturing should be licensed
- Consideration of the appropriateness of franchising as a distribution strategy for the Greencar

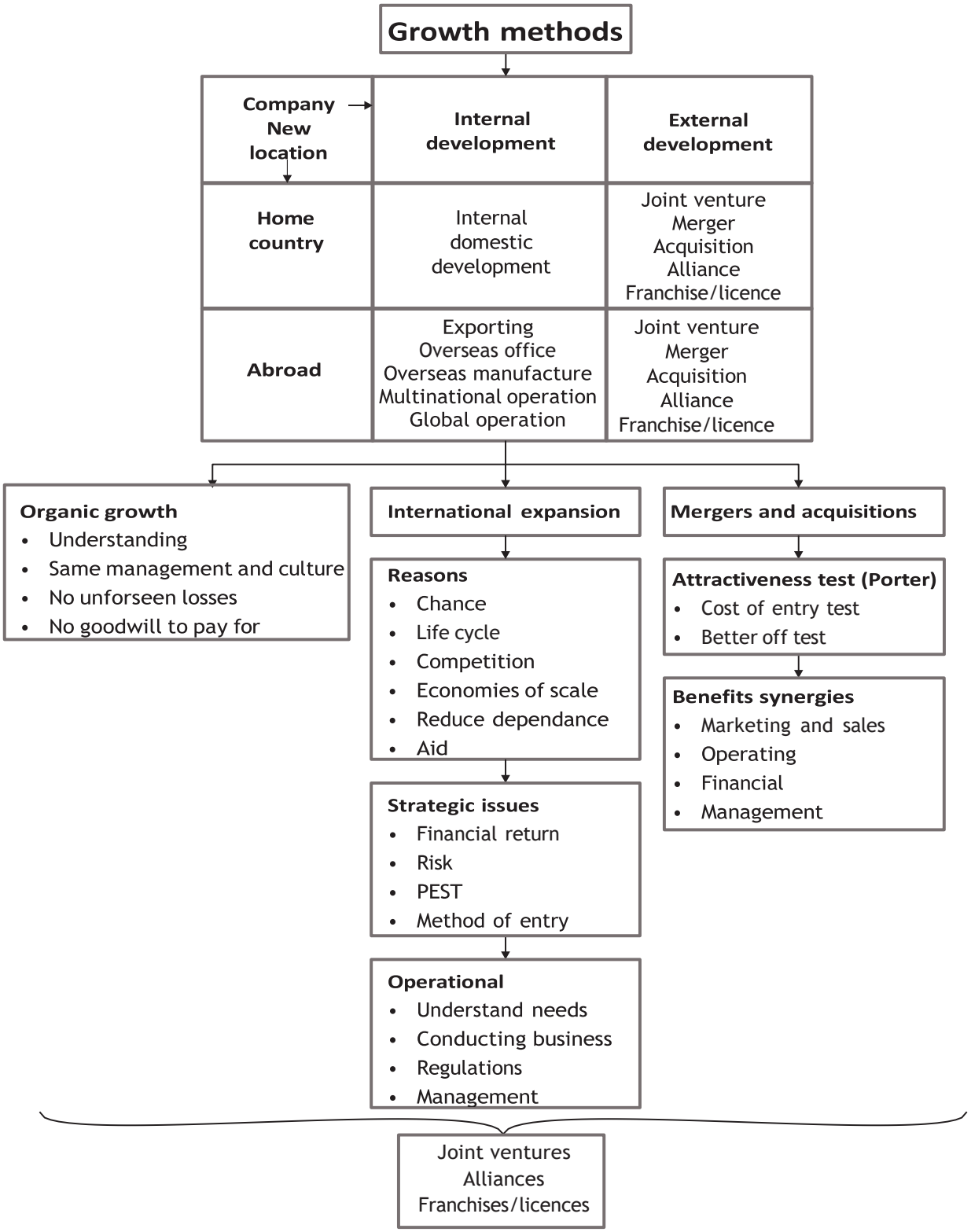
See **Answer** at the end of this chapter.



Professional skills focus: Concluding, recommending and communicating

Ensure that you produce your exam answer in the format requested. You may be asked to write a report or a briefing note, amongst others, but all answers should be presented professionally using a clear and concise style. It is important to avoid 'sitting on the fence' in your conclusion and strong candidates will state a clear recommendation based on the analysis in their report.

Summary


Growth methods
Company New location
Internal development
External development
Home country
Internal domestic development
Joint venture
Merger
Acquisition
Alliance
Franchise/licence
Abroad
Exporting
Overseas office
Overseas manufacture
Multinational operation
Global operation
Joint venture
Merger
Acquisition
Alliance
Franchise/licence
Organic growth
Understanding
Same management and culture
No unforseen losses
No goodwill to pay for
International expansion
Reasons
Chance
Life cycle
Competition
Economies of scale
Reduce dependance
Aid
Strategic issues
Financial return
Risk
PEST
Method of entry
Operational
Understand needs
Conducting business
Regulations
Management
Mergers and acquisitions
Attractiveness test (Porter)
Cost of entry test
Better off test
Benefits synergies
Marketing and sales
Operating
Financial
Management
Joint ventures
Alliances
Franchises/licences

Further question practice

1 Knowledge diagnostic

Before you move on to question practice, confirm you are able to answer the following questions having studied this chapter. If not, you are advised to revisit the relevant learning from the topic indicated.

Confirm your learning	
(1)	Do you know the possible expansion methods summarised by Lynch? (Topic 1)
(2)	What are the main benefits and drawbacks of organic growth? (Topic 2)
(3)	What are the reasons for expanding internationally? (Topic 3)
(4)	What are potential reasons behind the failure of acquisitions? (Topic 4)
(5)	Do you know the difference between joint ventures, alliances and franchising? (Topic 5)

2 Question practice

Aim to complete all self-test questions at the end of this chapter. The following self-test questions are particularly helpful to further topic understanding and guide skills application before you proceed to the next chapter.

Question	Learning benefit from attempting this question
2	This short question tests your technical knowledge gained from the chapter, regarding mergers and acquisitions.
4	This is another technical knowledge test, assessing whether you can list the disadvantages of a joint venture.

Once you have completed these self-test questions, it is beneficial to attempt the following questions from the Question Bank for this module. These questions have been selected to introduce exam style scenarios that will help you improve your knowledge application and professional skills development before you start the next chapter.

Question	Learning benefit from attempting this question
Balad Ltd (Sept 2018)	Part 3 of the question requires a discussion of the advantages and disadvantages of a joint venture for the company in the scenario. It is a good example of combining technical knowledge with application.
Ten Ten Burgers Inc (June 2018)	Part 3 of the question has a practical and applied element, asking you to consider the relevant factors in choosing on one of two expansion strategies; franchising or organic growth.

Refer back to the learning in this chapter for any questions which you did not answer correctly or where the suggested solution has not provided sufficient explanation to answer all your queries. Once you have attempted these questions, you can continue your studies by moving on to the next chapter.

Self-test questions

Answer the following questions.

1 What is the primary method of growth for most organisations?

A Acquisitions
B Organic growth
C Merger
D Franchising

2 Distinguish a merger from an acquisition.

3 Fill in the blanks in the statement below using the following words:

- acquisitions
- cost
- market
- market-share
- resources
- risk

[ ] provide a means of entering a [ ] or building up a [ ], more quickly and at a lower [ ] than would be incurred if the company tried to develop its own [ ] Corporate planners must however consider the level of [ ] involved.

4 Define a joint venture. What are the chief disadvantages of a joint venture for a company?

5 Fill in the blanks.

Particularly important questions in a buyout decision are:

Can the buyout team [ ] to pay for the buyout?

Can the bought-out operation generate [ ] ?

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

Memorandum

To:	The managing director, Ponda plc
From:	F Smith
Date:	Today
Subject:	The strategic aspects of the Greencar project

Government policy

According to research present policy favours the US firms over foreign competition. Import tariffs for non-resident companies in the sector, preferential taxation treatment relating to R&D expenditure, grants to aid the purchase of equipment from US firms, and heavy government expenditure on Ponda's competitors all mean that Ponda has an inherent disadvantage compared to the US competition. The extent of the government support is well illustrated by the government contract to buy electric motorbikes for the state sector. This has resulted in the supplier making large profits, and thus being able to finance new developments internally, which may include the production of a Greencar-like product.

The planned legislation concerning increasing the redundancy payments in the high-technology sector effectively acts as an exit barrier, as firms will incur high costs if they attempt to close down manufacturing plant. Therefore, excess capacity in the industry is to be expected. Some firms may use this spare capacity to produce rival products to the Greencar. Alternatively, Ponda can use this resource to 'contract out' its productions.

The output tax treatment of electricity from the recharging network (at 0%) gives the electric car product an advantage over its petrol-engine rivals. This is an advantage for the entire industry, but is most significant for Ponda, which, as the pioneering manufacturer, will be the first to realise the benefits. Similarly, the government-funded recharging network will prove advantageous to Ponda, particularly as the technical capability (with Ponda) will inevitably force Ponda's rivals to alter their designs.

The manufacturing plant

The manufacturing facility, whether acquired or constructed, will only be required if the initial sales go well. This is expected because of the extraordinary technical specification of the Greencar.

Therefore, given a successful result in the test market, the company should proceed with a 'fast growth' strategy to capture the mass market before the competition has a chance to enter the market. Ponda could achieve this by:

- Buying an existing manufacturing firm (bearing in mind that the process of adoption may be onerous and lengthy)
- Allowing other firms to manufacture the Greencar under licence (essentially a joint venture)
- Setting up its own plant

The case in favour of Ponda setting up its own plant is weak: to build up the necessary capacity quickly by internal means could be difficult. However, the overwhelming case against Ponda setting up its own plant is found in the US Government policy of favouring US firms. Because of

increased redundancy payment legislation, many suitable firms will have spare capacity that they are anxious to use. It should be possible to enter into a very favourable joint venture with a US firm. (It is assumed that if a US firm were taken over by Ponda, it would lose its favoured position as a US firm.)

Franchising as a possible distribution strategy

Assuming that the test market is a success, the firm should attempt to reach the mass market as soon as possible. Assuming that Ponda has arranged sufficient production capacity to achieve this, distribution networks must be set up quickly. The main choice appears to be between established dealers and franchisees (since Ponda has expressed an intention not to set up its own networks).

There is an additional constraint: under either scenario Envirofriend must be included in the distribution mix.

Franchising is an arrangement whereby a company distributes products – usually of a 'famous' brand type (as Greencar is expected to become). For the privilege the franchisee pays a capital sum (at the start of the arrangement) plus a stream of royalty payments, dependent on the success of the franchise.

Franchising would be a suitable component of distribution for Ponda for the following reasons.

- The arrangement typically involves tight control over the marketing elements of the business such as brand name, unique selling points to employ, in-store promotions, etc. A unified marketing effort across the outlets is essential for the firm to penetrate the market, and this is arguably easiest to achieve by franchising.
- The franchisees are strongly motivated to sell a high volume of cars (since their success depends on it).
- Ponda does not have to invest in any capital (this is provided by the franchisee), thus the strategy is low risk (for Ponda) if the project fails no capital has been wasted, whereas if the project succeeds the franchisees suffer (and Ponda is not affected).
- Franchise systems have been associated with a high rate of growth – ideal for Ponda.

The disadvantages include the following.

- Selection of the franchisees may be difficult (given the new type of products).
- Since the Greencar has no track record, setting the balance between the initial fixed capital and the variable royalty payments could be difficult.
- The time spent in training the franchisees could result in a drain on resources.

Nevertheless, these disadvantages are not insurmountable, and therefore the franchising strategy is to be recommended.

Answers to Self-test questions

1 Correct answer(s):

B Organic growth

Most organisations expand via organic growth. This involves the use of internal resources.

2 A merger is the joining of two separate companies to form a single company. An acquisition is the purchase of a controlling interest in another company.

3 Acquisitions provide a means of entering a market or building up a market-share, more quickly and at a lower cost than would be incurred if the company tried to develop its own resources. Corporate planners must however consider the level of risk involved.

4 A joint venture is an arrangement where two firms (or more) join forces for manufacturing, financial and marketing purposes and each has a share in both the equity and the management of the business. The major disadvantage of joint ventures is that there can be conflicts of interest.

5 Particularly important questions in a buyout decision are:

Can the buyout team afford to pay for the buyout?

Can the bought-out operation generate earnings to pay the interest on the borrowings ?

Chapter 12

Evaluating and analysing performance and strategies

Introduction

Learning outcomes

Syllabus links

Examination context

Chapter study guidance

Learning topics

1 Choice of strategic options

2 Strategic control

3 Measuring performance

4 Divisional performance measurement

5 Balanced scorecard

6 Measuring performance in NFP organisations

7 Types of data

8 The data related requirements in the exam

9 Developing an approach to data analysis questions

10 Showing supporting calculations

Summary

Further question practice

Self-test questions

Answers to Interactive questions

Answers to Self-test questions



Introduction

Learning outcomes

- Explain and analyse an organisation's current position and performance using financial and non- financial data from internal and external sources, presented in different formats including visualisations.
- Explain and evaluate different types of data distributions and data trends using appropriate statistical tools, data analysis and spreadsheets, including consideration of cause and effect and the implications for business risk.
- Evaluate and interpret data provided, including identifying and explaining the causes and effects of different types of data bias, and applying an appropriate degree of professional scepticism.
- Assimilate, structure and analyse transactions and other granular data provided, using spreadsheets.
- Explain and demonstrate how to capture and analyse qualitative and quantitative data, presented in different formats, to provide relevant information for decision making at an appropriate level within an organisation.
- Explain and demonstrate how an organisation compares and selects from competing strategies in order to maximise the achievement of its key objectives.
- Select and justify a strategy or combination of strategies which will best achieve the organisation's objectives, taking account of known constraints and capabilities, including stakeholder risk preferences and the potential for developing technologies to increase or mitigate risks.
- Explain and demonstrate how management information can be used to evaluate an organisation's proposed strategies, including forecasting financial performance and the risk associated with forecasts.
- Evaluate the value of information and limitations of data, having regard to data variability, data bias and risk, and considering the ethical implications of the selection, capture, analysis and use of data.
- Organise, structure and assimilate historic and estimated data in appropriate ways, using available statistical tools, data analysis and spreadsheets, to support business decisions.
- Explain and demonstrate how qualitative and quantitative data, including forecasts, budgets and other management information presented in a range of formats, can be analysed to monitor the performance of an organisation's projects, divisions and other strategic units.
- Explain and analyse how an organisation can implement and monitor sustainability and ESG policies.
- Explain and demonstrate how data can be selected, structured, assimilated and analysed using spreadsheets to provide management with information to monitor or modify strategic or operating decisions having regard to data variability, data bias and risk, and applying an appropriate degree of professional scepticism.
- Explain and evaluate how an organisation can select, capture, analyse and use data, including big data, to provide management with information that enables it to implement, monitor and modify a strategy and to create or sustain competitive advantage.

Syllabus links

The techniques covered in this chapter were introduced in your Business, Technology and Finance and Management Information studies at Certificate Stage.

This chapter provides guidance on how to tackle the data related requirements in the exam. Specific consideration is given to answering the data analysis requirement and requirements where supporting calculations need to be shown.

As we discussed in 'Introduction to the Business Strategy and Technology exam,' question scenarios will feature both quantitative (often financial) and qualitative (often non-financial) data. It is important to recognise that both types of data are intended to provide you with different sources when considering the performance of, and issues affecting, organisations featured in questions.

Qualitative data in questions may take the form of customer feedback or comments made on social media about an organisation's products or services. Such data should be used in conjunction with any quantitative data to explain why an organisation has performed as it has.

The analytical skills covered at Professional level will help you to develop the necessary skills in preparation for Advanced level.

Examination context

A key point to note is that when you are presented with data in a question you are expected to use it in forming your answer. It is particularly important that you draw upon this data even if you are not presented with a separate data related requirement.

In the chapter, 'Introduction to the Business Strategy and Technology exam,' we highlighted that questions in the exam requiring the use of data commonly fall into one of four types:

- A data analysis requirement, which requires candidates to analyse a range of data in various exhibits, produce some relevant calculations and provide a commentary to go with their numerical analysis

Requirements such as this may ask candidates to evaluate an entity's performance over time, analyse its product portfolio and competitive position or evaluate whether a strategy has performed successfully against budget.

- Performing a specific calculation as specified by the question requirement, such as a break-even or profit forecast working
- A 'show supporting calculations' requirement where you are required to support your narrative answer with supporting calculations
- A requirement asking candidates to assimilate and analyse granular data provided in a pre- populated spreadsheet

Such data is intended to replicate the type of raw, unprocessed data that could be downloaded from a company's management information system. Candidates will need to use spreadsheet functionality to analyse the data to provide meaningful analysis and draw appropriate conclusions. Examinable spreadsheet functionality and example questions can be seen in the chapter Spreadsheet functionality.

We also flagged the importance of using data provided in the scenario to inform your answers, even where there is no specific requirement involving numbers. In such instances be sure to use spreadsheet software and appropriate spreadsheet functionality to perform calculations quickly and accurately.

In the chapter, 'Risk management,' we focused predominantly on the techniques that are most likely to be required in questions where candidates are asked to perform a specific calculation. In this chapter, greater consideration is given to attempting supporting calculations and data analysis style requirements, using spreadsheet functionality to perform calculations where appropriate (see chapter Spreadsheet functionality for more information).

While you need a good understanding of the different types of data related requirement and an ability to analyse the data using a combination of traditional calculations and spreadsheet functionality, you need to bear in mind the fact that the examiner is not just expecting you to display your numerical skills. Many of the data related requirements that you will see in exam require you to use the results from the calculations that you have performed in order to inform business decisions and provide appropriate advice and recommendations to the management of the organisations featured in the question scenario.

As such it is important that any data you use in your answers is linked to the wider strategy, position or issues facing the organisation featured in the scenario. Numbers will form part of your answer, but it is important to recognise that the numbers are only a peg on which to hang discussion: you must demonstrate an understanding of the story behind the numbers. Being selective about what numerical analysis to undertake and then explaining your numerical analysis is thus an important element of all related data requirements.

The examiners have repeatedly highlighted that those candidates who struggle, particularly with the data analysis requirement, tend to make the same mistakes. These include only calculating the numbers provided with little or no attempt to interpret the results, solely focusing on the issue of profitability with no consideration of liquidity or investment, failure to adopt a methodical approach (the so-called What-How-Why-When-So What analysis) when interpreting the results of any data analysis (see later in this chapter), failing to present calculations in tables.

It is important to note that data analysis requirements could examine more than your ability to carry out ratio analysis as questions could involve the analysis of forward-looking information in the form of forecasts (as opposed to analysing historic information). You also need to be alert to the fact that not all data provided will necessarily be for a whole year as a question could provide you with quarterly or even monthly trend data to review.

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this chapter.

Topic	Practical significance	Study approach	Exam approach	Interactive questions
1	**Choice of strategic options** Choosing the right strategic option is essential for commercial success and survival of the organisation.	**Approach** Ensure you understand how to evaluate a strategy, taking into consideration the benefits and risks, the financial considerations, and the impact on stakeholders, before concluding on the most appropriate strategy for the organisation. **Stop and think** Having the perfect strategy on paper may not lead to the perfect strategy in	An important skill when evaluating strategies in the exam is the ability to assess them in light of the organisation's current position, mission and stakeholder objectives. Carefully consider each strategy and the benefits and risks it delivers for the company in question.	

Topic	Practical significance	Study approach	Exam approach	Interactive questions
		practice. Management must be able to separate good from bad strategic ideas. What strategies can you think of that went wrong?		
2	**Strategic control** Strategic control indicates the need for a review of strategic performance by comparing plans with actual performance. An annual budgeting system ensures an organisation is regularly conducting such a review.	**Approach** Understand the stages involved in setting up a system of strategic control. Also be aware of the benefits and behavioural impacts of budgets. **Stop and think** Consider an organisation you are familiar with. How is performance controlled and monitored?	Questions may ask you to prepare budgets and forecasts using the data provided; evaluate the differences between budgeted and actual results; or use data to understand the impact of strategic decisions.	
3	**Measuring performance** Effective performance measurement helps a business to understand and communicate whether it has achieved its mission and objectives. The approach to performance measurement varies between the private and public sectors.	**Approach** Remind yourself of key performance measures seen in earlier studies and other papers. Review the three E approach adopted by non-profit organisations and remind yourself of CSFs and KPIs seen in previous chapters. **Stop and think** Can you give examples of performance measures for a school, utilising the three Es approach?	Questions on performance measurement enable you to use your finance studies in a different context. Focus on control as well as the numbers themselves. You are expected to demonstrate your ability to use the results from your calculations to draw conclusions.	
4	**Divisional performance measurement** Divisionalised organisations may	**Approach** Revise the divisional performance measures of ROI and RI. Ensure you	Be prepared to perform calculations in the exam but ensure you comment on what	**IQ1: ROI** Practise the ROI calculations and comment on any problems they

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	choose to decentralise decision making to each division. In this case it is important to track the performance of each division using a range of comparable performance measures.	understand the benefits and problems of each measure. **Stop and think** Why should divisional managers only be held accountable for factors within their control?	the results of your calculations are telling you.	reveal. **IQ2: Asset disposal** Review calculations that have already been performed to help make a decision about asset disposal. **IQ3: ROI or RI?** Perform basic calculations to answer the questions. **IQ4: Comparing divisions** Use RI and ROI to assess and compare divisional performance. Explain your calculations.
5	**Balanced scorecard** Reliance on financial performance metrics ignores non-financial strategic goals. The balanced scorecard attempts to integrate financial and non- financial performance measures in order to link strategic and operational goals.	**Approach** Review the four perspectives of the balanced scorecard, looking at the examples of core outcome measures given for each perspective. Understand the problems and recent developments in the balanced scorecard. **Stop and think** Consider the firm you work for. What non-financial goals and objectives do they have? How are these currently measured?	It is important to recognise that the data featured in questions may take the form of both qualitative and quantitative data. In such instances a balanced scorecard approach enables a holistic assessment of performance.	
6	**Measuring performance in NFP organisations**	**Approach** Read quickly through this short	Exam questions are just as likely to focus on NFPs as they are	

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	Not-for-profit organisations are not motivated by traditional financial performance measures, such as profit. Also, NFPs tend to have a more diverse set of stakeholders to manage. Both of these factors make performance measurement more challenging.	section. Revisit the 3Es approach if required. **Stop and think** Consider a charity with which you are familiar. Who are the key stakeholders and what are their needs?	on profit seeking organisations. Ensure you understand the challenges of managing different stakeholders and of measuring performance in an NFP organisation.	
7	**Types of data** An organisation has access to many sources of data. In order to understand and improve organisational performance this data must be processed into meaningful information that can be analysed and compared. To avoid being overwhelmed by the large quantities of data available, an organisation should focus on assessing the achievement of its Critical Success Factors.	**Approach** Read quickly through the definitions and examples of quantitative and qualitative data. Spend time understanding how data will be incorporated in the exam. **Stop and think** Why should you apply judgement and professional scepticism when analysing data?	Be prepared to use quantitative and qualitative information in the exam to understand an organisation's performance. Data may be presented in different formats including graphs and charts.	
8	**The data related requirements in the exam** An essential skill, as an accountant and in the exam, concerns your ability to analyse and interpret both quantitative and qualitative data	**Approach** Good exam performance means using data when and where it is relevant to support your discussion. Read through this topic to understand the skills needed and the common	Common weaknesses highlighted by the examiner include; calculating the numbers with little or no attempt to interpret the results; only considering profitability and ignoring liquidity or	**IQ5: Staff bonus** Data can be presented in many ways in the exam and spreadsheet functionality enables you to analyse this data. In this question, practise using

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	(financial and non-financial data) and relate them to the organisation's strategy.	weaknesses in candidates' scripts, observed by the examiners. **Stop and think** How would you best present data and information that is to be used by the Board?	investment; and failing to present calculations professionally using tables.	RANK to sort the data and IF to assess which employees should receive a bonus.
9	**Developing an approach to data analysis questions** Candidates often perform poorly in the exam when it comes to data analysis. To improve your chances, you should develop an approach to tackling numerical and data- based questions.	**Approach** Read through the five stage approach and see how it has been applied to the worked example. **Stop and think** Review a business news article. Have they used data to explain implications for the business and to draw conclusions?	Be selective in the numerical analysis you undertake. Numbers should be well presented and act as a peg on which to hang your discussion. There is no value in performing a calculation that you do not discuss and draw conclusions from. Use spreadsheet functionality to perform calculations quickly and accurately.	
10	**Showing supporting calculations** One of the data related requirements in the exam will require you to prepare an answer which is complemented by supporting calculations.	**Approach** Read carefully through the worked examples to understand how to deal with supporting calculations in the exam. **Stop and think** Your exam answers should be presented in a professional manner. What does this mean to you?	It is good practice in the exam to highlight other factors or pieces of information that may be useful in assessing a proposal or strategy. You should also use spreadsheet functionality where appropriate to help you produce your calculations.	

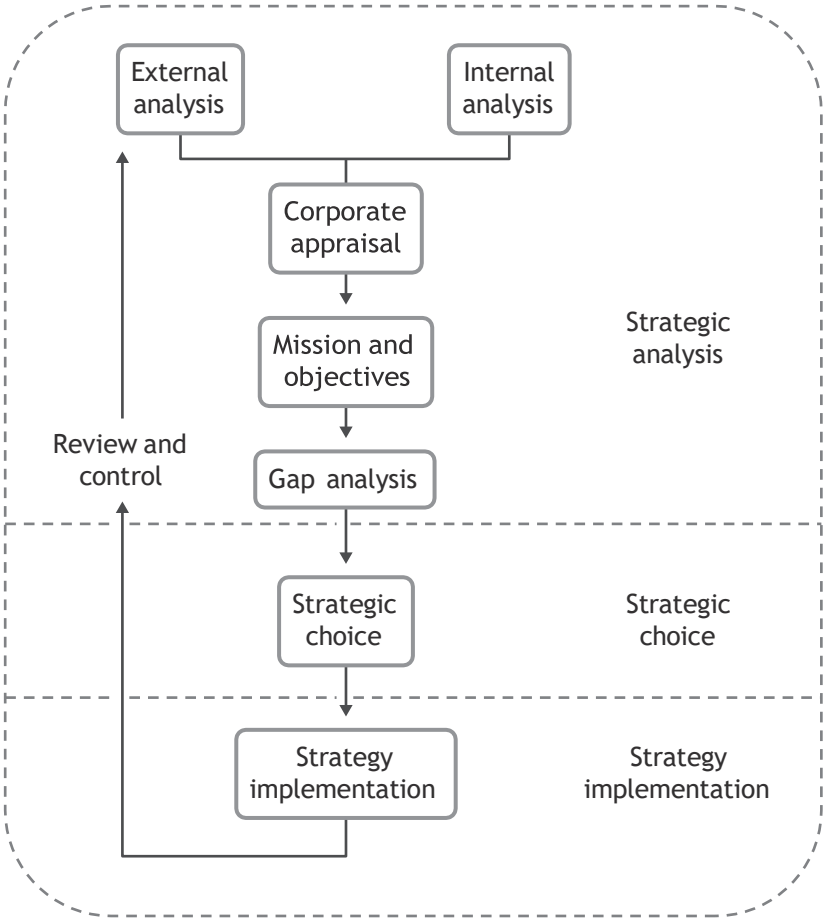
1 Choice of strategic options

Section overview

- Strategic choices are evaluated to determine whether they meet the needs of the organisation and its current situation, by delivering benefits and mitigating risks. It should be determined whether the strategy is financially and technically feasible and whether stakeholders are in support of the strategy.
- Using this criteria, a strategy can be selected and then be broken down into functional strategies to be implemented.

1.1 Recap of the rational approach

The rational model was described in the chapter Strategy and organisations. Figure 12.1: Diagram: The rational planning model



If the business is facing a gap, then devising strategies to fill the gap is a two-step process:

(a) Option generation – creative thinking to generate options

(b) Evaluation and selection – assessment of options against goals of the business and commitment to some and not to others

CHAPTER 12

In the chapters, 'Strategic options' and 'Methods of development,' we considered how a business might develop strategic options, using models such as Ansoff.

Evaluation of these options to decide which of the options to implement occurs as the second phase of **strategic choice**.

In the next section we will look at how the business can use strategic control systems to monitor the success of the strategy and its impact on performance, once implementation has taken place.

1.2 Evaluating strategies

As we highlighted in the chapter, 'Introduction to the Business Strategy and Technology exam,' you need to be prepared for the fact that some questions in the Business Strategy and Technology exam will outline a proposed strategy or series of strategies that you are then required to evaluate. It is important recognise however, that this may not always the case as at least one question requirement in the exam is likely to be 'open ended.' This type of situation will require you to 'think on your feet.' Requirements such as this may require you to identify options for resolving a problem facing the entity featured in the scenario, or they may require you to suggest and evaluate a number of strategies to capitalise on an opportunity.

Strategies can be evaluated in a number of different ways, and these include:

- Considering benefits and risks that the strategy brings
- Considering the strategic and financial considerations associated with the strategy
- Discussing the quantitative and non-quantitative issues associated with the strategy
- Considering the viewpoints held by different stakeholder groups in relation to a proposed strategy

1.3 *Strategy selection and implementation*

After evaluating the various strategic options, the business must make a decision as to which strategy (or strategies) to implement. It may be the case that some of the strategies are mutually exclusive.
Others may be implemented in parallel. To implement the chosen strategy, it needs to be broken down into the relevant functional strategies (marketing, operations, human resources, finance etc). This is considered further in the chapter Business planning and functional strategies.

1.3.1 *The role of the finance function as a business partner*

Today, the role of the Chartered Accountant has evolved from one of simply reporting historic financial information, to one of providing commercial insights which can help organisations in achieving their objectives. This is evident as the number of finance professionals working as 'business partners' providing support and advice to departments and business units at an operational level has increased significantly.



Definition

The business partner model: *Has resulted in finance professionals adopting a more commercially- focused, action-orientated approach. Increasingly, finance professionals are being embedded within operational departments and business units so that they can provide financial insights in a more integrated manner than if they were based in the finance department alone.*

The experience and specialist expertise offered by Chartered Accountants makes them well-placed to assist operational teams in the evaluation of quantitative and qualitative data, and to

add value to management by helping them to make more informed strategic and operational decisions, for example, about whether or not to undertaken potential new projects or investments.

The role of the finance function as a business partner is considered in greater detail in the chapter Business planning and functional strategies.

2 Strategic control



Section overview

- A business's planning and control cycle is designed to ensure its mission and objectives are met by setting plans, measuring actual performance against plans, and taking control action.
- Control at a strategic level means asking the question: 'is the organisation on target to meet its overall objectives, and is control action needed to turn it around?'
- The existence of an annual budgeting process compels planning and enables the establishment of a system of control by comparing budgeted and actual results.

2.1 Strategic control systems

Steps in setting up formal systems of strategic control:

Step 1 **Strategy review**: review the progress of strategy.

Step 2 Identify milestones of performance (**strategic objectives**), both quantitative (eg, market share) and qualitative (eg, quality, innovation, customer satisfaction):

- Milestones are identified after critical success factors have been outlined.
- Milestones are short-term steps towards long-term goals.
- Milestones enable managers to monitor actions (eg, whether a new product has been launched) and results (eg, the success of the launch).

Step 3 **Set target achievement levels**. These need not be exclusively quantitative:

- Targets must be reasonably precise.
- Targets should suggest strategies and tactics.
- Competitive benchmarks are targets set relative to the competition.

Step 4 **Formal** monitoring of the strategic process. Reporting is less frequent than for financial reporting.

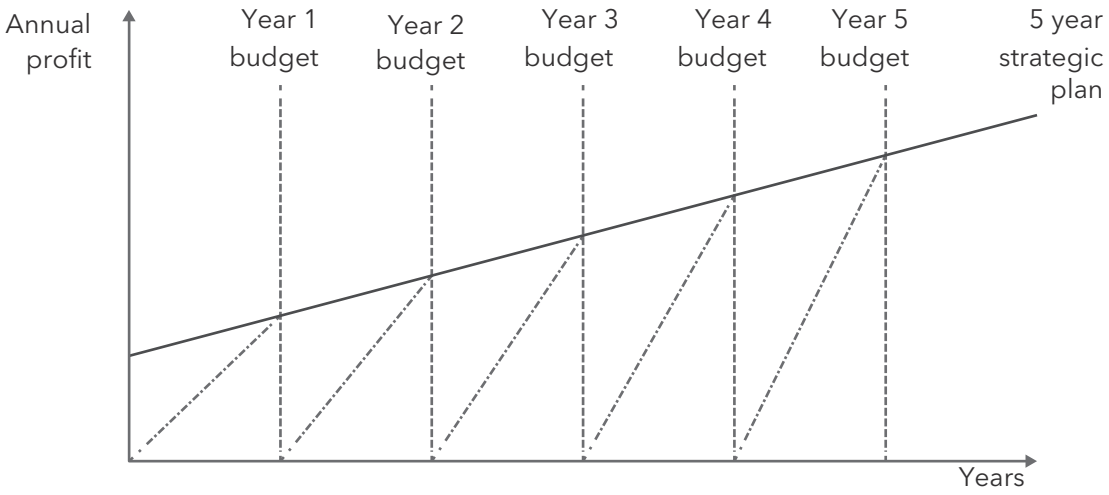
Step 5 **Reward**: for most systems, there is little relationship between the achievement of strategic objectives and the reward system, although some companies are beginning to use measures of strategic performance as part of the annual bonus calculations.

2.2 Budgets and budgetary control

2.2.1 Short-term planning and budget preparation

A budget is a plan expressed in financial terms. Short term plans attempt to provide short-term targets within the framework of longer-term strategic plans. This is generally done in the form of a budget.

Figure 12.2: An example of a five year budget



The diagram shows that the five-year strategic plan is to grow annual profits. This gives annual profit milestones and these are taken as the starting point for each annual budget.

2.2.2 Budgets, long term plans and corporate objectives

Budgets are an important planning tool for the organisation and are directly related to the mission, objectives and business strategy of an organisation.

A budget is a business plan for the short term, usually one year. It is likely to be expressed in financial terms and its role is to convert the strategic plans into specific targets.

It therefore fits into the strategic planning process as follows:

- The mission sets the overall direction.
- The strategic objectives illustrate how the mission will be achieved.
- The strategic plans show how the objectives will be pursued.
- The budgets represent the short-term plans and targets necessary to fulfil the strategic objectives.

These budgets will then have to be **controlled** to ensure the planned events actually occur and is therefore a key element in assessing the effectiveness of the business strategy. This is as much a part of the budgeting process as actually setting the budget.

A budget is a **plan** not a forecast. You plan to meet the targets in the budget, a forecast is a prediction of future position. Forecasts are helpful to budget setters and planners.

A **periodic budget** is a budget covering a period (eg, a year). A **continual**, or **rolling budget**, is continually updated.
Different budgets are prepared for each specific aspect of the business, and the contents of the individual budgets are summarised in **master budgets**. The contents of each of the individual budgets affect, and are influenced by, the contents of the others; they are linked together.

The sales budget is usually the first budget prepared as the level of sales determines the overall level of activity for the period and therefore most of the cost budgets.

Planning (including budgeting) is the responsibility of managers, not just accountants – although management accountants help by providing relevant information to managers and contributing to decision making.

2.2.3 Benefits of budgets

There are five main benefits of budgets:

(a) **Promotes forward thinking.** Potential problems are identified early, therefore giving mangers time enough to consider the best way to overcome that problem.

(b) **Helps to co-ordinate the various aspects of the organisation.** The activities of the various departments and sections of an organisation must be linked so that the activities complement each other.

(c) **Motivates performance.** Having a defined target can motivate managers and staff in their performance. Managers should be able to relate their own role back to the organisational objectives, seeing as the budgets are based on these objectives.

(d) **Provides a basis for a system of control.** Budgets provide a yardstick for measuring performance by comparing actual against planned performance.

(e) **Provides a system of authorisation.** Allows managers to spend up to a certain limit. Activities are allocated a fixed amount of funds at the discretion of senior management, thereby providing the authority to spend.

These five uses may however conflict with each other. For example, a budget used as a system of authorisation may motivate a manager to spend to the limit even though this is wasteful. This is particularly likely where the budget cannot be rolled over into the next period. In addition to this, budgets have some other limitations that could potentially arise:

(a) Employees may be **demotivated** if they believe the budget to be unattainable.

(b) **Slack** may be built in by managers to make the budget more achievable.

(c) Focuses on the **short-term results** rather than the underlying causes.

(d) Unrealistic budgets may cause managers to make decisions that are **detrimental** to the company.

2.2.4 Using budgets for control

Budgets are useful for exercising **control** over an organisation as they provide a yardstick against which performance can be assessed. This means finding out where and why things did not go to plan, then seeking ways to put them right for the future.

If the budget is found to be unachievable, it may need to be revised. Only realistic budgets can form the basis for control, and therefore they should be adaptable.

Budgets enable managers to manage by exception, that is focus on areas where things are not going to plan (ie, the exceptions). This is done by comparing the actual performance to the budgets to identify the **variances.** We will look at variances later in this chapter.

2.2.5 Making budgetary control effective

Successful budgetary control systems tend to share the same common features.

- **Senior management** take the system seriously. They pay attention to, and base decisions on, the monthly variance report. This attitude cascades down through the organisation.
- **Accountability.** There are **clear responsibilities** stating which manager is responsible for each business area.
- **Targets** are **challenging but achievable**. Targets set too high, or too low, would have a de-motivating effect.
- **Established data collection, analysis and reporting routines** which involve looking at actual versus budgeted results to calculate and report variances. This should be done automatically on a monthly basis.
- **Targeted reporting**. Managers receive specific rather than general purpose reports so that they do not have to wade through information to find the relevant sections.

- **Short reporting periods**, usually a month, so that things can't go too wrong before they are picked up.
- **Timely reporting.** Variance reports should be provided to managers as soon as possible after the end of the reporting period. This is so they can take action to prevent the same problems occurring in the next reporting period.
- **Provokes action.** Simply reporting variances does not cause change. Managers have to act on the report to create change.

2.2.6 Behavioural aspects of budgets

Budgets can be very effective in motivating managers. In particular, budgets have been found to:

- Improve performance
- Be most effective when they are demanding, yet achievable

  However, if they are unrealistically demanding, performance may deteriorate.
- Are most effective when the managers have participated in the setting of their own targets

2.2.7 Informal control

Many companies do not define explicit strategic objectives or milestones that are regularly and formally monitored as part of the ongoing management control process:

- Choosing one objective (eg, market share) might encourage managers to ignore or downgrade others (eg, profitability), or lead managers to ignore wider issues.
- Informality promotes flexibility.
- Openness of communication is necessary.
- Finite objectives overlook nuances especially in human resource management. In other words, an objective like 'employee commitment' is necessary for success, but hard to measure quantitatively.

Informal control does not always work because it enables managers to skate over important strategic issues and choices.

2.3 Budgets and forecasts in the Business Strategy and Technology exam

Budgets and forecasts could easily feature in Business Strategy and Technology questions. These types of question can take many forms, and may require you to the use budgeted/ forecast data provided in a number of different contexts. As such you may encounter question scenarios which require you to:

- Prepare budgets and forecasts using the data provided
- Evaluate the significance of differences which exist between budgeted and actual results
- Use the data provided to isolate the financial impact of certain decisions or events on performance

The section which follows provides some examples of how the three types of requirement discussed above have been examined in past Business Strategy and Technology exams.

2.3.1 Forecast financial and operating data

Question 2 in the March 2022 exam ('Xoug Ltd') featured a small company involved in the development of a device to aid sleep. The company is now seeking finance to launch the device. Candidates were provided with assumptions for profitability forecasts, including sales volume and pricing assumptions. For 12 marks, candidates were required to calculate the yearly profits and the total profit for the first four years of operation based on these assumptions. In addition, candidates were required to discuss the impact on the original level

of total profit by applying changes to sales volumes in a worst-case and best-case scenario.

The examiner noted that most candidates performed well on the calculations, presenting them in a clear and understandable format using the spreadsheet software. However, despite this, many candidates failed to perform well on the discussion element of the question simply repeating in words the results of the calculations rather than offering any insight into the figures and the relationships between them. It is important to remember that the Business Strategy and Technology exam requires candidates to use their calculations to draw conclusions and make decisions.

2.3.2 Achieving budgeted results

Question 1 in the December 2022 exam ('Blade Ltd') featured a company which produced bottled soft drinks at its UK factory. The exhibit information included financial and operating data including budgeted figures for 20X3 and actual figures for 20X2. Candidates were required to use the data provided to answer a range of questions posed by the CEO in respect of the company performance, such as: why is ROCE expected to reduce in 20X3 when profits are forecast to increase in 20X3 compared to 20X2? The requirement was worth 18 marks in total and there were five queries to respond to.

The examiner noted that, overall, the performance for most candidates was good but some of the five queries revealed gaps in candidates' knowledge. Errors included incorrect calculations as well as discussion that revealed some basic misunderstanding about the impact of the numbers. The Business Strategy and Technology exam requires candidates to be technically proficient in terms of being able to perform basic calculations, but there is also a requirement to be able to interpret the meaning of calculations. Failure to do this will undermine the quality of the candidate's answer and lose marks.

2.3.3 Quantifying the financial impact of events on performance

Question 1 in the September 2016 exam ('Guayaporto Ltd') featured a company based in South America, which manufactured discs (CDs and DVDs) for businesses in the entertainment and education sectors. The scenario identified, that as a result of technological developments, the market for discs was declining, and the company's financial performance had deteriorated. The question scenario also included a summary statement of profit or loss for 20X5 and 20X6, as well as operating data which included details relating to the number of orders placed, and the total number of discs sold.

The second requirement asked candidates to use the data provided to identify and explain the key causes of the decline in the company's performance, and to analyse the impact that each cause had had on its financial results. Candidates were required, where feasible, to provide calculations to quantify the financial impact and set out any further information that was required to make a fuller assessment. This requirement was worth 18 marks.

As this question illustrates, requirements in the Business Strategy and Technology exam are often multifaceted. In this question, candidates were asked to do a number of different things: analyse of the company's performance; explain the causes behind this deterioration; and to quantify the financial impact of the decline by providing relevant supporting calculations. The answer to this requirement analysed the three main elements that had contributed to the company's falling operating profit: sales; cost of sales; and overheads. Better candidates used the operating data relating to the number of discs being sold to provide supporting calculations to illustrate that the company's production facility was only operating at 46.7% capacity. As such the company was not able to realise economies of scale, which in turn was having a detrimental impact on its profits. The examiner noted that very few candidates answered the entire requirement by quantifying the financial impact of the company's declining performance. As such candidates restricted the number of marks that they could earn.

It is important to note that the examples provided here should only be taken as a guide as to the potential types of requirement that might appear in the exam. This is not a definitive list of all of the ways in which budgets and forecasts might be examined. To ensure that you are prepared for the different ways in which budgets and forecasts might be examined you are strongly advised to ensure that you attempt as many of the questions as possible which cover this topic in the Business Strategy and Technology Question Bank.

2.4 Guidelines for a strategic control system

The characteristics of strategic control systems can be measured on two axes:

- How formal is the process?
- How many milestones are identified for review?

As there is no optimum number of milestones or degree of formality, Goold and Quinn (*Strategic Control*) suggest these guidelines:

Guideline	Comment
Linkages	If there are linkages between businesses in a group, the formality of the process should be low, to avoid co-operation being undermined.
Diversity	If there is a great deal of diversity, it is doubtful whether any overall strategic control system is appropriate, especially if the critical success factors for each business are different.
Criticality	Firms whose strategic stance depends on decisions which can, if they go wrong, destroy the company as a whole (eg, launching a new technology) need strategic control systems which, whether formal or informal, have a large number of milestones so that emerging problems in any area will be easily and quickly detected.
Change	Fashion-goods manufacturers must respond to relatively high levels of environmental turbulence, and have to react quickly. If changes are rapid, a system of low formality and few measures may be appropriate, merely because the control processes must allow decisions to be taken in changed contexts.
Competitive advantage	• Businesses with few sources of competitive advantage – control can easily focus on perhaps market share or quality with high formality. • Businesses with many sources of competitive advantage – success over a wider number of areas is necessary and the firm should not just concentrate on one of them.

2.5 Using CSFs for control

In the chapter, 'Strategic capability,' we considered critical success factors (CSFs) as the areas that are vital if the business is to achieve competitive advantage. CSFs focus management attention on what is important. The advantages of building a control system based around CSFs are:

- The process of identifying CSFs will help alert management to the things that need controlling (and show up the things that are less important).

- The CSFs can be turned into key performance indicators (KPIs) for periodic reporting in the same way as conventional budgetary control might focus on, and report costs against standard costs.
- The CSFs can guide the development of information systems by ensuring that managers receive regular information about the factors that are critical to their business.
- They can be used for benchmarking organisational performance internally and against rivals (benchmarking was also discussed in the chapter Strategic capability).

3 Measuring performance



Section overview

- Effective performance measurement helps a business ascertain whether it has achieved its strategic and operational objectives.
- This section reminds you of some of the key performance measures you will have seen in your previous studies:
  - Profitability, liquidity and gearing
  - Economy, efficiency, effectiveness (the three Es)
  - CSFs and KPIs
- Performance measurement is only effective if there are yardsticks against which to compare.
- The three Es is an approach widely used in the public sector and by NFP organisations to measure value for money.
- Performance measurement could feature in any of the data related requirements in the exam.

3.1 Measuring performance

Actual performance follows on from setting objectives and developing plans and targets.

Performance measurement is a fundamental part of the control process as it allows the business to consider how it has performed in comparison to those plans and to take corrective action where necessary.

Performance can be measured in different ways:

- **Financial performance** can be measured in terms of growth, profitability, liquidity and gearing.
- **Resource use** can be measured in terms of effectiveness, economy and efficiency.
- **Competitive advantage** can be measured by identifying critical success factors (CSFs) and then measuring achievement via appropriate key performance indicators (KPIs).

You should be familiar with some of these approaches from your Certificate level studies.

A performance measure is useful if it is given meaning in relation to comparatives such as budgets or targets, trends over time and benchmarks from within or outside the business.



Professional skills focus: Assimilating and using information

It is easy to get caught up in numerical analysis and lose sight of the key issues as well as wasting valuable exam time. Ensure that you prioritise calculations that focus on key business

issues. If, for example, expenses are static year on year there is little value to be added in performing a calculation to demonstrate this.

3.2 Measures of financial performance

3.2.1 Measures of growth

An organisation's growth may be expressed in a number of ways, for example:

- Sales revenue (a growth in the number of markets served)
- Market share
- Profitability
- Number of goods/services sold
- Number of outlets/sites
- Number of employees
- Number of countries in which the business operates (sales and/or production)

3.2.2 Measures of profitability

A company should be profitable, and there are obvious checks on profitability:

- Whether the company has made a profit or a loss on its ordinary activities
- By how much this year's profit or loss is bigger or smaller than last year's profit or loss

It is probably better to consider separately the profits or losses on exceptional items if there are any. Such gains or losses should not be expected to occur again, unlike profits or losses on normal trading.

Common measures of profitability include **gross profit margin** and **net profit margin** (measured in relation to revenue) and **gross profit mark up** (measured in relation to cost of sales).

Profitability can also be measured in terms of return, via the **return on capital employed**.

3.2.3 Measures of liquidity and gearing

These are used to consider the financial health of the business.

The availability of short-term cash and working capital can be considered using the **current ratio** and ratios such as **inventory turnover** and **receivable/payable days**.

The longer-term financial structure can be assessed by considering **gearing ratio** and **interest cover**.

These measures of performance are covered in more detail in Management Information and Financial Management.

3.3 Economy, efficiency and effectiveness

Organisations need to ensure that the resources they employ are put to optimum use and that they are achieving an effective ratio of outputs to inputs. One way of doing this is to consider the three Es.



Definitions

Economy: A measure of the actual resources used (inputs) eg, cost of labour per employee or the rent per square metre.

Efficiency: A measure of productivity. It considers the relationship between the goods or services produced (outputs) and the resources used to produce them (inputs) eg, output per person or per £ of labour, sales per square metre.

Effectiveness: A measure of the impact achieved and considers outputs in relation to objectives. It looks at the extent to which the processes used by the business deliver the right results, eg, percentage of products delivered without faults, market share.

The three Es can be used to measure performance of support functions eg, when assessing the performance of the IT department, the organisation could consider:

- Economy – the total cost of the IT function
- Efficiency – the amount spent on IT per user
- Effectiveness – the competence of the users (measured for example by a survey or via an online assessment)
- The 3Es are often used as an approach for measuring 'value for money' (VFM) in the public and not-for-profit sectors.

3.4 Critical success factors and KPIs

Once the business has identified its critical success factors CSFs, it must set performance standards to be achieved in these areas. This is done by identifying appropriate KPIs. CSFs represent the areas an organisation needs to focus on to achieve its objectives. KPIs are the indicators that allow managers to monitor whether those objectives are being achieved and are different to specific measures and targets. For example, a CSF may be to grow market share. To track the achievement of this a KPI could be 'achieve growth in revenue for all four product lines.' The specific measure could be 'rate of growth of volume and revenue from year to year' and the targets could be 'at least x% growth in revenue and y% growth in volume for product line 1 in 20X4.'

For example, if high quality is a critical success factor, a KPI might be 'achieve improved quality in all products.' To develop appropriate measures and targets the business needs to ascertain the quality expected by customers. Then specific measures should be set which might include the percentage of products returned; the amount spent on warranties etc , with targets clearly stating an acceptable or anticipated percentage of products returned or amount spent on warranties.

Some examples of KPIs which can be used to measure the effectiveness of different areas of the organisation and which cover both financial and non-financial criteria are outlined here:

Activity	Key performance indicators
Marketing	Sales volume Market share Gross margins
Production	Capacity utilisation Quality standards
Logistics	Capacity utilisation Level of service
New product development	Proportion of revenue from new products (or services) Patents registered
Sales programmes	Contribution by region, salesperson Controllable margin as percentage of sales Number of new accounts Travel costs

Advertising programmes	Awareness levels Attribute ratings Cost levels
Pricing programmes	Price relative to industry average Price elasticity of demand
Management information	Timeliness of reports Accuracy of information

3.5 Strategic performance measures

Desirable features of strategic performance measures are shown below.

Role of measures	Comment
Focus attention on what matters in the long term	For many organisations this might be shareholder wealth.
Identify and communicate drivers of success	Focus on how the organisation generates shareholder value over the long term.
Support organisational learning	Enable the organisation to improve its performance.
Provide a basis for reward	Rewards should be based on strategic issues not just performance in any one year.

Characteristics of strategic performance measures:

- Measurable
- Meaningful
- Defined by the strategy and relevant to it
- Consistently measured
- Re-evaluated regularly
- Acceptable

4 Divisional performance measurement

Section overview

- Return on capital employed and residual income are two methods of measuring divisional performance.
- Divisional performance calculations could be examined in any of the three data related requirements in the exam.

4.1 Return on capital employed (ROCE)

Return on capital employed (ROCE) is also called return on investment (ROI or return on net assets (RONA). This divisional performance measure is calculated as:

$$\frac{\text{Profit for the period} \times 100\%}{\text{Average capital employed during the period}}$$

The economic principle behind this measure, developed by Du Pont in the early 1900s, is that the return derived should be in excess of the cost of capital of the firm to provide a suitable return to investors.

In practice this measure has achieved popularity because:

- It can **lead to a desired group ROCE** ie, if all divisions return a 15% ROCE then, assuming all central costs and assets are charged back to divisions, the group as a whole will make 15%. Improvements in group ROCE can also feed into the EPS of the group and so into the share price.
- It **enables comparisons** to be made between divisions of different sizes for the purposes of identifying where group value is being created or destroyed and also for the identification of high and low performing divisional managers.
- It is **readily understood by management** due to its similarity to an interest rate or other yield on assets.
- It is **cheap to calculate** given that the financial reporting system will be calculating profits and asset values already.

Definition

Return on capital employed (ROCE): Is a divisional performance measure whose principle is that the return (profit) should be in excess of the cost of capital, in order to increase shareholder wealth.

4.2 Residual income (RI)

This measure was developed in 1950s by GE to avoid a dysfunctional consequence of ROCE/ROI, namely that managers who are evaluated and rewarded against ROCE improvements may choose to forego investments which are actually in the investor's interest.

Context example: XYZ Corporation 1

XYZ Corporation has a cost of capital of 10% which is the rate that all new investments are expected to achieve as a minimum (the 'hurdle' rate), and so is the minimum benchmark for divisional performance.

Division A presently has profits of £158 million on assets of £610 million.

A proposed capital investment of £60 million will yield net cash flows of £13 million per year for the next 10 years after which the asset will be worthless.

Will management of Division A undertake the project?

According to the NPV approach to project appraisal, they should:

ie, £13m for 10 years at 10% = £13 × 6.145 = £79.89m NPV = (£79.89m – £60m) £19.89m
But the division's present ROI is £158/£610 = 26% whilst the project has an ROI of only £13/£60 = 22%

It will drag the divisional ROI down to (£158 + £13)/(£610 + £60) = 25.5%. So managers who are evaluated against ROI targets may decide against it.
The board of XYZ Corporation would not know that this valuable project had been foregone. The decision is dysfunctional and so potentially is the use of ROI as a divisional performance measure.

Definition

Residual income (RI): Is a divisional performance measure. It is calculated as: Divisional profit – Net assets of division (required rate)

Context example: XYZ Corporation 2

Returning to the example of XYZ Corporation above:

Present divisional residual income is:

	£m
Profit	158
Cost of assets (£610 × 10%)	(61)
Residual income	97
Divisional residual income with new project	
Profit (£158 + £13)	171
Cost of assets (£610 + £60) × 10%	(67)
New residual income	104

Residual income is theoretically superior to ROI for the reasons shown above. However, it is not widely used because:

- It is conceptually more complex than a simple percentage yield.
- It does not allow easy comparison of divisions of different sizes.
- It requires that a required rate be assessed and this may be different between divisions according to risk.
- It lacks the clear link to the ROCE of the group. Although **in financial theory** the market capitalisation of the firm will be the NPV of all its business units rather than being influenced by one year's ROCE and EPS, the fact remains that, for all its shortcomings, ROCE is still closely monitored by the investment analysts who determine share prices with their buy/sell recommendations.

4.3 Problems of using ROCE/RI

Problems of both ROCE/ROI and RI in the evaluation and control of business divisions are:

- **Short-termist** – being based on annual profit figures both disregard the future earnings of the division. Using BCG terminology from the chapter Strategic capability, a cash cow might present a high ROI and a star a low one, which would be a misleading guide to their true financial value if assessed as the NPV of future earnings.
- **Discourage investment in assets** – to boost ROI or RI, assets with low book values will be used in preference to new assets. This could reduce the prestige of the organisation (eg, shabby fittings) or lead to risk (eg, leaking vessels). It may also lead to inappropriate outsourcing to avoid having the assets required to provide the service on the division's statement of financial position.
- **Lack of strategic control**: unless the corporation is acting merely as a super financial controller (eg, as defined by Goold and Campbell) it will wish to co-ordinate and integrate operations of its divisions to gain group synergies. Financial control measures alone cannot do this.

Interactive question 1: ROI

An asset costs £100,000, has a life of four years, and no scrap value.

It generates annual cash flows of £34,000. Depreciation is calculated on the straight-line basis.

Requirements

1.1 Calculate annual ROI using opening carrying amount.

1.2 Calculate annual ROI using historic cost.

1.3 Comment on any problems identified by these calculations.

See **Answer** at the end of this chapter.

Interactive question 2: Asset disposal

A manager has the following data:

	Year 1	Year 2	Year 3	Year 4
Profit (£'000)	20	15	10	5
Historic cost (£'000)	100	100	100	100
ROI	20%	15%	10%	5%

Manager's target ROI = 12% per annum.

Requirement

When would the manager dispose of the asset and what problem might this cause?

See **Answer** at the end of this chapter.

Interactive question 3: ROI or RI?

A division manager has the following data:
Target ROI 20%
Divisional profit £300,000
Capital employed £1 million

Requirements

3.1 Would the division manager accept a project requiring capital of £100,000 and generating profits of £25,000, if the manager were paid a bonus based on ROI?

3.2 Would the decision change if the manager's pay were based on RI?

See **Answer** at the end of this chapter.

Interactive question 4: Comparing divisions

A company has two divisions. Target ROI = 20%

	Division 1	Division 2
Capital	£1m	£100k
Profits		
Year 1	£200k	£20k
Year 2	£220k	£40k

Requirement

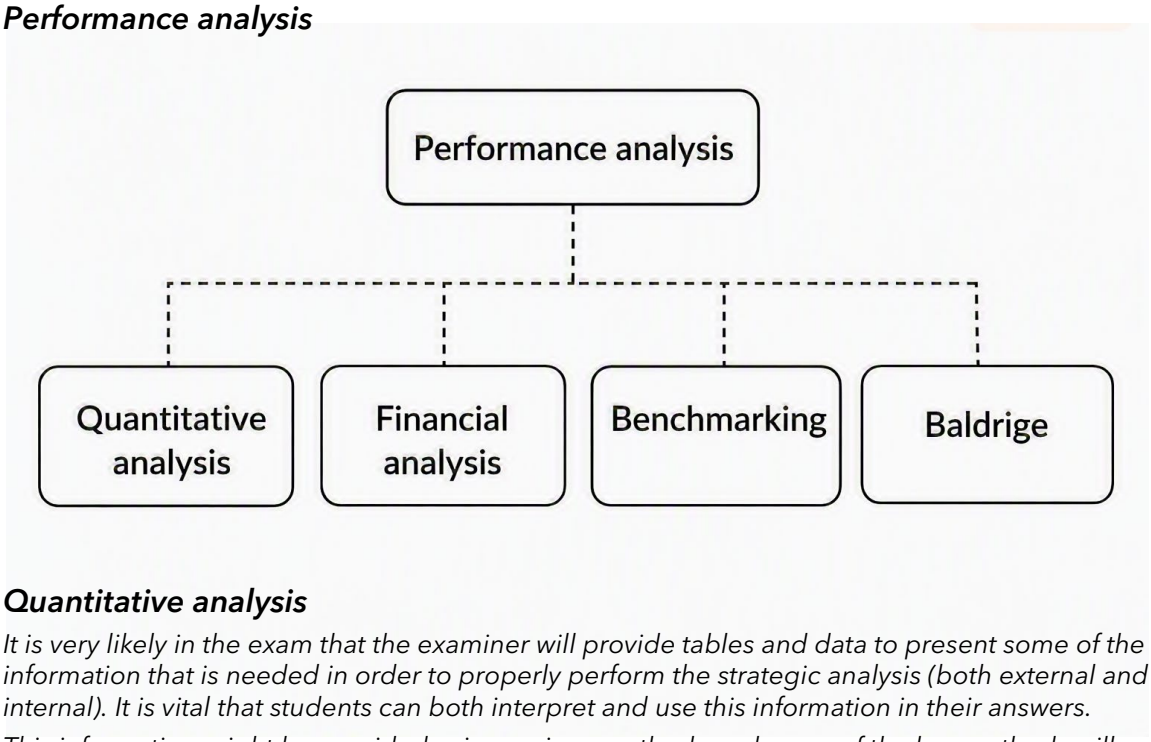
Which division is performing better:

(a) using RI?

(b) using ROI?

See **Answer** at the end of this chapter.

Performance analysis



Quantitative analysis

It is very likely in the exam that the examiner will provide tables and data to present some of the information that is needed in order to properly perform the strategic analysis (both external and internal). It is vital that students can both interpret and use this information in their answers.

This information might be provided using various methods and some of the key methods will be:

Tables of data

To reduce the amount of text in a scenario the examiner will often provide tables of data to provide part of the story. It is important that a student can understand what the table is trying to explain, and that this part of the story is used in the answer to this part of the examination.

Test your understanding 1

The following data is given for sales of cinema tickets in a large country over the last two years.

Company	***20X7 Sales ($bn)***	***20X8 Sales ($bn)***
A	*2.0*	*2.0*
B	*1.8*	*2.0*
C	*2.2*	*2.1*
D	*2.1*	*1.9*
Others	*1.6*	*1.7*
Total industry sales	***9.7***	***9.7***

Required:
Analyse the competitive nature of the cinema industry in this country.

Professional skill:
Illustrate analysis skills in investigating the competitive nature of the cinema industry.

Financial statements

The examiner might provide sets of financial statements, and a student must use these to pull out the key messages and issues. There will be some important technique points to this:

- *choose three or four key ratios*
- *there is no need to illustrate the formula or the calculation*
- *only one comparator should be needed*
- *focus on the cause of any changes and what this might tell us about the organisation's position.*

Explanation of the financial statement analysis technique

- ***choose three or four key ratios***

  *The aim is to try to pick some ratios that best **tell the organisation's "story"**.*

  Key ratios might be its sales growth (which can be linked to the PESTEL), its margins (which can be linked to the 5 forces model), and its gearing (which will give an idea of the organisation's risk profile and its ability to raise finance for future opportunities).

  Ratio analysis has already been examined at the fundamentals level and we are not trying to show how well we can calculate ten or twelve different ratios.

  A ratio should not be included if it adds nothing to the story (for example, there is little point in calculating receivable days if it has not changed during the year and the business does not have a problem with debt collection).

- ***there is no need to illustrate the formula or the calculation***

  This has been examined at the fundamentals level and would only waste time at this level. There is also no need to explain what the ratio means (for example, comments such as "The gross profit percentage shows how much profit the business makes per $ of turnover" are unlikely to gain any marks).

- ***only one comparator should be needed***

  The examiner will often provide four or five years' worth of financial statements. However, to get the key messages for the company's story we do not need to calculate the ratios for every year.

Normally we simply need to compare this year's results with last year's, or occasionally this year's results to the first year's results if we're assessing how performance has changed over time. The average growth could also be calculated.

- ***focus on the cause of any changes and what this might tell us about the organisation's position***

  *The key to gaining marks will be to **analyse** the data that has been calculated (simply performing the calculation and not discussing it will not achieve all of the marks that are available). We need to explain **why** a ratio has changed (for example, is it due to changes in the external or internal environment) and **what these changes mean** for the business (for example, does it need to react to these changes or can the position be improved in the future).*

Revision of financial ratios

Ratios

The mechanics of ratio analysis are repeated here for revision purposes.

Profitability ratios

- *ROCE = (Operating Profit / Capital Employed) × 100*
- *Gross margin = (Gross profit / Sales) × 100*
- *Operating margin = (Operating profit / Sales) × 100*
- *ROE = (Profit after tax and preference dividends / Shareholders' funds) × 100*

Efficiency ratios

- *Asset turnover = Sales / Capital Employed*
- *ROCE = Operating margin × asset turnover*
- *Receivables days = (Receivables balance / Credit sales) × 365*
- *Payables days = (Payable balance / Credit purchases) × 365*
- *Inventory days = (Inventory / Cost of sales) × 365*
- *Revenue per employee = Sales / Number of employees*

Liquidity ratios

- *Current Ratio = Current Assets / Current Liabilities*
- *Quick Ratio (acid test) = (Current Assets – inventory) / Current liabilities*

Gearing ratios

- *Financial Gearing = Debt / Equity*
- *Financial Gearing = Debt / (Debt + Equity)*

Investor ratios

- *Dividend Cover = PAT / Total Dividend*
- *Interest Cover = Operating profit / Interest*
- *EPS = Profit after tax and preference dividends / Number of shares × 100*
- *PE ratio = Share price / EPS*

Inter-firm comparisons

It is possible (through use of financial ratios) to compare and contrast the performance of one entity within an industry with that of another within the same industry. It is also possible to compare the performance of one firm with that of the whole industry, or segment of that industry.

However, these comparisons may suffer from one or more of the following limitations:

- *Different accounting methods may be used by individual firms making up the industry sample, or by the firm being compared.*
- *The industry figures may be biased by one or a few very large firms within the sample.*
- *Conversely, an industry mean may be misleading for a small or large firm being compared with the mean. Ratios may vary for different sizes of firms.*
- *The companies within the industry sample may span more than one industry classification.*
- *The industry figures may be relevant for a different financial period and could possibly be out-of-date.*

Non-financial performance measures

Although profit cannot be ignored as it is the main objective of commercial organisations, performance analysis should not focus on profit alone.

As discussed in Chapter 3, an organisation needs to identify the most important things it needs to 'do well at' in order to achieve success in its long- and short-term goals and objectives.

These are an organisation's ***Critical Success Factors (CSFs).***

To measure performance against the CSFs an organisation will usually set targets and measure performance against a number of ***Key Performance Indicators (KPIs)****.*

KPIs are quantifiable metrics or measures that are used to track and assess the performance of an organisation, department, or specific process. They provide objective data and are used to evaluate how effectively an organisation is achieving its objectives.

A range of performance indicators (KPIs) should be used, and these should ideally be a mix of financial and non-financial measures.

Using Key Performance Indicators (KPIs) in scenarios

KPIs may be presented in the scenario (such as customer return rates, % of repeat business, market share, age of products etc.) alongside financial information (or in an accompanying exhibit).

It will be important to react to this data in the same way that you would react to data that is presented in tables or financial statements – that is, to interpret what they are trying to tell you, and to link these into the analysis of the organisation's position.

Examples of Non-financial CSFs and KPIs

The table below shows a number of non-financial key performance indicators (KPIs) grouped against CSFs. The organisation will formulate its own, specific KPIs which best suit its business.

CSFs	*KPIs*
Competitiveness	• *sales growth by product or service* • *measures of customer base* • *relative market share and position*
Resource utilisation	• *efficiency measurements of resources planned against consumed* • *measurements of resources available against those used* • *productivity measurements*
Quality of service	• *quality measures in every unit* • *evaluate suppliers on the basis of quality* • *number of customer complaints received* • *number of new accounts lost or gained*

Customer satisfaction	• *speed of response to customer needs* • *informal listening by calling a certain number of customers each week* • *number of customer visits to the factory or workplace* • *number of factory and non-factory manager visits to customers*
Quality of working life	• *absence days* • *labour turnover* • *overtime* • *measures of job satisfaction*
Innovation	• *proportion of new products and services to old* • *new product or service sales levels*
Responsiveness (lead time)	• *order entry delays and errors* • *wrong blueprints or specifications* • *minimal set-up times* • *defect count* • *machines break downs*
Quality of output	• *returns from customers* • *reject rates* • *reworking costs* • *warranty costs*
Flexibility (ability to react to changing demand and a changing environment)	• *product/service introduction flexibility* • *product/service mix flexibility* • *volume flexibility* • *delivery flexibility* • *time to respond to customer demands*

Problems with non-financial performance indicators

The use of NFPI measures is now common, but it is not without problems:

- *Setting up and operating a system involving a wide range of performance indicators can be time-consuming and costly*
- *It can be a complex system that managers may find difficult to understand*
- *There is no clear set of NFPIs that the organisation must use – it will have to select those that seem to be most appropriate*
- *The scope for comparison with other organisations is limited as few businesses may use precisely the same NFPIs as the organisation under review.*

Test your understanding 2

Explain how your approach to performance appraisal would differ if you were asked to assess the performance of a not-for-profit organisation.

Multi variable performance analysis

As discussed above, organisations should combine financial and non-financial performance measures to provide a broader view of the organisation's performance and position. There are several tools that can be used to do this. Two of these are explored below.

The balanced scorecard

The balance scorecard, developed by Kaplan and Norton, suggests that managers should appraise performance from four perspectives:

- *a financial perspective – this considers whether an organisation is achieving its financial targets and meeting the needs of shareholders*
- *a customer perspective – this considers the organisation from a customer point of view to determine whether the organisation is meeting customer needs*
- *an innovation perspective – this considers whether the organisation is continuing to improve and develop*
- *an internal business process perspective – this considers whether the organisation's processes are efficient as well as whether employees are satisfied and motivated*

Further details

The aim of the model is to ensure that managers are only deemed to have been a success if they have succeeded from all perspectives.

So, for example, a manager who has increased financial results (and thus scored well from a financial perspective) will not be rewarded if this success has been achieved by cutting back on staff training (a poor internal business perspective) or cutting back on research and development (a poor innovation perspective).

Critical success factors and associated key performance indicators will be attributed to each perspective. For example, from a customer perspective an organisation might want to ensure that it has a quick response to customer queries and might therefore measure the average response time.

Companies often use benchmarking (see section 3) to set realistic targets by comparing their performance with industry leaders.

Successful organisations score well across all these areas, not just one or two. Focusing solely on financial results might lead to long-term issues like unhappy customers or outdated products.

A balanced approach helps companies meet their strategic goals and maintain a strong position over time.

Baldrige performance excellence

An alternative approach is to use the Baldrige performance excellence model. This was developed by Malcolm Baldrige as a way of measuring and rewarding those organisations that have performed better than their contemporaries.

Baldrige developed criteria for performance excellence in the belief that organisations that incorporate them into their organisational practices can expect to achieve performance superior to their competitors.

The Baldrige model assesses an organisation across seven categories:

- ***Leadership** – how the organisation's leadership guides, governs and sustains the organisation's performance.*
- ***Strategy** – the ability to successfully plan, develop and implement strategies.*
- ***Customers** – the success in building and sustaining strong, lasting relationships with customers.*

- ***Workforce*** *– how the organisation enables and empowers its workforce to achieve organisational goals.*
- ***Operations*** *– the design and effectiveness of organisational processes and whether these are improving and meeting strategic needs.*
- ***Results*** *– the performance and improvement of the organisation relative to competitors, in the key categories of the model.*
- ***Measurement, analysis, and knowledge management*** *– how data is stored, managed, analysed and used within the organisation.*

Further details

Leadership *will consider an organisation in areas such as:*

- *whether its leaders provide clear, well communicated goals and lead by example*
- *its corporate governance – for example, in whether best practice is being applied and all stakeholder views are considered*
- *whether it has considered its wider social responsibilities*

Strategy *will consider an organisation in areas such as:*

- *whether it has clear objectives*
- *its ability to create strategic plans*
- *whether strategic plans fit with the strategic objectives*
- *how strategies are developed and implemented*
- *whether suitable and sufficient resources are provided for strategic implementation*
- *whether it measures success or failure of strategic objectives*

Customers *will consider an organisation in areas such as:*

- *the level of customer support*
- *customer complaint management*
- *the level of customer interaction and engagement*
- *whether customers are segmented, and products/services personalised to segments*

Workforce *will consider an organisation in areas such as:*

- *recruitment and retention*
- *workforce career progression*
- *workforce training and development*
- *reward schemes and controls*
- *appraisals*
- *workforce engagement*

Operations *will consider an organisation in areas such as:*

- *whether processes are reviewed and assessed*
- *whether processes are improved*
- *how effective processes are in delivering customer value and strategic value*

Results *will consider organisational improvement in all key areas such as:*

- *product and process results*
- *customer results*
- *workforce results*
- *leadership and governance results*
- *financial and market results*

The category asks about performance levels relative to those of competitors and other organisations with similar product offerings.

***Measurement, analysis and knowledge management** will consider an organisation in areas such as:*

- *whether organisational knowledge is managed and shared*
- *whether benchmarking is used*
- *which performance measures are used and how data is collected*
- *whether action is taken as a result of performance measurement*

From this it can be seen that the Baldrige Values include:

- *visionary leadership*
- *customer-driven excellence*
- *organisational and personal learning*
- *valuing employees and partners*
- *agility*
- *focus on the future*
- *managing for innovation*
- *management by fact*
- *social responsibility*
- *focus on results and creating value*
- *systems perspective*

An organisation that scores well in the areas of the model should be better at:

- *achieving its strategic goals*
- *competing*
- *improving its results*
- *improving and learning*
- *meeting customer needs and retaining customers*

Test your understanding 3
Two of the Baldrige values are valuing people and agility. Explain how these might contribute to the strategic success of an organisation.

Benchmarking

Why benchmark?
Benchmarking is the process of systematically comparing a service, practice, or process. Its goal is to set a target to improve an organisation's competitive position.

The strategic role of benchmarking
Benchmarking is integral to the entire strategic planning process.

- *In strategic analysis, benchmarking can be used as part of value chain analysis to compare one company's values to another. By comparing strategic capabilities with those of rivals, a company can identify its strengths and weaknesses. For example, if a company considers its staff a key asset, it might compare its level of staff turnover and sick days to industry averages to see if there is a problem in this area.*
- *In making strategic choices, we have seen in this chapter that a company can gain a competitive advantage if it has unique strategic capabilities in areas such as innovation, cost efficiency, and knowledge management. For instance, if a company is pursuing a low-cost*

strategy, it might benchmark its production cost per unit against industry standards to ensure it is cost-efficient.

- *In putting strategy into action (explored in detail later in the text), benchmarking can be used in many ways to determine, for example, which processes need to be redesigned or whether staff are being utilised in the best way. It can be used to help set budgets and targets, and to assess the efficiency and effectiveness of IT solutions, etc.*

For example, a financial services company might benchmark the efficiency of its IT systems against industry standards. If it finds that its systems are less efficient, it might invest in new technology or training for its staff to improve performance.

Benchmarking is closely related to CSFs and KPIs. It helps determine which areas to measure and sets targets against which KPIs are measured. For example, if customer satisfaction is a CSF, a company might benchmark its customer service ratings against industry leaders to set a realistic and challenging KPI.

Types of benchmarking

There are various types of benchmarking such as:

- *internal*
- *competitive/industry*
- *functional/activity*
- *generic*

Types of benchmarks explained

Internal benchmarking
This method examines past performance over a period of time to determine trends and best performance. Alternatively, a range of processes might be assessed to determine internal best practice, which can then be used as the benchmark for other processes.

There is a danger however that this will result in the continuance of poor habits and that competitors are ignored.

Competitive benchmarking
This method compares performance of the process against other firms in the same industry or sector.

Major automakers, for example, will buy cars made by their competitors then reverse engineer those cars to see how to improve their own product.

However, there is a danger that, if this is only carried out on a local level, it may not promote performance that is good enough to match wider (e.g. international) rivals.

Functional benchmarking
In functional benchmarking, comparisons are made with a similar function (for example selling, order handling, despatch) in other organisations that are not direct competitors.

For example, a fast food restaurant operator might compare its buying function with buying in a supermarket chain.

Generic benchmarking
For some activities, the process might be so unique that there may not be competitive or activity benchmarks available. In these cases, a conceptually similar process is sought as a benchmark. For example, when building a rail tunnel connecting Aomori Prefecture on the Japanese island of Honshū and the island of Hokkaidō which travels under sea, the construction company would have no similar activities against which to benchmark (the under-sea tunnel between England and France was yet to be built). However, the tunnelling was conceptually similar to explorations into volcanic crusts and this process was used as the benchmark.

Illustration 1 – Benchmarking

Since 2018, Walmart has been actively benchmarking Amazon's e-commerce and logistics strategies to enhance its own online retail operations.
Observing Amazon's success with fast delivery through Prime, Walmart launched its own subscription service, Walmart+, in September 2020, offering free shipping and other perks.
Walmart also benchmarked Amazon's extensive use of automation and robotics in warehouses, investing in similar technologies to improve efficiency and reduce delivery times.
Additionally, Walmart expanded its online grocery services, inspired by Amazon's acquisition of Whole Foods and its integration with Amazon Fresh.
These strategic moves have significantly boosted Walmart's e-commerce growth, helping the company compete more effectively in the digital marketplace and demonstrating the impact of benchmarking on business transformation.

Test your understanding 4
Outline the advantages of internal benchmarking compared with the other types.

Benefits and dangers of benchmarking

The main benefits include:

- *Improved performance and added value – benchmarking identifies methods of improving operational efficiency and product design and helps companies focus on capabilities critical to building strategic advantage.*
- *Improved understanding of environmental pressures.*
- *Improved competitive position – benchmarking reveals a company's relative cost position and identifies opportunities for improvement.*
- *A creative process of change.*
- *A target to motivate and improve operations.*
- *Increased rate of organisational learning – benchmarking brings new ideas into the company and facilitates experience sharing.*

Dangers of benchmarking:

- *'You manage what you measure' – managers may learn to direct attention at what gets benchmarked rather than at what is important strategically.*
- *Benchmarking does not always reveal the reasons for good/poor performance.*
- *Managers need to be aware that a benchmarking exercise can appear to threaten staff where it appears that benchmarking is designed to identify weaknesses in individual performance rather than how the process itself can be improved. To alleviate this fear, managers need to be involved in the benchmarking exercise and provide reassurance to staff regarding the aims and objectives of benchmarking.*
- *In today's environment, the more innovative companies are less concerned with benchmarking numbers (for example, costs or productivity) than they are with focusing on the processes. If a company focuses on the processes, the numbers will eventually self-correct.*

*To rigorously **assess performance, detect bias, and support strategic decisions using statistical tools and spreadsheets**, you'll want a structured workflow that combines quantitative analysis with visualization. Here's how you can approach it:*

Step 1: Performance Assessment

- ***Descriptive Statistics**: Use mean, median, variance, and standard deviation to summarize performance data.*
- ***Trend Analysis**: Apply time-series charts in Excel to detect growth, decline, or seasonality.*

- ***Benchmarking**: Compare KPIs (e.g., ROI, market share, productivity) against industry averages.*
- ***Dashboards**: Build interactive dashboards with pivot tables and slicers for real-time monitoring.*

Step 2: Bias Detection

- ***Sampling Bias**: Check whether data represents all relevant segments (e.g., customer demographics, product lines).*
- ***Regression Diagnostics**: Use residual plots to detect model bias or omitted variable bias.*
- ***Correlation vs. Causation**: Apply correlation matrices to ensure relationships aren't misinterpreted.*
- ***Visualization**: Scatter plots and box plots reveal anomalies or skewed distributions.*

Step 3: Strategic Decision Support

- ***Scenario Analysis**: Model different pricing, positioning, or investment scenarios using Excel's "What-If Analysis" (Data Tables, Goal Seek).*
- ***Forecasting**: Apply regression or exponential smoothing to predict demand and revenue.*
- ***Optimization**: Use Solver in Excel to maximize profit or minimize cost under constraints.*
- ***Risk Analysis**: Monte Carlo simulations (via add-ins or Python/R integration) quantify uncertainty in strategic choices.*

Recommended Statistical Tools

- ***Excel/Google Sheets**: For dashboards, pivot tables, Solver optimization.*
- ***R / Python (Pandas, NumPy, SciPy)**: For regression, bias detection, and advanced simulations.*
- ***Power BI / Tableau**: For interactive visualization and stakeholder reporting.*

4.4 Other considerations in divisional assessment

When assessing divisional performance, the following issues must also be considered:

- The division manager should only be held accountable for factors within their control. The divisional profit figure may be distorted by arbitrarily allocated head office costs and thus performance measurement should focus on traceable profit.
- Where there is inter-divisional trade, careful consideration should be given to any transfer pricing mechanism in place, which may under or over-state the profits of a particular division.
- Divisions operating in different marketplaces and facing differing levels of competition cannot be expected to produce similar returns. Thus, in addition to comparing one division against another, external comparisons should be made via benchmarking (eg, to industry leader).
- Wider strategic issues need to be taken into account such as any interdependence between divisions eg, shared distribution systems, shared customers, the impact that a division has on the portfolio of the business and its brand.
- In assessing the future strategic direction of the business it is not just the historic performance of the division but also its future potential that is relevant.
- Focus on a narrow set of financial measures is unlikely to give a true picture of performance (see the section below).

5 Balanced scorecard

Section overview

An approach that tries to integrate the different measures of performance is the balanced scorecard, where key linkages between operating and financial performance are brought to light. This offers four perspectives:

- Financial
- Customer
- Innovation and learning
- Internal business processes

Definition

Balanced scorecard: An integrated set of performance measures linked to the achievement of strategic objectives. The balanced scorecard consists of four perspectives: customer, internal business, innovation and learning, financial.

5.1 Financial measures of performance

There are a number of limitations where management rely solely on financial performance measures.

- **Encourages short-termist behaviour** – focusing on hitting monthly, quarterly and annual targets may be inconsistent with longer term strategic business development.
- **Ignores strategic goals** – financial control cannot enable senior management to affect other important determinants of commercial success such as customer service and innovation.
- **Cannot control persons without budget responsibility** – operative and other staff cannot be controlled by financial targets if they have no responsibility for financial results.
- **Historic measures** – financial measures of performance are essentially **lagging indicators** of competitive success ie, they turn down after competitive battles have been lost. Management needs **lead indicators** of where problems are occurring to avoid losing such battles.
- **Distorted**: financial measures can be distorted by creative accounting. Also some of the conventions of financial reporting mean that internal financial control measures are of limited value for decision-making, eg, lack of valuation of intangibles such as brands, failure to value inventory and transactions at opportunity cost and so on.

Earlier in this chapter, we discussed divisional performance measures. The research of Goold and Campbell revealed the practice by the boards of divisionalised firms to utilise a **strategic control style** that combined financial and non-financial performance measures.

5.2 The origins of the balanced scorecard

Kaplan and Norton's **balanced scorecard** (BSC) was developed in the early 1990s following research into performance measures in high-performing US firms, notably in the IT industries of Silicon Valley, Northern California.

According to their research these firms conducted regular assessments of **four different perspectives**, as follows:

Perspective	Question	Explanation
Customer	What do existing and new customers value from us?	Gives rise to targets that matter to customers: cost, quality, delivery, inspection, handling and so on.
Internal business	What processes must we excel at to achieve our financial and customer objectives?	Aims to improve internal processes and decision making.
Innovation and learning	Can we continue to improve and create future value?	Considers the business's capacity to maintain its competitive position through the acquisition of new skills and the development of new products.
Financial	How do we create value for our shareholders?	Covers traditional measures such as growth, profitability and shareholder value but set through talking to the shareholder or shareholders direct.

Performance targets are set once the key areas for improvement have been identified, and the balanced scorecard is the **main monthly report**.

Kaplan and Norton claimed that the scorecard is **balanced** in the sense that managers are required to think in terms of all four perspectives, to **prevent improvements being made in one area at the expense of another**.

A decade later and the BSC was being proposed as more than a performance measurement system. In *The Strategy Focused Organization*, Kaplan and Norton state that the setting of performance measures in the four areas can drive strategic change (a topic returned to in the chapter Strategies for change).

5.3 Developing a balanced scorecard

Kaplan (*Advanced Management Accounting*) offers the following 'core outcome measures.'

Perspective	Core outcome measure
Financial	Return on Investment Profitability Revenue growth/revenue mix Cost reduction Cash flow
Customer	Market share Customer acquisition Customer retention Customer profitability Customer satisfaction On-time delivery
Innovation and learning	Employee satisfaction Employee retention Employee productivity Revenue per employee Percentage of revenue from new services Time taken to develop new products
Internal business	Quality and rework rates Cycle time/production rate Capacity utilisation

Kaplan and Norton suggest the following process for setting the BSC:

(a) Senior executives decide strategy.

(b) Budgets and information systems are linked to the measures in the BSC. This allows divisional and operational management to monitor the performance of their areas of responsibility.

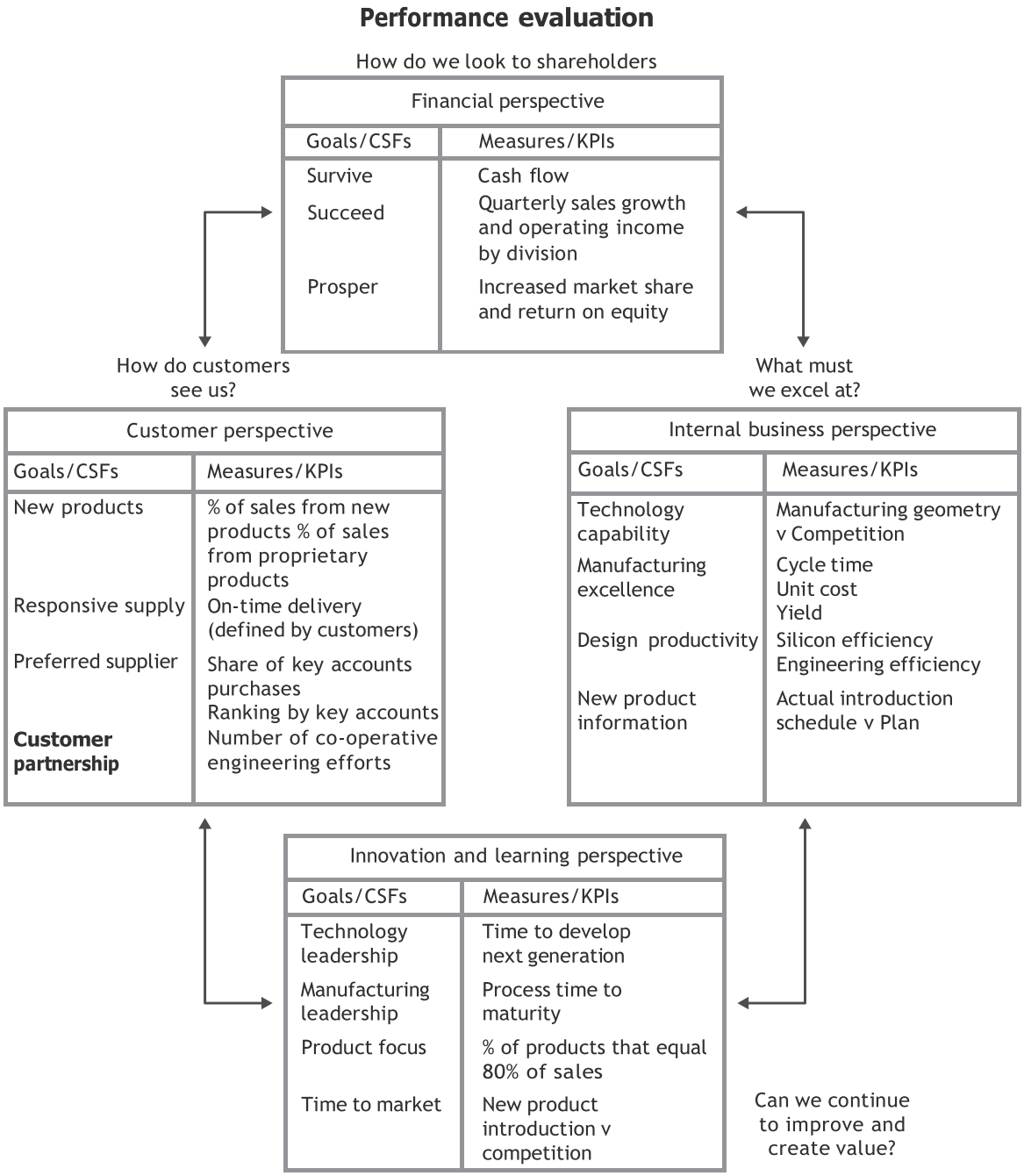
(c) Personal scorecards are developed and, through performance management become the basis for staff development and incentive payments. (These are discussed in the chapter Business planning and functional strategies.)

(d) Collaborative working occurs as many targets require team work to achieve.

(e) Therefore strategy is 'operationalised' through being turned into day-to-day operations.

Kaplan and Norton recognise that the four perspectives they suggest may not be perfect for all organisations: it may be necessary, for example, to add further perspectives related to the environment or to employment.

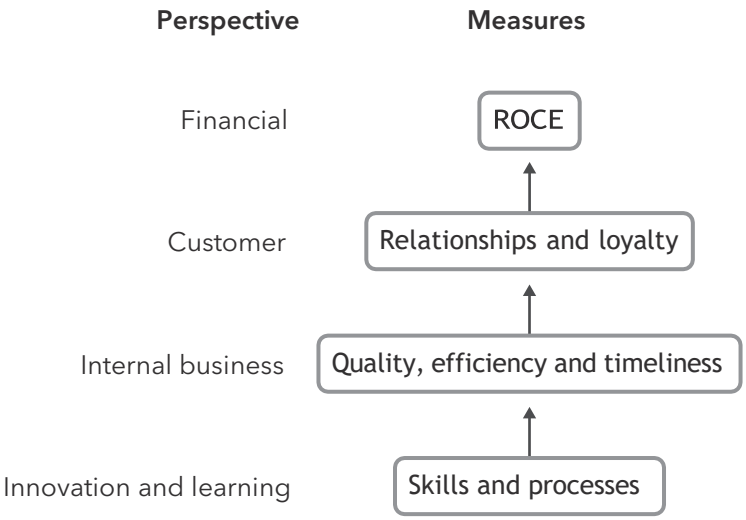
Figure 12.3: A balanced scorecard for a computer manufacturing company



5.4 The vertical vector

Kaplan and Norton's original perspectives may be viewed as hierarchical in nature, with a **vertical vector** running through the measures adopted, indicating that an organisation's financial performance is underpinned by its performance in the other three perspectives.

Figure 12.4: The vertical vector



The balanced scorecard only measures performance. It does not indicate that the strategy is the right one. 'A failure to convert improved operational performance into improved financial performance should send executives back to their drawing boards to rethink the company's strategy or its implementation plans.'

5.5 Problems

As with all techniques, problems can arise when it is applied.

Problem	Explanation
Conflicting measures	Some measures in the scorecard such as research funding and cost reduction may naturally conflict. It is often difficult to determine the balance which will achieve the best results.
Selecting measures	Not only do appropriate measures have to be devised but the number of measures used must be agreed. Care must be taken that the impact of the results is not lost in a sea of information. The innovation and learning perspective is, perhaps, the most difficult to measure directly, since much development of human capital will not feed directly into such crude measures as rate of new product launches or even training hours undertaken. It will, rather, improve economy and effectiveness and support the achievement of customer perspective measures.
Expertise	Measurement is only useful if it initiates appropriate action. Non-financial managers may have difficulty with the usual profit measures. With more measures to consider, this problem will be compounded.
Interpretation	Even a financially-trained manager may have difficulty in putting the figures into an overall perspective.

5.6 Using the balanced scorecard

- Like all performance measurement schemes, the balanced scorecard can influence behaviour among managers to conform to that required by the strategy. Because of its comprehensive nature, it can be used as a wide-ranging driver of organisational change.

- The scorecard emphasises **processes** rather than **departments**. It can support a competence- based approach to strategy, but this can be confusing for managers and may make it difficult to gain their support.
- Deciding just what to measure can be particularly difficult, especially since the scorecard **vertical vector** lays emphasis on customer reaction. This is not to discount the importance of meeting customer expectations, purely to emphasise the difficulty of establishing what they are.
- The scorecard should be used **flexibly**. The process of deciding **what to measure** forces a business to clarify its strategy. For example, a manufacturing company may find that 50%–60% of costs are represented by bought-in components, so measurements relating to suppliers could usefully be added to the scorecard. These could include payment terms, lead times, or quality considerations.

The scorecard can be used both by profit and not-for-profit organisations because it acknowledges the fact that both financial and non-financial performance indicators are important in achieving strategic objectives.

5.7 Developments in the balanced scorecard

The original balanced scorecard of Kaplan and Norton was initially developed as a measurement and control tool for managers. Essentially, its focus was 'strategic control' rather than 'management control.' It has, however, been developed over time in terms of both its design and application.

The **second-generation** balanced scorecard attempted to develop further the guidance on how the measures should be selected (filtered) and linked to the overall objectives of the organisation. It also began to look at more complex linkages and causality between the perspectives (sometimes called a 'strategic linkage model'). As a consequence, the balanced scorecard evolved from an improved measurement system to a core management system.

More recently, what has been termed a **third-generation** balanced scorecard, has been developed. This involves enhancing the links between the balanced scorecard and the strategic objectives to be achieved within a given timeframe. This has been reflected in a 'destination statement' which sets out in some detail what the organisation is trying to achieve. Its purpose is to check the extent to which the objectives, targets and measures chosen, have been attained after a given period of time. It has also supported the development of multiple balanced scorecards in complex organisations.

5.8 *Reporting on Environmental, social and governance (ESG) issues*

Increasingly, stakeholders (from regulators to investors, customers and the public) are requiring organisations to monitor and report on ESG metrics. For example, investors want financial returns but they also look for companies that are sustainable and have a positive impact on society and the environment.

Following the UN Climate Change Conference of the Parties (COP 26) in 2021 there has been an increasing emphasis on climate related reporting. The conference saw the announcement of the International Sustainability Standards Board (ISSB) which is tasked with developing sustainability disclosure standards to satisfy the needs of investors and wider stakeholder groups. From April 2022 it is mandatory for the largest UK companies to disclose climate-related financial information.

The EU also requires large and listed companies to report on the social and environmental risks they face and how their business activities impact the environment and people. On January 5th 2023 the Corporate Sustainability Reporting Directive (CSRD) came into force, which is the EU's latest initiative to improve the comparability and quality of corporate ESG disclosures.

The CSRD incorporates the concept of 'double materiality.' This means that companies have to report on:

Financial materiality: *Sustainability issues which might create financial risks for the company. Such risks influence the enterprise value, in other words the market value of a company's shares and debt.*

Impact materiality: *How an organisation's activities impact on people and the environment (impact materiality). Impact materiality reflects the risks created for multiple stakeholder groups such as consumers, business partners, employees, local communities and governments.*

For example, a company may report on how severe weather events caused by climate change are affecting its ability to manufacture (financial materiality). It may also report on its carbon emissions which affect the environment and people's health (impact materiality).

Examples of performance indicators which could be used to measure ESG factors might include:

ESG factor	***Performance Indicator example***
Environmental – taking actions to protect the environment	• *Energy consumption* • *Total water consumption* • *Greenhouse Gas (GHG) emissions, absolute emissions and intensity of emissions* • *Waste generated* • *Pollutants and effluents released* • *Recycling: recycled materials used in productions, recycled end- of-life products*
Social – building and maintaining relationships with stakeholders	• *Employee turnover overall, by department, by age, by gender, by length of service* • *Average training hours per employee split by gender, seniority and role* • *Health and safety: number of fatalities, high consequence injuries, recordable injuries and recordable work-related ill health cases* • *Fair working and pay conditions* • *Incidences of child and forced labour* • *Number of supply chain sustainability initiatives*
Governance – ensuring leadership are transparent and accountable in their stewardship of the organisation.	• *Diversity of board by age or gender* • *Composition of board by area of expertise* • *Number of risks effectively identified and mitigated* • *Incidences of bribery and corruption*

Effective measuring and benchmarking require the timely publication of information. Accountants have a role in supporting measurement of climate change and its consequences by providing relevant and reliable information in an accessible, meaningful and comparable way. Accountants will also have a role in interpreting the results of measures against targets. This is likely to include understanding the different bases used to be able to compare and analyse the results.

Context example: The World Economic Forum (WEF) International Business Council (IBC) metrics

The World Economic Forum (WEF) International Business Council (IBC) metrics provide a useful starting point when considering what ESG metrics may be relevant for an organisation.

The WEF paper 'Measuring Stakeholder Capitalism' was written with the aim of developing a core set of common metrics and disclosures on non-financial factors, for sustainable value creation. It highlights the key themes significant to the planet, society and business, and was produced in collaboration with Deloitte, EY, KPMG and PwC in 2020. The ESG metrics are categorised into four pillars:

(a) *Principles of Governance*

(b) *Planet*

(c) *People*

(d) *Prosperity*

Each metric aligns with relevant United Nations' Sustainability Development Goals (SDGs). For example, the People metric is aligned to the following six SDGs:

- *Goal 1 – No Poverty*
- *Goal 3 – Good Health and Well-being*
- *Goal 4 – Quality Education*
- *Goal 5 – Gender Equality*
- *Goal 8 – Decent Work and Economic Growth*
- *Goal 10 – Reduced Inequalities*

*(**Source**: https://www.weforum.org/reports/measuring-stakeholder-capitalism-towards-common- metrics-and-consistent-reporting-of-sustainable-value-creation/)*



Interactive question 5: ESG and sustainability key performance indicators

Panoramic Hotels (PH) is a large, privately-owned hotel chain based in the UK with a reputation for owning hotels situated in areas of outstanding natural beauty. PH is an aspirational brand with a target market of individuals with high levels of disposable income aged between 28–49 years. Each hotel aims to offer a first-class experience for guests with luxurious spa facilities, signature spa treatments, state of the art gyms and highly rated restaurants open to hotel residents and non- residents. There are now eight hotels in the PH portfolio with a strategy to double this in three years.

The board of PH has recently commissioned a hotel guest survey. One of the main findings was that the company could improve its communications around sustainability. In response, the board has agreed on nine new environmental, social and governance (ESG) objectives to be published on its website, together with quarterly sustainability reports which will disclose PH's progress in meeting these objectives.

Environmental objectives

(1) *Climate-change related emissions: Ensure compliance with relevant laws and regulations relating to air and greenhouse gas emissions (such as carbon dioxide, methane, nitrous oxide) and water pollutants (such as the use of cleaning chemicals in hotel laundry and cleaning.)*

(2) *Consumption of resources: Ensure PH's operations make efficient use of resources, including energy and water.*

(3) *Waste minimization: Ensure that actions are taken to identify and reduce waste across all PH operations.*

Social objectives

(1) *Employee wellbeing: Ensure compliance with relevant laws and regulations relating to rewards and benefits; recruitment, promotion and dismissal; working hours and rest periods; equal opportunity, diversity and anti-discrimination.*

(2) *Guest and employee safety: Ensure compliance with relevant laws and regulations relating to providing a safe working environment for employees and protecting hotel guests from hazards.*

(3) *Employee personal and professional development: supporting personal and professional development for PH employees.*

Governance objectives

(1) *Supply chain sustainability: Improve procurement policies and processes to increase sustainability practices in PH's supply chain management.*

(2) *Local community support: Improve community engagement to understand the needs of the communities where PH hotels operate and ensure PH takes into consideration the communities' interests.*

(3) *Anti-bribery and corruption: Eliminate bribery, money laundering and other corruption practices from all PH operations and remove the potential for such incidences to occur.*

Requirement

For each sustainability objective, recommend one suitable key performance indicator (KPI) which will improve management and disclosure of ESG related risks. For each KPI, suggest practical steps and initiatives that PH could implement to help the company meet its ESG objectives.

*See **Answer** at the end of this chapter.*



Context example: Asda Environmental, Social and Governance report

In 2021, Asda launched its first ESG report, setting out its commitments to a sustainable business strategy. To prepare the report Asda consulted with more than 3,000 customers to understand the key ESG matters that mattered most to them.

Some of the key ESG performance targets outlined in the report are:

- Becoming a net zero carbon business by 2040
- Removing 3 billion pieces of plastic from products by 2025
- Ensuring the top 20 commodities, such as cotton and cocoa, are sustainably sourced
- Cutting food waste by 20% by 2025
- Delivering 120 million meals by the end of the year to communities across the UK

(**Source**: https://corporate.asda.com/newsroom/2021/05/06/asda-launches-first-esg-report [Accessed 1 June 2022]).

6 Measuring performance in NFP organisations

Section overview

- Measuring the performance of not-for-profit organisations (NFPs) can be challenging.
- Traditional financial measures of performance may not apply, and outcomes and outputs may be harder to measure or quantify.

Measuring the performance of NFPs can be more challenging because they tend to have a more diverse set of stakeholders and therefore a wider set of objectives. Also, the traditional financial measures of performance may not apply in the absence of a meaningful bottom-line profit figure. Thus, the outcomes or outputs may be harder to measure or quantify.

If we consider a school for example, performance at a very basic level could be measured by considering the pass rates in external examinations.

However, if the school is felt to have a wider role, eg, in developing non-academic talents, or in preparing children for independent lives, then progress towards these objectives also need to be measured.

Since many NFP organisations have to deal with limited resources, the concept of 3Es is particularly relevant.

7 Types of data

Section overview

- It is important that you understand the key differences between quantitative and qualitative types of data as both are mentioned in the Business Strategy and Technology syllabus and could easily appear in your exam. You need to be prepared to use both types of data in order to understand an organisation's performance.
- Data related requirements in the exam may require you to consider data presented in a range of different formats.

Definitions

Qualitative data: Is fundamentally concerned with trends in opinions, thoughts and motivations, as a result it tends to be subjective. Qualitative data cannot be calculated or computed.

Quantitative data: Is focused on numerical data which can be calculated and computed.

7.1 Qualitative and quantitative data

Feedback received from customers about an organisation's new product which allows them to express their opinion is an example of qualitative data. Such feedback may be captured through the use of customer surveys; discussion groups or social media posts: as a result, it is likely to be of an unstructured nature. Qualitative data is of a non-financial nature.

Quantitative data is presented in numerical form. The quantity of products sold in the year is an

example of quantitative data. Quantitative data is likely to be easier to structure than qualitative data. It is typically financial but can also include operating data eg, the number of customer returns.



Context example: Quantitative and qualitative data in action

Euro-Shop is a low-price supermarket chain which operates in a number of developed European countries across much of central Europe. It sells a narrow selection of grocery and household products. Customer choice is limited to one or two lines per product, allowing greater quantities of inventory to be stored on the shop floor. Many of the items sold are well-known brands in Eastern European countries and are not as well known in central Europe. Inventory is stacked on wooden pallets on the shop floor for customers to access. Each store operates with the minimum number of staff required by European law. This is part of Euro-Shop's strategy to keep costs low.

Two years ago Euro-Shop established a number of stores in Country X, a European country in which it had previously not operated. Customers in Country X expect a high degree of personalised service when they visit the supermarket, and demand lots of choice in the products available for them to purchase. Euro-Shop's managing director in Country X has gathered the following data based on the first two full years of trading:

	20X4	20X3
	£	£
Sales	8,294,000	8,513,000
Profit for period	531,000	596,000

Using social media analytics tools Euro-Shop's marketing manager in Country X has collated the most frequently used words that customers use when tweeting on the Euro-Shop Country X Twitter feed. He has specifically looked at the terms used when customers tweet complaints, and those words which signify what customers most value when shopping at Euro-Shop:

Most common complaints between 20X3 and 20X4: poor customer service/lack of product choice What customers value the most between 20X3 and 20X4: low prices

Types of data

In this example the sales and profit figures represent quantitative data as they are in numeric form. 'Poor customer service', 'lack of product choice' and 'low prices' are examples of qualitative data as they embody the opinions of Euro-Shops customers. Although, the use of social media analytics makes capturing and collating the tweets easier, it is likely that less reliance can be placed on these findings, as it will only be based on those comments made by shoppers using the Euro-Shop Twitter account.

Based on the limited scenario detail and data provided we can make the following assessment of Euro-Shops performance over the period.

Firstly, it is evident that there has been a fall in sales between the two years of 219,000 (or 2.6%), and a reduction in profitability of 65,000 (10.9%) over the period. This may reflect the fact that customers in Country X have not responded as well as may have been hoped when Euro-Shop first set up operations two years ago. This is likely to have been caused by the fact that customers expect a higher degree of personalised service when they go shopping whereas Euro-Shop operates its stores using the minimum of workers required by law. This point is supported by the fact that one of the most common complaints between 20X3 and 20X4 related to poor customer service. The drop in sales and profits potentially indicates, that after the initial novelty of Euro-Shops arrival in Country X, shoppers are now gradually returning to the supermarkets that they used prior to Euro-Shops entry in the market. This may

in part be supported by the fact that customers are unwilling to sacrifice the lack of choice of products available when shopping at Euro-Shop. This point is reinforced by the complaint made via Twitter that there is a lack of choice in Euro-Shop's stores. Despite this, it is evident that Euro-Shops Twitter users do value the opportunity to complete their shopping at lower prices.

Clearly, more detail would be needed in order to make a full assessment of Euro-Shops performance.

7.2 Format of data in the exam

You need to be prepared for the fact that, in the Business Strategy and Technology exam, quantitative and qualitative data may be presented in a range of formats. Examples may include:

- Extracts from financial statements
- Extracts from balanced scorecards
- Extracts of posts on social media
- Data dashboards
- Diagrams (eg, value chain, BCG matrix)
- Graphs
- Pie charts (eg, analysis of sales revenue by product)
- Detailed granular data in a pre-populated spreadsheet to be assimilated and analysed using the spreadsheet functionality in the exam software

You must ensure that you are comfortable analysing and interpreting data, regardless of the format in which it is presented. A key part of this involves not solely focusing on financial data to the detriment of other non-financial or operating data which may also appear.

7.2.1 Using spreadsheet software

Spreadsheet software provided in the Business Strategy and Technology exam can be used in a variety of ways. Candidates may be presented with question exhibits containing extracts of financial statements or dashboard data. By copying the data into the spreadsheet area of the exam candidates can quickly perform calculations and analysis to support their discussion and conclusions.
Alternatively, some questions may provide more granular data in pre-populated spreadsheets. Again, spreadsheet functionality should be used to interrogate and analyse this data.
More information on the use of spreadsheets and examinable spreadsheet functions can be found in the chapter Spreadsheet functionality. Example questions demonstrating the style of exam questions and the application of spreadsheet functionality can be found in the ICAEW practice software and in the chapter Spreadsheet functionality.

The functionality used in each question is as follows:

Question name	Format	Spreadsheet functionality used
Pico Fogo Ltd	Pre-populated data	SUM
Zinnia Ltd	Non-pre-populated data	SUM
By Camplight	Pre-populated data	AVERAGE, IF, RANK, SUM
Chrono Ltd	Pre-populated data	IF(incl. nested IF statements), SUMIF
HGF Ltd	Pre-populated data	AVERAGE, COUNTIF, COUNTIFS, IF, STDEV

7.2.2 Use all relevant data

If qualitative data in the form of posts on social media are provided in the question scenario, then the examiner is expecting you to use it when constructing your answer. Extracts from social media should be used in conjunction with the other detail provided in the question scenario and any relevant financial data to give your answers greater depth and insight. Adopting this approach is likely to be useful when, for example, explaining why an organisation may have performed better or worse than had been expected in a given period. The last worked example which featured Euro-Shop, the discount supermarket chain, illustrated this interaction.

Professional skills focus: Structuring problems and solutions

This paper is not designed to test your numerical skills in the same way as, for example, Financial Management. Instead, you need to use the results from calculations performed or graphs provided in order to structure your response and evaluate business problems.

7.2.3 Follow question requirements carefully

It is worth noting that some exam questions may only provide operating data with no corresponding financial information. The December 2014 question, Norgate Bank plc, focused on a bank providing services to customers in the UK and France via its two telephone call centres (one of which was based in the UK and the other based in Vietnam). Three key performance indicators were used by the bank to assess the performance of its call centres. One of the requirements asked candidates to explain the benefits and problems of Norgate using benchmarking to evaluate its performance.
Candidates were expected to refer to different types of benchmarking, and were directed to use the operational data provided in the form of a data dashboard:

Exhibit: Data for the year ended 30 November 20X4

	UK call centre (UK customers)		Vietnam call centre (French customers)	
	Business	**Individuals**	**Business**	**Individuals**
Number of calls in the year ('000s)	120	1,200	90	600
Number of call operators	20	100	12	30
Time to answer a call (minutes)	2	2	1	1.2
Length of call (minutes)	10	4	8	3
Average customer satisfaction score	3.9	4.1	3.1	3.3

Here, candidates were expected to draw upon their knowledge of the different types of benchmarking when discussing advantages and disadvantages, and to comment on the operational data provided in relation to its applicability to different types of benchmarking. Failure to use the data provided, especially when it is specified in the requirement, will cost you marks when attempting your exam.

7.2.4 Understand the interaction between strategy models and data

Some exam questions may require you to combine your data analysis skills with knowledge of key strategy models. Past exam questions for example, have required candidates to construct

a BCG matrix using product sales data provided in the question scenario (and the sales data for the organisation's largest competitor) in order to determine the market share of the products sold by the featured organisation.

7.2.5 Levels of data in exam questions

You need to be prepared for question scenarios in the Business Strategy and Technology exam to feature Exhibits where the data included will be at a more detailed, granular level than only consisting of high-level figures derived from the financial statements. For example, sales data or customer satisfaction scores may be provided, which relate to performance on a daily or weekly basis. Equally, sales data could be provided for individual stores, or individual products, rather than as an overall total.

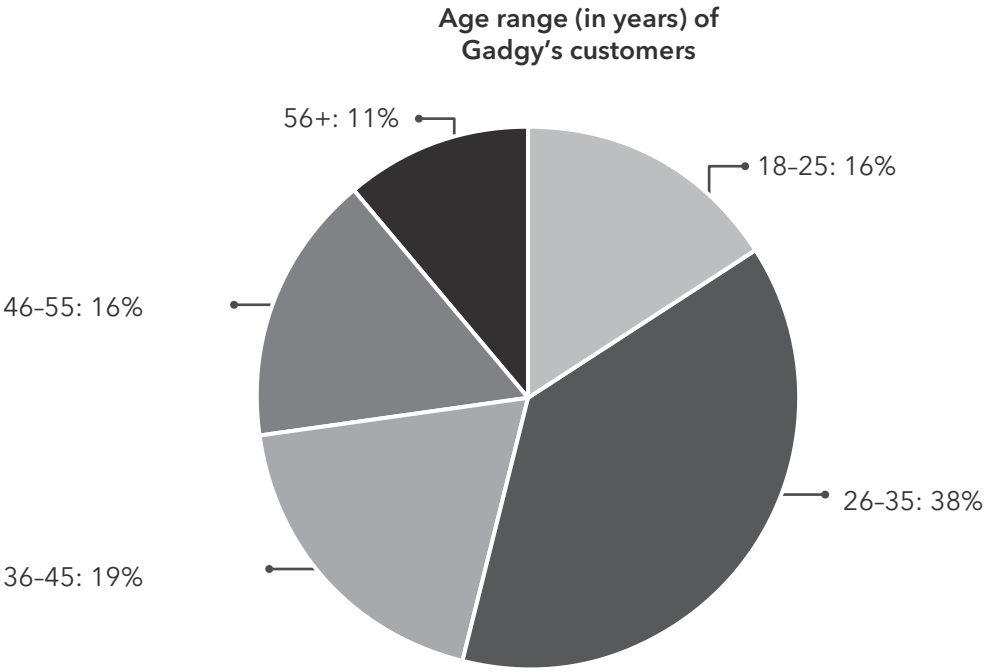
It is also important that you are alert to the fact that data may be provided which only covers the performance results for an entity for part of the financial year, for example, covering a nine month period as opposed to the full year. Some questions may require you to analyse the performance of the entity featured using for example, six months actual performance data against a full 12 month budget. To get the most from your analysis you need to be comfortable adjusting the actual performance data for the six months to make your comparison with the budget meaningful.

7.2.6 Interpreting visualisations

To ensure that you are able to use data presented in graphs, charts and diagrams (visualisations) in the exam effectively, you are advised to take the time to review the following section as this provides some helpful pointers in understanding data presented in different formats.

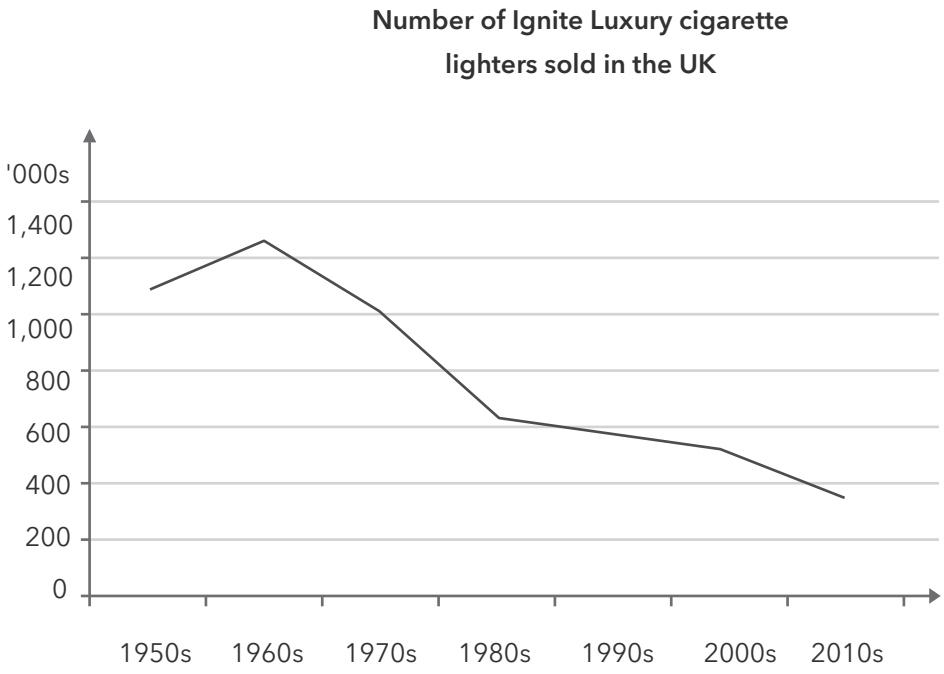
- As we have mentioned throughout this Workbook, if you are provided with data presented in visualisations in questions, it is vital that you use it. Ignoring such data will limit the number of marks that you will be able to earn.
- You should aim to read through the question scenario first before reviewing any visualisations as this will help to put the data presented into context. Once you have an understanding of the scenario you should then move on to review the requirements as this will drive the tasks that you are required to complete. The requirements may also direct you to the Exhibit featuring the visualisation that you are expected to use when constructing your answer.
- Having reviewed the requirements, you should then take a moment to review the information presented in Exhibits. At this point there are a number of key things that you should aim to do:
  - Carefully review the headings of the visualisation as this will make it clear what data is being presented.
  - In the case of pie charts, where leader lines are present (ie, those lines directed at segments on the pie chart) you should review them in conjunction with the detail presented on the pie chart itself as this is crucial to understanding the data being presented. The following pie chart shows leader lines in respect of the age ranges of customers purchasing products from the company featured in Question 1 of the March 2019 exam (Gadgy Ltd). The segments on the pie chart represent the percentages of customers falling into the stated age ranges.

Figure 12.5: Age range (in years)



- You should aim to identify the key elements presented in pie charts, graphs and diagrams. This is likely to involve identifying the largest segments on a pie chart or considering the most dramatic movements on a graph or diagram. This is particularly important as it is unlikely that you will have time to analysis all aspects of the data presented due to the time pressures of the exam. Moreover, prioritisation is an important professional skill, so it is important that you demonstrate your ability to identify, and prioritise, the key issues or features revealed in the data.
- In the case of line graphs or scatter graphs you need to ensure that you have identified what the axes of the graph are showing, as this will be central to understanding the data presented. You also need to review the graph to identify any trends presented in the data provided. The aim here is to pick out relevant points from the graph which will support your narrative answer. The March 2017 exam featured a company called Ignite plc, which made a range of products including cigarette lighters, multi-purpose lighters, and lighter fuel. Candidates were provided with a line graph (in addition to company background information and financial performance data) which showed the sales performance of Ignite's luxury cigarette lighters over a 60-year period:

Figure 12.6: Luxury cigarette lighters sold



- Candidates were required to use the graph, alongside the other data in the question to identify the key changes in demand for Ignite's products over the period. The graph clearly shows the downward trend in the sales volumes of Ignite's luxury cigarette lighters, from their peak in the 1960s. This analysis could be extended further by drawing upon the additional information provided in the scenario which highlighted the growing health concerns of smoking cigarettes. Reduced demand for cigarettes had in turn reduced demand for cigarette lighters. As this example illustrates, you need to be comfortable combining data presented in a visual format with the supporting narrative in the question, to produce a good answer.
- As we highlighted in the chapter Risk management when we considered the pie charts from Question 1 in the March 2019 exam (Gadgy Ltd), it is important that you pay careful attention to what it is that you are expected to do per the question requirement as this will direct you as to how any pie charts, graphs and diagrams should be used. This is particularly important where more than one chart, graph or diagram has been provided, as you will be expected to integrate them in your answer. The first requirement in the March 2019 exam required an analysis of a company's product portfolio and competitive positioning. The company in the question scenario manufactured two different types of wearable technology, and the scenario included two pie charts which detailed the market share of competing firms which sold the two types of product. This should have been a clue that drawing out the key points from both charts was necessary when analysing the competitive position of the company.
- However, where charts, graphs, diagrams and tables are provided in questions, the examiners do not want you simply to repeat or reformat the same data. Following the March 2019 sitting, the examiner highlighted that a number of weaker candidates converted the data that had been provided in the pie charts into tables, or recalculated the market share figures which were given. As such these candidates wasted their time and earned no marks. Instead, the examiner wants you to analyse and refer to the data to help you provide valuable insights to an organisation's management, about the organisation's performance.

- When presented with data in different formats it is important that you give some consideration to the party that prepared it. In the chapter, 'Introduction to the Business Strategy and Technology exam,' we highlighted the fact that in the exam you will need to display some professional scepticism when evaluating the quality of the charts, graphs and diagrams provided. For example, if a pie chart has been produced by an inexperienced member of staff, without any guidance or review, then it would be reasonable to question the quality of the chart and the extent to which you will be able to rely on it when preparing your answer.

7.2.7 Developing data analysis skills

Developing your skills in analysing and interpreting data in different formats will be of particular importance as you progress to the Advanced Level of your ACA studies. The Business Strategy and Technology Question Bank contains a variety of questions with data presented in a range of different formats for you to attempt. As such you are strongly advised to attempt as many of these as possible in the lead up to sitting your exam.

8 The data related requirements in the exam



Section overview

- As we discussed in the chapter Introduction to the Business Strategy and Technology exam, every exam may contain up to four different data related requirements: data analysis requirements; specific calculations, supporting calculations and a requirement asking candidates to assimilate and analyse granular data provided in a pre-populated spreadsheet.
- Questions may include financial data or alternatively they may include some management data which may require analysis and some supporting argument or judgement to comment on the situation.
- Regardless of the type of data related requirement that you get in the exam your answer should always be linked to the question scenario.
- This section looks at the types of requirement you may have to deal with, the key skills you need to develop and the mistakes to avoid to be successful.

8.1 Data related requirements

As we highlighted in the chapter, 'Introduction to the Business Strategy and Technology exam,' data and data analysis are key features of the exam. As such you need to be comfortable in handling the data related question requirements which will be tested in every exam.

These data related question requirements can be broadly grouped into four types:

- A data analysis requirement, requiring candidates to analyse a range of data in various exhibits, produce some relevant calculations and provide a commentary to go with their numerical analysis. This type of requirement might include, for example an analysis of a company's product portfolio and competitive positioning, an analysis of performance over two financial periods, or a comparison of the featured entity's actual financial results against its budgeted results. Such questions may also require the analysis of performance with reference to operational, non- financial key performance indicators. Questions may include data which cover different periods of time, for example, an entity's quarterly results rather than its full year results. Data may also relate to the whole entity featured in the question scenario or it may focus on particular business units, product streams, individual stores or branches.

- Performing a specific calculation as specified by the question requirement. Common examples of this type of calculation include computing a break-even working for a particular project, or preparing a profit forecast for a proposed business venture. We looked at the types of specific calculation that you may encounter in the chapter Risk management.
- A 'show supporting calculations' requirement, where you are required to support your narrative answer with supporting calculations. Such questions may require an analysis of the financial merits of two proposed strategies, or an assessment of the performance of a given strategy. Requirements which request supporting calculations require candidates to use their judgement in identifying the most appropriate type of calculation to be performed. Again, we looked in the chapter Risk Management at the types of supporting calculations that you might need to perform when faced with such a requirement. Further consideration is given to this in the section Showing supporting calculations of this chapter.
- A requirement to assimilate and analyse granular data provided in a pre-populated spreadsheet. You will need to use spreadsheet functionality to analyse the detailed granular data before commenting on the data and drawing appropriate conclusions. Examinable spreadsheet functionality and example questions can be seen in the chapter Spreadsheet functionality.

It is also important to note that some question requirements in the Business Strategy and Technology exam will not specify how the data provided in the scenario is to be used. In such cases you are required to use your judgement to determine how to make the best use of the data provided when constructing your answer. However, a key point to reiterate is that if you are provided with data in a scenario, then the examiner expects you to use it. Failure to draw upon the data provided will restrict the number of marks that can be earned. In all types of data question, you should make use of the spreadsheet software and functionality to improve the speed and accuracy of the calculations performed. All analysis should be copied from the spreadsheet area of the exam and pasted into the question response area for marks to be awarded.

This section provides a useful insight into types of data that you are likely to encounter in your exam, and the skills that you will need to display to be able to use it effectively.

8.2 Using data in the exam

A key skill that you will need to demonstrate Business Strategy and Technology exam is your ability to interpret financial and other data provided in question scenarios and to relate this to the featured entity's situation. This skill is relevant to answering the different types of data related requirement discussed in section Data related requirements above.

As we highlighted in this chapter, data in the exam may be presented in a number of formats and relate to a number of different issues including:

- The financial position of the business (eg, basic groups of key financial ratios (profitability/liquidity/gearing))
- The performance of the business – financial and non-financial (eg, KPIs/balanced scorecard)
- The use of resources – economy, efficiency, effectiveness
- Product/market positioning (eg, market growth/share/BCG/life cycle)
- Risk (eg, break-even calculations, best- and worst-case scenarios, sensitivity analysis, expected values)
- The financial impact of a new project or strategy (eg, analysing the effect by considering incremental contribution)
- Evaluating the overall impact of a strategy in both quantitative and qualitative terms

- Operational and resource data over time (eg, units of output, number of branches, floorspace, number of employees)
- Forecasts and budgets – forward looking data which, for example, may reflect the expected impact of a strategy or be used to monitor a strategy after implementation

Interactive question 6: Staff bonus

Xtra Recruitment is a small firm specialising in law and finance recruitment.

It pays a monthly bonus to the three highest revenue generating employees, based on placement fees.

The bonus is calculated as 1% of their monthly fee earnings. The sales data for January is as follows:

	A	B	C
1	**Employee ID**	**Monthly placement fees**	
2	693	13,860	
3	392	17,820	
4	894	25,097	
5	774	13,120	
6	761	27,526	
7	432	62,411	
8	757	59,547	
9	180	17,838	
10	661	46,831	
11	340	79,724	
12	237	42,310	
13	624	65,458	
14	810	55,881	
15	349	53,290	
16	803	44,031	

Requirement

Using the extract from January's sales data, determine which employees will receive the bonus and what the value of the bonus is.

*See **Answer** at the end of this chapter.*

8.3 The skills required to use data effectively

When you are provided with data in exam questions, the examiners are looking for you to demonstrate the following skills:

- Choosing analytical tools that are appropriate in the context of the question – eg, financial ratios, KPIs, BCG analysis, break-even calculations
- The ability to use spreadsheet functionality, where appropriate, to assimilate and analyse spreadsheet data quickly and accurately

- Identifying analysis that would be appropriate/useful and carrying out the relevant calculations
- Interpreting the resulting information to demonstrate an understanding of the story behind the numbers and communicating that analysis succinctly
- Exercising judgement to draw conclusions and/or produce sensible recommendations The examiners are looking for you to justify any conclusions or recommendations that you provide. This means avoiding just stating your recommendations, as you should be confident in briefly explaining why you have recommended a particular course of action.
- Looking beyond the information provided at what additional information may be useful to generate a better analysis/understanding and at any reservations regarding the data/techniques/assumptions applied Central to this is the need for a degree of professional scepticism when analysing the data provided in a question. You should recall from your Audit and Assurance studies that professional scepticism requires a questioning mind and to not simply accept information at face-value. When detailing additional information that you could request, you need to ensure such information could be realistically obtained.
- Linking different pieces of data to explain trends or outcomes
- Highlighting weaknesses or omissions in the data provided
- Analysing cause-and-effect relationships – eg, identifying underlying causes of changes in the data

You need to be mindful of the advice given in the chapter, 'Introduction to the Business Strategy and Technology exam,' concerning the use of the exam software when presenting your numerical and narrative answers. Remember that the examiner can only see your answers as they appear on the screen. The examiners cannot see the formulae that you have used. It is therefore important that you present your numerical workings in an appropriate format ie, in an easy-to-follow table with appropriately labelled rows and columns, and ensure that all narrative and copied spreadsheet workings are clearly visible so that the examiners can fully review your answer. Furthermore, you will not be expected to reproduce calculations and figures already provided in the question scenario.
Failure to follow this advice in the exam may cost you marks.



Professional skills focus: Applying judgement

Be prepared to question the data provided. Apply professional scepticism and judgement. Question the source of the data to ascertain whether it is complete and accurate or could be subject to bias.



Worked example: XY data

XY is a small company that makes and sells two specialist components the A and the B. Sales areas are in Spain, Germany and Italy but production occurs in the UK. Two years ago, a sales management system was introduced in an attempt to improve management information.

Sales are made in Euros but are recorded in the sales management system in £. The sales manager for each area receives an annual bonus in September (after the European summer holidays have finished) based on these £ revenues.

XY's production team estimate monthly production requirements by reviewing last year's sales data. They keep zero inventory. Recently the production team has been criticised because components are out of stock when the sales team try to ship them to customers. The

production team complain that this is the fault of the sales team because the sales information is not accurate.

Data available in the sales system is shown below:

	A	B	C	D
1	Month	Country	Product	Sales revenue £'000
2	Jun	Spain	B	54
3	Aug	Germany	B	72
4	Jun	Germany	B	12
5	Jun	Italy	B	23
6	Aug	Italy	B	78
7	Jun	Italy	A	32
8	July	Germany	A	3
9	Jun	Spain	B	31
10	Aug	Germany	A	105
11	July	Italy	A	6
12	Aug	Spain	A	44
13	Aug	Italy	A	79
14	Jun	Germany	A	31
15	Jun	Italy	B	23
16	Aug	Spain	B	104
17	July	Spain	A	7
18	Aug	Germany	B	13

Requirement

Analyse the data provided and assess the problems with the current approach to recording and using sales data.

Identify additional information that would be useful in assessing the sales data.

Solution

An analysis of the data has been carried out using the SUMIFS functionality.

To ensure that formulae have captured all the relevant data it is good spreadsheet practice to reconcile spreadsheet numbers. For example total sales revenue in D19 should reconcile to Cell K8.

The formula in Cell H2 looks at the sales revenue (column D) for June (column A), for Spain (column B) and for Product A (column C).

=SUMIFS($D$2:$D$18,$A$2:$A$18,"Jun",$B$2:$B$18,"Spain",$C$2:$C$18,"A")

	A	B	C	D	E	F	G	H	I	J	K
1	Month	Country	Product	Sales revenue £'000		**Country**	**Product**	**June**	**July**	**Aug**	**Total**
2	Jun	Spain	B	54		**Spain**	A	0	7	44	51
3	Aug	Ger	B	72			B	85	0	104	189
4	Jun	Ger	B	12		**Italy**	A	12	6	79	117
5	Jun	Italy	B	23			B	46	0	78	124
6	Aug	Italy	B	78		**Ger**	A	31	3	105	139
7	Jun	Italy	A	32			B	12	0	85	97
8	July	Ger	A	3		**Total**		206	16	495	**717**
9	Jun	Spain	B	31							
10	Aug	Ger	A	105							
11	July	Italy	A	6							
12	Aug	Spain	A	44							
13	Aug	Italy	A	79							
14	Jun	Ger	A	31							
15	Jun	Italy	B	23							
16	Aug	Spain	B	104							
17	July	Spain	A	7							
18	Aug	Ger	B	13							
19	**Total**			717							

Problems with XY's sales data

(1) Incomplete data

Some products appear to have no sales at all in July suggesting that the sales data recorded by the sales managers could be incomplete. For example, in Spain there are no sales revenues for product B in July despite there being £85,000 and £104,000 in June and August respectively. The lack of sales may be attributable to the annual European holidays but could also be caused by delays in sales managers recording the sales in the system.

(2) Overstated data

The sales figures in August are significantly higher than in the months of June and July. Whilst sales teams may have made more sales in August because the summer holidays are over, it could also suggest that as many sales as possible are being pushed through in August in order to secure a higher bonus.

(3) Production problems

The production team are relying on data that is potentially inaccurate. For example, if sales occur in July but are not recorded on the system due to sales manager holidays,

components will be out of stock when customers require them. Inaccurate and incomplete data could result in the loss of customers in the long term.

(4) Lack of integrated objectives

The system has been set up to meet the needs of the sales team but does not address the requirements of the production team who are struggling to forecast production requirements. The current system produces inaccurate management information affecting decision making and may also result in dissatisfied customers if stock is not available when required.

Additional information

- Annual sales figures for the last three years will help XY to understand overall sales trends to assess whether the sample data provided is in line with previous years. For example, are sales always lower in July than in August?
- Exchange rate information used to record the sales revenue would allow XY to assess whether all the sales managers are recording their sales at the same Euro/£ exchange rate. This could influence the bonus received.
- Clarity should be sought over the different objectives and responsibilities in relation to the system. For example, are sales managers measured on the accuracy and timeliness of the information they input to the system? Is the production team penalised for failing to meet customer demand?

Tutorial Note

Be prepared to question the data provided and apply professional scepticism and judgement. After using spreadsheet functionality to analyse the data assess whether it appears complete and accurate. For example, is it realistic that some products have no sales in July even though they have large sales in June and August? Consider why sales suddenly peak in August. Could the data be manipulated in any way? The scenario hints at problems with the sales data so use these to guide your answer.

8.4 Key weaknesses in candidate scripts

The following list highlights the weaknesses most commonly identified in exam answers which require the use of data and suggests ways to address these:

(a) Restating facts or numbers without applying them to the context of the question. A common failing is to explain **what** has happened rather than **why**, eg, stating that sales have grown by 15% in the period but not indicating why they have done so.

Solution: Use the '**because rule**' to add value.

In most cases a good answer passes the 'because' test, eg, 'Market share has fallen from 35% to 29% **because** of the entry of a new lower-cost competitor.'

Including the word 'because' in your answer changes the 'what' into a 'why.'

The 'why' should be related to specific information in the scenario demonstrating that you have understood the relationship between the financial/quantitative data and information and the business issues.

(b) Failure to use the additional data and information from the scenario in formulating answers resulting in generic answers that could apply to any company rather than the one in question.

Solution: Make specific points, focusing on the particular organisation and relating to the circumstance and data set out in the scenario.

For example, 'falling R&D expenditure may be a problem for XYZ Ltd because it has built market share on the basis of innovative products.'

(c) Interpreting data/figures/results in isolation

Solution: Link the data and analysis, eg, if market share has increased but gross margins have decreased, the company may have made a decision to reduce the selling price as part of a market penetration strategy. For example, a company is considering outsourcing production as a result of rising raw material prices. This might be expected to lead to a decrease in gross margin if the company is unable to pass on the price increase to customers or a loss in sales/market share if, as a result of passing on the price increase, the company's product is no longer competitive. Outsourcing mat also lead to a reduction in fixed costs and hence lower operating gearing (risk), but this would depend on the nature of the outsourcing contract.

(d) Focusing on a narrow range of measures

Financial measures alone will not provide the full picture and are often the result of other factors, not the cause.

Solution: In addressing performance issues your answer should, where possible, address a variety of performance indicators. Consider using the balanced scorecard headings to help you cover a wider range of measures. Remember that these measures will often help you understand what is causing the strategy to succeed or fail.

(e) Failure to use data analysis elsewhere to support your answer

Often the data analysis element is one of the first requirements. Conclusions that you draw from this will help in answering later parts of the question.

Solution: Consider where else in your answer the analysis may be relevant/how you can 'make the numbers talk.'

For example, if you have been asked in another requirement to assess the appropriateness of a strategy that is being considered, you can refer to earlier calculations of profitability and other measures in your assessment of returns and/or break-even calculations when looking at risk.

For example, if the data provided shows that the business is currently loss making and that sales and profitability are forecast to decline further, then the business cannot afford to do nothing.

Any new strategy that is expected to address this decline or increase returns should be acceptable to the shareholders.

(f) Failure to explain trends in the data by identifying **cause and effect relationships**. This may, for example, include analysing information into ratios or percentages based on the data provided.

For example, where sales revenues are growing by (say) 10% per year but the number of branches/outlets is growing by 15% per year then calculating the sales per branch/outlet will show that sales per branch are, on average, falling. Growth in overall sales revenue may therefore be due to greater investment (ie, more outlets) rather than generic growth in sales per branch because of improved efficiency or stronger market conditions.

For example, an alternative analysis of sales revenue growth might be in relation to volume growth, changes in selling prices and changes in sales mix. Analysis of this data may reveal the relative causes (quantitatively) of sales revenue growth from each of these underlying factors.

(g) Failure to get a **reasonable balance** between numerical analysis and descriptive analysis. Some weak answers are almost entirely descriptive, while other weak answers go to the opposite extreme, producing reasonable numbers but descriptive analysis that is little more than briefly stating which numbers have gone up and which have gone down.

Solution: Both numerical and descriptive analysis are important and need due emphasis. Two approaches by candidates in this respect are:

(1) to set out a comprehensive numerical analysis at the beginning of the answer with workings (eg, in a table) then produce the descriptive interpretation of these numbers; or

(2) to mix the numerical analysis with the descriptive analysis by producing calculations as each issue arises.

In general, approach (1) tends to produce better answers with a more systematic evaluation of the issues. Specifically, with approach (1) the numerical analysis tends to be more comprehensive and better thought out with clearer workings.

(h) Failure to use spreadsheet functions accurately and spreadsheet functionality fully. It is also important to copy spreadsheet work from the spreadsheet software and paste it into the word processing area of the exam so that marks can be awarded.

Solution: Ensure you understand and practise using the spreadsheet software and examinable functionality. Refer to the chapter Spreadsheet functionality to understand the functions and types of questions you are likely to see. Familiarise yourself with the software using the ICAEW practice software.

8.5 Additional information required

The examiner may ask you to highlight any additional information that might be necessary for a more meaningful analysis and on occasions you may be required to specify possible data sources. Even if this is not part of the requirement, a comment about the limitations of your data analysis and any additional information required will demonstrate higher skills. It is important however that requests for additional data are realistic, relevant and obtainable.

- Data and information in the scenario is likely to be in the form of a summary.

  More detailed data/information may be required – eg, a breakdown by product, country, business unit. If you have been provided with historic information for analysis purposes (eg, at five-yearly intervals) data for the years in between would help assess the trend more accurately. If average figures have been given, then actual prices/volumes/data would be necessary to properly identify which products/business units are under-/over-performing for instance.

- The source of data/information can be both external as well as internal.

  To properly interpret performance, it is necessary to make comparisons. Comparisons may be made between:

  - Current results and past results/budgets
  - One business unit and another
  - Industry/sector information
  - Market leader/other competitors

  Thus, any data obtainable on the industry and competitors will help benchmark the performance of the business.

- Set out any assumptions made and explain any limitations of these.

  To carry out break-even analysis for example you will have to make certain assumptions about the split of costs between fixed and variable, and about the sales price and costs remaining constant.

9 Developing an approach to data analysis questions

Section overview

- To help you develop an approach and a range of skills when attempting data analysis style exam questions, this section leads you through a sample question 'The Legume.' The intention is to demonstrate the thought processes to be adopted when tackling such a question and the attributes of an answer that will score well in the exam.
- Once you have worked through this section, you will find further examples of data analysis questions contained in the Business Strategy and Technology Question Bank.

9.1 Recommended approach to data analysis requirements

The following is a suggestion for the approach to adopt when tackling questions:

Step 1 Initial review (of scenario and requirements).

Step 2 Decide on appropriate analysis to undertake.

Step 3 Produce the necessary calculations (to form the appendix) making use of spreadsheet functionality as much as possible to speed up your work and increase accuracy. Be careful to use absolute cell referencing when copying and dragging formulae across cells. Also be consistent and sensible with the number of decimal places you work to, and format cells appropriately using accounting £, $ or %.

Step 4 Interpret your analysis.

Step 5 Identify additional information required.

9.1.1 What-How-Why-When-So What analysis

The parts that cause most problems are Steps 3 and 4. It is difficult to produce a universal approach but one tool which draws together many of the issues already noted above is: **What-How-Why- When-SoWhat analysis** (WHWWsW), which includes both numerical and descriptive elements.

- **What** – looks at WHAT has happened overall (eg, revenue has increased by 10.5%).
- **How** – attempts to analyse the available data in more detail to obtain a better understanding of HOW the WHAT element occurred (eg, sales prices have risen by 10% and monthly sales volumes have risen by 10% following a price change).
- **Why** – looks for the underlying causes of the HOW element, which may be part of the story or strategy provided in the question (eg, there have been significant product improvements introduced during the year with additional features compared to competitors which has enabled a higher price to be charged but also an increase in demand despite this increased price).
- **When** – in assessing the impact of changes in strategy over time or in making comparisons it is important to know WHEN changes occurred. Thus, for example, in assessing the

impact of a price change it is important to know when during the year the price change took place. If the price change occurred, for example, halfway through the year, then the impact on the current year may be reduced, but may be more significant on next year's trading. For example, using the above numbers if the change in the product, the price and the sales volumes only occurred halfway through the year it would only impact on two of the four quarters resulting in a sales revenue increase of only 10.5%, rather than the 21% that might be expected for a full year.

- **So What** – the above steps analyse and interpret the nature of the data provided and attempt to identify and explain the underlying causes of any changes in the data. This step goes to the next stage by attempting to ask the question 'SO WHAT are the consequences of our analysis for deciding on the future business strategy'? (For example, what are the consequences for profit of the 10% increases in sales price and sales volume after considering the variable cost increases arising from the product improvements and volume increases? How have competitors responded with price changes and improvements in their own products which may make the consequences next year different from those which occurred this year?)



Worked example: Snowdon Ltd

This is an example with a very simplified application of WHWWsW analysis.

Snowdon Ltd is the sole UK importer of a branded cast iron cooker. To penetrate the market, it reduced its selling prices by 5% on 30 June 20X5. There have been no changes in total fixed costs or variable costs per unit. Prior to the change in price, monthly demand had been constant for some time.

The Snowdon board is attempting to assess the impact of this change in strategy for the year ended 31 December 20X5 and provides the following data:

	20X5	20X4
	£'000	£'000
Sales	21,400	20,000
Cost of sales	(11,000)	(10,000)
Gross profit	10,400	10,000

Requirement

Analyse the change in performance of Snowdon in 20X5.

Solution

What: Gross profit has increased by £400,000 in absolute terms, but gross margins have fallen from 50% in 20X4 to (10,400/21,400) = 48.6% in 20X5.

How: Sales revenues have risen by 7%. This includes a 5% price reduction so there must have been a significant increase in sales volumes to compensate and cause an overall increase in sales revenue.

Note: There is only one product, so sales mix is not a factor in explaining HOW the change occurred. Also, see the **when** question below, as the price change only occurred halfway through the year.

Why: Sales prices have been reduced to sell greater volumes to pursue a strategy of penetrating the market.

When: As the price change only occurred half way through the year the impact is restricted to this period. Given we are told that the first half of the year was similar to 20X4 we could do some analysis to isolate the second half effect in 20X5. If we take the first half of 20X5 as similar to half of 20X4 and strip this out we obtain the following:

	Actual full year 20X5	Half of 20X4 (ie first half of 20X5)	Imputed second half 20X5
	(A)	(B)	(A)–(B)
	£'000	£'000	£'000
Sales	21,400	10,000	11,400
Cost of sales	11,000	5,000	6,000
Gross profit	10,400	5,000	5,400

This table enables us to compare the two half years of 20X5. This shows that sales revenues have risen by 14% from £10m for six months to £11.4m. Given prices have fallen by 5% this implies that sales volumes have risen by 20% (ie 1.14/0.95 = 1.2). This 20% is consistent with the increase in cost of sales.

Note: (This is because Snowdon is a retailer and hence cost of sales are variable costs. If Snowdon had been a manufacturer, then cost of sales would not increase in a linear relationship with output as there would be an element of fixed costs.)

So What: More information is needed to assess the impact of this strategy in the longer term (eg, market share). The response of competitors needs to be assessed. Snowdon's price reduction may not have been immediately followed but if the company is taking market share then competitors may decrease prices after a delay. In extrapolating the consequences of the price change forward care therefore needs to be taken in allowing for different market conditions. Also price penetration is a temporary policy, but if competitors follow the price decrease of Snowdon it may be more difficult to increase them in future.

Summary

It is too early to judge the new strategy yet but in terms of short-term financial performance there appears to have been a benefit in higher absolute profit despite the fall in percentage margin.

Note, however, that any tool needs to be used flexibly. In this case the HOW and WHEN questions could have been dealt with jointly.

The following is a longer question with more detailed analysis and shows that an understanding of the issues is needed and that any tool needs to be applied to purpose. It should not dominate thought and application to the scenario.



Worked example: The Legume

The business profile

'The Legume' is a prize-winning vegetarian restaurant located near the centre of Barchester, a cathedral city with a population of 100,000.

The restaurant was opened in January 20X0 by its proprietors, John and Eleanor Bold, who share responsibility for preparing the food and managing the restaurant. The staff comprises a full time chef, a kitchen assistant, four full-time and two part-time waiting staff.

'The Legume' seats 60 customers and is open six days a week, both at lunchtimes and in the evenings. It is licensed to sell alcohol and has established a reputation for having good wines.

Business is traditionally best at weekends, but activity is seasonal. Sales are above average in the four weeks in the year when there are meetings at the local racecourse, and the restaurant is equally busy before Christmas and during the summer tourist season.

The history of the business

John Bold commenced the business after investing £50,000 that he had inherited from his parents. Located on the ground floor of a Georgian building near the cathedral close, the restaurant attracted customers from among the many tourists visiting the city, but in addition its clientele included a small core of regular customers, local businesspeople and shoppers.

Up until 20X5, the business was doing well, turning in good profits. However, in 20X6 there was a 20% decline in the number of overseas tourists visiting Barchester as a result of a strong UK pound. In addition, the local economy suffered when one of the city's largest employers decided to move its production overseas and was forced to lay off sizeable numbers of staff.

Although there was some revival in the number of overseas tourists visiting Barchester in 20X7, this coincided with a gradual erosion of the popularity of 'The Legume' amongst its local clientele. This was largely because rival restaurants and pubs had begun to include vegetarian dishes on their menus, but also because a new and inferior chef had been appointed, the decor was becoming rather faded, and the novelty value of the restaurant had worn off. Another factor, however, was the closure of the city's two cinemas, which were replaced by two multi-screen complexes sited in new out-of-town shopping centres.

As a result, there was a further sharp fall in trade in 20X7, forcing John Bold to turn to the bank for increased funding in the form of an increased overdraft facility. Initially support was forthcoming, since it was widely assumed that the setback would be temporary, and the bank took a second mortgage on the Bold's home as security. However, results in 20X8 were only slightly better and the Bolds were forced to cut their drawings significantly to keep the overdraft within reason.

Extracts from the management accounts for 'The Legume' restaurant over the last four years:

	20X5	20X6	20X7	20X8
	£	£	£	£
Sales				
Food	170,000	149,750	155,400	154,000
Drink	85,000	74,875	81,400	87,500
Total	255,000	224,625	236,800	241,500
Cost of sales				
Food	68,000	59,900	62,160	61,600
Drink	42,500	35,940	36,630	35,000
Total	110,500	95,840	98,790	96,600
Gross profit				
Food	102,000	89,850	93,240	92,400
Drink	42,500	38,935	44,770	52,500
Total	144,500	128,785	138,010	144,900
Other costs				
Wages	74,000	71,000	74,000	76,000

Rent	14,000	15,000	15,000	15,000
Depreciation	3,000	3,000	3,000	3,000
Interest	0	3,000	4,000	2,000
Other expenses	17,000	22,000	20,000	24,000
Total	108,000	114,000	116,000	120,000
Net profit	36,500	14,785	22,010	24,900
Average spend per customer				
Food	£10	£10	£10.50	£11
Drink	£5	£5	£5.50	£6.25
	20X5	**20X6**	**20X7**	**20X8**
	£	**£**	**£**	**£**
Non-current assets at carrying amount	21,000	18,000	15,000	12,000
Bank overdraft	1,000	14,000	18,000	10,000
Drawings	33,000	30,000	25,000	20,000

The Bolds' dilemma

Somewhat belatedly the Bolds have begun thinking about how they might turn the business round. John Bold has been considering various options, each of which would be feasible under the existing 25-year rental agreement for the premises and under the current licensing arrangements:

Options

(1) The restaurant could be converted into a wine bar serving hot and cold vegetarian and fish dishes. This might attract a younger clientele which at present largely patronises the local pubs and fast food restaurants.

(2) By redesigning the layout a fast food, but vegetarian, menu could be offered at lunchtimes, while in the evenings the restaurant could revert to its more traditional role. However, to try to increase the volume of trade, prices would be cut and further discounts offered for regular customers, those eating midweek, and those starting their meals before 7pm.

As part of the reorganisation, the restaurant would have to be redecorated and new kitchen equipment installed. Option 1 would make it possible to dispense with full-time waiting staff and – apart from the chef – only employ part-time staff. Likewise, Option 2 would make it possible to dispense with one of the full-time waiting staff and instead employ two part-timers at lunch times.

The Bolds estimate that their pre-tax joint income in alternative employment would total £35,000 a year. Any additional finance requirement would be provided by a loan from Eleanor Bold's family.

Requirements

You are a business consultant. The Bolds have asked for your help in assessing the current situation and deciding on a new strategic direction for The Legume.

(a) Using the data and other information provided, explain how the business has performed since 20X5.

(b) State what additional information would be useful to provide a more detailed assessment of performance.

(c) Evaluate the risk and potential rewards associated with the strategic options being considered.

Solution

Step 1 Initial review

Read the requirements and then read through the question. As you read, you should be thinking about:

- Context
- Requirements
- Key issues
- Purpose of analysis
- Additional information

Context

You have been employed as a business consultant by the owner managers of the restaurant.

Requirements

Part (a) requires an analysis of performance since 20X5. The phrase 'using the data and other information provided' indicates the need to include both quantitative analysis and information drawn out of the scenario (see key issues) in formulating your answer.

In part (b) you are required to state additional information that might be useful to you. Jot down anything that comes to mind as you read through the scenario.

Part (c) asks for an evaluation of the strategic options being considered under the heading of risks and rewards. This evaluation will need to be undertaken in the light of your answer to (a). Thus if your performance analysis identifies issues such as a lack of sales volume or high fixed costs then it will be important to evaluate whether the options address this.

Key issues

- As a prize-winning vegetarian restaurant The Legume appears to be operating a focused differentiation strategy.
- Whilst initially profitable it now appears to lack competitive advantage (inferior chef, faded décor, other competitors offer vegetarian food etc).
- The business is highly seasonal and also very dependent on the state of the economy and the level of tourist trade.
- The high level of fixed costs, combined with falling sales volumes, have led to deteriorating profitability.
- The two key stakeholders could earn more elsewhere (£35k) than they are currently earning from the business. Also their home is at risk as a result of the bank taking a second mortgage.
- A new strategy is urgently needed for the business to survive.
- As a small business, sources of finance are presumably limited and if Eleanor's family are unable/unwilling to provide the necessary funds, the bank will need convincing of the Bolds' ability to turn the business round.

Purpose of analysis

Here we are told in the scenario, and we can see clearly from the accounts extracts, that in terms of profitability, the business performance has deteriorated since 20X5. The analysis should seek to address the underlying reasons for the decline and identify areas that need to be addressed by the new strategic proposals.

Additional information

The information provided is mainly financial, in the form of a statement of profit or loss and a few highlights from the statement of financial position. We also have figures for average spend by customer.

There is little by way of non-financial performance indicators and no external information on competitors/the market sector.

Step 2 Decide on appropriate analysis to undertake

Much of the WHAT analysis is already provided in the question but in answering the HOW question and in deciding on what analysis to undertake and which calculations to perform it is important to consider the following:

- How does the analysis help to identify the issues the company is facing?
- How does the analysis demonstrate an understanding of the scenario?

If we look at the information from the statement of profit or loss extract, the following analysis should help to confirm the reasons behind the declining performance:

- Sales mix
- Sales volumes (using the information on average customer spend)
- Sales growth
- Margins
- Operating gearing (level of variable costs in relation to fixed costs)

We might also want to analyse the other financial information provided:

- Non-current assets
- Drawings
- Bank overdraft

Finally, there is sufficient information to undertake a break-even analysis, which would help to quantify the risk of the business becoming loss making. Operating gearing is also a good measure of risk; if a high proportion of total costs are fixed, any reduction in activity takes the business closer and closer to being loss-making.

Step 3 Produce the necessary calculations

This will form the appendix to your answer. Appendix 1:

	20X5	20X6	20X7	20X8
Sales mix	**%**	**%**	**%**	**%**
Food	66.7	66.7	65.6	63.8
Drink	33.3	33.3	34.4	36.2
Sales growth				
Food	n/a	–11.9	+3.8	–0.9

	20X5	20X6	20X7	20X8
Sales mix	**%**	**%**	**%**	**%**
Drink	n/a	-11.9	+8.7	+7.4
Total	n/a	-11.9	+5.4	+2.0
Gross profit margin				
Food	60	60	60	60
Drink	50	52	55	60
Total	56.9	57.5	58.4	60.0
% Increase in fixed costs	n/a	5.6	1.8	3.4
Net profit margin	14.3	6.6	9.3	10.3
Operating gearing %	49.4	54.3	54.0	55.4
Other analysis				
No. of customers served (Sales/average spend)	17,000	14,975	14,800	14,000
Reduction in customers		11.9%	11.7%	5.4%
Avg. contribution per customer (GP/Customers)	£8.5	£8.60	£9.33	£10.35
Break-even no. of customers	12,706	13,256	12,440	11,594
(Other costs/average contribution)				

When faced with a requirement which asks you to perform some type of data analysis it is important to not only focus on the financial figures provided, as these will only tell part of the story.

There are three ways in which ratios and calculations can be considered when attempting questions which require some form of data analysis:

(1) **Using two financial figures** together, for example, the sales and gross profit figures, to determine the gross profit margin for a period. As the appendix above illustrates 'The Legume' has seen its gross profit margin rise from 56.9% in 20X5 to 60% in 20X8. This has been caused by annual rises in the gross profit margin of drink.

(2) **Comparing one financial and one operating figure** with one another, for example, the average contribution per customer, which could be determined using the gross profit figure divided by the number of customers served by the organisation. Over the period the average contribution per customer has risen from £8.50 in 20X5 to £10.35 in 20X8. This is promising, especially as the number of customers has in fact fallen over this period.

(3) **Comparing two operating figures**, for example, the number of customers served per year. As mentioned above, the number of customers served between 20X5 and 20X8 has fallen each year. This would match the detail provided in the scenario.

Performance analysis requirements such as this provide you with the opportunity to show that you are confident in using all types of data in constructing a well-balanced answer. In the following section we consider the importance of interpretation in greater detail.

Step 4 Interpretation

In conducting the WHY analysis it is important to look at the reasons behind the performance. Here are a number of areas that you might find it useful to consider:

- Trend in revenue – mix/volume/price. How achieved if rising /cause if falling?
- Pattern of costs against revenue – rising or falling; are they fixed or variable?
- Is the business seasonal? Does this place a strain on cash flow or lead to periods of under-capacity and a lack of recovery of fixed overheads?
- What is the level of operating gearing? Is this an issue for the business, eg, because of the seasonality of sales or in view of falling demand?
- Is the level of profit satisfactory/stable/increasing/decreasing?
- Is the company earning an appropriate return for its stakeholders? Is it paying this out in the form of dividends?
- What is the organisation's long-term finance strategy and is it appropriate for the business?
- What are the strategic implications of the level of investment in R&D/HR and training, marketing and non-current assets? Is this consistent with the organisation's generic strategy? For example, if a company is following a differentiation strategy, it is likely to need to invest in at least some of these areas.

Interpreting the analysis from Step 3

This will form the text of your answer to requirement (a).

Sales growth/volumes

Sales growth figures indicate that sales have fallen by almost 12% between 20X5 and 20X6 and then risen by 5% and 2% respectively in the following two years. It is important to recognise that these movements could be the result of a price or volume change or a combination of the two.

Here we can use the figures provided on average spend per customer to deduce that the drop in revenue between 20X5 and X6 must be due to a loss of customers, because the average spend in the period remained constant.

In fact, we can go a step further and calculate the number of customers served in the year, which allows us to see that the number of people served fell from 17,000 to 14,975. This is consistent with the information in the scenario which explains that there was a 20% decline in the number of overseas tourists visiting Barchester as a result of a strong UK pound. In addition, the redundancies at the large local company are likely to have had an impact on the volume of local business.

The growth in sales in the following two years came at a time when the number of customers continued to fall. Although the volume of tourists increased, The Legume is unlikely to have benefited because of a loss of its competitive edge/differentiating factors (faded décor/inferior chef etc). It will also have suffered as a result of the reduction in customers due to the new out-of-town cinemas. Thus the overall increase in revenue in this period must be down to an increase in the average spend per customer.

Pricing and margins

Average food spend per customer, which was constant in 20X5 and 20X6, increased by approximately 5% per year in each of 20X7 and 20X8. This could be because customers were buying more eg, having a starter or dessert in addition to a main course, or it could be that the Legume is now charging more for the same food. If this has led to The Legume charging more than other restaurants in the town it could explain some of the loss of business.

Food margins have remained constant at 60%, which could be a result of good control of purchasing/ wastage etc. Alternatively, it may be that The Legume's pricing policy is to charge a fixed mark up on costs incurred.

The average spend on drinks has followed a similar trend although the increases have been higher at 10% and 14% respectively. Again, this could be that the price of the drinks has increased or that, attracted by the reputation of The Legume's wines, customers are buying higher quality bottles of wine.

The margins on drinks have increased from 50 to 60% over the period. This may be because the more expensive wines can be sold at higher margins or might reflect better purchasing policies which have led to reduced costs.

The sales mix appears to have remained relatively constant over the period, with a marginal increase in the significance of drinks sales (33% in 20X5, increasing to 36% in 20X8).

In 20X8 the margins on both products were the same at 60% so the sales mix is irrelevant to profitability. However, if the trend for increasing margins in the drinks side of the business continues, then increasing the sales mix in favour of drinks will have a positive effect on profits.

Other costs

The fixed costs have been increasing both in total and as a proportion of the total cost base.

This means the business has high and increasing operating gearing (moved from 49.4% fixed costs in 20X5 to 55.4% in 20X8) and as a result has profits that are more risky.

The restaurant is open twice a day, six days a week, and can seat 60 people.

Hence the maximum number of people eating per week is 720 which, assuming it is open 50 weeks of the year, amounts to a total potential volume of 36,000 customers pa.

Clearly, with 14,000 customers in 20X8, it is operating significantly below capacity but despite the downturn in business, the restaurant is unable to reduce its cost base significantly, since it only has two part-time staff.

There are other clues in the question that operating gearing might be an issue:

- The seasonality of trade
- The new suggestions for reducing the number of full-time staff and increasing the level of part-timers, thus attempting to make the cost base more flexible. (Notes that in other questions, companies may be attempting to do this by outsourcing either production or services.)

Investment in the business/returns to the owners

The question states that the Bolds could together earn £35k pa in alternative employment.

The profits of the business in all but 20X5 have been considerably less than this, forcing the owners to overdraw in both 20X6 and 20X7, presumably to maintain their personal lifestyle. The total drawings taken over the four-year period amount to £108k which exceed the £98,195 profits made.

Thus, none of the profits have been reinvested in the business and after the initial £50k investment it does not appear that the Bolds have provided any further capital. This may explain the state of the restaurant and the lack of investment in non-current assets.

As a small business, sources of finance are presumably limited. If the Bolds are unable to invest further in the business and if Eleanor's family are unwilling to provide the necessary funds, then the bank will need convincing of the Bold's ability to turn the business round.

Any new strategy will need to generate better returns or the Bold's might be advised to consider closing the restaurant. Having said that, it may be that the owners are prepared to sacrifice some of this alternative income as a result of the benefits they perceive from being their own boss, so it would be useful to ascertain the minimum level of income they need to live off.

Risks

Having signed a second mortgage, the Bold's house is at risk if the business fails or the bank withdraw the overdraft facility.

The break-even analysis indicates that a further 17% drop in sales (14,000 – 11,594/14,000) would cause The Legume to become loss making.

Conclusions

The high level of fixed costs, combined with the fall in the volume of business, has sharply cut into profits so that earnings are below what the owners could earn outside.

The reduction in the volume of business is not just a result of reduced consumer spending but appears to be a result of a change in the competitive environment. Other restaurants now offer similar food to the Legume, which appears to have lost its competitive advantage by not maintaining its differentiating factors.

In addressing the SO-WHAT question we can conclude that, if the business is to survive, it urgently needs to improve performance by:

- Reducing fixed costs or finding ways to convert these to variable costs
- Increasing the volume of trade
- Addressing the finance requirement

Note: In applying WHWWsW analysis, the WHEN question is implicit here in the wide time span of the data provided, so intra-year analysis is not really required.

Step 5 Additional information

This will form the text of your answer to requirement (b).

The following is indicative of the type of information that might help better assess the performance to date and indicate which of the proposals are likely to be more successful:

- Split of sales revenue by midweek/weekend/month/lunchtime/evening and between food and drink at these times. Also mix of meals and drinks served (eg, sales of wine split by price range: under £10/bottle; £10 – £15; over £20). This may help identify which of the new proposals is more suitable; for example, it may be that the lunchtime trade is typically food only, whereas in the evenings more drink is sold.
- Analysis of capacity utilisation/throughput of customers, ie, covers served v available tables at different times in the week and at different points in the year. The Bolds might be able to save money by closing the restaurant on certain days or at certain times of the year when trade is normally slow. Alternatively, this may indicate when to implement special offers.
- The figures quoted are for average spend per customer. This is likely to vary between lunchtime and evening and by type of customer, eg, business v tourist. A breakdown of the average spend by lunchtime and evening would be helpful, as well as details of any discounts offered at off-peak times.
- Whether the spend per customer has increased as a result of Legume increasing prices or whether they are trading up the sales eg, by persuading more customers to buy a coffee at the end of the meal.
- Competitor information regarding prices charged, average spend per customer, number of customers (although not all of this will be easily available).
- Data regarding the total potential customers, their demographics and spending habits.

For example, number of tourists visiting the town's restaurants each year, number of business customers, average age of local population, to help assess market share and suitable target markets.

- Minimum income level required for the Bolds to live off
- Maximum overdraft limit
- Cash flow showing the pattern of receipts/payments during the year to help better assess the short- and long-term financing requirement

Step 6 Sample answer for (c)

This will form the text of your answer to requirement (c). Strategic options for the business

Option 1: Conversion to a wine bar

Rewards

- Will capitalise on the good name built up for the wine cellar and it may be harder for others such as pubs to compete with this strategy.
- May increase profitability since margins earned on wines have increased to 60% in 20X8.
- The addition of fish to the menu may increase trade by attracting non-vegetarian customers.
- The business will not be totally reliant on vegetarian meals. However, there will be still no meat dishes on the menu.
- It may reduce reliance on the tourist trade and lessen the effect of uncontrollable factors such as the exchange rate.
- Only having part-time waiting staff should reduce operating gearing and hence profit volatility.

Risks

- If Barchester has an ageing population then this approach may not generate more trade ie, without cinemas etc, there may be nothing to attract young people to Barchester in the first place.
- The Legume may alienate its existing clientele by changing the décor/music/menu etc to appeal to young people.
- This strategy may simply replace one type of clientele with another, rather than generate additional trade. Alternatively, it may increase the sales of drinks but reduce the sales of food as customers purchase snacks to go with their wine, rather than full meals.
- The wine bar image may be more suitable to evening trade only and still leave the restaurant empty at lunchtimes during the week.
- The Legume may not be able to change its image sufficiently to appeal to its target clientele.
- Running a successful wine bar rather than a restaurant may require different skills.
- The strategy may lead to a loss of the good name built up amongst its existing clientele.

Option 2: Offering fast food at lunchtime and varying prices

Rewards

- There should be little impact on the existing trade, which is likely to be mainly evening based.
- It is consistent with the existing reputation for vegetarian food and the Bolds have the necessary experience.
- It is in line with business customers need for fast food at lunchtime.
- By varying the pricing some attempt is made to address the seasonality and offering discounts may increase the trade at off-peak times.
- There will be some reductions in staffing levels and by employing part-timers more of the costs become variable.

Risks

- There is a possibility that their evening clientele may not approve of the lunchtime operations.
- The fast-food concept may be seen as cheaper and detract from the image of a prize- winning restaurant.
- The business will be entering a highly competitive lunchtime market, one it knows little about.

Conclusion

The additional information noted in (b) above will help identify whether it is appropriate to increase lunchtime/evening trade and which food and drinks items on the menu are more profitable.

The Bolds should undertake some market research to assess the size of the potential customer base for each option and the forecast returns, before a decision can be made.

Whichever option is chosen, The Legume need to replace the inferior chef and improve the décor of the restaurant so that it addresses its current lack of competitive advantage.

10 Showing supporting calculations

Section overview

- This section considers the different types of supporting calculation requirements that you may encounter in the Business Strategy and Technology exam.
- It concludes with some general advice that you are recommended to read in advance of attempting the exam.
- The intention is to demonstrate the thought processes to be adopted when tackling such requirements.

10.1 Supporting calculations in the Business Strategy and Technology exam

As we have highlighted throughout the Workbook, one of the data related requirements in the Business Strategy and Technology exam will require candidates to prepare an answer which is complemented by supporting calculations. In this section, we look at some worked examples which contain exam standard question requirements which feature supporting calculations. The worked examples in this section illustrate the types of workings and calculations that you may be expected to produce in the exam. The worked examples discussed in this section focus on supporting calculation requirements in three different contexts:

- Evaluating a contract with supporting calculations
- Forecasting profit and variability with supporting calculations. This illustration also has a separate requirement asking for specific calculations (breakeven analysis).
- Pricing strategies with supporting calculations

To ensure that you are as well prepared as possible in tackling requirements which ask for supporting calculations in the Business Strategy and Technology exam you are strongly advised to take the time to read through the following worked examples carefully.

To understand how supporting calculations can be examined using spreadsheet software, please refer to the following questions in the chapter Spreadsheet functionality; Pico Fogo Ltd and By Camplight Ltd.

These examples require candidates to prepare supporting calculations in the form of 'Whatif analysis', with no specific calculations being stated in the requirement.

Worked example: Evaluating a contract with supporting calculations

Dreamy Potato Products Ltd (DPP) is a food processing company based in the west of England. DPP purchases potatoes from local farmers then peels, washes and processes them into two uncooked forms: whole or cut. It sells these products to food manufacturers which choose to buy from companies, like DPP, that benefit from economies of scale in procurement and processing.

In the year ended 31 December 20X5, DPP generated an operating profit of £352,000 on sales of £10.8 million. DPP has been approached by a national fast-food chain that wants to buy uncooked, cut potatoes. The initial contract is intended to be for a period of three years, with DPP being paid a fixed price of £210 per tonne of potatoes.

To implement this strategy, DPP will have to acquire new machinery to grade and cut the potatoes and also rent a further 9,000 square metres of premises. Working assumptions for the contract are provided in the **Exhibit**.

Requirement

Evaluate the fast-food contract proposal and discuss whether it should be undertaken. Show supporting calculations. **(10 marks)**

Exhibit: Working assumptions for Proposal – fast-food contract

(1) Expected sales volume: 9,000 tonnes pa at £210 per tonne, for three years.

(2) Direct labour and variable costs of processing are expected to be £65 per tonne pa.

(3) In 20X5 DPP's fixed overhead absorption rate was 12% of revenue pa. Fixed operating expenses will increase by £25,000 pa as a result of this contract.

(4) For the first year of the contract DPP has agreed, with its existing suppliers, a purchase price of
£125 per tonne of potatoes. Industry price forecasts assume that Year 3 potato prices are conditional on market prices in Year 2. The predicted average purchase prices are as follows:

	Probability	Year 2 Price/tonne £		Probability	Year 3 Price/tonne £
Low price	0.40	125	Low price	0.45	115
			High price	0.55	135
High price	0.60	160	Low price	0.65	150
			High price	0.35	170

Solution

Evaluation of fast food contract Appendix of possible calculations
Potato prices: EV calculation

Year 1 price = £125

Year 2 price: (0.40 × £125) + (0.60 × £160) = £146

Year 3 price: 0.40 × [(0.45 × £115) + (0.55 × £135)] + 0.60 × [(0.65 × £150) + (0.35 × £170)] = £144.60

Annual profits from contract

	Year 1	Year 2	Year 3	Year 4
	£	£	£	£
Revenue: 9,000 × £210	1,890,000	1,890,000	1,890,000	5,670,000
Potatoes: 9,000 × £125/£146/ £144.60	1,125,000	1,314,000	1,301,400	3,740,400
Labour and other VC: 9,000 × £65	585,000	585,000	585,000	1,755,000
Contribution: 9,000 × £20/(/£1)/ (£0.40)	180,000	(9,000)	3,600	174,600
Other operation cost	25,000	25,000	25,000	75,000
Profit	155,000	(34,000)	(21,400)	99,600

Sensitivity to price of potatoes:

The annual revenue of £1,890,000 less labour and increased operating costs of £610,000 generates £1,280,000 per annum to cover the costs of purchasing 9,000 tonnes of potatoes.

This amounts to £142.20 per tonne. Anything paid above this price will be loss-making for DPP. The EV is a long–term average and in fact the purchase price of potatoes will vary from £115 to £170 per tonne. The profitability at different purchase prices can be considered:

Price	115	125	135	150	160	170
	£	£	£	£	£	£
Net revenue	1,280,000	1,280,000	1,280,000	1,280,000	1,280,000	1,280,000
Costs of potatoes	1,035,000	1,125,000	1,125,000	1,350,000	1,440,000	1,530,000
Profit/loss	245,000	155,000	65,000	(70,000)	(160,000)	(250,000)

Evaluation

At the proposed fixed price of £210 per tonne for the three-year contract, DPP will make an additional £99,600 of operating profit overall, so on the face of it the contract is financially viable. However, this is an operating margin of only 1.8% (99,600/5,670,000), which is less than DPP's current margin (352k/10.8m = 3.3%). Also the profit is not spread evenly. The key issue affecting profitability is the price that DPP is going to have to pay for potatoes.

On the basis of the agreed price of £125 for Year 1 of the contract, there will be a profit in Year 1 of £155,000. However, based on the expected value calculations, the contract will make losses of £34,000 and £21,400 in the subsequent two years. Potato prices are dependent on crops planted and harvested and market supply. It is not clear how accurate or reliable the price forecasts are. The average price used in the contract profitability calculation will not actually apply. There is a 40% chance that the actual price in Year 2 is £125 and DPP will make a £155,000 profit in Year 2. However, it is more likely (p=0.6) that the price will be £160, and the actual loss in Year 2 will be £160k.

The maximum that DPP can afford to pay for potatoes, while remaining profitable, is £142.20 per tonne. Thus in Year 3 the contract will be profitable if purchase prices are £115 or £135 and loss making at £150 or £170.

Other factors to consider

(1) Are all the additional costs of the machine and premises space reflected in the £25,000 incremental fixed costs?

(2) If DPP turns the contract down, will it go to a competitor?

(3) If the fast food chain does not renew the contract after three years will DPP be able to use the machine and the additional warehouse space it has rented?

(4) Is there scope to add a clause to the contract for variation in prices depending on the market conditions?

(5) Can DPP pass the risk on to their suppliers by agreeing three-year supply prices now?

Conclusion: Further research should be undertaken before a final decision to go ahead is taken.

Tutorial Note

As this worked example shows, although not specified in the requirement it is good practice to highlight other factors or pieces of information which would be helpful in evaluating the merits of the proposal. Including points like these shows the examiner that you are comfortable identifying those issues which would support a fuller analysis of the proposal. To produce a good answer to this requirement you would need to bring together your supporting calculations with your narrative discussion of the proposed contract.

Worked example: Forecasting profit and variability with supporting calculations

The following worked example illustrates how separate requirements asking for 'specific calculations' (which we discussed in the chapter Risk management), and 'show supporting calculations' could appear in Business Strategy and Technology exam questions.

Bigville Council is a local government body responsible for providing community services to residents of Bigville, a large city in the UK. The Council is financed by central government, together with local taxes paid by residents and businesses in the area. Bigville Council is considering constructing a community stadium on a new site, with a shared pitch suitable for the city's local rugby and football teams. The proposal for building a new stadium was put forward in recognition of the recent success that the local rugby club had enjoyed. The rugby team's success meant that its current ground had reached its full capacity. The council also recognised the fact that the local football team was in need of newer facilities and ground. The following forecast has been prepared in respect of the annual match revenues and costs for the new stadium.

Requirements

1 Using the quantitative data in Exhibit 1 relating to the new stadium, estimate the break-even attendance figures for each rugby and football match.

2 Explain the implications of your calculations in (1) for the forecast match profits and discuss the potential variability in the annual attendance for each club, including the impact this would most likely have on the forecast in Exhibit 1. Show any supporting calculations.

Total: 13 marks

Exhibit: Forecast annual match revenues and costs for the new stadium:

	Expected attendance per match(1)	Total annual attendance	Total annual revenue(2)	Total annual contribution	Traceable annual fixed operating costs	Forecast annual profit/loss
Football	3,200	80,000	£800,000	£640,000	(£644,000)	(£4,000)
Rugby	3,000	45,000	£450,000	£270,000	(£214,000)	£56,000

1 Based on club increasing capacity by moving to new stadium and remaining in existing league.

2 Based on expected average total spend per visitor on tickets, refreshments and merchandising.

Raw data from Bigville's management information system recording each home match played

	A	B	C	D
1	**Home matches played**			
2	Q1	Q2	Q3	Q4
3	Football	Rugby	Football	Rugby
4	Football	Rugby	Football	Football
5	Football	Football	Rugby	Rugby
6	Rugby	Football	Football	Football
7	Rugby	Rugby	Rugby	Rugby
8	Football	Rugby	Rugby	Football
9	Rugby	Football	Football	Football
10	Football	Football	Rugby	Football
11	Football	Football	Football	Football
12	Football	Rugby	Football	Football
13				
14				
15				

The average home match attendance for the city's teams at their current stadiums is:

Football: 2,863

Rugby: 2,234

Comparable data for similar clubs is as follows:

	Average attendance per match for a typical club in the:		
	League below	**Existing league**	**League above**
Football	1,800	3,050	5,000
Rugby	1,900	2,840	4,000

Solution

1 Break-even attendance figures:

=COUNTIF(A3:D12,"Football")

	A	B	C	D
1	**Home matches played**			
2	Q1	Q2	Q3	Q4
3	Football	Rugby	Football	Rugby
4	Football	Rugby	Football	Football
5	Football	Football	Rugby	Rugby
6	Rugby	Football	Football	Football
7	Rugby	Rugby	Rugby	Rugby
8	Football	Rugby	Rugby	Football
9	Rugby	Football	Football	Football
10	Football	Football	Rugby	Football
11	Football	Football	Football	Football
12	Football	Rugby	Football	Football
13				
14	Football	25[1]		
15	Rugby	15[2]		

[1] =COUNTIF(A3:D12,"Football")
[2] =COUNTIF(A3:D12,"Rugby")

Break even

The break-even attendance is as follows:

Contribution per visitor

Football £640,000/(25 × 3,200) = £8

Rugby £270,000/(15 × 3,000) = £6

Break-even

Football £644,000/£8 = 80,500 visitors p.a which with 25 matches = 3,220 per match

Rugby £214,000/£6 = 35,667 visitors p.a which with 15 matches = 2,378 per match

2 **Discussion**

Note: The suggested solution shown here is intended to illustrate the range of relevant points you could have made in your discussion. However, you would not need to include all of these points to score the 9 marks available for this requirement.

Implications of calculations

The break even (BE) attendance for the football club is marginally higher (by 20 visitors per match) than the attendance figure used in the forecast, which explains the predicted loss (20 × £8× 25 = £4,000).

The BE figure is 12.5% higher than the current attendance enjoyed by the football club and 5.6% higher than the average for the league. Thus, the projections may be quite optimistic and if the initial attendance is below this figure then the football club will make a bigger loss than predicted. (It can be seen from the appendix that there is a loss of £71,400 based on current attendance and a loss of £34,000 based on the average for the current league.) It would be useful to know what the capacity of the existing football ground is and whether, like the rugby club, it is already at full capacity.

For the rugby club, the BE figure is 6.4% higher than the current attendance (which would result in a loss of just under £13k) but 16.3% lower than the average for the league. As the club's existing ground appears to be at capacity, there is likely to be an implied waiting list of fans wishing to attend and so the estimated attendance level of 3,000 seems reasonable and provides a margin of safety of 20.7% (3,000 – 2,378/3,000).

Variability of attendance levels

The attendance and financial performance of any sports club is heavily dependent on its team's performance. Thus, the financial success of both the football and rugby club will depend on their ability to retain their current league position or the chances of being relegated/promoted, which can be seen from the data provided to significantly affect attendance levels.

(On average, attendance levels for football clubs are 64% higher in the league above and 41% lower in the league below that which Bigville's team currently plays in.)

As the break-even levels of attendance for both clubs are above the average attendance for either club in the lower league, then relegation of a club would lead to significant losses or the need to curtail costs. The financial impact of promotion/relegation is set out in the Appendix, based on the figures for a typical club in each league. These calculations make it clear that the profitability of the football club is much more at risk than the rugby club because of the high level of traceable fixed costs (discussed further below). As a result, the potential results for football range from a loss of £284k if relegated to a profit of £356k if promoted, which compares to a loss of £43k and a profit of £146k for the rugby club.

In addition to variations in attendance, success in additional competitions during the year would lead to more matches, further increasing profits. Each additional football match would contribute an additional £25,600, based on projected attendance levels, with an extra rugby match generating £18,000. Using the expected attendance levels provided, it can be seen from the appendix that 26 football matches and 12 rugby matches would be more than sufficient to break- even.

To better assess the accuracy of the forecasts and the likelihood of profitability, it would be useful to know each club's position in the current league to assess the likelihood of promotion/relegation. Also, the attendance levels and number of matches for the previous two or three seasons would indicate any trend – in reality the attendance is likely to vary between matches depending on the recent success of the team and who the opposition is.

Revenues and costs

The calculations of break-even attendance depend on assumptions about the average spend per visitor, contribution and traceable fixed costs. If the forecasts have been based on existing ticket prices and spending on merchandise/catering, these may increase with the new stadium, in which case the contribution per visitor might increase and the required attendance would be lower. A break-down of the average spend per visitor between ticket price and merchandising and details of the different ticket prices, which presumably include concessions for the young and elderly, would be useful.

One of the reasons the football club is predicting a loss is because its traceable fixed costs are significantly higher than those of the rugby club leading to a greater variability of profits if

relegated/promoted. This may be because of the level of wages in the marketplace for footballers but more information is required here.

Appendix: Financial analysis

Estimated profits at various attendance levels: £'000

	Current attendance	If relegated	League average	If promoted	Per forecast
Football					
Attendance	2,863	1,800	3,050	5,000	3,200
	£	£	£	£	£
Contribution	572.6	360	610	1,000	640
Fixed costs	644	644	644	644	644
Profit	(71.4)	(284)	(34)	356	(4)
Rugby					
Attendance	2,234	1,900	2,840	4,000	3,000
	£	£	£	£	£
Contribution	201.06	171	255.6	360	270
Fixed costs	214	214	214	214	214
Profit	(12.94)	(43)	41.6	146	56

Sensitivity to number of matches

	Fixed cost	Forecast contribution per match	BE no of matches (fixed cost/contribution)
Football	£644,000	640,000/25 = £25,600	25.16
Rugby	£214,000	270,000/15 = £18,000	11.89

Tutorial Note

As this worked example illustrates, to earn the marks on offer in part (b) you would need to make use of your answer in part (a). As the answer above highlights the key issue in part (b) concerned the fact that the break-even attendance for football was in excess of the current forecast, whereas rugby had some margin of safety, although it was not clear how the traceable fixed costs (which were considerably higher for football than they are for rugby) had been arrived at.

Worked example: Pricing strategies with supporting calculations

Questions in the Business Strategy and Technology exam may require you to evaluate existing or proposed pricing strategies for different products and services, and to provide supporting calculations to justify your answer.

ElectroInfo Ltd is a company which provides electrical and IT services. The company's board are in the process of evaluating its pricing strategy. ElectroInfo Ltd charges all of its services, to all its customers, at a single rate of £32 per chargeable hour. The average variable cost to the company for both IT and electrical services is £20 per chargeable hour. In a bid to increase profits, some of the directors have suggested that the company should raise the prices that it charges customers for both types of service.

The finance director, Denise Jones, was concerned about this, and highlighted that, as many of the company's customers were on low incomes, they would be resistant to price increases and some may switch to lower priced competitors if the company raised its prices. Denise Jones instead proposed that the company should consider charging different prices for each of the two different services it provides to customers.

The following market research has been completed to show the different prices that customers could be charged and the impact that different prices would have on the levels of demand, as indicated by the number of chargeable hours.

Segmented by service type

Selling price per chargeable hour	**£30**	**£32**	**£34**	**£36**
Demand (chargeable hours per year)				
– IT services	125,000	120,000	110,000	100,000
– Electrical services	125,000	110,000	90,000	60,000

Requirement

Evaluate Denise Jones' suggestion that a different price should be charged for the two different service types, IT and electrical. Provide supporting calculations. Assume the same single uniform price would continue to be charged to both low income and all other customers.

Total: 9 marks

Solution

Using the market research information about levels of demand for the two services at different prices, we can calculate the maximum contribution that could be earned by charging different prices.

Service type

IT services

Price per hour	£30	£32	£34	£36
Variable cost per hour	£20	£20	£20	£20
Contribution per hour	£10	£12	£14	£16
Demand	125,000	120,000	110,000	100,000
Total contribution	£1,250,000	£1,440,000	£1,540,000	**£1,600,000**

Electricals

Price per hour	£30	£32	£34	£36
Variable cost per hour	£20	£20	£20	£20
Contribution per hour	£10	£12	£14	£16
Demand	125,000	110,000	90,000	60,000
Total contribution	£1,250,000	**£1,320,000**	£1,260,000	£960,000

With price differentiation based on service type, the optimal prices per hour would be:

- IT services £36
- Electrical £32

This would generate a total contribution of:

IT services	£1.60m
Electrical	£1.32m
Total	**£2.92m**

Evaluation

IT services

The price for IT services should be increased by £4 from the current level of £32 per hour to £36 per hour. The IT services market appears more resilient to higher prices than electrical services. This may be because they are more highly valued by customers or because of the pricing policies and availability of competitors. This higher price is advantageous to ElectroInfo Ltd because although it leads to a fall in demand for IT services of 16.7% (from 120,000 hours to 100,000 hours), the 25% increase in contribution per hour (from £12 to £16) more than compensates for this, and means that the contribution from this service type increases by 11.1% (from £1,440,000 to £1,600,000).

Electrical services

The table also shows that the current price of £32 per hour is optimal for electrical services. As a result, the contribution from electrical services would be unchanged.

Total contribution

Overall contribution from differential pricing of the two types of service would lead to an increase from £2.76 million to £2.92 million (ie, 5.8%). Even if there was homogeneous pricing for both services, an additional contribution could be gained, compared to the current situation, by increasing the price from £32 to £34 per hour. This will increase total contribution from £2.76 million to £2.8 million (£1.54m from IT services and £1.26m from Electricals).

Tutorial Note

As the question scenario included the details of a range of different prices to be charged to customers, the expected hourly demand per service, and the average variable cost per chargeable hour, it should have been clear that the total contribution per service at different prices could be determined. This was central to producing a meaningful evaluation of the finance director's proposal.

10.2 Advice on attempting supporting calculation requirements

As the worked examples in this section have illustrated, supporting calculation requirements will always be specific to the context of the question scenario. Therefore, the calculations that you include in your answer will need to be driven by the detail and data provided in the scenario.

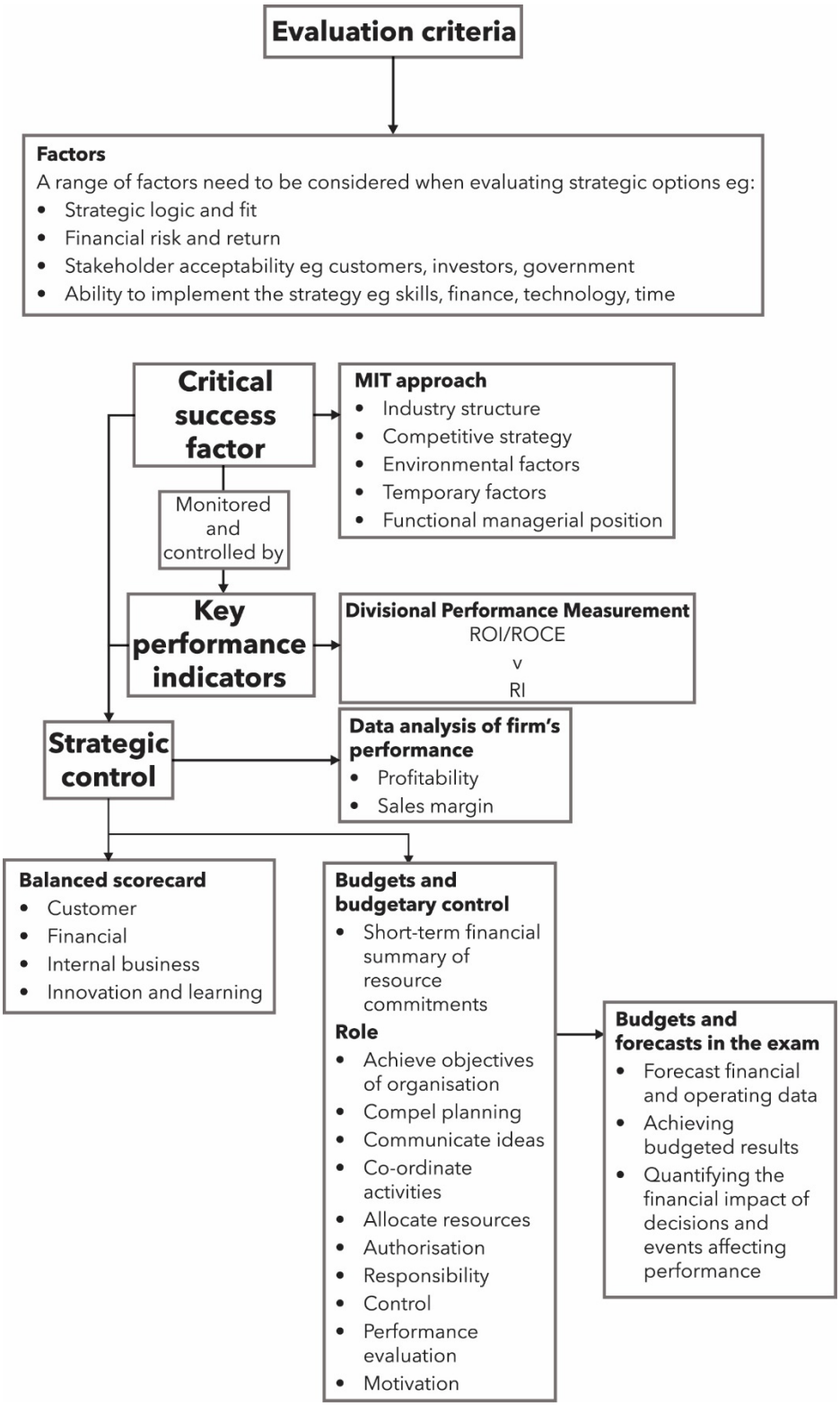
When you are asked for supporting calculations it is important to remember that the examiner is not only interested in your ability to display your numerical skills, but wants to see that you can integrate your workings into your narrative answer. Therefore, it is critical that you are comfortable picking out the key figures from your workings and are able to then explain their significance in your answer.

Effective communication of the calculations that you have performed is an important skill that you will need display as you progress through your career.

As the worked examples in this section illustrate, issues relating to sensitivity and probabilities (such as expected values which were discussed in the chapter Risk management) are common themes in supporting calculation requirements. As such you need to ensure that you take the time to revise these areas carefully. Nonetheless, it is important to recognise that the worked examples outlined in this section only serve as a guide as to the types of supporting calculations that could potentially appear in your exam. They are not indicative all of the different ways that supporting calculations might be examined in the future.

It is important to note that the supporting calculations that you will be expected to produce will not be overly technical. The real difficulty lies in determining, under exam pressure, which calculations will add the most value to your answer. These skills can only really be learned by attempting plenty of questions from the Business Strategy and Technology Question Bank.

Summary


Evaluation criteria
Factors
A range of factors need to be considered when evaluating strategic options eg:
• Strategic logic and fit
• Financial risk and return
• Stakeholder acceptability eg customers, investors, government
• Ability to implement the strategy eg skills, finance, technology, time
Critical success factor
MIT approach
• Industry structure
• Competitive strategy
• Environmental factors
• Temporary factors
• Functional managerial position
Monitored and controlled by
Key performance indicators
Divisional Performance Measurement
ROI/ROCE
v
RI
Strategic control
Data analysis of firm's performance
• Profitability
• Sales margin
Balanced scorecard
• Customer
• Financial
• Internal business
• Innovation and learning
Budgets and budgetary control
• Short-term financial summary of resource commitments
Role
• Achieve objectives of organisation
• Compel planning
• Communicate ideas
• Co-ordinate activities
• Allocate resources
• Authorisation
• Responsibility
• Control
• Performance evaluation
• Motivation
Budgets and forecasts in the exam
• Forecast financial and operating data
• Achieving budgeted results
• Quantifying the financial impact of decisions and events affecting performance


Data related requirements in the exam
Data analysis requirement
Specific calculation requirement
Show supporting calculation requirement
Requirement to assimilate and analyse transactional data in a spreadsheet
If data is present in question scenarios use it
Specific calculation questions
The calculations needed will be directed by the requirement
Calculations could include break-even, NPVs, Expected Values, RI and ROCE and others
Show supporting calculations questions
Supporting calculations can take many forms
The calculations to perform are driven by the scenario detail
Need to integrate findings from supporting calculations into narrative answers
Use spreadsheet functionality to perform calculations
Format of data in the exam
Extracts from financial statements
Extracts from balanced scorecards
Extracts of posts on social media
Data dashboards
Diagrams (eg, value chain, BCG matrix)
Graphs
Pie charts (eg, analysis of sales revenue by product)
Detailed transaction data in a spreadsheet
Types of data
Qualitative data
Quantitative data
Skills
Choosing analytical tools
Calculations
Interpreting results
Conclusions/recommendations
Looking beyond the information given
Linking data/explaining trends
Highlighting weaknesses/omissions in the data provided
Using spreadsheet functionality to analyse spreadsheet data
Approach to data analysis questions
1 Initial review
2 Decide on appropriate analysis
3 Perform calculations – make use of spreadsheet functionality
4 Interpret your analysis
5 Additional information required
Additional information required
More detailed information
Data for prior years
Missing data
Actual figures (if given estimates)
Other data from the industry
Market leader data
Think of all information you may need, from internal and external sources
To interpret data properly, you need to make comparisons
Avoid common mistakes
Use the What-How-Why-When-So What analysis
Use 'because'
Relate to scenario
Be specific
Link the figures/analysis
Use two financial figures together in your analysis
Compare one financial figure and one operating figure in your analysis
Compare two operating figures in your analysis
Use financial and non-financial indicators
Identifying cause and effect relationships
Understand and practise using spreadsheet software and functionality

Further question practice

1 Knowledge diagnostic

Before you move on to question practice, confirm you are able to answer the following questions having studied this chapter. If not, you are advised to revisit the relevant learning from the topic indicated.

Confirm your learning	
(1)	Do you know how to evaluate a strategy? (Topic 1)
(2)	What are the main benefits of budgets? (Topic 2)
(3)	Can you calculate key performance measures relating to profitability, liquidity and gearing? (Topic 3)
(4)	What are the main divisional performance measures and can you calculate them? (Topic 4)
(5)	Do you understand how to develop a balanced scorecard? (Topic 5)
(6)	Do you know how to measure performance in NFPs? (Topic 6)
(7)	Do you know the recommended approach to data analysis requirements? (Topic 9)

2 Question practice

Aim to complete all self-test questions at the end of this chapter. The following self-test questions are particularly helpful to further topic understanding and guide skills application before you proceed to the next chapter.

Question	Learning benefit from attempting this question
Greens Ltd	This is a knowledge-based question asking you to suggest two measures for each of the four scorecard areas.
Maysize Ltd	Using the information in the scenario you are required to suggest appropriate performance measures. The example requires you to prioritise and focus only on the six most important areas for the company in the question.

Once you have completed these self-test questions, it is beneficial to attempt the following questions from the Question Bank for this module.

These questions have been selected to introduce exam style scenarios that will help you improve your knowledge application and professional skills development before you start the next chapter.

Question	Learning benefit from attempting this question
Bagaj Lux Ltd (March 2018)	Part 3 of the question asks you to assess the financial and non-financial factors relevant in deciding on a new contract. The question asks for supporting calculations but does not specify which or how many calculations to perform. Use your judgement and the data provided, as well as being guided by the number of marks available in the question.
Ten Ten Burgers Inc (June 2018)	Part 2 of the question requires you to evaluate company performance. You are also asked to evaluate the effect of a new app on revenues. Data is presented in a variety of formats, including graphs and extracts from financial statements and management accounts.

Refer back to the learning in this chapter for any questions which you did not answer correctly or where the suggested solution has not provided sufficient explanation to answer all your queries. Once you have attempted these questions, you can continue your studies by moving on to the next chapter.

Self-test questions

Answer the following questions.

1 What do the three Es stand for?

2 What are the four perspectives of the balanced scorecard?

3 Identify the role that CSFs can play in strategic control and performance measurement.

4 A firm's ROCE is the product of which two ratios?

5 Define four liquidity ratios.

6 List four limitations of relying on financial measures alone to manage divisional performance.

7 Greens Ltd

Greens Ltd is a growing firm providing organic fruit and vegetables for delivery via phone or internet order. It has decided to measure performance for the coming year using the balanced scorecard approach.

Requirement

Suggest two measures in each of the four areas covered by the scorecard.

8 Maysize Ltd

Maysize Ltd owns a department store located in a prime site in a regional city in the UK. The store has 16 different departments, selling such diverse items as shoes, fabrics and children's toys, and it has an accounts office for administration. The store is run by a store manager and there is a manager for each department within the store. Selling margins are set by the store manager, although there is scope to flex the margin after consultation with departmental managers. All suppliers are paid by the accounts office. Each departmental manager is responsible for product sales, employee costs and any sale events (eg, the January sales).

Management reporting

The management reporting system is very simple. The accounts office prepares a monthly cash flow statement, statement of financial position and cumulative statement of profit or loss. The statement of profit or loss and statement of financial position are compiled from the 16 departmental statement of profit or loss and statement of financial positions. The departments prepare their statements of profit or loss by calculating cost of goods sold by reference to the selling margins.

Maysize Ltd is now in the process of changing to a computerised management reporting system. This system will be able to generate reports in as much detail as is necessary for management.

The managing director has come to you for advice about the new reporting system. They would like to know what financial and non-financial performance indicators should be generated by the new reporting system so that they can monitor in more detail and improve the profitability and liquidity of the department store.

Requirement

Prepare a memorandum for the managing director suggesting the six most useful performance indicators that the new system should generate for the departments. For each performance indicator which you recommend, specify what information it should contain, its frequency and why it would be useful for the managing director.

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

1.1 ROI using opening carrying amount:
Year 1 £34k – £25k ÷ £100k = 9%
Year 2 £34k – £25k ÷ £75 = 12%
Year 3 £34k – £25k ÷ £50 = 18%
Year 4 £34k – £25k ÷ £25k = 36%

ROI improves despite constant annual profits thus divisional managers might hang on to assets for too long.

(RI would also improve, giving the same problem.)

1.2 ROI using historic cost:

Years 1–4 £34k – £25k ÷ £100k = 9%

(RI would also be constant under these circumstances.)

1.3 ROI using historic cost overcomes the increasing return problem of using the carrying amount. However, it is not perfect.

Answer to Interactive question 2

The manager would dispose of the asset after two years, ie, they might get rid of the asset too quickly. (The same problem occurs with RI with interest at 12%.)

Answer to Interactive question 3

3.1 ROI

Divisional ROI pre-project	=	£300k / £1m
	=	30%
Divisional ROI post-project	=	£325k / £1.1m
	=	29.5%

Although the project ROI is acceptable to the company, the manager would not be motivated to accept a project which lowers divisional ROI.

3.2 In this particular circumstance, RI would lead to the right decision as the absolute figure for the division would increase.

RI pre project 300,000 – 20% (1m) = £100k

RI post project 325,000 – 20% (1.1 m) = £105k

CHAPTER 12

Answer to Interactive question 4

(a) **Residual Income**

	Division 1	Division 2
Year 1		
£200k – 20% (£1m)	–	
£20k – 20% (£100k)		–

	Division 1	Division 2
Year 2		
£220k – 20% (£1m)	20k	
£40k – 20% (£100k)		20k

(b) **Return on investment**

		Division 1	Division 2
Year 1			
£200k/£1m	20%		
£20k/£100k			20%
Year 2			
£220k/£1m	22%		
£40k/£100k			40%

It is much easier for the larger division to generate a further £20k of residual income; hence using RI to compare divisions of different sizes is misleading. ROI gives a better indication of performance.

Answer to Interactive question 5

Climate-change related emissions

KPI: *Consumption of electricity, gas and oil. For example, electricity, gas or oil usage per hotel per month (kWh).*

PH could increase the usage of renewable energy and install solar panels on hotel land and roof tops. PH could also limit the minimum air-conditioning temperatures and reduce the temperature of heated areas and pools. Energy efficient lightbulbs, timers on light switches and turning off excessive external lighting will contribute to reduced energy consumption.

Consumption of resources

KPI: *Reduction in annual water use. For example a suitable measure could be % reduction in annual water use per hotel with a target of 10% reduction per annum.*

PH could install smart water timer systems to reduce showering times or to limit the watering of hotel gardens and lawns to when it is required. PH could and limit the quantity of towels per room for each new guest and encourage hotel guests to reuse towels, as this will reduce consumption of energy used by each hotel laundry.

Waste minimisation

KPI: *Reduction in food waste. A suitable measure could be percentage reduction in food purchased per month per hotel with a target of a reduction in food waste of 5% per year by weight.*

PH could improve meal planning to reduce the quantity of food prepared. Menus could be redesigned to reduce choice so that less is wasted. PH could also encourage the consumption of healthier choices or plant-based meals which result in fewer greenhouse gases. Where

possible PH should aim to eliminate the use of plastic in the hotels, such as plastic bottles, straws and single use toiletry packaging, by using water fountains, paper straws and refillable toiletry dispensers.

Employee wellbeing

KPI: *Ensure gender balance and equality. For example a measure could be total employees by gender or employee turnover by gender and age group, overall and by hotel.*

Assuming that mothers continue to bear the greatest responsibility for childcare, PH could improve gender balance in its hotels by supporting women to remain in the workplace. For example, policies and support such as paid parental leave, provision of childcare arrangements, flexible hours or agreed absence during school holidays will help parents to manage childcare and work responsibilities. Monitoring the number of employees by gender and employee turnover by gender will help PH and its stakeholders to evaluate the success of these policies.

Guest and employee safety

KPI: *Improvement in guest and employee safety. Suitable measures could include the number of working days absent per month due to injury, or the number of hotel guest accidents per month.*

To ensure consistent application of safety policy at all its hotels, PH could implement an onsite safety team to focus on identifying and resolving potential safety issues at each hotel site. Frequent safety inspections and asking for staff feedback in relation to unsafe practices they have observed can help to promptly identify safety issues which can then be addressed by on site-safety teams.

Employee personal and professional development

KPI: *Improve employee professional development. A suitable measure could be the average days of training undertaken per employee per annum, in total and split by gender, age group, and employment category.*

PH could increase investment in training and development for all employee categories which would provide PH with the knowledge, skills and experiences it requires to meet any skills gaps identified. This training could include specific areas to improve performance in ESG and sustainability areas, such as compliance with food hygiene and safety regulations. Monitoring the investment in training and development will help PH determine clusters of specific groups of employees who are not benefitting from the training.

Supply chain sustainability

KPI: *Greater use of local suppliers. A measure of this KPI could be the value of supplier orders by distance in kilometres (km), with a target of, for example, 60% of orders within a 30km radius.*

A supply chain management initiative could prioritise local suppliers as much as possible, limiting the overall delivery kilometres and thereby reducing energy consumption and related greenhouse gas emissions. Also, PH could adopt a sustainability charter which sets out expected ESG practices to guide its suppliers. PH can encourage its suppliers to sign up for compliance with a sustainability charter in exchange for preferred supplier status. The take-up of this charter can then by monitored to track its success.

Local community support

KPI: *Increased engagement with local community. A measure of this KPI could be the frequency of PH representatives meeting with the local community representatives each month, or the volume of queries received from the local community.*

PH could set up a local community 'give back' scheme where it hosts sessions with local community representatives to listen to their concerns and understand how PH can be more supportive to local community actions (eg, education, health, environmental concerns).

Anti-bribery and corruption

KPI: *Prevention of bribery, money laundering or other corruption cases. A suitable measure could be the percentage bribery, money laundering or other corruption cases reported, investigated and resolved within a specified timeframe.*

PH can set up an anti-corruption and compliance team to allow employees to anonymously report incidences of potential corruption, thus providing whistleblowing protections. This KPI can help PH measure the effectiveness of its anti-corruption measures and the efficiency of its investigative processes. By tracking the KPI over time, PH can identify areas for improvement in its anti-corruption efforts, such as the need to streamline its investigative process or to provide additional resources to its anti-corruption and compliance team.

Answer to Interactive question 6

	A	B	C	D	E	F
1	**Employee ID**	**Monthly placement fees**	**Rank**	**Bonus?**	**Bonus value (£)**	**Bonus %**
2	693	13,860	14[1]	No[2]		1%
3	392	17,820	13	No		
4	894	25,097	11	No		
5	774	13,120	15	No		
6	761	27,526	10	No		
7	432	62,411	3	Yes	624.11[3]	
8	757	59,547	4	No		
9	180	17,838	12	No		
10	661	46,831	7	No		
11	340	79,724	1	Yes	797.24	
12	237	42,310	9	No		
13	624	65,458	2	Yes	654.58	
14	810	55,881	5	No		
15	349	53,290	6	No		
16	803	44,031	8	No		

[1] =RANK(B2,$B$2:$B$16,0) Enter the formula into Cell C2. Ensure the range (B2:B16) is fixed using the $ sign.
The list will be sorted in descending order because 0 has been entered at the end of the formula. Drag the formula down to Cell C16 to complete the ranking
[2] =IF(C2<4,"Yes","No")

The formula selects the employees who rank in first to third position and who will receive a bonus.

*[3] =B7*$F$2 The formula can be copied in cells where there is a bonus eg E11, E13*

Column C uses the RANK function to rank employees from 1 to 15 based on the value of their monthly placement fees (highest to lowest). The IF function is applied to Column D to identify the three employees who are entitled to a bonus. Column E calculates the value of the bonus by multiplying the value in Column B by 1% (cell F2).

Answers to Self-test questions

1 Economy, Efficiency, Effectiveness

2 Perspectives:
- Financial
- Customer
- Innovation and learning
- Internal business processes

3 CSFs identify the areas that an organisation must excel at if it is to gain competitive advantage. Having identified its CSFs a business can determine suitable KPIs to measure the achievement of those CSFs. These can be used as part of the strategic control system and for performance measurement.

4 Profit margin × asset turnover = ROCE

5 Current ratio: current assets/current liabilities

Quick ratio: (current assets less inventories)/current liabilities Receivable days: trade receivables/revenue × 365
Payable days: trade payables/cost of sales × 365

6 Four of:
- Encourages short-termist behaviour
- Ignores strategic goals
- Cannot control persons without budget responsibility
- Historic measures
- Distorted

7 Greens Ltd

Financial perspective

Increase revenue

Market share

Customer perspective

Time taken from order to delivery

Freshness of products (measured in terms of days since harvesting)

Internal business process perspective

Time taken to process an individual order

Speed with which product availability updates register on the website

Innovation and learning perspective

Number of different products in the range offered Geographical areas covered by delivery teams

8 Maysize Ltd

Memorandum

Subject - Departmental performance indicators

This memorandum considers the new computer management reporting system for Department Store X. As requested six specific performance indicators for departments are identified as being of critical importance.
The relative importance of each of these indicators will vary over time as organisational priorities change (eg, if liquidity were to become more important than profitability).

The specific indicators identified are, as requested, local in nature. Global considerations, such as market share and capital expenditure, will also be of relevance but are not discussed in this report. In each instance the reporting system should incorporate information on prior period and (flexed) budgetary comparatives, and be analysed between individual departments and the store as a whole. Where possible, appropriate industry averages should also be incorporated (eg, in margin analysis or area utilisation).

Suggested indicators are as follows:

Customer feedback summaries Content
A summary of customer feedback, including details of the number and nature of complaints. These should include feedback received at the customer service desk, and a log of matters noted by store and department managers.

Frequency

Periodically (monthly – depending on the extent and gravity of matters arising).

Benefits

Levels of service quality will be critical to the success of the store. Weaknesses in customer relations management need identification to ensure that the appropriate client culture is adopted.

Sales margins Content
Gross profit margins should be calculated, expressing gross departmental profitability as a percentage of departmental revenue.

Frequency

Margins should be calculated periodically (probably weekly). In the event of specific queries, further detailed analyses could be provided on demand (eg, a summary of exceptional margins for product lines relating to a specific department).

Benefits

Since authority to fix margins is delegated to the store manager (and in some cases department managers), some control will be vital.

Weak margins may indicate poor purchasing policies, or highlight high or low profitability for specific departments that may warrant a reallocation of resources.

Margins will also be relevant when considering strategic pricing policies (eg, price wars). High level review should also facilitate overall goal congruence, eg, one area targeting economy purchases (low margin), with another targeting the other end of the market.

Area utilisation Content
A measure of departmental revenue against space occupied should be prepared. (If possible, a more elaborate measure of departmental profitability by floor space would provide further benefit.)

Frequency

Periodically (monthly) or even as sales seasons change.

Benefits

Store space is a key scarce resource; maximising returns from available space will be critical to success.

Measures yielding higher utilisation of existing space, once identified, could also be replicated within other departments.

Potential improvements in profitability from devoting more space to areas yielding the highest return will need to be balanced with interdependencies between the departments. Shoppers may, for example, be using the store primarily for convenience food shopping, but may make additional impulsive purchases while on the premises.

Staffing

Content

A simple calculation of departmental turnover against head count should be performed.

Frequency

This should be produced on a periodic basis (monthly).

Benefits

Staff costs will represent a significant determinant of profitability. Given the extent of delegation of responsibilities, they are also an important resource.

The measure should highlight areas of possible over/under staffing and may assist in staff planning.

An 'equitable' allocation of the workload should also avoid demotivating the staff and reduce staff turnover (training costs, etc).

Inventory holding

Content

Average inventory holding days by department calculated as:

$$\frac{\text{Average inventory of finished goods}}{\text{Annual cost of sales}} \times 365$$

Frequency

Periodically (possibly monthly) with fuller analyses of individual product lines available on demand in the event of exceptional inventory holdings.

Benefit

Inventory will be the key element of working capital. Inventory levels may also have a direct impact on sales levels (eg, if shelves are bare).

A simple comparison of inventory levels between departments will identify excessive or minimal inventory levels, assisting in the review of purchasing policies, or handling of slow-moving inventory lines and the use of sales.

Inventory counts may also be modified to allow investigation of unusual inventory levels, which may have been a result of inventory losses or inaccuracies in the recording of inventory movements.

Purchasing efficiency

Content

Value of trade and settlement discounts, in both absolute and relative (as a percentage of purchases) terms.

Frequency

Periodically (monthly), with the option of specific exception reporting (eg, discounts over £1,000).

Benefits

The ability to purchase quality goods at a reasonable price will be a key factor in the success of the store.

The organisation of purchasing responsibilities between departments and a central purchasing department will be particularly relevant.

A review of discounts obtained may assist in consideration of strategic purchasing policy (central buying from key suppliers giving rise to potential economies of scale) and a consideration of payables management issues.

Conclusion

The new system offers the opportunity for a review of the full range of operational and tactical management information. The six indicators suggested have considered measures that may have more strategic relevance. I would be happy to offer further advice on other information enhancements at your instruction.

Chapter 13

Business planning and functional strategies

Introduction

Learning outcomes

Syllabus links

Examination context

Chapter study guidance

Learning topics

1 Business planning

2 Implementing strategy

3 The finance function

4 Marketing planning

5 Human resources planning

6 Research and development, and innovation planning

7 Operations planning and management

8 Purchasing

Summary

Further question practice

Self-test questions

Answers to Interactive questions

Answers to Self-test questions



Introduction

Learning outcomes

- Evaluate how an organisation's overall strategy can be achieved by implementing appropriate functional strategies, including the role of the finance function as business partner.
- Explain and evaluate the form and content of an organisation's business plan.
- Prepare a business plan, or extracts, which will achieve given or implied objectives.

Syllabus links

The development of operational plans to implement corporate strategy received brief coverage in your Business, Technology and Finance exam.

Examination context

There are two key elements in this chapter. Firstly, in the exam you might be expected to draft elements of a business plan for a client or to identify weaknesses and omissions in a given business plan and suggest improvements. Secondly, the chapter looks at functional strategies. These would normally be examined in the context of the overall objectives and strategy of an organisation. So for example you might be expected to look at how the organisation could develop its functional strategies to achieve its objectives.

It is therefore important that you are comfortable analysing the issues affecting the functional strategies of those organisations featured in exam questions.

It is perfectly acceptable to draw upon your own work experiences as they may help you to make practical points when attempting questions such as this. It is, however, crucial that you link the points you make to scenario detail, and failure to apply your answer will cost you valuable marks.

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this chapter.

Topic	Practical significance	Study approach	Exam approach	Interactive questions
1	**Business planning** Banks and other lenders will require a business plan with key information about the organisation, its financial position and its future plans, before they are prepared to invest.	**Approach** Read through the section paying attention to the worked example business plan. The interactive question will help you to critically appraise a business plan. **Stop and think** Who, other than lenders, is a	In the exam you might be expected to draft elements of a business plan for a client or to identify weaknesses and omissions in a given business plan and suggest improvements	**IQ1: Computers and Training** This question asks you to write a report looking at how a business plan can be improved.

Topic	Practical significance	Study approach	Exam approach	Interactive questions
		business plan useful for and why?		
2	**Implementing strategy** Functional strategies are essential for carrying out the broader organisational strategy. So far, detailed discussion has focused on marketing only. Later chapters will consider the key functions of IT/IS and change management. This chapter considers the role of the finance function in strategic planning and its evolving role as a business partner.	**Approach** Read quickly through this short introduction to functional strategies. **Stop and think** What role do accountants in industry play in carrying out organisational strategies? What role do other functions like HR, Procurement and Operations play?	In the exam you may be expected to analyse the issues affecting different functions and suggest improvements to functional strategy to support corporate strategy.	
3	**The finance function** The finance function plays a critical business partnering role, supporting the achievement of strategic goals and objectives.	**Approach** Ensure you are clear on the role of the finance function as business partner to other functions in the organisation. **Stop and think** Consider a firm you are familiar with. Can you identify business partnering activities within the finance function?	Exam questions may ask you to suggest how the finance function can become more involved with supporting the strategic planning process, by acting as a business partner.	
4	**Marketing planning** There is a close link between marketing and strategic management. Without products and services the corporate strategic plan would not exist. Marketing planning focuses on the direction an organisation will take, with regard to markets, as well as outlining the tasks and activities necessary to move in the right direction.	**Approach** Understand the relationship between the corporate and marketing strategies. Review the marketing control process. **Stop and think** The strategic marketing plan requires that an organisation identifies and selects its target markets. Can you recall how to segment a market?	In the exam you may be expected to assess how the marketing strategy supports the corporate strategy and how the marketing approach may be improved to drive strategic change.	**IQ2: Marketing plan** This question asks you to explain the importance of marketing planning and to identify the main steps.

Topic	Practical significance	Study approach	Exam approach	Interactive questions
5	**Human resources planning** Human resource management involves identifying an organisation's future staff needs, finding the right people to fill those needs and then training, developing and rewarding personnel to ensure performance.	**Approach** Read through the worked example to understand HRM implications in the airline industry. Look at HR strategies to close the gap between the demand and supply of personnel. The way we work is changing as a result of technology. Read through the worked examples to see what changes are occurring. **Stop and think** For many organisations, people are a major asset. How can employees be motivated to perform better?	An exam question may ask you to develop or critically appraise and improve the human resource plan in order to achieve the corporate strategy.	**IQ3: ScannerTech** This question demonstrates how the human resource plan is linked to the company's growth strategy.
6	**Research and development and innovation planning** Research and development is essential for product improvement; innovation can be a differentiator when it comes to strategy development. Organisations also have to consider the huge cost of research and development and balance this against the benefits.	**Approach** Understand the types of research and development including innovation planning. **Stop and think** Think of organisations that have successfully used R&D to differentiate themselves and gain competitive advantage.	An exam question may ask you to assess the potential benefits and risks of research and development for the organisation in the scenario.	
7	**Operations planning and management** Operations management is concerned with transforming business inputs into some kind of output, be that goods or services, to satisfy customer needs.	**Approach** Read through the concepts of operational planning. Ensure you understand the approaches to capacity planning and how the JIT and TQM philosophies can be applied. **Stop and think**	Improved operations planning can lead to greater efficiency and reduced costs. In the exam, where appropriate, recommend changes to operations management strategy to improve business	**IQ4: Lean manufacturing at Toyota** This question is based on Toyota so will develop your awareness of functional strategies in real companies. Look at the changes Toyota

Topic	Practical significance	Study approach	Exam approach	Interactive questions
		JIT manufacturing and JIT purchasing can bring many benefits. However, when is JIT purchasing not appropriate for an organisation?	performance.	made and assess their impact on the business.
8	**Purchasing** Purchasing strategies are a major influence on an organisation's costs and quality. The purchasing manager has to balance the need for profitability with ensuring that goods and services are produced to the required standard, in order to maintain competitiveness.	**Approach** Review the purchasing mix and understand the different supply sourcing strategies. Also consider the impact of technology on procurement. **Stop and think** Purchasing managers have the dual challenge of procuring the desired quality at the lowest possible cost. What ethical dilemmas may this result in?		**IQ5: PicAPie Ltd** This question requires you to review TQM from topic 7 as well as assessing the current sourcing strategies and recommending changes to the approach. **IQ6: Beau Ltd** This question revisits sourcing strategies as well as examining supplier review approaches.

Once you have worked through this guidance, you are ready to attempt the further question practice included at the end of this chapter.

1 Business planning

Section overview

- Business planning converts longer-term business strategies into actions to be taken now.
- Business plans are also used to apply for funding and are a critical document for a potential investor.
- In the exam, you must be able to critically appraise a business plan provided by a client, recognising weaknesses and omissions and making recommendations for improvement.

1.1 Planning

The chapter, 'Strategy and organisation,' covered strategic planning as a way of determining the long-term success of the business.

This chapter looks at planning as an activity concerned with implementation of strategy.

The long-term corporate plan serves as the long-term framework for the organisation as a whole, but for operational purposes it is necessary to **convert the corporate plan into a series of short-term plans relating to sections, functions or departments**, perhaps covering one year.

Business planning may assist with:

- Co-ordinating the activities of different functions behind the achievement of the strategic goals for the year
- Putting the case for finance to funding sources (eg, small businesses may approach a bank with a business plan or a charitable organisation will approach potential donor organisations)
- Gaining the approval of the board (eg, a national car dealership requires the manager of each showroom to submit an annual business plan for its approval)
- Winning contracts where the potential client wishes to be convinced that the firm will fully support the product or service being offered
- The development of the annual budget

Context example: Planning at an airline

A major global airline operates two levels of planning.

Strategic planning: This considers the development of the business over the coming 10 to 15 years, a long period coinciding with the lifespan of its major capital investments. Here management will consider the development of emerging markets, the airline's market position, issues such as carbon and noise pollution and consolidation in the airline industry.

Business planning: This may concern the coming 12 months and is driven by route planning, ie, which planes will fly which routes and where route schedules will be increased or cut-back. From this route-planning will come estimates of staffing needs, the number of aircraft required, fuel and maintenance requirements and the number of passengers, and the promotional activity to be undertaken. These will then form the basis of the annual budget of costs and revenues.

The annual business plan is in effect an annual instalment of the airline's strategic plan. Of course, given the inherent uncertainties of the industry, the strategic plan is very flexible.

1.2 Creating the business plan

Definition

Business plan: Is a document which is usually prepared as part of the process of applying for funding. It is a critical document for the potential investor. The contents of a business plan may include: a statement of purpose, a description of the business, financial data and supporting documents.

The process of creating a business plan from a bigger picture strategy leads to questions being asked, and issues raised, which require detailed resolution. Frequently, business plans are created using a pro-forma supplied by the approving body, say a bank or government development agency.

Professional skills focus: Concluding, recommending and communicating

If you are asked to draft elements of a business plan in the exam, it is important that you can recall the format and layout. As professionals it is expected that your answers are presented professionally, in accordance with instructions provided in the question.

Context example: Small business plan template The following is a suggested outline for a business plan.

Elements of a business plan

(a) Cover sheet

(b) Statement of purpose

(c) Table of contents

(1) The business

- Description of business
- Marketing
- Competition
- Operating procedures
- Personnel
- Business insurance
- Financial data

(2) Financial data

- Loan applications
- Capital equipment and supply list
- Statement of financial position
- Break-even analysis
- Pro-forma income projections (forecast income statements)

Three-year summary

- Detail by month, first year
- Detail by quarters, second and third years
- Assumptions upon which projections were based

- Pro-forma cash flow
- Follow guidelines for letter (e)

(3) Supporting documents

- Tax returns of the business and owners for last three years
- Personal financial statement (all banks have these forms)
- In the case of a franchised business, a copy of franchise contract and all supporting documents provided by the franchisor
- Copy of proposed lease or purchase agreement for building space
- Copy of licences and other legal documents
- Copy of resumes of all owners and senior managers
- Copies of letters of intent from suppliers, etc

Small businesses frequently request the help of their accounting advisers in the preparation of these business plans.



Interactive question 1: Aldine Computers & Training

A new client, Aldine Computers & Training (Aldine), has approached you for assistance in preparing a business plan to obtain bank funding. Aldine have drafted the following business plan and have requested your comments on its viability and on how it might be improved to maximise the chance of the bank giving them funding.

BUSINESS PLAN

For: Aldine Computers & Training May 20X0
For: South East Bank plc Index

(1) **Summary of business plan**

(a) Introduction to Aldine Computers & Training & Co
(b) Product: Computer sales
(c) Product: Training services
(d) Critical success factors which will make Aldine succeed
(e) Financial summary

(2) **Management**

(a) Proprietor: Employment record
(b) Junior Partners: Brief summary
(c) How and why the business started

(3) **Product/service**

(a) Computers sales
(b) Printer sales
(c) Training sales

(4) **Marketing**

(a) The market size
(b) Future growth
(c) Market sector
(d) Expected client profile

(e) Competitors

(5) **Sales**

(a) Advertising

(b) Promotion

(c) Who will sell

(d) The unique selling points

(e) Setting the price

(6) **Operational**

(a) Location

(b) Equipment and costs to-date

(c) Equipment required

(d) Staff

(7) **Short-medium term trading**

(a) Objectives short and medium-term

(b) What if? – contingency plans

(8) **Financials**

(9) **Contact details**

Summary of business plan

Introduction to Aldine Computers & Training

Aldine Computers & Training (Aldine) was established in March to trade on a regional basis. There are currently two partners: Vinay Kamath (37) and Kate Schofield (26). Both have extensive experience of computers and training.

Product: Computer sales

Vinay Kamath has been building computers for five years for the well-known firm PCNow.com and can use their contacts from there to obtain computer parts at favourable prices. Aldine will aim its computers at the corporate market and will aim to sell them with service contracts as an add-on. They may also offer leasing facilities.

Product: Training services

Kate Schofield has spent the past five years in computer training with the last two years as an independent consultant. Kate's area of expertise is in the most modern cutting-edge technology. The planned courses will run for two – three days, with quarterly one-day progress and update courses to keep those under training up to date with the latest technology. Advanced update courses will include training in the fields of intelligent system design, and digital asset management. It is intended that corporate clients will pay for an initial training course for a group of employees with an annual fee payable to entitle members of this group to attend update courses and also to receive technical e-bulletins.

Critical success factors which will make Aldine succeed

Vinay and Kate will make a good team. Vinay worked for the family manufacturing firm for four years as an office manager in a small company and has demonstrated strong managerial skills. These will compliment Kate's ability to easily master new technology and their proven skills in technology training. Further, both bring existing business and network contacts and both believe that they could increase their own business areas by 200% within 6 – 12 months if they could delegate some of the more day-to-day activities in their current work schedules to a team within the expanded Aldine business.

Financial summary

Initial forecasts suggest mark-up will be 40% and will yield expected profits of £90,000 in the first year. Vinay and Kate have, between them, £20,000 capital to invest in the company. They require a further £40,000 in funds from a lender – secured on Vinay's home. An overdraft facility of £15,000 is also required to finance larger than expected growth if cash flow is limited due to increased costs.

Management, Page 2

Proprietor: Employment records

Vinay Kamath– CV attached
Aged 37, Vinay holds a City & Guilds qualification in Computer Studies, and has always held responsible positions in their past career. Vinay worked for their family's manufacturing company as office manager for five years where their interest in computers began. Since then, Vinay worked for PCNow.com as computer services manager. They gained much experience of 'blue chip' requirements. Vinay has been married for four years and has two young children.

Kate Schofield – CV attached

Aged 26, Kate has three A Levels in English, History and Art. Kate has always sought positions that allow them to use their skills as a communicator and organiser. Kate worked for the leading firm APB Training Systems for three years before branching out as a successful independent training consultant for the past two years.

Other brief summary

Vinay Kamath's father, Aneesh, will be assisting with the initial setup of the manufacturing outlet. They retired four years ago after running their own successful manufacturing business for 40 years. They have promised to be available as an ongoing source of help and advice.

Two other suitable individuals have been sounded out for management positions and are keen to be involved with this venture.

How and why the business started

Vinay and Kate met about a year ago whilst working for the same client. Vinay later used Kate's services on an in-house training need at PCNow.com. Both came to believe that actively sourcing business for each other as a combined service will give them 'first sight' of opportunities.

Product/service, Page 3

Computer sales
Three models of all-in-one touchscreen computers will be built:

- RS 01 2.6 GHz 6MB 8GB 23.8" Screen
- RS 02 2.7 GHz 6MB 8GB 23.8" Screen
- RS 03 2.8GHz 8MB 8GB 25" Screen

Each machine can have a number of specified add-ons: CD-RW, DVD-RW, WiFi, Network Card, extra USB ports, and Memory Card readers. The machine was built in Aldine's own manufacturing unit, with service for the first year at the client's site. Thereafter an additional service contract will be offered which will feature a back-to-base repair service.

Servers will be purchased externally, as required to the specifications of the client.

Printer sales

A number of leading laser and ink jet printers will be available, with the anticipated service contracts as a major source of continuing income.

Training sales

Four levels of training will be available:

(1) Trainee: Up to eight persons with no, or little, computer experience

(2) Basic: Up to eight persons with basic computer skills

(3) Intermediate: Up to eight persons who have a good working knowledge of computer systems

(4) Advanced: Up to four persons who understand technical computer systems

Marketing, Page 4

Market outlook

Corporate computer sales are rising at 8% per quarter. It is expected to continue at that rate for another two years at which point the forecast economic slowdown may reduce sales growth to 4%. This will also be affected by the availability of new applications requiring corporations to upgrade their IT equipment. The ability to configure and give service support to virtual networks of computers to facilitate remote (home) working will be an important requirement.

Future growth

The growth of PCs took place over the last two decades to a point that penetration of the UK corporate market is complete. The market is saturated. However new applications require larger memory capacity and higher processing (and co-processing) speeds. It seems likely that concerns over data security will make future applications less PC based and instead require much greater reliance on access to corporate 'cloud' networks for data and applications.

This will create a further wave of PC spending as corporations upgrade to the new 'portal' technologies.

Market sector

Based in Runcorn, North-West England, the need for computer hardware and training is always available. SME Office space is increasing by 10% per annum. There are currently 95,000 SMEs within a 50-mile radius of Runcorn.

Expected client profile

- Small medium enterprise (SME) computer sales
- Medium to large companies for training services
- Commerce and industry
- Revenue £2m +
- Initially: Manchester and districts
- Prefer client to be computer literate

Competitors

Computers

Three main companies have been identified as true competitors ie, computer and peripheral sales and service direct to corporate clients on-site.

- **PowerShed Computers (Liverpool)** – established six years, revenue £0 (not known), six staff, no growth in past two years
- **Kato & Son PCs (Cheadle)** – established four years, revenue £2 million, 10 staff, specialist computer systems, steady growth

- **Solutions.com (national company)** – established two years, revenue £7.5 million, 40 staff, mainly top end clients, always stretched

Summary

Service complaints are common with all competitors, and hardware upgrades are long overdue. Solutions.com is looking to revamp their national operation in the spring.

Training

Two companies currently service the above client profile in computer training.

- **TrainOnline (Runcorn)** – established 15 years, revenue £900,000, computer training staff four, no plans for rapid growth
- **SolutionsTraining.com (National company – sister firm to Solutions.com)** – established eight years, revenue £6.5 million, computer training staff 35, no local office

Summary

At present demand exceeds resources. No effective competition is challenging the top companies, who offer only limited cutting-edge technology training skills.

Sales, Page 5

Advertising
An advert has been placed in:

- The 'computer sales and training' sections of local telephone directories
- The Tuesday business section of the syndicated regional newspapers covering Runcorn, Manchester and Liverpool

  (This appears each week and is booked to the end of the present calendar year.)
- *Computing* magazine – classified advertisement for Aldine in each issue of this weekly trade newspaper for IT specialists (available on free subscription)

  The advertisements have also been posted on the *Computing* magazine website and social media platforms.

Promotion

- Targeted mail shot within three months of receipt of funding to SMEs within 50 miles radius of Runcorn
- Telesales company to start promotion within one month of receipt of funding
- Two regional computer shows booked – first in two months, second in six months

Who will sell?

Both Vinay and Kate are competent sales negotiators.

The unique selling points (USPs)

- Dual capability of hardware supplier and training provider
- Latest technology training package
- Availability within seven working days

Setting the price

Computers
PC computers:

Prices include one year on-site total cover

- RS 01 2.6 GHz 6MB 8GB 23.8" Screen: £699.00

- RS 02 2.7 GHz 6MB 8GB 23.8" Screen: £999.00
- RS 03 2.8 GHz 8MB 8GB 25" Screen: £1,199.00

PC Service contracts are:

- £150 per annum for one PC and then £85 for each additional PC

Printers will be leased and serviced:

- £50 per annum for one, then £30 for each additional unit. This excludes consumables (paper, cartridges etc) which will be sold at cost + 30%

Training

- £110 Per Day **Trainee**: Up to 8 persons with no, or little, computer experience
- £135 Per Day **Basic**: Up to 8 persons with basic computer skills
- £180 Per Day **Intermediate**: Up to 8 persons, a good working knowledge of computer systems
- £295 Per Day **Advanced**: Up to 4 persons who understand technical computer systems

20% reduction for in-house training

Operation, Page 6

Location
New office-warehouse development on Industrial Estate located three miles from Runcorn town centre. Development has three meeting/conference rooms of high standard.

Three-year lease – no payment for first three months, then £3,000 per quarter in advance.

Equipment and costs to-date

Office	**£**
4 × Tables	475
4 × Chairs	350
4 × Filing cabinet	210
2 × Computer (at cost)	875
1 × Laser printer	375
1 × Ink jet printer	300
1 × Photocopier	1,450
1 × Answer phone	100
2 × Phone lines (installation)	150
Stationery and printing	1,100
Sub Total	5,385

Warehouse	£
4 × Work bench	2,000
6 × Stack shelving	1,500
Lighting	750
Sub Total	4,625

Inventory	£
Computer parts	3,750
Total	13,860

Staff

A mature office manager, and a trainee computer assembler will be sufficient for the initial three months – between three and nine months two training staff (to cover Trainee and Basic training) and a junior office clerk will be employed. At nine months a review of staff will need to take place.

Short-medium term trading, Page 7

Objective short-term

Computers

An average of five computer sales per week, and two service contracts. The trainee assembler to be competent at assembly within six months, and to service a computer on site within nine months.

Training

Kate training for average of three days per week within three months. New training staff to be working an average of four days per week by nine months.

At the 12-month plus period, a number of Kate and Vinay's tried and tested ex-colleagues and friends will be approached to join the company with equity available.

What if? – Contingency plans

Both Vinay and Kate are confident (with good reason) that a contingency plan is unlikely to be needed. However, Vinay has contacts who can provide him with computer fitting and servicing without notice, and likewise Kate has her training contacts.

In a real emergency, say, Vinay or Kate became ill, it is anticipated that the new staff could continue with 'basic' computer servicing or training – the requirement of a mature office manager is a strategic and important position.

Financials, Page 8

Monthly statement of profit or loss – projected for two years

- 20X1 to 20X2 Net Profit £90,000
- 20X2 to 20X3 Net Profit £175,000

As you can see from the Statement of Profit or Loss the business is expecting to make a net profit of £90,000 in the first year. The figures include the salary of the full complement of staff as outlined on page 6 above. The second year shows a net profit of £175,000: if this does happen, they will purchase their own premises for £180,000. The total complement of staff at the end of 20X2 is 25.

Statement of financial position projected for two years

- 20X1 to 20X2 Net Assets £130,000
- 20X2 to 20X3 Net Assets £260,000

The projected figures show the anticipated reinvestment of all profits for the first two years: this trend will continue. Inventory at the end of the second year is high but the discounts on bulk purchase is a profitable option in the computer industry, if technology allows.

Monthly cash flow forecast for two years

Scenario

- 20X1 to 20X2 Year-end Net Cash Flow £25,000
- 20X2 to 20X3 Year-end Net Cash Flow £120,000

In the first year, the cash flow will remain positive unless an opportunity to buy bulk computer parts presents itself. Likewise, the entire second year is planned to be in the black and consideration will be given at that time as to the purchase of assets.

Contact details

Vinay Kamath		Kate Schofield
0XXXX 9876543	**Office** Unit 3 The Trading Estate The Avenue Runcorn WA7 4SA	0XXXX9876543
019671 3546387	**Home**	01234 1234567
09876 1234567	**Mobile**	017790 9876544
96 Avenue Hill Runcorn WA7 5RR	**Home Address**	12 High Road Runcorn WA7 3AA

Requirement

Draft a report assessing the viability of the business proposal and making recommendations on how the document may be supplemented to improve the chances of Aldine securing funding.

See **Answer** at the end of this chapter.

2 Implementing strategy



Section overview

- Implementing strategy requires the corporate strategy to be broken down into functional strategies and operational plans.
- The business's choice of competitive strategy (cost leadership or differentiation) has implications for its functional strategies.

To implement the chosen corporate strategy, the business's competitive, investment and financial strategies need to be broken down further into functional strategies and operational plans.

The operational plan sets out what is expected of each function of the business and how actions will be taken to meet those expectations.

2.1 Functional strategies and competitive advantage

In the chapters, 'Strategic capability' and 'Strategic options,' we considered how businesses can achieve competitive advantage over their rivals. We examined the concept of the value chain and discussed how a business can create and sustain competitive advantage using cost or value drivers. The business's choice of competitive strategy determines whether the focus of the chain should be cost (cost leadership) or value (differentiation).

This has implications for an organisation when it is implementing its functional strategies, because these functional strategies need to be aligned to the organisation's overall competitive strategy.

The rest of this chapter will discuss functional strategies for:

- Finance
- Marketing
- Human resources
- Research and development, and innovation planning
- Operations/production
- Purchasing

Strategies for information and technology are considered in the chapter, 'Strategies for information.' In the chapter, 'Strategic capability,' we considered the emergence of workforce flexibility and the increasing use of developing technologies (such as artificial intelligence). Both of these trends can have a significant impact on the strategies that organisations implement to achieve their objectives.

3 The finance function



Section overview

- The finance function plays an important role in the strategic planning process, helping to set goals and objectives, and then monitoring progress towards them.
- Many finance functions have been transformed to become business partners.

3.1 The finance department and strategic planning

The role of finance is three-fold:

(a) Finance is a resource, which can be deployed so that objectives are met.

(b) A firm's objectives are often expressed in financial or semi-financial terms.

(c) Financial controls are often used to plan and control the implementation of strategies. Financial indicators are often used for detailed performance assessment.

As a planning medium and tool for monitoring, the finance function makes a variety of strategic contributions.

- Ensuring that resources of finance are available. Issues of raising equity or loan capital are important here. The amount of resources that the strategy will consume needs to be assessed, and the likely cost of those resources established. Cash flow forecasting will also be necessary.
- Integrating the strategy into budgets for revenues, operating costs and capital expenditure over a period. The budgeting process serves as a planning tool and a means of financial control and monitoring.

- Establishing the necessary performance measures, in line with other departments for monitoring strategic objectives.
- Establishing priorities, if, for example, altered conditions make some aspects of the strategy hard to fulfil.
- Assisting in the modelling process. Financial models are simplified representations of the business. It is easier to experiment with models, to see the effect of changes in variables, than with the business itself.

3.2 The role of the finance function as a business partner

The need for finance professionals and the finance function to provide insights not only at a strategic level, but also at a departmental level, has resulted in many organisations adopting functional strategies which have transformed the role of the finance function. As we highlighted in the chapter Evaluating and analysing performance and strategies, many finance functions have evolved to become business partners, providing advice and support to operational departments and business units, rather than simply reporting historical financial information.

This change has resulted in finance professionals being embedded within operational departments, so that they can provide financial insights at a local level. Proponents of the business partner model argue that embedding the finance function within operational teams enables finance professionals to take account of the issues and business decisions facing departments and operational teams, and therefore enables them to provide better focused support.

Acting as a business partner enables finance professionals to:

- Provide 'real time' support in the form of detailed data and information specific to the needs of given business units, departments, and projects
- Assist departmental managers in analysing both financial and non-financial performance data relevant to their area of responsibility, and to devise appropriate strategies to improve performance
- Help with the preparation of departmental business cases for new projects by robustly challenging the financial credentials of proposals before they are reviewed by senior management
- Support departmental heads in understanding investment appraisals, by going beyond simply identifying whether a proposal meets a financial criteria to help develop and refine business proposals
- Collaborate with departmental managers in the preparation of departmental budgets and forecasts
- Assist departmental managers in designing departmental information systems so that the financial requirements and outputs needed from the system are captured and incorporated into the final design



Professional skills focus: Assimilating and using information

When discussing and analysing the functional strategies of an organisation it is, as always, important to make use of the scenario detail provided. However, you may also make use of your own work experience and understanding to help generate practical ideas and solutions to problems.

4 Marketing planning

Section overview

- Marketing planning is one way by which corporate strategy is implemented.
- This requires a detailed plan of implementation and also of control.

4.1 The marketing plan

The main concepts and tools of marketing received detailed coverage in the chapter Strategies for products and markets.

The implementation and control of the marketing effort might take the form of a **marketing plan**:

Section	Content
The executive summary	This is the finalised planning document with a summary of the main goals and recommendations in the plan.
Situation analysis	This consists of the SWOT (strengths, weaknesses, opportunities and threats) analysis and forecasts.
Objectives and goals	What the organisation is hoping to achieve, or needs to achieve, perhaps in terms of market share or 'bottom line' profits and returns.
Marketing strategy	This considers the selection of target markets, the marketing mix and marketing expenditure levels.
Strategic marketing plan	• Three to five (or more) years long • Defines scope of product and market activities • Aims to match the activities of the firm to its distinctive competences
Tactical marketing plan	• One-year time horizon • Generally based on existing products and markets • Concerned with marketing mix issues
Action plan	• This sets out how the strategies are to be achieved. • The Marketing mix strategy should cover the 7 Ps discussed in the chapter Strategies for products and markets. • The mix strategy may vary for each segment.
Budgets	These are developed from the action programme.
Controls	These will be set up to monitor the progress of the plan and the budget.

4.2 Corporate strategy and marketing strategies

So, what is the relationship between marketing and overall strategic management? The two are closely linked since there can be no corporate plan which does not involve products/services and customers.

Corporate strategic plans aim to guide the overall development of an organisation. Marketing planning is subordinate to corporate planning but makes a significant contribution to it and is concerned with many of the same issues:

- The **strategic** component of marketing planning focuses on the direction which an organisation will take in relation to a specific market, or set of markets, to achieve a specified set of objectives.
- Marketing planning also requires an **operational** component that defines tasks and activities to be undertaken to achieve the desired strategy. The **marketing plan** is concerned uniquely with **products** and **markets**.

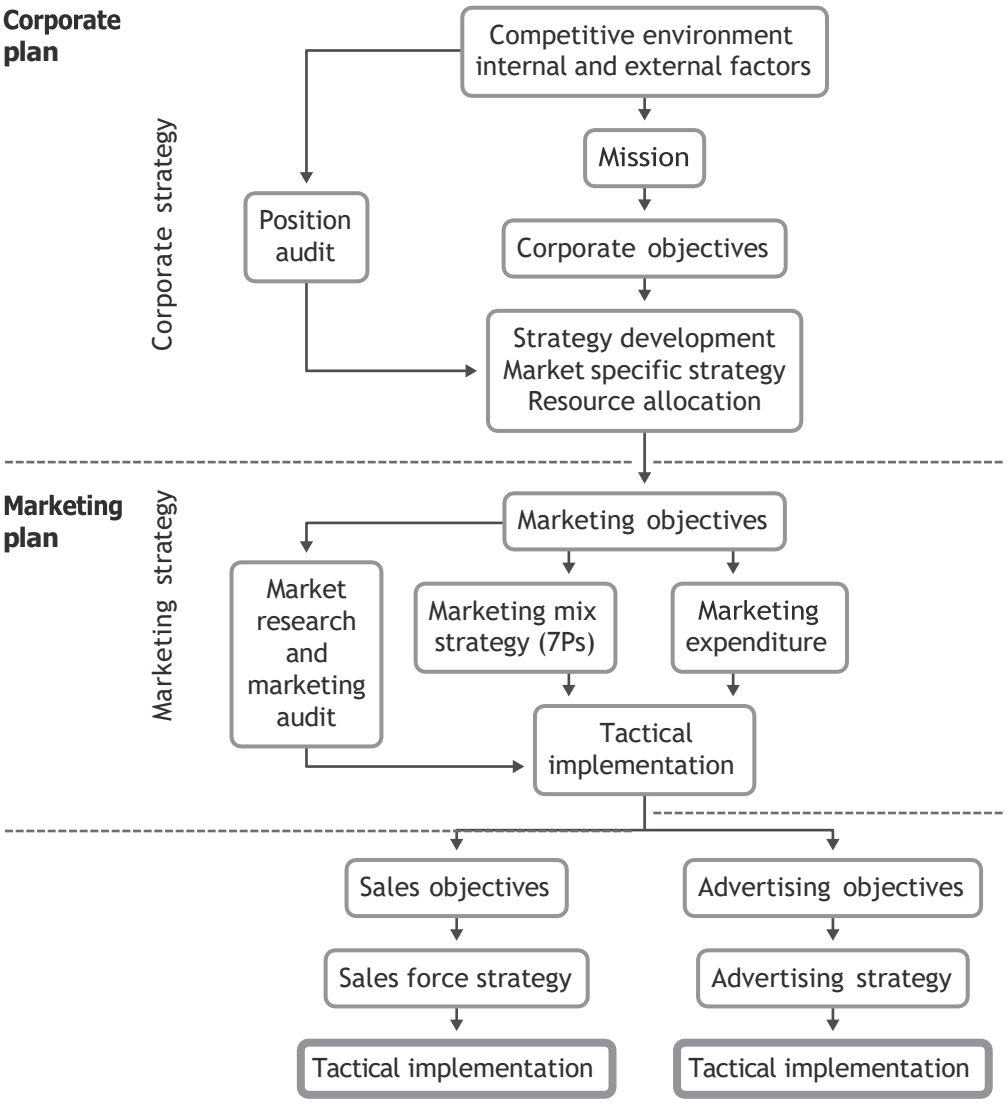
The process of corporate planning and the relationship with marketing strategy is shown in the following table:

	Corporate	Marketing
Set objectives	For the firm as a whole: eg, increase profits by X%.	For products and market: eg, increase market share by X%; increase revenue.
Internal appraisal (strengths and weaknesses)	Review the effectiveness of the different aspects of the organisation.	Conduct a marketing audit: a review of marketing activities. Does the firm have a marketing orientation?
External appraisal (opportunities and threats)	Review political, economic, social, technological, ecological factors impacting on the whole firm.	Review environmental factors as they affect customers, products and markets.
Gaps	There may be a gap between desired objectives and forecast objectives. How to close the gap.	The company may be doing less well in particular markets than it ought to. Marketing will be focused on growth.

	Corporate	Marketing
Strategy	Develop strategies to fill the gap: eg, diversifying, entering new markets.	A marketing strategy is a plan to achieve the organisation's objectives by specifying: • Resources to be allocated to marketing • How those resources should be used. In the context of applying the marketing concept, a marketing strategy would: • Identify target markets and customer needs in those markets • Plan products which will satisfy the needs of those markets • Organise marketing resources, so as to match products with customers.
Implementation	Implementation is delegated to departments of the business.	The plans must be put into action, eg, advertising space must be bought.
Control	Results are reviewed and the planning process starts again.	Has the firm achieved its market share objectives?

The following diagram summarises the relationship of marketing planning to the corporate plan.

Figure 13.1: Marketing and corporate planning



4.3 Controlling marketing activities

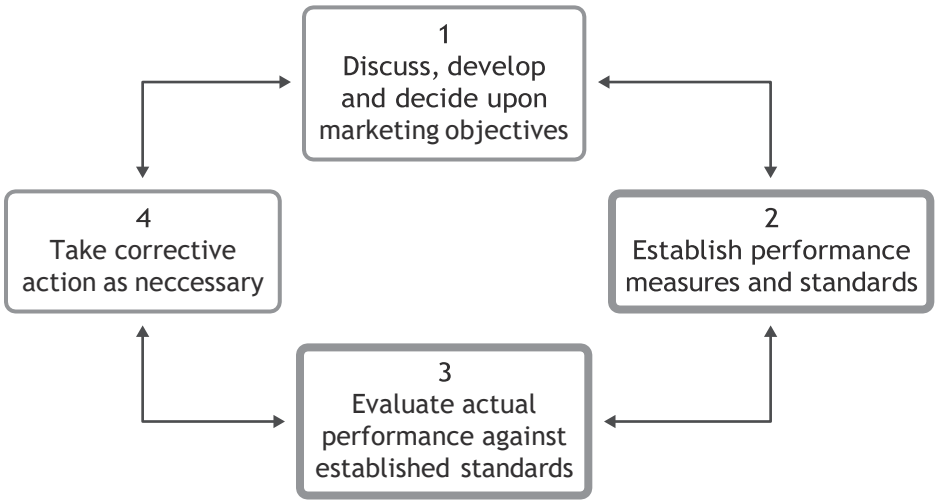
Once marketing strategies are implemented, there needs to be **control** and **performance measures** in place to support the purpose of the plan. **Marketing strategies** are developed to satisfy corporate objectives and may reflect the results of the marketing audit.

The **marketing control process** can be broken down into four stages:

- Development of objectives and strategies
- Establishment of standards
- Evaluation of performance
- Corrective action

Part of the corrective action stage may well be to adjust objectives and strategies in the light of experience.

Figure 13.2: The marketing control process



Typical quantitative performance levels might be as follows:

- Market share, perhaps by comparison with a major competitor
- Operational targets may also be relevant to marketing performance, for example having the right products available
- Other measures can include measures of customer satisfaction, if these are regularly monitored

Performance is evaluated by comparing actual with target. Control action can be taken.

4.4 The marketing audit

A marketing audit is a wide ranging **review of all activities associated with marketing** undertaken by an organisation.

To exercise proper strategic control a marketing audit should satisfy four requirements:

- It should take a comprehensive look at every product, market, distribution channel and ingredient in the marketing mix.
- It should not be restricted to areas of apparent ineffectiveness such as an unprofitable product, a troublesome distribution channel, or low efficiency on direct selling.
- It should be carried out according to a set of predetermined, specified procedures.
- It should be conducted regularly.

Interactive question 2: Marketing plan

2.1 What is a SWOT analysis and how does it lead to an understanding of realistic market opportunities?

2.2 Explain the importance of marketing planning for a new consumer product to be launched in your country.

2.3 Using examples, identify the main steps involved in the marketing planning process.

See **Answer** at the end of this chapter.

4.5 *Digital marketing*

Definition

Digital marketing: *This refers to any marketing methods conducted through electronic devices, including online marketing via the internet. A business may use websites, search engines, blogs, social media, video, email and similar channels to reach customers.*

Traditional marketing results in one-way communication with the intended recipient, since it is not possible to create interaction with, for example, a television or print advert. Digital marketing facilitates two-way communication between the business and its current or prospective customers. For example, customers are able to comment on products or services via social media and companies can respond directly to the customer, offering a fast and tailored response. Successful digital marketing strategies promote goods and services using the internet via online advertising, search engine optimisation (SEO), social media marketing and content creation. Benefits of digital marketing include:

- *Global reach – traditional marketing is restricted to specific geographic areas unlike internet marketing.*
- *Low entry cost – methods such as a website, social media promotion and blogs are accessible for all business regardless of size and availability of financial resources.*
- *Improved targeting – it is possible to identify and target specific market segments, improving the success rate of the marketing campaign.*
- *Adaptability – if a marketing campaign is not working the business will get quicker feedback and can make rapid changes to tailor the strategy to its audience.*
- *Relationship building – customer interaction with the business, via social media, builds longer- term customer loyalty.*

5 Human resources planning

Section overview

- Human resource management (HRM) is the process of evaluating an organisation's human resource needs, finding people to fill those needs, and getting the best work from each employee by providing the right incentives and job environment – with the overall aim of helping achieve organisational goals.
- This requires planning resource needs for the future and succession planning for existing staff.
- Staff appraisals are a vital part of this process.
- The increased use of technology in the workplace raises some interesting considerations in respect of human resource planning.

5.1 Scope of human resource management (HRM)

Definition

Human resource management (HRM): 'A strategic and coherent approach to the management of an organisation's most valued assets: the people working there who individually and collectively contribute to the achievement of its objectives for sustainable competitive advantage' (Armstrong).

5.1.1 Goals of strategic HRM

- Serve the **interests of management**, as opposed to employees
- Suggest a **strategic approach to personnel issues**
- Link **business mission to HR strategies**
- Enable **human resource development to add value** to products and services
- Gain **employees' commitment** to the organisation's values and goals

The HR strategy has to be related to the business strategy.



Context example: HR issues in the airline industry

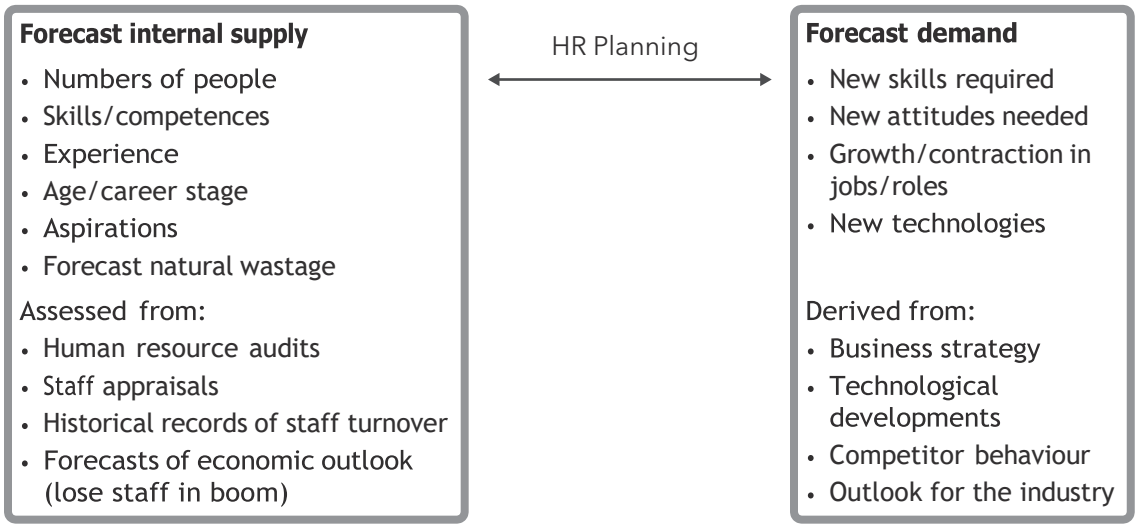
Most airlines are trying to become global companies to avoid dependence on one country.

Business strategy	HR implications	Airline example
What business are we in?	What people do we need?	Air transportation requires pilots, cabin crew, ground crew etc.
What products/markets, level of output and competitive strategy, now and in future.	Where do we need people, what are they expected to do, and how many? Location and size of workforce? Productivity expected and output?	The airline is going global and therefore it needs cabin crew who are skilled in languages and are sensitive to cultural differences.
What is the culture and value system? Is it the right one?	The need to change culture and values.	A cultural change programme; recruiting people to fit in with the right value system; attitudinal assessments.
Tomorrow's strategies, demands and technologies.	Tomorrow's personnel needs must be addressed **now**, because of lead times. New technology requires training in **new skills**.	Recruitment, training, cultural education.
Critical success factors.	How far do these depend on staff?	Service levels in an aircraft depend very much on the staff, so HRM is crucial.

5.2 Human resources planning

HR must keep a **balance** between the **forecast supply** of human resources in the organisation and the organisation's **forecast demand** for human resources.

Figure 13.3: HR planning



5.3 Closing the gap between demand and supply: the HR plan

The HR plan is prepared on the basis of personnel requirements, and the implications for productivity and costs. The HR plan breaks down into subsidiary plans.

Plan	Comment
Recruitment plan	Numbers; types of people; when required; recruitment programme
Training plan	Numbers of trainees required and/or existing staff needing training; training programme
Redevelopment plan	Programmes for transferring, retraining employees
Productivity plan	Programmes for improving productivity, or reducing manpower costs; setting productivity targets
Redundancy plan	Where and when redundancies are to occur; policies for selection and declaration of redundancies; re-development, re-training or re- location of redundant employees; policy on redundancy payments, union consultation
Retention plan	Actions to reduce avoidable labour wastage

The plan should include budgets, targets and standards. It should allocate responsibilities for implementation and control (reporting, monitoring achievement against plan).

5.4 Succession planning

Succession planning should be an integral part of the HR plan and should support the organisation's chosen strategy. The developed plan should also be compatible with any changes that that are foreseen in the way the organisation operates. It is likely that strategic objectives will only be obtained if management development proceeds in step with the evolution of the organisation.

5.4.1 Benefits of succession planning

- The development of managers at all levels is likely to be improved if it takes place within the context of a succession plan. Such a plan gives focus to management development by suggesting objectives that are directly relevant to the organisation's needs.
- Continuity of leadership is more likely, with fewer dislocating changes of approach and policy.
- Assessment of managerial talent is improved by the establishment of relevant criteria.

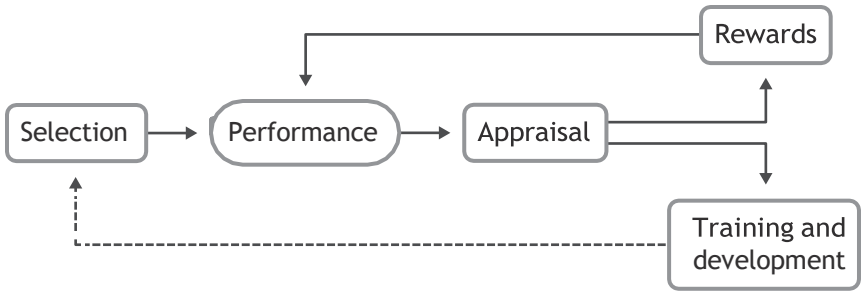
5.4.2 Features of successful succession planning

- The plan should focus on future requirements, particularly in terms of strategy and culture.
- The plan should be driven by top management. Line management also have important contributions to make. It is important that it is not seen as a HR responsibility.
- Management development is as important as assessment and selection.
- Assessment should be objective and preferably involve more than one assessor for each manager assessed.
- Succession planning will work best if it aims to identify and develop a leadership team rather than merely to establish a queue for top positions. A pool of talent and ability is a flexible asset for the organisation.

5.5 The human resource (HR) cycle

A relatively simple model that provides a framework for explaining the nature and significance of HRM is Devanna's human resource cycle.

Figure 13.4: Devanna's human resource cycle



Selection is important to ensure the organisation obtains people with the qualities and skills required.

Appraisal enables targets to be set that contribute to the achievement of the overall strategic objectives of the organisation. It also identifies skills and performance gaps, and provides information relevant to reward levels.

Training and development ensure skills remain up-to-date, relevant, and comparable with (or better than) the best in the industry.

The **reward system** should motivate and ensure valued staff are retained.

Performance depends upon each of the four components and how they are co-ordinated.

5.5.1 The role of staff appraisals

The setting up of a systematic approach to staff appraisal (otherwise called a performance review) is essential to good human resources management. It has the following benefits.

- A forum for agreeing objectives for the coming year that ensure the individual pursues goals that are congruent with the business strategy

- An opportunity to outline or respond to difficulties affecting the individual's performance
- Provision of feedback will motivate and develop the individual
- Identifies personal development needs of the individual, eg, for future roles
- Identifies candidates for succession and development

Interactive question 3: ScannerTech

ScannerTech is a fast growing hi-tech company with expertise in electronic scanners. It has 100 employees and aims to double in size over the next three years. The company was set up by two researchers from a major university who now act as joint managing directors. They are intend leaving ScannerTech once the growth objective is achieved and it is large enough to be sold.

ScannerTech makes sophisticated imaging devices used by the airline security and health industries. These two markets are very different in terms of customer requirements but use the same basic technology. Because of growing sales from exports the current strategic plan anticipates a foreign manufacturing plant being set up within the next three years. Present managers are staff who joined in the early years of the company and have their expertise in research and development. Further growth will require additional staff in all parts of the business, particularly in manufacturing and sales and marketing.

Olivia Marcuse is HR manager at ScannerTech. They are annoyed that HR is the one management function not involved in the strategic planning process shaping the future growth and direction of the company. They feel trapped in a role traditionally given to HR specialists, that of simply reacting to the staffing needs brought about by strategic decisions taken by other parts of the business. They feel it is time to make the case for a strategic role for HR at ScannerTech to help it face its challenges.

Requirement

Prepare a short report for Olivia Marcuse to present to ScannerTech's board of directors on the way a human resource plan could link effectively to its growth strategy.

See **Answer** at the end of this chapter.

5.6 Human resources and technology

In the following section, we explore the impact that technological change is having on people working within organisations and its implications for human resource planning.

5.6.1 Human resources and big data

In the chapter, 'Strategic capability,' we discussed the emergence of big data and developing technologies, and the role that they are starting to play in organisations. It is important to note that advances in wearable data capturing technologies are now being used by organisations as way of improving the work life balance of employees. Wearable data capturing technologies are clothing accessories which incorporate computer and electronic technologies, such as sensors. They can be used to track and monitor the physical movements of the wearer.

5.6.2 Human resources and home working

Earlier in the Workbook we explored the concept of the virtual organisation and the growth in workforce flexibility. The drive among employers to allow their employees to work from home has provided many obvious benefits, including reductions in the amount of physical office space required and the costs associated of running premises, whilst also giving workers greater flexibility. Home working however is not without its challenges.

Context example: The perils of home working

The outbreak of the coronavirus forced a move towards home working for many companies. Post- pandemic, remote and hybrid working are now seen as key aspects of flexible working and therefore well-being. Home working can bring work-life benefits to employees but there are also financial benefits for companies. According to a report commissioned by Cigna Healthcare (2022), such benefits include: smaller space requirements and lower real estate costs; the ability to look anywhere in the world to find the best talent; and reduced attrition as employees do not have to leave the company if they want to move. In fact a 2022 survey by the flexible office space provider Regus found that 69% of companies are planning to reduce their office footprints, indicating the impact of hybrid working on office space.

However, home working also brings challenges. It becomes harder to maintain company culture and keep employees connected and engaged. A survey of 65,000 employees, conducted by management consultancy firm McKinsey found that working from home increased burnout rates, as employees struggle to juggle careers and family. The survey also found that the gap between male and female burnout rates nearly doubled, with 42% of women reporting burnout compared to a third of men. Experiences at Finastra, a global fintech company, have been similar and the company has been actively encouraging employees to block time off in their calendars for things that they value such as exercise, school runs and time with family.

Dr Lynda Gratton, professor of management practice at London Business School, explains how home working can also lead to loneliness which is an 'enormous trigger of mental health problems.' Whilst home and hybrid working are here to stay, firms are looking for more creative ways to replicate the office experience. For example, Japanese firm Fujitsu has been experimenting with virtual reality to create virtual office environments for its employees. According to Andrew Davis, head of future workplace strategy and growth at Fujitsu, such virtual environments are 'a great way of trying to replicate the informal connections you get from a traditional physical office environment, which I think helps with mental health.'

Sources:

- Cigna Healthcare (2022) Re-writing the Rules: Building a Healthy Hybrid Workplace. *The Economist* [Online] Available from: https://impact.economist.com/projects/rewriting-the-rules/rewriting-the-rules-building-a-healthy-hybrid-workplace/ [Accessed 18 July 2023]
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6 Research and development, and innovation planning

Section overview

- Research and development (R&D) is essential for product and process improvement.
- The need to ensure that R&D has a commercial application forms a link between R&D and other disciplines such as marketing, operations and finance.
- Closely connected to the need for R&D planning is the need for innovation planning.

6.1 Product and process research

Research may be intended to improve **products** or **processes**. R&D should support the organisation's strategy, be properly planned and be closely co-ordinated with marketing.

Many organisations employ **specialist staff** to conduct research and development (R&D). They may be organised in a separate functional department of their own. In an organisation run on a product division basis, R&D staff may be employed by each division.

There are two categories of R&D.

Product research – new product development

The new product development process must be carefully controlled; new products are a major source of competitive advantage but can cost a great deal of money to bring to market. A screening process is necessary to ensure that resources are concentrated on projects with a high probability of success.

Process research

Process research involves attention to how the goods/services are produced. Process research has these aspects:

- **Processes** are crucial in service industries (eg, fast food) as part of the services sold.
- **Productivity** – efficient processes save money and time.
- **Planning** – if you know how long certain stages in a project are likely to take, you can plan the most efficient sequence.
- **Quality management** for enhanced quality.

R&D should be closely co-ordinated with marketing

- Customer needs, as identified by marketers, should be a vital input to new product developments.
- The R&D department might identify possible changes to product specifications so that a variety of marketing mixes can be tried out and screened.

6.2 Strategic role of R&D

Despite the evident costs and uncertainties of R&D expenditure, its strategic importance can be understood by reference to some of the strategic models discussed earlier:

- **Porter's generic strategies**: Product innovation could be a source of **differentiation**. Process innovation may enable differentiation or **cost leadership**.
- **Porter's value chain**: R&D is included within the support activities of **technology development**. It can be harnessed in the service of lower costs or improved differentiation.
- **Ansoff matrix**: R&D supports all four strategic quadrants. Strategies of Market Penetration and Market Development can be served by product refinement. Product Development and Diversification will require more significant innovations to product.
- **Industry and product lifecycles**: The obsolescence of existing products can be accelerated by product R&D and so R&D is required to provide the firm with replacements.

6.3 Innovation planning



Definition

Innovation: Is concerned with the generation of new ideas of how to do business. It is primarily a creative activity.

Closely linked to the need for R&D planning is the need for innovation planning. The desire to reduce costs and build a basis for differentiation are key drivers for innovation. Generating and maintaining a creative environment within the organisation is a key feature of innovation, establishing such an environment can be created through a number of dimensions:

- **Leadership:** Senior management need to set and communicate a vision which encourages the exploration of new ideas which challenges existing ways of doing things. Leadership is important in giving direction.
- **Culture:** Encouraging a creative culture requires managers to change their ways of thinking, for example viewing failure as an important learning process to be expected. 'Learning organisations' don't view failure as something negative, but instead they learn from their mistakes.
- **People:** Organisations need to encourage a team-based approach that creates a sense of ownership and encourages participation.
- **Structure:** Managers need to embrace new ways of working, part of which may involve developing increasingly flexible organisational structures which encourage innovation.
- **Communication:** Management need to encourage greater openness in communication between workers so that ideas can be shared more freely.

7 Operations planning and management



Section overview

- Operations management is concerned with the transformation of 'inputs' into 'outputs' that meet the needs of the customer.
- It is characterised by the four Vs of volume, variety, variation in demand, and visibility.
- Capacity planning and some of the modern IT/IS applications supporting them are reviewed.
- Quality assurance and TQM are essential components of many modern manufacturing approaches.

7.1 Formulating operations strategy



Definition

Operations management: Is concerned with the **design**, **implementation** and **control** of the **processes** in an organisation that transform inputs (materials, labour, other resources, information and customers) into output products and services.

In broad terms, operational planning will include many of the following concepts.

- Setting **operational objectives** that are consistent with the overall business strategy of the organisation
- Translating **business strategy** or **marketing strategy** into **operations strategy**, by means of identifying key competitive factors (referred to perhaps as order-winning factors or critical success factors)
- Assessing the relative importance of different **competitive factors**
- Assessing current operational performance by **comparison** with the performance of **competitors**

- Using the idea of a 'clean-slate' or 'green-field' approach to strategy selection

  Managers are asked to consider how they would ideally design operations if they could **start again from scratch**. The ideal operations design is then compared with actual operations, and important differences identified. Strategy decisions are then taken to move actual performance closer towards the ideal.
- Formulating strategy

  This could be based on other types of **gap analysis**, such as comparing what the market wants with what the operation is actually achieving, and taking decisions aimed at closing the significant gaps.
- Emphasising the iterative process of strategy selection

  Strategies should be **continually reviewed, refined and re-developed** through experience and in response to changes in the environment.

Six items that should be incorporated into an organisation's operations strategy:

Item	Comment
Capability required	What is it that the organisation wants to 'do' or produce?
Range and location of operations	How big does the organisation want to be – or can it be? How many sites and where should they be located?
Investment in technology	How will processes and production be performed?
Strategic buyer-supplier relationships	Who will be key strategic partners?
New products/services	What are the expected product life-cycles?
Structure of operations	How will staff be organised and managed?

7.2 Operations: The four Vs

In Business Technology and Finance. you will have learnt about the **four Vs** of operations: **volume**, **variety**, **variation** in demand and **visibility**. These affect the way in which an operation will be organised and managed.

	High	Low
Volume	A high-volume operation lends itself to a capital-intensive operation, with specialisation of work and well- established systems for getting the work done. Unit costs should be low.	Low-volume operations mean that each member of staff will have to perform more than one task, so that specialisation is not achievable. There will be less systemisation, and unit costs of output will be higher than with a high-volume operation.
Variety	When there is large variety, an operation needs to be flexible and capable of adapting to individual customer needs. The work may therefore be complex, and unit costs will be high.	When variety is limited, the operation should be well defined, with standardisation, regular operational routines and low unit costs.

	High	Low
Variation in demand	When the variation in demand is high, an operation has a problem with capacity utilisation. It will try to anticipate variations in demand and alter its capacity accordingly. For example, the tourist industry takes on part-time staff during peak demand periods. Unit costs are likely to be high because facilities and staff are under- utilised in the off-peak periods.	When demand is stable, it should be possible for an operation to achieve a high level of capacity utilisation, and costs will accordingly be lower.
Visibility	Many services are highly visible to customers. High visibility calls for staff with good communication and inter- personal skills. They tend to need more staff than low-visibility operations and so are more expensive to run. When visibility is high, customer satisfaction with the operation will be heavily influenced by their perceptions. Customers will be dissatisfied if they have to wait, and staff will need high customer contact skills. Unit costs of a visible operation are likely to be high.	When visibility is low, there can be a time lag between production and consumption, allowing the operation to utilise its capacity more efficiently. Customer contact skills are not important in low-visibility operations, and unit costs should be low. Some operations are partly visible to the customer and partly invisible, and organisations might make this distinction in terms of front office and back-office operations.



Professional skills focus: Structuring problems and solutions

When assessing the efficiency and effectiveness of business operations, use the 4Vs of operations to help you generate a structured response. However, as with all frameworks, do not confine yourself to only examining the 4Vs and ensure you use the scenario detail to identify and assess a broad range of operational issues.

7.3 Capacity planning

Various types of capacity plan may be used.

- **Level capacity plan** is a plan to maintain activity at a constant level over the planning period, and to ignore fluctuations in forecast demand. In a manufacturing operation, when demand is lower than capacity, the operation will produce goods for inventory. In a service operation, such as a hospital, restaurant or supermarket management must accept that resources will be under-utilised for some of the time, to ensure an adequate level of service during peak demand times. Queues will also be a feature of this approach.
- **Chase demand plan** aims to match capacity as closely as possible to the forecast fluctuations in demand. To achieve this aim, resources must be flexible. For example, staff numbers might have to be variable and staff might be required to work overtime or shifts. Variations in equipment levels might also be necessary, perhaps by means of short-term rental arrangements.
- **Demand management planning** – reduce peak demand by switching it to the off-peak periods such as by offering off-peak prices. Price discrimination is discussed in the chapter Strategies for products and markets.
- **Mixed plans** – capacity planning involving a mixture of level capacity planning, chase demand planning and demand management planning.

7.4 Capacity control

Capacity control involves reacting to actual demand and influences on actual capacity as they arise. IT/IS applications used in manufacturing operations include:

- **Materials requirements planning (MRP I)**: Converts estimates of demand into a materials requirements schedule.
- **Manufacturing resource planning (MRP II):** A computerised system for planning and monitoring all the resources of a manufacturing company: manufacturing, marketing, finance and engineering.
- **Enterprise resource planning (ERP)** software includes a number of integrated modules designed to support all of the key activities of an enterprise. This includes managing the key elements of the supply chain such as product planning, purchasing, stock control and customer service including order tracking.

7.5 Just-in-time systems

Definition

Just-in-time manufacturing: An approach to planning and control based on the idea that goods or services should be produced only when they are ordered or needed. Also called lean manufacturing.

Interactive question 4: Lean manufacturing at Toyota

Japanese car manufacturer Toyota was the first company to develop JIT (JIT was originally called the Toyota Production System). After the end of the world war in 1945, Toyota recognised that it had much to do to catch up with the US automobile manufacturing industry. The company was making losses. In Japan, however, consumer demand for cars was weak, and consumers were very resistant to price increases. Japan also had a bad record for industrial disputes. Toyota itself suffered from major strike action in 1950.

The individual credited with devising JIT in Toyota from the 1940s was Taiichi Ohno, and JIT techniques were developed gradually over time. The Kanban system for example, was devised by Toyota in the early 1950s, but was only finally fully implemented throughout the Japanese manufacturing operation in 1962.

Ohno identified wastes and worked to eliminate them from operations in Toyota. Measures that were taken by the company included the following.

(1) The aim of reducing costs was of paramount importance in the late 1940s.

(2) The company should aim to level the flow of production and eliminate unevenness in the work flow.

(3) The factory layout was changed. Previously all machines, such as presses, were located in the same area of the factory. Under the new system, different types of machines were clustered together in production cells.

(4) Machine operators were re-trained.

(5) Employee involvement in the changes was seen as being particularly important. Teamwork was promoted.

(6) The Kanban system (production on demand) was eventually introduced, but a major problem with its introduction was the elimination of defects in production.

Requirement

Can you explain how each of the changes described above came to be regarded as essential by Toyota's management?

See **Answer** at the end of this chapter.

7.5.1 Three key elements in the JIT philosophy

Element	Comment
Elimination of waste	Waste is defined as any activity that does not add value. Examples of waste identified by Toyota were: • Overproduction – ie, producing more than was immediately needed by the next stage in the process. • Waiting time – measured by labour efficiency/machine efficiency. • Transport – moving items around a plant does not add value. Waste can be reduced by changing the layout of the factory floor so as to minimise the movement of materials. • Waste in the process – some activities might be carried out only because there are design defects in the product, or because of poor maintenance work. • Inventory – inventory is wasteful. The target should be to eliminate all inventory by tackling the things that cause it to build up. • Simplification of work – an employee does not necessarily add value by working. Simplifying work reduces waste in the system (the waste of motion) by eliminating unnecessary actions. • Defective goods are quality waste. This is a significant cause of waste in many operations.
The involvement of all staff in the operation	JIT is a cultural issue, and its philosophy has to be embraced by everyone involved in the operation if it is to be applied successfully. Critics of JIT argue that management efforts to involve all staff can be patronising.
Continuous improvement	The ideal target is to meet demand immediately with perfect quality and no waste. In practice, this ideal is never achieved. However, the JIT philosophy is that an organisation should work towards the ideal, and continuous improvement is both possible and necessary. The Japanese term for continuous improvement is *kaizen*.

7.5.2 JIT purchasing

With JIT purchasing, an organisation establishes a close relationship with trusted suppliers, and develops an arrangement with the supplier for being able to purchase materials only when they are needed for production. The supplier is required to have a flexible production system capable of responding immediately to purchase orders from the organisation.

7.5.3 JIT and service operations

The JIT philosophy can be applied to service operations as well as to manufacturing operations. Whereas JIT in manufacturing seeks to eliminate inventories, JIT in service operations seeks to remove queues of customers.

Queues of customers are wasteful because:
- They waste customers' time.
- Queues require space for customers to wait in, and this space is not adding value.
- Queuing lowers the customer's perception of the quality of the service.

The application of JIT to a service operation calls for the removal of specialisation of tasks, so that the work force can be used more flexibly and moved from one type of work to another, in response to demand and work flow requirements.



Context example: JIT in a postal service

A postal delivery has specific postal workers allocated to their own routes. However, there may be scenarios where, say, Route A is overloaded whilst Route B has a very light load of post.

Rather than have letters for Route A piling up at the sorting office, when the person responsible for Route B has finished delivering earlier, this person might help out on Route A.

Teamwork and flexibility are difficult to introduce into an organisation because people might be more comfortable with clearly delineated boundaries in terms of their responsibilities. However, the customer is usually not interested in the company organisation structure because they are more interested in receiving a timely service.

In practice, service organisations are likely to use a buffer operation to minimise customer queuing times. For example, a hairdresser will get an assistant to give the client a shampoo to reduce the impact of waiting for the stylist. Restaurants may have an area where guests may have a drink if no vacant tables are available immediately; such a facility may even encourage guests to plan in a few drinks before dinner thereby increasing the restaurant's revenues.

7.6 Quality management



Definitions

Quality assurance: Focuses on the way a product or service is produced. Procedures and standards are devised with the aim of ensuring defects are eliminated (or at least minimised) during the development and production process).

Quality control: Is concerned with checking and reviewing work that has been done. Quality control is focused on detecting defects in the output produced, as a result it has a narrower focus than quality assurance.

7.6.1 Cost of quality

The **cost of quality** may be looked at in a number of different ways. For example, some may say that producing higher quality output will increase costs – as more costly resources are likely to be required to achieve a higher standard. Others may focus on the idea that poor quality output will lead to customer dissatisfaction, which generates costs associated with complaint resolution and warranties.

The demand for better quality has led to the acceptance of the view that quality management should aim to **prevent** defective production rather than simply detect it because it reduces costs in the long run.

Most modern approaches to quality have therefore tried to assure quality in the production process, (quality assurance) rather than just inspecting goods or services after they have been produced.

7.6.2 Total quality management (TQM)

Total quality management (TQM) is a popular technique of **quality assurance.** Main elements are:

- **Internal customers and internal suppliers**: All parts of the organisation are involved in quality issues, and need to work together. Every person and every activity in the organisation affects the work done by others. The work done by an internal supplier for an internal customer will eventually affect the quality of the product or service to the external customer.
- **Service level agreements**: Some organisations formalise the internal supplier-internal customer concept by requiring each internal supplier to make a **service level agreement** with its internal customer, covering the terms and standard of service.
- **Quality culture within the firm**: Every person within an organisation has an impact on quality, and it is the responsibility of everyone to get quality right.
- **Empowerment**: Recognition that employees themselves are often the best source of information about how (or how not) to improve quality.

8 Purchasing



Section overview

- Purchasing is a major influence on a firm's costs and quality. It is often called procurement.
- Sourcing strategy is developing from the use of many suppliers to get a better price to the fostering of strategic procurement relationships with just a few.
- E-procurement systems can help to improve the efficiency of the process and reduce costs.



Definition

Purchasing: Is the acquisition of material resources and business services for use by the organisation.

8.1 The importance of purchasing

Cost: Raw materials and subcomponents purchases are a major cost for many firms.

Quality: The quality of input resources affects the quality of outputs and the efficiency of the production function.

Strategy: In retailing, buying goods for resale is one of the most important activities of the business.

8.2 The purchasing mix

The purchasing manager has to obtain the best purchasing mix.

8.2.1 Quantity

The size and timing of purchase orders will be dictated by the balance between two things:

- Delays in production caused by insufficient inventories
- Costs of holding inventories: tied up capital, storage space, deterioration, insurance, risk of pilferage

A system of inventory control will set optimum reorder levels (the inventory level at which supplies must be replenished so as to arrive in time to meet demand) to ensure economic order quantities (EOQ) are obtained for individual inventory items.

8.2.2 Quality

The production department will need to be consulted about the quality of goods required for the manufacturing process, and the marketing department about the quality of goods acceptable to customers. Purchased components might be an important constituent of product quality.

8.2.3 Price

Favourable short-term trends in prices may influence the buying decision, but purchasing should have an eye to the best value over a period of time – considering quality, delivery, urgency of order, inventory-holding requirements and so on.

8.2.4 Delivery

The lead time between placing and delivery of an order can be crucial to efficient inventory control and production planning. The reliability of suppliers' delivery arrangements must also be assessed.

8.3 Sourcing strategies

There are a range of possible strategies open to an organisation when deciding who they will purchase their supplies from.

Supply sourcing strategies	
Option	**Comment**
Single supplier	**Advantages** Stronger relationship with the supplier Possible source of superior quality due to increased opportunity for a supplier quality assurance programme Facilitates better communication Economies of scale Facilitates confidentiality Possible source of competitive advantage **Disadvantages** Vulnerable to any disruption in supply Supplier power may increase if no alternative supplier The supplier is vulnerable to shifts in order levels
Multiple suppliers	**Advantages** Access to a wide range of knowledge and expertise Competition among suppliers may drive the price down Supply failure by one supplier will cause minimal disruption **Disadvantages** Not easy to develop an effective quality assurance programme Suppliers may display less commitment Neglecting economies of scale
Delegated	A supplier is given responsibility for the delivery of a complete sub-assembly. For example, rather than dealing with several suppliers a 'first tier' supplier would be appointed to deliver a complete sub-assembly (eg, a PC manufacturer may delegate the production of keyboards). **Advantages** Allows the utilisation of specialist external expertise

Supply sourcing strategies	
Option	**Comment**
	Frees-up internal staff for other tasks The purchasing entity may be able to negotiate economies of scale **Disadvantages** First tier supplier is in a powerful position. Competitors may utilise the same external organisation so unlikely to be a source of competitive advantage.

Interactive question 5: PicAPie Ltd

PicAPie Ltd employs a total quality management program and manufactures 12 different types of pie from chicken and leek to vegetarian. The directors of PicAPie are proud of their products, and always attempt to maintain a high quality of input at a reasonable price.

Each pie has four main elements:

- Aluminium foil case
- Pastry shell made mainly from flour and water
- Meat and/or vegetable filling
- Thin plastic wrapping

The products are obtained as follows:

- The aluminium is obtained from a single supplier of metal related products. There are few suppliers in the industry resulting from fall in demand for aluminium related products following increased use of plastics.
- The flour for the pastry shell is sourced from flour millers in four different countries – one source of supply is not feasible because harvests occur at different times and PicAPie cannot store sufficient flour from one harvest for a year's production.
- Obtaining meat and vegetables is difficult due to the large number of suppliers located in many different countries. Recently, PicAPie obtained significant cost savings by delegating sourcing of these items to a specialist third party.
- Plastic wrapping is obtained either directly from the manufacturer or via an internet site specialising in selling surplus wrapping from government and other sources.

Requirements

5.1 Explain the main characteristics of a total quality management (TQM) programme.

5.2 Identify the sourcing strategies adopted by PicAPie and evaluate the effectiveness of those strategies for maintaining a constant and high-quality supply of inputs. Your answer should also include recommendations for changes you consider necessary.

See **Answer** at the end of this chapter.

8.4 Strategic procurement

The traditional supply chain model (see diagram below) shows each firm as a separate entity reliant on orders from the downstream firm, commencing with the ultimate customer, to initiate activity.

The disadvantages of this are:

- It slows down fulfilment of customer order and so puts the chain at a competitive disadvantage.
- It introduces possibility of communication errors delaying fulfilment and/or leading to wrong specification products being supplied.
- The higher costs of holding inventories on a just-in-case basis by all firms in chain.
- The higher transactions costs due to document and payment flows between stages.

Strategic procurement is the development of a true **partnership** between a company and a supplier of strategic value. The arrangement is usually long-term, single-source in nature and addresses not only the buying of parts, products, or services, but product design and supplier capacity.

This recognises that, increasingly, organisations are recognising the need for and benefits of establishing **close links** with companies in the supply chain. This has led to the **integrated supply chain** model (the second model in the following diagram) and the concept that it is **whole supply chains** which compete and not just individual firms.

Traditional and integrated supply chain models

Figure 13.5: Traditional supply chain

Traditional supply chain

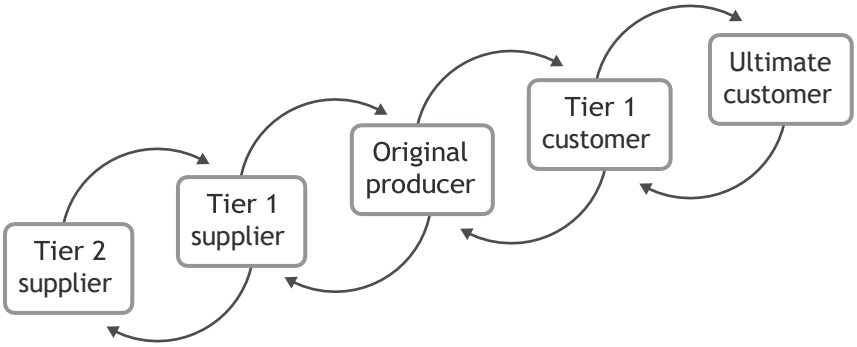
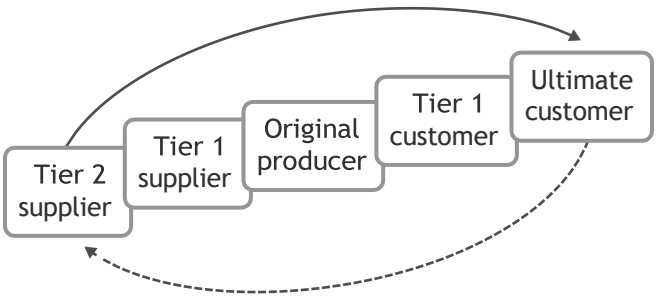


Figure 13.6: Integrated supply chain

Integrated supply chain



The integrated supply chain shows that the order from the ultimate customer is shared between all the stages in the chain and that the firms overlap operations by having integrated activities as business partners. This is consistent with the idea of a **value system** and the concept of **supply chain networks** discussed in the chapter Strategic capability.

Professional skills focus: Concluding, recommending and communicating

When making recommendations, you should incorporate a wide range of knowledge. You must always apply your answer to the scenario but you can take inspiration from your own work experience or from the wider business world. To help formulate realistic recommendations, ensure you read high quality business news on a regular basis.

Interactive question 6: Beau Ltd

Beau Ltd is a medium sized maker of specialised components which are used in the manufacture of tablet computers. It has three major customers, each of which are major players in the global personal tablet market. The three customers are very demanding, and constant advances in tablet technology mean that Beau Ltd invests heavily in R&D to keep the components it produces at the cutting edge of technology.

Beau Ltd purchases the parts it uses in production from a supplier in China, Xu Co. Xu Co sends deliveries from its factory in China to Beau Ltd's production facility in the UK. Xu Co is in daily communication with Beau Ltd's procurement staff regarding quantities. Recently, there have been delays in supply, so Beau Ltd has taken to holding 20 days' worth of parts in inventory. Xu Co charges high prices; the company's MD explains that this reflects Xu Co's focus on improving product quality and service delivery.

Beau Ltd is currently in the process of updating the main component that it sells, the POD52. The POD52 is regarded as being 'state of the art' technology. The updated version will be launched next year and is expected to be highly sought after by the three customers.

Beau Ltd has identified two issues in respect of its procurement and supply chain management:

Issue 1: Beau Ltd is unsure whether to procure parts for the mass production of POD52 from (i) one supplier, Xu Co; or (ii) a range of seven new suppliers, producing similar parts for the POD52 component. One of the suppliers would be Xu Co, three would be American, and three would be German.

Issue 2: POD52 is constructed from 37 individual parts. In a bid to reduce costs, Beau Ltd is undertaking an assessment of its procurement activities. It is looking to review all of its direct 'Tier 1' suppliers to improve efficiencies. The head of procurement at Beau Ltd has mentioned the need to review the company's own supplier's suppliers (Tier 2) to ensure that they are being cost efficient.

Requirements

6.1 Issue 1

Outline the main advantages and disadvantages to Beau Ltd of using one supplier or more than one supplier for the components needed to manufacture POD52. You should recommend a course of action.

6.2 Issue 2

Review the head of procurement's comment that Beau Ltd should conduct a Tier 2 inspection of its supplier's suppliers. You should recommend whether a Tier 1 or Tier 2 supplier review would be most appropriate.

See **Answer** at the end of this chapter.

Professional skills focus: Applying judgement

When asked to outline the main advantages and disadvantages of using one or more suppliers, you should apply judgement to identify the key issues. Prioritising the issues facing a business is an important exam skill. Use the marks available for each requirement to determine how many points to include in your response.

8.5 Suppliers and e-procurement

E-procurement involves using technology to conduct business-to-business purchasing over the internet. As we explored earlier in the chapter Strategic capability, organisations need to carefully consider cyber security issues when linking their own computer systems to those of suppliers. A key aspect of this concerns whether a supplier's systems are sufficiently robust enough to securely store and handle sensitive data. This issue is further compounded when data breaches occur, as it may be difficult to determine which party (ie, the organisation or supplier) is responsible for resolving the matter. This issue is particularly topical as organisational supply chains are becoming increasingly global.

There are however huge savings to be had, especially for large corporate organisations with vast levels of procurement. Siemens believes that, since it embarked on its fully-integrated e-procurement system, this purchasing strategy saved $15 million from material costs and $10 million from process costs in the one year alone, close to a 1,000% increase in savings from the previous year and only the second year into implementation.

8.5.1 Advantages of e-procurement for the buyer:

- Facilitate cost savings
- Easier to compare prices
- Faster purchase cycle
- Reductions in inventory
- Control indirect goods and services
- Reduces off-contract buying
- Data rich management information to help reduce costs and predict future trends
- Online catalogues
- High accessibility
- Improved service levels
- Control costs by imposing limits on levels of expenditure

8.5.2 E-procurement from a supplier's perspective

Traditionally, the business of supplying goods has been about branding, marketing, business relationships, etc. In the expanding e-procurement world the dynamics of supplying are changing and, unlike the expectations of companies implementing e-procurement systems for cost savings, suppliers are expecting to feel profit erosions due to the e-procurement mechanism.

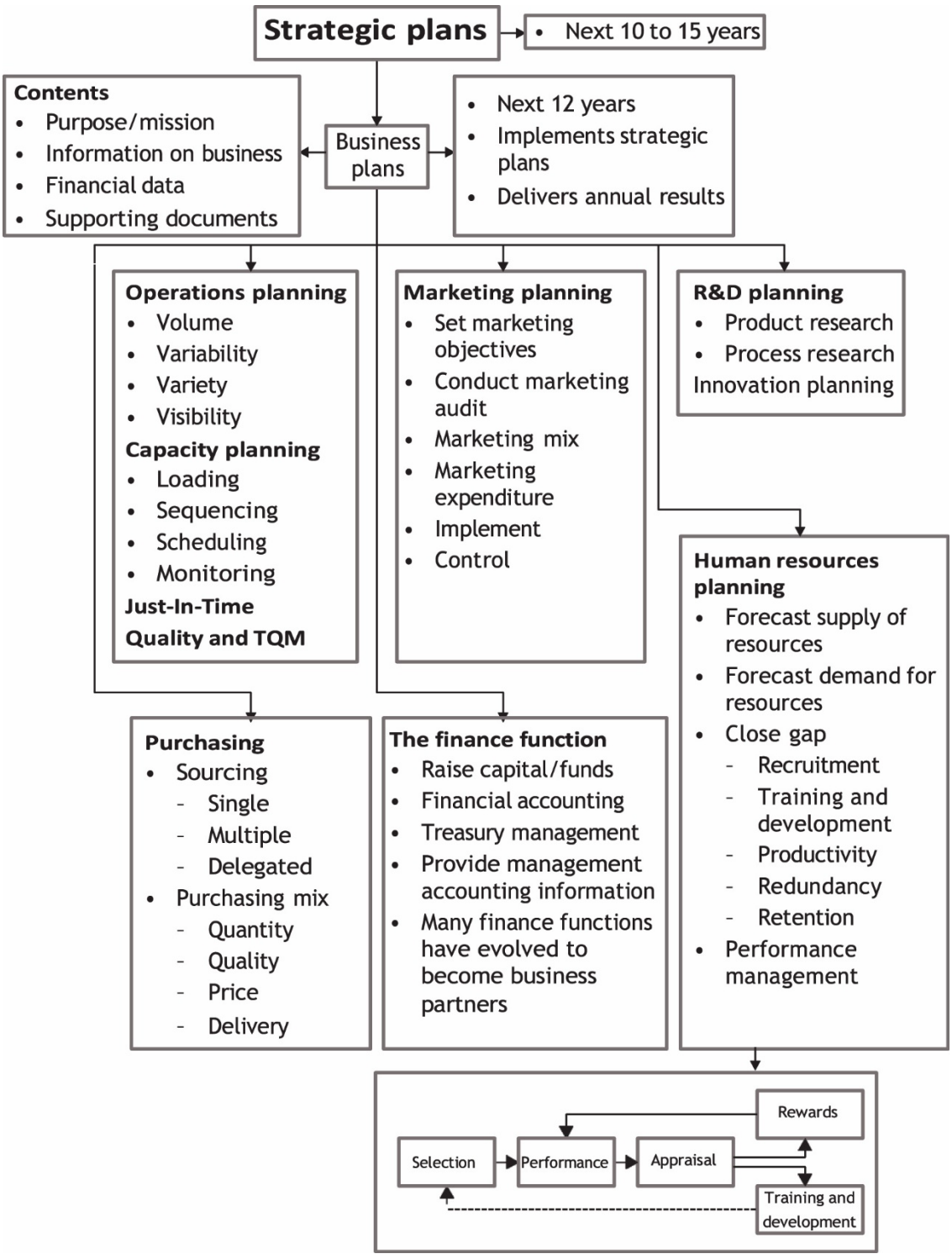
Nevertheless, there are obvious advantages to suppliers:

- Faster order acquisition
- Immediate payment systems
- Lower operating costs
- Non-ambiguous ordering
- Data rich management information
- 'Lock-in' of buyers to the market
- Automate manufacturing demands

8.6 Ethical aspects of purchasing and procurement

Many firms seek to fulfil their own CR commitments by demanding similar commitments from their suppliers. This gives rise to **ethical procurement** which is examined in more detail in the chapter, 'Ethics, sustainability and ESG.'

Summary


Strategic plans
Next 10 to 15 years
Contents
Purpose/mission
Information on business
Financial data
Supporting documents
Business plans
Next 12 years
Implements strategic plans
Delivers annual results
Operations planning
Volume
Variability
Variety
Visibility
Capacity planning
Loading
Sequencing
Scheduling
Monitoring
Just-In-Time
Quality and TQM
Marketing planning
Set marketing objectives
Conduct marketing audit
Marketing mix
Marketing expenditure
Implement
Control
R&D planning
Product research
Process research
Innovation planning
Human resources planning
Forecast supply of resources
Forecast demand for resources
Close gap
Recruitment
Training and development
Productivity
Redundancy
Retention
Performance management
Purchasing
Sourcing
Single
Multiple
Delegated
Purchasing mix
Quantity
Quality
Price
Delivery
The finance function
Raise capital/funds
Financial accounting
Treasury management
Provide management accounting information
Many finance functions have evolved to become business partners
Rewards
Selection
Performance
Appraisal
Training and development

Further question practice

1 Knowledge diagnostic

Before you move on to question practice, confirm you are able to answer the following questions having studied this chapter. If not, you are advised to revisit the relevant learning from the topic indicated.

Confirm your learning	
(1)	Do you know how to format a business plan? (Topic 1)
(2)	What is the role of finance as a business partner? (Topic 3)
(3)	How can marketing planning contribute to strategic planning? (Topic 4)
(4)	How do you close the gap between HR demand and supply? (Topic 5)
(5)	How does an organisation create a culture of innovation? (Topic 6)
(6)	What procurement sourcing strategies are available to an organisation? (Topic 8)

2 Question practice

Question	Learning benefit from attempting this question
Operations strategy	This is a knowledge-based question asking you to consider the factors to take into account when devising an operations strategy.
Performance management and appraisal system	This question looks at the role of appraisal systems and performance management in the strategic management process.

Once you have completed these self-test questions, it is beneficial to attempt the following questions from the Question Bank for this module. These questions have been selected to introduce exam style scenarios that will help you improve your knowledge application and professional skills development before you start the next chapter.

Question	Learning benefit from attempting this question
Air Services UK Ltd (Sept 2017)	Part 2 of the question asks how the HR and the R&D functional strategies could help to achieve the company vision. It is important to apply your technical knowledge of functional strategies to the scenario. If required you can draw on your own work experiences to help you make practical points, so long as they are applied to the scenario.
Maya plc (June 2018)	Part 2 of the question requires you to select the appropriate sourcing strategy using the information provided in the scenario. Technical knowledge is required but, as always, good answers will be applied to the scenario provided.

Refer back to the learning in this chapter for any questions which you did not answer correctly or where the suggested solution has not provided sufficient explanation to answer all your queries.

Once you have attempted these questions, you can continue your studies by moving on to the next chapter.

Self-test questions

Answer the following questions.

1 Here are four characteristics of long-term planning information. List the corresponding characteristics of short-term planning information.

(1) Used by top management

(2) Broad in scope rather than deep in detail

(3) External

(4) Looks to the future and lacks certainty

2 List the 10 purposes of using budgets.

3 Which of the following is not a use of budgetary control?

A To define the objectives of the organisation as a whole

B To ensure that resources are used as efficiently as possible

C To provide some central control when activities are decentralised

D To provide the basis for the preparation of future budgets

4 What is the main intention behind R&D?

5 What are the elements of the purchasing mix?

6 What is the role of the production function?

7 Define marketing.

8 How does the finance function relate to strategic planning?

9 What are the four main objectives of HRM?

10 Operations strategy

Six factors that should be taken into account when devising an operations strategy are:

- Capacity
- Range and location of operations
- Investment in technology
- Strategic buyer-supplier relationships
- New products/services
- Structure of operations

Requirement

Briefly describe what each of the six factors identified above mean in the context of operations strategy.

Illustrate your answer with examples related to a retail supermarket chain.

11 Performance management and appraisal system

The recently appointed HR manager in a medium sized accounting firm is struggling. The senior partner of the firm is unconvinced about the benefits of appraisal systems. They argue that accountants, through their training, are self-motivated and should have the maximum freedom to carry out their work. Their experience of appraisal systems to date has shown them to lack clarity of purpose, be extremely time consuming, involve masses of bureaucratic form filling and create little benefit for the supervisors or their subordinates. They refuse to have their own performance reviewed through an appraisal system.

The HR manager is convinced that performance management and an appraisal system are integral elements in helping the firm achieve its ambitious strategic growth objectives. This reflected their experience of introducing an appraisal system into the corporate finance unit for which they were responsible. The unit had consistently outperformed its growth targets and individual members of the unit were well motivated and appreciative of the appraisal process.

Requirements

11.1 Evaluate the extent to which an effective appraisal system could help the accounting firm achieve its goals.

11.2 Assess the contribution of performance management to the strategic management process.

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

To: Aldine Computers & Training

From: An Accountant

Date: Today

Subject: Evaluation of business plan and application for funding

Terms of reference

We have been asked to comment on the viability of the business proposal by Aldine Computers & Training (Aldine) to secure funding for an extension to its business and to propose improvement to the proposal document.

Disclaimer

This report is based on the information given us by the management of Aldine and its accuracy is restricted to the accuracy of that data. No opinion is expressed on the accuracy of that data. This report is intended solely for the management of Aldine and is not for any third parties. The authors cannot be held liable for any loss incurred by third parties relying on this report.

Introduction

Aldine has been established three months and is trading. We understand that some initial capital investment has taken place and that premises have been secured. The business plan therefore is to secure funding for the expansion of the business.

The business model

A business proposals should be evaluated to assess whether it is a good strategic fit for the business, whether the proposal is considered acceptable to stakeholders and whether the business has the skills and ability to implement the proposal.

Strategic fit

To assess the strategic fit, a good starting point would be to review the strengths, weaknesses, opportunities and threats of the business and the industry in which it operates.

The **strengths** of Aldine include:

- Relevant experience of management in the proposed areas of business
- Industry links with suppliers and corporate customers

The **weaknesses** of Aldine are:

- Very high reliance on its principal managers, Vinay and Kate for both direct making and provision of training and also for business development and client prospecting
- Relatively small size will deny Aldine economies of scale in the purchasing of components compared to larger PC makers like PCNow and Dell
- Its inability to offer a national coverage for servicing and training means it will not be able to gain access to national corporate clients
- Low brand and business profile compared to established PC providers
- Many aspects necessary to run its business have not been considered in its business plan (see below)

The **opportunities** for Aldine include:

- Continued sales of PCs encouraged by new applications and the desire by firms to encourage flexible working
- The perceived service failings and backlogs of rival providers

The **threats** facing Aldine include:

- Competition in the PC construction market from larger firms able to access cheaper manufacturing resources overseas
- Development of more sophisticated computer-based training and virtual learning, reducing demand for the face-to-face training proposed by Aldine
- Increased outsourcing of IT by firms, reducing the demand for Aldine's services
- Forecast economic slowdown in two years, reducing demand for PCs and hence leave Aldine with excess capacity and costs

In summary, its business model poses significant risks to Aldine. By focusing on SMEs it is possible that it may access a market segment overlooked by larger rivals but at the expense of high marketing to sales costs and it will be serving markets where low IT literacy will necessitate considerable and costly support.

Stakeholder acceptance of the proposal

Kate and Vinay are clearly in favour of the business proposal. There is no research presented to help us form a judgement about client attitudes to the services proposed and to dealing with Aldine. The remaining key stakeholder will be the bank. The present report will need some modification before it will secure the banks agreement to lend the money (see below).

Ability to implement the plan

The production plan seems well-thought through. However, questions must be raised over the feasibility of the 40% mark-up behind the forecast first year profit of £90,000 which implies a revenue of £225,000 (100/40 × £90,000) supporting the two founders, the mature office manager and the trainee assembler as well as the actual costs of production and service provision. Margins in a competitive industry like IT are likely to be lower and it is an industry with a high failure rate amongst start-ups like Aldine.

The other issue which could affect the success of the business seems to be Aldine's reliance on supplier relations built up by Vinay whilst at PCNow. Faced with competition from Aldine there is a danger that PCNow would pressurise its suppliers for preferential terms over Aldine.

In summary, the business model is potentially viable but the founders need to satisfy themselves, and the bank, on the issues above.

Critique of the business plan

The main omissions from the business plan are:

- A statement of the legal entity: It is not clear whether Aldine is to be a company or a partnership.
- Detailed financial workings to back up the ambitious forecasts on p. 8: The business mix between sales, service and training is not clear and nor are the volumes and costs of each. These should be included as an appendix together with a sensitivity analysis of profits against price, costs and volumes.
- The purposes for which the money is required: The initial equipment has been detailed and purchased. What is not clear is what the further £40,000 is required for inventory, working capital, marketing etc. It is not clear. A bank will wish to ensure it is adequate and will not be impressed by the £15,000 additional commitment it is being asked to pledge.

- Any evidence of demand for its services or the preferential supply terms: These are key elements in the business model and so letters of reference and indications of willingness to buy/supply should be included.
- Job roles: Aldine intends to employ office staff, trainees etc. It is not clear what they will do.
- Market information: It is not appropriate to leave details of competitors at the level they are. Evidence is needed of their size, strengths and weaknesses and also their likely responses to Aldine's arrival in the market.
- The proposals for the loan: The business plan does not indicate the likely timing for take-up of the borrowing nor the repayment proposals.
- Plans beyond 20X3: The plan ceases at the year the economic slowdown is forecast to begin. There is no indication on how Aldine intends to develop its business then. Most business plans will be for five years.
- Detailed operational information such as how clients will be prospected and dealt with.

Suggested improvements to the business plan

As indicated above, there are several elements of the business plan that need

improvement. As a minimum, Aldine should include:

- A cash flow forecast broken down by month indicating the likely timing for take-up of the borrowing and the repayment proposals
- Evidence of supplier and client intentions
- Details of the security being offered for the loan
- A better strategic analysis utilising the SWOT analysis in this report indicating how the weaknesses and threats may be addressed
- Better competitor analysis
- Job descriptions for the roles being recruited and also how work will be allocated to cover the manufacture and servicing of the PCs

Conclusions

Providing the above issues are addressed, we believe that Aldine will be successful in business and in the application for its loan.

If there are any questions on this report, or further assistance is required, please revert to the author.

Answer to Interactive question 2

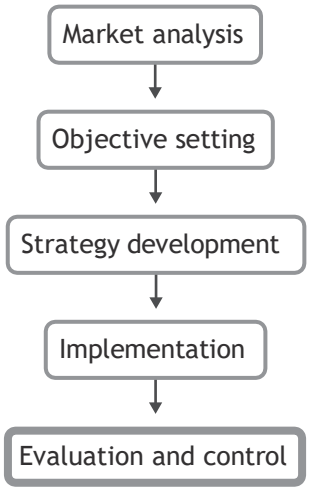
2.1 A **SWOT analysis** identifies the strengths and weaknesses of the organisation relative to the opportunities and threats it faces in its marketing environment. The SWOT analysis leads to an understanding of **realistic market opportunities** through the process of a detailed **marketing audit**, covering market and environment analysis, competitor and supplier analysis, customer analysis and internal analysis.

This analysis will highlight potential market gaps, new customer needs, marketing channel developments, competitor strategies and their strengths and weaknesses. The organisation can evaluate market opportunities against its strengths and weaknesses and identified threats and determine what actions to take to exploit the opportunity.

2.2 To successfully launch a new customer product into the UK market the organisation needs to have clear and realistic **objectives**, and identification and understanding of its **target market segment**. Its channel of distribution, branding, packaging and communications activity need to be in place to support the launch. Forecasts of future demand, and return on investment need to be assessed and projected.

A formalised **marketing planning system** that involves all departments, customers, agents and suppliers, the complexity and timing of activities is required to support the launch to prevent lack of coordination, resource and ultimate failure. A **monitoring system** is also required to evaluate the effectiveness of the launch and take actions as required. A successful launch requires a planning process that pulls all the people and activities together, co-ordinates what is done, by who, when and with what resource.

2.3 Marketing planning involves the following stages:



Market analysis: This phase involves establishing an audit process that assesses the macro and micro market environment, market segment analysis, customers, competitors and development strategy. Without a clear understanding of these issues it is difficult to set objectives and develop strategy.

Objective setting: Once the issues arising from market analysis have been understood, objectives can be set. Objectives should be consistent with the overall mission of the organisation and goals, and they must be realistic.

Strategy development: This phase can begin once the objectives have been agreed. In this process alternative strategic options will be evaluated to determine the best way forward for the organisation. Strategy evaluation should consider the organisation's current strengths and weaknesses, market attractiveness, resource requirements, and profitability.

Implementation: This is frequently the hardest part of the marketing planning process. Effective implementation requires co-ordination between different organisations, people and departments. An organisation structure and culture should support this co-ordination, provide good communication and access to information and appropriate levels of resources. In reality, many issues, conflicts and trade-offs occur within organisations that act as barriers to effective implementation.

Evaluation and control: The final phase of the process involves setting an effective system of monitoring and control to measure and evaluate performance.

Answer to Interactive question 3

To: Olivia Marcuse: HR manager, ScannerTech

From: Board of directors: ScannerTech

Date: Today

Human resource planning and strategy

ScannerTech's strategy calls for the company to double in size over the next three years. This will require the employment of extra staff, particularly in marketing, sales and manufacturing. The ambitious planned rate of growth and the high technology base of ScannerTech's business mean that these extra staff must be of very high quality. Human resource (HR) management is thus an **essential component** of the company's business strategy and so should be **integrated with its development**. The alternative is increased potential for serious shortages of staff and mismatches between job requirements and staff availability. The establishment of a foreign manufacturing plant will complicate all HR issues significantly and will demand very careful consideration.
Human resource planning follows a logical sequence, echoing the rational model of strategy. This is not necessarily linear and some of the activities involved in establishing a satisfactory plan can overlap chronologically. There will also be occasions where the various activities influence one another, as, for example, when the persistence of staff shortages in important areas leads to a change in reward policy.

An **audit of existing staff** should reveal those with potential for promotion or employability in new specialisations. It would also indicate where shortages already exist.

Concurrently, an analysis of **likely future staff requirements** could be carried out. We anticipate the need to employ more staff in the areas already mentioned, but we do not really know how many will be required, whether other functions will need to be increased in size or if more support and administrative staff will be needed. There are also the related and sensitive issues of **management succession** and **internal promotion** to consider. In particular, we must consider the eventual replacement of our existing joint managing directors, who are likely to leave once the current growth objective has been achieved.

These two studies should enable us to identify the gaps that we need to fill if we are to have the staff required for our overall strategy.

Recruitment, in the sense of attracting applicants, and **selection** from within the pool of applicants are the logical next steps. This work is often **outsourced** and it will be necessary to decide whether the **expertise** and **economies of scale** offered by outsourcing outweigh the need for deep familiarity with our operations on the part of the recruiters.

Reward policy must be considered. At the moment, ScannerTech's staff profile is heavily biased towards people with a background in research and development. Different types of people will be required in the future and their expectations must be expected to show some differences. A doubling in size to, say, 200 employees is likely to take the company into an area of HR complexity in which a formal reward policy and structure is required. Informal decisions about pay and benefits will not be satisfactory. It may be necessary to establish a more formal scheme of **employee relations**, possibly along the lines of a works council.

Increasing size is also likely to require the establishment of a policy on **appraisal and performance management**. This should be linked to a programme of **training and development**. No doubt ScannerTech will continue to hire well-qualified technical staff, but there will be a need for development of staff in other functions and for management development in particular.

Answer to Interactive question 4

Toyota's management:

(1) **Cost reduction**: Toyota was losing money, and market demand was weak, preventing price rises. The only way to move from losses into profits was to cut costs, and cost reduction was probably essential for the survival of the company.

(2) **Production levelling**: Production levelling should help to minimise idle time whilst at the same time allowing the company to achieve its objective of minimum inventories.

(3) The **change in factory layout** was to improve the work flow and eliminate the waste of moving items around the work floor from one set of machines to another. Each cell contained all the machines required to complete production, thus eliminating unnecessary materials movements.

(4) With having **cells of different machines**, workers in each work cell would have to be trained to use each different machine, whereas previously they would have specialised in just one type of machine.

(5) A **change of culture** was needed to overcome the industrial problems of the company. Employee involvement would have been an element in this change. Teamwork would have helped with the elimination of waste: mistakes or delays by one member of a team would be corrected or dealt with by others in the team. The work force moved from a sense of individual responsibility/blame to collective responsibility.

(6) The **Kanban system** is a 'pull' system of production scheduling. Items are only produced when they are needed. If a part is faulty when it is produced, the production line will be held up until the fault is corrected. For a Kanban system to work properly, defects must therefore be eliminated.

Answer to Interactive question 5

5.1 In a nutshell, **total quality management** (TQM) is a management philosophy, aimed at **continuous improvement** in all areas of operation.

A TQM initiative aims to achieve continuous improvement in quality, productivity and effectiveness. It does this by establishing management responsibility for processes as well as output.

Principles of TQM

- **Prevention:** Organisations should take measures that prevent poor quality occurring.
- **Rightfirsttime:** A culture should be developed that encourages workers to get their work right first time. This will save costly reworking.
- **Eliminate waste:** The organisation should seek the most efficient and effective use of all its resources.
- **Continuous improvement:** The Kaizen philosophy should be adopted. Organisations should seek to improve their processes continually.
- **Everybody's concern:** Everyone in the organisation is responsible for improving processes and systems under their control.
- **Participation:** All workers should be encouraged to share their views and the organisation should value them.
- **Teamwork and empowerment:** Workers across departments should form team bonds so that eventually the organisation becomes one. Quality circles are useful in this regard. Workers should be empowered to make decisions as they are in the best position to decide how their work is done.

  Note: This is a question that may appear daunting at first, but if you go through and deal with each element in turn it should not prove too difficult to earn a pass. Ensure you provide justification for the changes you recommend.

5.2 **Aluminium foil** is obtained from a single supplier – a sourcing strategy termed '**single sourcing**.' The advantages of this strategy include:

- Easy to develop and maintain a relationship with a single supplier – which is especially beneficial when the purchasing company relies on that supplier.
- A supplier quality assurance program can be implemented easily to help guarantee the quality of products – again mainly because there is only one supplier.

Economies of scale may be obtained from volume discounts. However, the **disadvantages** of this strategy are:

- PicAPie is dependent on the supplier – providing significant supplier power. Issues such as quality assurance may not be addressed quickly because the supplier is aware that there are few alternative sources of supply.
- PicAPie is vulnerable to any disruption in supply.

Given that there are few suppliers in the industry, this strategy may be appropriate. However, there is no guarantee that the current supplier will not go out of business so the directors of PicAPie could look for alternative sources of supply to guard against this risk.

The **pastry shell flour** is obtained a number of suppliers – a strategy known as multi-sourcing. The advantages of this strategy include:

- Ability to switch suppliers should one fail to provide the flour. Having suppliers in different countries is potentially helpful in this respect as poor harvests in one country may not be reflected in another.
- Competition may help to decrease price.

Disadvantages include:

- It may be difficult to implement a quality assurance program due to time needed to establish it with different suppliers.
- Suppliers may display less commitment to PicAPie depending on the amount of flour purchased making supply more difficult to guarantee.

PicAPie appears to have covered the risk of supply well by having multiple sources of supply. The issue of quality remains and PicAPie could implement some quality standards that suppliers must adhere to keep on supplying flour.

A third party is given the responsibility for obtaining **meat and vegetables** – this is termed **delegated sourcing.** Advantages of this method include:

- Provides more time for PicAPie to concentrate on pie manufacture rather than obtaining inputs. Internal quality control may therefore be improved.
- The third party is responsible for quality control checks on input – again freeing up more time in PicAPie. Where quality control issues arise, PicAPie can again ask the third party to resolve these rather than spending time itself.
- Supply may be easier to guarantee as the specialist company will have contacts with many companies.

Disadvantages are:

- Quality control may be more difficult to maintain if the third party does not see this as a priority.
- There will be some loss of confidentiality regarding the products that PicAPie uses, although if there are no 'special ingredients' then this may not be an issue.

Given the diverse sources of supply, PicAPie are probably correct using this strategy.

The **plastic film** is obtained from two different sources utilising two different supply systems. This is termed **parallel sourcing**. The advantages of this method include:

- Supply failure from one source will not necessarily halt pie production because the alternative source of supply should be available.
- There may be some price competition between suppliers.

Disadvantages include:

- PicAPie must take time to administer and control two different systems.
- Quality may be difficult to maintain, and as with multiple sourcing, it will take time to establish supplier quality assurance programmes. Given that some stock is surplus to

requirements from other sources, quality control programmes may not be possible anyway.

The weakness in the supply strategy appears to be obtaining film from the internet site – in that quality control is difficult to monitor. Changing to single sourcing with a supplier quality assurance programme would be an alternative strategy to remove this risk.

Answer to Interactive question 6

6.1 **Issue 1**

The key issues:

(1) Whether to have one supplier or many

(2) Whether to continue solely with a current supplier or to engage with new suppliers

One supplier: Xu Co

- The arrangement with Xu Co represents strategic procurement.
- This relationship can be beneficial as Beau Ltd and Xu Co can establish close links in the supply chain through introducing EDI facilities to reduce the need for daily communication.

Advantages to Beau Ltd from single sourcing from one supplier:

- Consistent parts from a sole supplier.
- Should make monitoring quality easier.
- Xu Co may be dependent on Beau Ltd for income, and therefore more responsive to Beau Ltd's needs.
- Collaboration in working together on new components is mutually beneficial.
- Beau Ltd already has a relationship with Xu Co.

Disadvantages to Beau Ltd from single sourcing from one supplier:

- Disruption to Xu Co's output will impact on Beau Ltd. Could damage relations with the three highly demanding customers if orders are not fulfilled on time. Furthermore, Beau Ltd is already having to hold higher levels of inventory than it might otherwise hold to overcome current problems.
- Xu Co may not to be able to support increases in demand.
- Xu Co may attempt to exploit relationship with Beau Ltd by increasing prices if a long-term supply agreement is entered into.

Advantages to Beau Ltd of having seven suppliers:

- Beau Ltd can encourage competition among suppliers to reduce prices.
- The costs of switching supplier are reduced.

Disadvantages to Beau Ltd of having seven suppliers:

- Increased scope for suppliers to lack commitment to supply as less revenue is generated per supplier from serving Beau Ltd
- Increase in costs and management time of communicating with seven suppliers
- Reduced economies of scale
- Still have long lead times if parts are being delivered from China and the US in particular This increases the uncertainty of delivery.

Recommendation:

Beau Ltd is advised to continue using existing supplier Xu Co due to the relationship that the two have already established. This approach helps increase Beau Ltd's buying power as the company will be better placed to negotiate on price. The threat of losing Beau Ltd altogether to other suppliers may incentivise Xu Co to improve its current terms of supply.

6.2 Issue 2

An integrated supply chain approach highlights that the supply chain of an organisation and its suppliers should be viewed as a whole. The desire for information about Beau Ltd's own suppliers (Tier 1) and its supplier's suppliers (Tier 2) is a feature of this approach.

In theory, if Tier 2 suppliers can find efficiencies and reduce costs these can be passed onto Tier 1 suppliers, which should result in lower purchase prices for customers, in this case the likes of Beau Ltd. Sound supply chain management is needed to attain such benefit.

Beau Ltd's supply chain review is focused on upstream supply chain management as there is no focus on its own customers, with the main emphasis being on realising cost efficiencies.

Beau Ltd as a customer does not have a legal right to request an inspection of its suppliers processes. However, it should be noted that such a review may help to improve transparency and the degree of openness between the parties. Suppliers for whom Beau Ltd is a major customer may be more inclined to participate in the review for fear of damaging existing relations.

Tier 2 suppliers have no legal obligations to participate in Beau Ltd's review as the two parties do not have any direct contractual obligations to one another. However, Tier 2 suppliers may be motivated to allow Beau Ltd to investigate their own operations due to the fact that Beau Ltd could cease to make purchases from a Tier 1 supplier, which in turn would result in fewer purchases from Tier 2 suppliers.

Recommendation:

Attempts to manage costs throughout the supply chain are in line with best practice in supply chain management. A focus on Tier 1 suppliers is most likely to be more value adding to Beau Ltd as these are the third parties which most closely interact with the company's own value chain.

Answers to Self-test questions

1 The corresponding characteristics are:

(1) Used at a lower level by those who implement plans

(2) Detailed

(3) Internal

(4) Definite

2 The 10 purposes of using budgets are:

- Ensure the achievement of the organisation's objectives
- Compel planning
- Communicate ideas and plans
- Co-ordinate activities
- Allocate resources
- Authorisation
- Provide a framework for responsibility accounting
- Establish a system of control
- Provide a means of performance evaluation
- Motivate employees to improve their performance

3 Correct answer(s):

C To provide some central control when activities are decentralised

4 To improve products or processes and so support the organisation's strategy.

5 Elements of purchasing mix:

- Quantity
- Quality
- Price
- Delivery

6 To control the necessary activities to provide products (or services) creating outputs which have added value over the value of inputs.

7 'The management process which identifies, anticipates and satisfies customer needs profitably' (*Chartered Institute of Marketing*).

8 The finance function relates to strategic planning by:

- ensuring that resources of finance are available;
- integrating the strategy into budgets;
- establishing the necessary performance measures;
- establishing priorities; and
- assisting in the modelling process.

9 Four main objectives of HRM:

- To develop an effective human component for the company
- To obtain, develop and motivate staff
- To create positive relationships
- To ensure compliance with social and legal responsibilities

10 Operations strategy

- **Capability required**: Any operations strategy will be influenced by what it is that the organisation does.

  For example, a supermarket chain sells food and other items to consumers.
- **Range and location of operations**: The operations strategy will be affected by the scale and geographical spread of the organisation's operations.

  For example, a supermarket chain with say 10 outlets in one region of a country will face different operation strategy issues than a nationwide chain.
- **Investment in technology**: Technology will impact upon operations and therefore operations strategy as it has the potential to change the processes associated with operations.

  For example, a supermarket chain operating using an EFTPOS system linked to their inventory (logistics/warehousing) systems will operate differently to a chain relying on less-automated systems.
- **Strategic buyer-seller relationships**: Who key strategic partners are will affect operations strategy.

  For example, a supermarket may have a preferred supplier for canned food items. Operations may then be designed to help facilitate this relationship. Relationships with 'buyers' (consumers) may be developed using loyalty card schemes – and operations changed based on what the scheme reveals.
- **New products/services**: This relates to how long the business will be able to do what it is currently doing (in the same way).

  A supermarket may find it also needs to offer online shopping and home delivery. It could also decide to move into non-traditional areas such as consumer electronics – or even consumer insurance or finance. These types of changes require changes to operations strategy.
- **Structure of operations**: Operations strategy will also be influenced by how staff are organised and managed.

  For example, will 'regional managers' have responsibility and complete control over all stores in one region – or will one national strategy apply?

  Issues such as staff levels, shift patterns and human resources policies will also affect operations strategy. For example, will stores be open 24 hours – and if so how will this be staffed?

11 Performance management and appraisal system

11.1 The senior partner and the HR manager emphasise the aspects of appraisal schemes that **support their own favoured policies**. Such schemes should support the organisation's overall objectives without incurring excessive administrative and management costs.

In an organisation such as an accounting practice, the professional staff should indeed be highly **self-motivated**, able to judge the effectiveness of their own performance and bring to their work a commitment to high professional standards. On the other hand, it is

inevitable that their **talents and performance will vary** and they will need **guidance and help with their future development**. Dealing with these issues would be the role of an appraisal scheme.

The overall aim of such a scheme would be to **support progress toward the achievement of corporate objectives** and it would do this in three ways: performance review, potential review and training needs review.

Performance review: Performance review should provide employees with an **impartial and authoritative assessment of the quality and effect of their work**. Individuals should have personal objectives that support corporate goals via intermediate objectives relevant to the roles of their work groups. A reasoned assessment of performance can have a **positive motivating effect**, simply as a kind of positive, reinforcing feedback. It can also provide an opportunity for analysing and addressing the **reasons for sub-optimal performance.**

Potential review: Any organisation needs to make the best use it can of its people; an accountancy practice is typical of many modern organisations in that its people are its greatest asset and its future success depends on managing them in a way that makes the best use of their skills and aptitudes. An important aspect of this is **assessing potential for promotion and moves into other positions of greater challenge and responsibility.**

Training needs review: A further aspect of the desirable practice of enabling staff to achieve their potential is the provision of training and development activities. The appraisal system is one means by which **training needs can be assessed** and training provision initiated.

The appraisal system: An appraisal system must be properly administered and operated if it is to make a proper contribution to the organisation's progress.

The appraisal cycle: Formal appraisal, with interviews and written assessments, is typically undertaken on an **annual cycle**. This interval is commonly regarded as too long to be effective because of the speed with which individual roles can evolve and their holders can develop, so the annual appraisal is often supplemented with a less detailed review after six months.
Sometimes the procedure is sufficiently simplified that the whole thing can be done at six monthly intervals. Much modern thinking on this topic is now suggesting that any frequency of periodic appraisal is unsatisfactory and that it should be replaced by a **continuous process of coaching and assessment**.

Objectivity and reliability: Appraisal involves an element of direct personal criticism that can be stressful for all parties involved. If the system is to be credible its outputs must be seen to be objective and reliable. This requires proper **training for appraisers**, the establishment of appropriate **performance standards** and, preferably, input into each appraisal from **more than one person**. Having reports reviewed by the appraiser's own manager is one approach to the last point; 360 degree appraisal is another.

Setting targets: Past performance should be reviewed against **objective standards** and this raises the question of the type of objective that should be set. Objectives set in terms of **results** or outcomes to be achieved can encourage **creativity** and **innovation** but may also lead to **unscrupulous**, **unethical** and even **illegal choice of method**. On the other hand, objectives designed to maintain and improve the quality of output by **encouraging conformity** with approved procedure and method may stifle the creativity and innovation widely regarded as a vital source of continuing competitive advantage.

11.2 **Performance management** involves the establishment of clear, agreed individual **goals and performance standards**, continuous leadership action to both **motivate and appraise subordinates** and a **periodic review** of performance at which the goals and performance standards for the next cycle are set.

Performance management is an application of the **rational model** of strategic management, in that individual goals are intended to form the lowest echelon of a **hierarchy of objectives** that builds up to support the **overall mission** of the organisation. It is an essential aspect of the system that individual goals should be **agreed and internalised** so that true **goal congruence** is achieved.

This overall approach was first described (as is so often the case) by Peter Drucker, in 1954, and is seen most clearly in the system of **management by objectives** (MbO). MbO as a management system has fallen somewhat from favour with the rise of quality management methods that emphasise processual and procedural conformance rather than the attainment of overall performance goals. Nevertheless, it has much to offer.

Under a formal MbO system, the process of setting goals is part of the **implementation phase** of strategic management and follows consideration of resources, overall objectives and SWOT analysis. Top level subordinate goals are agreed for heads of departments, divisions or functions: these goals should be specific, measurable, attainable, relevant and time-bounded (SMART). It is particularly important that the achievement of a goal can be established by objective **measurement**. There may be different timescales for different objectives, with short- term goals supporting longer-term ones.

Departmental heads then agree SMART goals for their subordinates in discussion with them, that support their own personal goals, and so on down the hierarchy to the level of the individual employee. All members of the organisation thus know what they are expected to achieve and how it fits into the wider fabric of the organisation's mission.

Periodic **performance review** is based on the objective appraisal of success against agreed goals, the agreement of goals for the next period and an assessment of the resources, including training, that the reviewee may require to reach those goals. The MbO system thus closes the **feedback loop** in the corporate control system.

Chapter 14
Strategies for information

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Introduction

Learning outcomes

- Explain and evaluate the risks attached to an organisation's present position, using all relevant qualitative and quantitative data, and considering attitudes to risk, security and cybersecurity.
- Explain and evaluate the risks attached to proposed courses of action in a given situation, considering all relevant factors and assumptions made and select and justify strategies for managing risk.
- Explain and demonstrate how qualitative and quantitative data, including forecasts, budgets and other management information presented in a range of formats, can be analysed to monitor the performance of an organisation's projects, divisions and other strategic units.
- Explain and demonstrate how data can be selected, structured, assimilated and analysed using spreadsheets to provide management with information that enables it to monitor or modify strategic or operating decisions having regard to data variability, data bias and risk and applying an appropriate degree of professional scepticism.
- Explain and evaluate how an organisation can select, capture, analyse and use data, including big data, to provide management with information that enables it to implement, monitor and modify a strategy and to create or sustain competitive advantage.

Syllabus links

You will have been introduced to the provision and use of financial information in your Management Information and Business, Technology and Finance studies.

The strategic value of IT/IS and big data complements earlier chapters on strategy. In the chapter, 'Risk management,' we explored the different types of risks that organisations are exposed to, one of the most significant of these concerned the threats now presented by cyber risks. The growing global dependency on information technology systems has increased the focus on cybersecurity. In this chapter we explore the implications of this in greater detail.

Examination context

In the examination, you are unlikely to get a whole question focusing on information systems. Instead you may be required to comment upon the information/data available for decision making in a scenario question – is there enough information available? What other information is needed? Where might the organisation source relevant information? Is the information system adequate to fulfil the functions required of it by the organisation? How can the organisation use information to generate competitive advantage?

In the context of the exam, a key aspect of information strategy is that it should provide the appropriate type and amount of information needed by management to select, implement and control its chosen overall strategy. The information strategy therefore needs to match the organisation's overall strategy in terms of the types of information available. Also, the level of detail, the form of the information and its timing should be appropriate to the role of the person(s) who receive it.

The Business Strategy and Technology exam may also test your understanding of issues related to cybersecurity and cyber risk. Therefore, you need to ensure that you are comfortable

with identifying potential cybersecurity issues, and providing recommendations as to how these could practically be addressed.

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this chapter.

Topic	Practical significance	Study approach	Exam approach	Interactive questions
1	**The role of information** We live in an era of data and information, where the ability to capture and harness corporate knowledge has become critical for organisations as they seek to adapt to changes in the environment.	**Approach** Read through the examples of information requirements at different levels in the organisation. It is also important to look at the information characteristics in various industry sectors so that you have practical examples for the exam. **Stop and think** What are the challenges of using qualitative data retrieved from customer feedback and social media sources?	In the examination you are unlikely to get a whole question focussing on data analytics and information systems. Instead, you may be required to comment upon the data and information available for decision making in a scenario question – is there enough information available? What other information is needed? Where might the organisation source relevant information?	**IQ1: Strategic decision making** This question looks at the sources and characteristics of information needed for strategic decision making.
2	**The strategic value of information technology** Information technology is playing an increasing role in helping an organisation to achieve competitive advantage. All functions are impacted by changing technology from operations and logistics through to sales and marketing.	**Approach** The emphasis of this topic is practical. Be sure to understand how information systems can deliver competitive advantage. The worked examples explain how real companies use IT/IS to gain competitive advantage. Big data is growing in importance so spend time understanding the advantages and disadvantages it can bring.	In the context of the exam, a key aspect of information strategy concerns the growing importance of big data and data analytics. You may be required to identify the benefits and challenges that big data can bring to an organisation. Also ensure you have examples of how technology can be implemented across business functions to deliver improvements in performance.	

Topic	Practical significance	Study approach	Exam approach	Interactive questions
3	**Knowledge management** Knowledge is a strategic asset and organisation-specific knowledge, which has been built up over time, is a core competence that cannot easily be imitated. Big data is linked to knowledge management and is concerned with how organisations manage data. Cloud computing plays a significant role in improving an organisation's knowledge management.	**Approach** Understand how knowledge management can deliver competitive advantage. Work through the interactive questions to develop your understanding. **Stop and think** How do supermarkets use the data gleaned from their customer loyalty cards to improve their ability to compete in the highly competitive supermarket industry?	You may be asked to identify how knowledge management can be used to enhance competitiveness. Knowledge management is an area that students have struggled with in previous exams so ensure you read through this section carefully.	**IQ2: Information and knowledge management** This question looks at how an organisation can develop a knowledge management strategy. **IQ3: Kid A Ltd** This is an exam standard question combining topics from previous chapters. Part 2 in particular focuses on how information and technology can provide competitive advantage.
4	**Risks associated with IT/IS** Information systems are time consuming and expensive to implement so risks need to be minimised and mitigated to avoid costly errors for the organisation.	**Approach** Read through and ensure you have plenty of practical examples of problems with information system implementations, user resistance and the risks to IT systems. **Stop and think** In 2023, British Airways experienced an IT failure resulting in the cancellation of more than 175 flights, over a national holiday weekend. The technical issue that caused the failure took more than two days to fix. What other risks do IT failures pose for British Airways?	A common theme when examining the reasons for information system failure is user resistance. Ensure you understand how and why users resist change.	

Topic	Practical significance	Study approach	Exam approach	Interactive questions
5	**Cybersecurity and IT controls** The issue of cybersecurity has become increasingly important as companies connect to their supply chains and customers using IT systems. Media reports of high-profile cybersecurity attacks are common so it is clear that cybersecurity must remain high on management's agenda.	**Approach** Read through the chapter concentrating, in particular, on the strategic implications of cybersecurity and practical measures for dealing with risks. **Stop and think** In January 2023, UK fashion retailer JD was hit by a cyberattack on all its brands. Names and billing addresses, phone numbers and order details of approximately 10 million customers were breached. How can retail organisations, especially those with an online presence, protect themselves from similar cyber-attacks?	Exam questions are likely to require a practical approach to combating IT and cybersecurity risks. To develop such practical solutions it can be helpful to keep up to date with business news and reports of how real life companies are managing IT and cybersecurity. Always use the scenario detail to apply your answer to the organisation in the question.	

Once you have worked through this guidance, you are ready to attempt the further question practice included at the end of this chapter.

1 The role of information

Section overview

Information takes many forms and has many roles within the organisation. It comes from **internal** and **external** sources.

Organisations require information for a range of purposes.

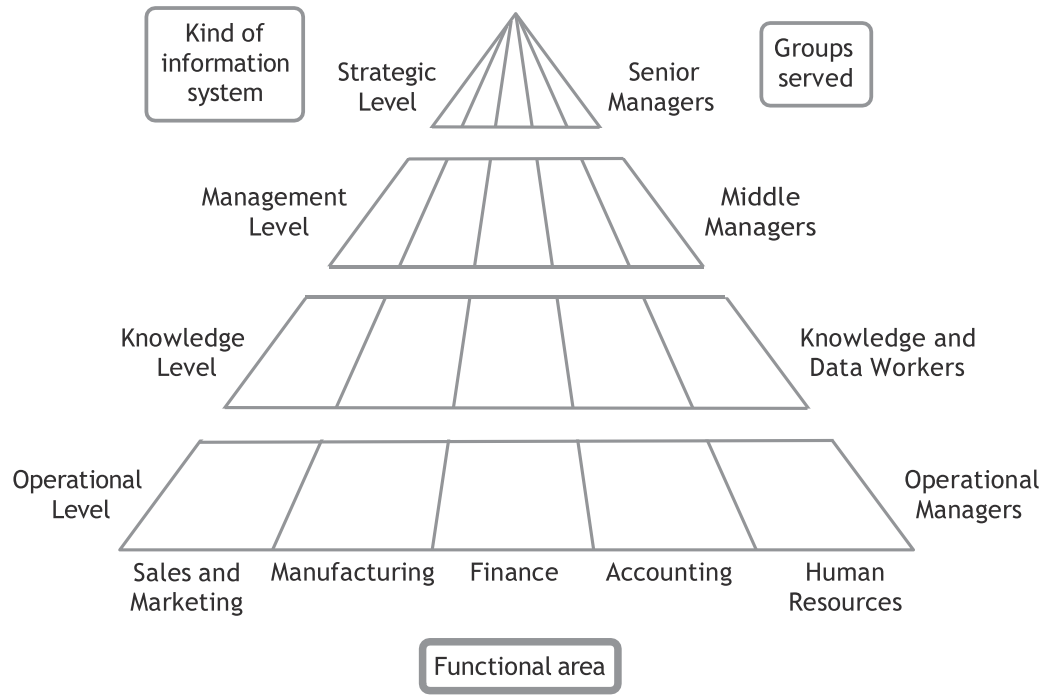
- Planning
- Controlling
- Recording transactions
- Performance measurement
- Decision making

Organisations require different types of information system to provide different levels of information in a range of functional areas, supporting the distinction between strategic, tactical and operational decision making.

1.1 Why do organisations need information?

A modern organisation requires a **wide range of systems** to hold, process and analyse information. Organisations require different types of information system to provide different **levels of information** in a range of functional areas. **Strategic planning**, **management control** and **operational control** may be seen as a hierarchy of decisions. This is sometimes called the Anthony hierarchy, after the writer Robert Anthony. One way of portraying this concept is shown on the following diagram.

Figure 14.1: Levels of information



The point to note from the above diagram is that the higher-level applications such as managerial information depend to a great extent on skimming data from the operational systems maintained by the different functional departments for their own purposes.

Strategic information	Tactical information	Operational information
Derived from both internal and external sources • Summarised at a high level • Relevant to the long term • Concerned with the whole organisation • Often prepared on an *ad hoc* basis • Both quantitative and qualitative • Uncertain, as the future cannot be accurately predicted	• Primarily generated internally (but may have a limited external component) • Summarised at a lower level • Relevant to the short and medium term • Concerned with activities or departments • Prepared routinely and regularly • Based on quantitative measures	• Derived from internal sources • Detailed, being the processing of raw data • Relevant to the immediate term • Task specific • Prepared very frequently • Largely quantitative

1.2 Information requirements in different sectors

Sector	Information type	Example(s)	General comments
Manufacturing	**Strategic**	Future demand estimates New product development plans Competitor analysis	The information requirements of commercial organisations are influenced by the need to make and monitor profit. Information that contributes to the following measures is important: • Changeover times • Number of common parts • Level of product diversity • Product and process quality
	Tactical	Variance analysis Departmental accounts Inventory turnover	
	Operational	Production reject rate Materials and labour used Inventory levels	
Service	**Strategic**	Forecast sales growth and market share Profitability, capital structure	Organisations have become more customer and results-oriented over the last decade. As a consequence, the difference between service and other organisation's

	Tactical	Resource utilisation such as average staff time charged out, number of customers per hairdresser, number of staff per account Customer satisfaction rating	information requirements has decreased. Businesses have realised that most of their activities can be measured, and many can be measured in similar ways regardless of the business sector.
	Operational	Staff timesheets Customer waiting time Individual customer feedback	
Public sector	**Strategic**	Population demographics Expected government policy	Public sector (and non-profit making) organisations often don't have one overriding objective. Their information requirements depend on the objectives chosen. The information provided often requires interpretation (eg, student exam results are not affected by the quality of teaching alone). Information may compare actual performance with: • Standards • Targets • Similar activities • Indices • Activities over time as trends
	Tactical	Hospital occupancy rates Average class sizes Percent of reported crimes solved	
	Operational	Staff timesheets Vehicles available Student daily attendance records	
Non-profit/ charities	**Strategic**	Activities of other charities Government (and in some cases overseas government) policy Public attitudes	Many of the comments regarding public sector organisations can be applied to not-for-profit organisations. Information to judge performance usually aims to assess economy, efficient and effectiveness. A key measure of efficiency for charities is the percentage of revenue that is spent on the publicised cause (eg, rather than on advertising or administration).
	Tactical	Percent of revenue spent on admin Average donation 'Customer' satisfaction statistics	
	Operational	Households collected from/approached Banking documentation Donations	

1.3 The qualities of good information

Regardless of the type of information or the sector in which an organisation operates it is important that organisational information has certain qualities.

The **qualities of good information** can be summarised by the mnemonic **accurate**: **accurate**, **complete**, **cost**-beneficial, **user**-targeted, **relevant**, **authoritative**, **timely** and **easy** to ese.

Professional skills focus: Assimilating and using information

Good quality information is vital for organisations but with the rise of big data it is easy to become overwhelmed with data and information. When faced with this situation, it is important to focus on the key strategies and business priorities for which the information is needed.

Interactive question 1: Strategic decision making

Decision making at the strategic level in organisations needs to be supported by information systems that are flexible and responsive.

Requirements

1.1 Describe the characteristics of information flows at the strategic level.

1.2 Describe the sources of information required for strategic decision making and the characteristics of an information system used to provide strategic information.

See **Answer** at the end of this chapter.

CHAPTER 14

2 The strategic value of information technology

Section overview

- Business analysis views the information system in the context of the organisation's operations and strategy.
- As the importance of big data and information has increased, organisations have realised the role that IT/IS can play in achieving a competitive advantage.

Definitions

Information systems (IS): All systems and procedures involved in the collection, storage, production and distribution of information.

Information management strategy: Strategy specifying who controls and uses the technology provided.

Information systems strategy: Strategy specifying how hardware, software and telecommunications can achieve delivery of the information systems strategy.

Information technology strategy: Strategy specifying the systems that will best enable the use of information to support the business strategy.

2.1 The benefits of a proposed information system

The benefits from a proposed information system should be evaluated against the costs. To quantify the benefits, several factors need to be considered.

- **Increased revenue**

  Improved data collection, storage and analysis tools may indicate previously unknown opportunities for sales. Such tools may include **data-mining and data analytics** software which allow relationships to be discovered between previously unrelated data. Later in this chapter we consider how big data and data analytics can help organisations achieve a competitive advantage.

- **Cost reduction**

  In the chapter, 'Strategic capability,' we explored the impact that new developing technologies are having on organisations' ability to automate previously manually intensive work. Increasing automation saves staff time and may result in a smaller workforce being required.

  Systems such as inventory control can benefit as losses from obsolescence and deterioration are reduced.

- **Enhanced service**

  Computerised systems that create a more prompt and reliable service will increase customer satisfaction. Earlier in the Workbook we considered the increasing use of 'chatbot' technologies to handle customer queries. As discussed in the chapter 'Strategic capability,' the use of computer systems isn't solely restricted to improving the customer experience, as they can also help to enhance the productivity of the workforce, through the use of digital asset management systems.

- **Improved decision making**

  Providing decision makers with the most accurate and up-to-date information that is possible can have substantial benefits. The main areas of benefit are:

  - **Forecasting**

    Predictive data analytics can be used to forecast sales trends, and their effect on costs. Organisations that can make accurate forecasts are in a better position to plan their structure and finances to ensure long-term success.

  - **Developing scenarios**

    Organisations facing uncertain times, or those which operate in dynamic, evolving environments, need to make complex decisions (often quickly) to take advantage of opportunities or to avoid threats. Scenario planning models enable a wide range of variables to be changed (such as inflation rates or sales numbers), the overall effect on the business to be identified and a business plan to be constructed.

  - **Market analysis**

    Modelling can be extended into the market that the organisation operates in. Trends such as sales volumes, prices and demand can be analysed. Relationships between price and sales volume can be identified. These can be used by an organisation when deciding on a pricing strategy. Setting the best price for a product can help drive up sales and profitability. The use of information systems to identify trends in social media posts is becoming common practice in business, particularly in relation to pricing decisions.

  - **Project evaluation**

    Organisations will benefit from improved decision making where systems can accurately evaluate a wide range of projects. Investment decisions often involve large capital

outlays and if the system prevents bad decisions it can prevent the organisation wasting large sums of money.

Systems can also prevent an organisation agreeing 'bad' deals. Tenders for suppliers or other long- term contracts can prove costly if the wrong choice is made.

2.2 Strategic implications

When formulating an overall IS/IT strategy the following aspects should be taken into consideration:

- What are the key business areas which could benefit most from an investment in information technology, what form should the investment take, and how could such strategically important units be encouraged to use such technology effectively? Will maintenance of the new system be provided in-house or be outsourced?
- How much would the system cost in terms of software; hardware; management commitment and time; education and training; conversion; documentation; operational manning; and maintenance? The importance of lifetime application costs must be stressed – the costs and benefits after implementation may be more significant than the more obvious initial costs of installing an information technology function.
- What criteria for performance should be set for information technology systems. Two areas can be considered: the technical standard the information system achieves and the degree to which it meets the perceived and often changing needs of the user.
- What are the implications for the existing work force – have they the requisite skills; can they be trained to use the systems; will there be any redundancies? The decision to automate existing processes through the use of IT and intelligent systems often underlines the need for business leaders to possess effective change management skills. A good change management process is likely to be critical in overcoming any resistance that may arise from those individuals affected by the introduction of a new IT system.



Professional skills focus: Applying judgement

When developing an IT/IS strategy the costs, in terms of hardware/software, implementation and training, need to outweigh the benefits such as faster processes, improved staff morale and competitive advantage. Judgement is required to assess the benefits since many are intangible and cannot be quantified.

2.3 Information systems and competitive advantage

Porter identified three generic strategies that an organisation can use to compete and depending on the chosen strategy, IT can be used in different ways to enhance the business's competitive advantage.

This if the business has chosen cost leadership, IT can be used within the supply chain to improve scheduling and, thus, reduce the costs of inventory.

A differentiator may use IT to make the customer ordering process as easy and flexible as possible.

A business that focuses on a particular niche market may use IT to capture data electronically via a loyalty card and use it to generate information about customer preferences and buying habits.

Value chain analysis, also covered in the chapter, 'Strategic capability,' can be used to assess the impact of IS/IT, and to identify **processes where IT could be used to add value**.

IT can be used to **automate** and improve physical tasks in the **manufacturing** sector. It also provides **extra information** about the process.

2.3.1 Operations

- **Process control**: Computer systems enable tighter control over production processes.
- **Machine tool control**: Machine tools can be automated and, it is hoped, be made more precise.
  - **Numerical control**: Information to operate the machine tool is prepared in advance to generate a set of instructions.
  - **Computer numerical control** is where the computer produces the instructions.
  - **Direct numerical control** is where the computer is linked directly to the machine tool.
- **Robots** can automate some of the process.
- **Computer aided manufacturing** (CAM) involves a variety of software modules such as:
  - Production control, supervisory systems
  - Materials requirement planning (MRP I) and manufacturing resources planning (MRP II)

    These are automated component ordering and automated production scheduling systems respectively.
  - Capacity requirements planning
- **3D printing** involves producing a three-dimensional object using a 3D printer. This technique uses a robotic printer laying successive layers of material on one another, and these are then etched into a three-dimensional shape using light. The increasing publicity surrounding the commercial benefits of using 3D printing has attracted interest from various industries, including clothing manufacturers and even the construction industry.
- **Computer integrated manufacturing** (CIM) integrates all aspects of an organisation's manufacturing activities. 'IT cannot solve basic organisational problems, but the essence is the use of the IT to provide integration though communication, effectiveness and efficiency.' Flexible manufacturing systems include:
  - Machine tools
  - Materials handling conveyor sets
  - Automatic guided vehicles
- **Enterprise resource planning** (ERP) systems take MRP II systems a step further and are not restricted to certain types of organisation. ERP systems are used for identifying and planning the **enterprise-wide** resources needed to record, produce, distribute, and account for customer orders.

2.3.2 Logistics

In both **inbound** logistics and **outbound** logistics, IT can have an impact.

- The use of IT in **inbound logistics** includes stock control systems such as MRP, MRPII, ERP and IT.
- **Warehousing** – the use of barcodes can increase knowledge about the quantity and nature of stock in hand.
- It is possible to create computer models, or **virtual warehouses**, of stock actually held at **suppliers**. For example an organisation with several outlets might have each connected to a system which indicates the total amount of stock available at different sites.

2.3.3 Marketing

Marketing activities can be made more effective by the use of **customer databases** and the use of **social media:**

- Many large organisations have redirected their marketing effort through **social media** ie, Facebook and X (formerly, Twitter), as a means of achieving competitive advantage. Organisations are increasingly using social media to find out what their customers are saying about their products or services, and to respond to complaints.
- Buying data about **customer purchasing habits** and interrogating it is providing a more precise method of targeting particular groups of consumers than television advertising.
- A variety of market research companies use IT/IS as a way of **interacting** with consumers. Examples include: the use of online viral marketing campaigns and online questionnaires.
- Supermarkets can use automated **EPOS** (electronic point of sale) systems to have precise real time data about how products are selling to enable speedy ordering and replenishments.

However, as the following worked example illustrates, organisations need to take special care when embracing social media.



Context example: When social media goes wrong

In April 2019, online fashion retailer, ASOS, hit the headlines for the wrong reasons when it posted a promotional picture on its X feed of one of its latest dresses. The picture featured a model wearing a dress which clearly showed two bulldog clips attached to it which gave the appearance of giving the wearer a better fitting garment. Pictures posted after the error was identified by members of the public showed the same dress with the bulldog clips removed.

The incident led to a debate online about retailers misleading consumers over the sizes of the clothes they sell and how well they actually fit the wearer in reality, with some accusing the company of deceiving customers. ASOS was swift to acknowledge the error and issued an apology.

(**Source**: BBC. (2019) *ASOS bulldog clipped dress causes a fuss online.* [Online] Available from: www.bbc.co.uk [Accessed 18 July 2023]).

2.3.4 Service

Customer relationship management (CRM) describes the methodologies, software, and usually internet capabilities that help an enterprise manage customer relationships.

For example, an enterprise might build a database about its customers that described relationships in sufficient detail so that management, salespeople, service staff, and maybe the customer, could access information, match customer needs with product plans, remind customers of service requirements and know what other products a customer had purchased. CRM consists of:

- helping an enterprise to identify and target their best customers, manage marketing campaigns with clear goals and objectives, and generate quality leads;
- assisting the organisation to improve telesales, account, and sales management by **optimising information shared**, and streamlining existing processes (eg, taking orders using mobile devices);
- allowing the formation of relationships with customers, with the aim of improving customer satisfaction and maximising profits; identifying the most profitable customers and providing them with the highest level of service; and
- providing employees with the information and processes necessary to know their customers, understand their needs, and effectively build relationships between the company, its customer base, and distribution partners.

2.3.5 Support activities

As far as **support** activities are concerned IT has some impact.

- **Procurement:** IT can automate some purchasing decisions. Paperwork can be saved if the organisation's purchase systems are linked directly to the sales order systems of some suppliers (eg, by electronic data interchange) and costs can be reduced through the use of internet based marketplaces (see following worked example).
- **Technology development:** Computer automated design (**CAD**) is, in a number of areas, an important influence.
  - **Drafting**: CAD produces engineer's drawings, component design, layout (eg, of stores, wiring and piping) and electronic circuit diagrams in complex systems.
  - **Updating**: It is easy to change design in CAD systems and to assess ramifications of any changes. Some CAD systems have archive data (eg, for reference).
  - CAD enables modelling to be **checked** without the necessity of producing working prototypes.
    Some 'stress testing' can be carried out on the model.
- As we explored earlier in the Workbook, IT has had an impact on **human resources** as it has changed the way in which people in certain industries now work. This is evident given the rise in workforce flexibility, which has resulted in growing numbers of people now working from home. In addition there are now a number of HR applications which organisations can use to assist in managing their human resources these include: the maintenance of a skills database, staff planning (eg, using network analysis), computer-based training, time attendance systems, payroll systems, pension systems.



Context example: Alibaba

Alibaba is the world's largest e-commerce company which predominantly operates in China. According to the company's website its mission is to 'make it easy to do business anywhere' (Alibaba, 2018).

Alibaba was set up in 1999 by a Chinese English teacher (Jack Ma), with the aim of helping small Chinese businesses trade globally. As Yueh (2013) explains, Alibaba is effectively a 'combination of eBay and Amazon. It is an online company with multiple revenue streams that are more conventional than a social network site. Alibaba.com is a B2B, or business-to-business, website. It links up businesses around the world looking for suppliers. For instance, they link wholesalers to distributors around the world, from the UK to China to the US.' Businesses can trade almost anything from olive oil to computer components.

(**Sources**: Alibaba (2018) *About Alibaba.* [Online] Available from: www.alibaba.com; Yueh, L. (2013)

Alibaba: The next facebook? [Online] Available from: www.bbc.co.uk [Accessed 18 July 2023]).

2.4 Strategic importance of big data

So far in this chapter we have explored some of the more general ways in which IT systems can help organisations. You may recall that in the chapter, 'Strategic capability' we introduced the concept of big data, and the increasingly important role that data plays in business. We explored the growth in the so-called 'internet of things' whereby organisations embed sensor technologies into their operations to collect data on a range of activities, including gathering data about their customers.

In the next section, we consider through the use of real-life examples how organisations can analyse and use the data they collect. Consideration is also given to how big data and data analytics can help organisations to monitor performance and modify strategies in order to achieve a competitive advantage.

A report by the McKinsey Global Institute, 'Big data: The next frontier for innovation, competition and productivity' highlighted that there five broad ways in which big data and data analytics can help organisations to compete in increasingly competitive environments:

- Enhanced data transparency
- Enhanced performance
- Market segmentation and customisation
- Improved decision-making
- New products and services

Each of these points is considered in more detail in the following section.

2.4.1 Enhanced data transparency

Big data can create value by making the meaning held within the data more transparent and usable. For example, new insights might be gained by data which was previously too expensive or complex to process. Transparency may also add value especially if data generated by other parts of the organisation can be integrated with externally obtained data. Improved transparency should assist the organisation to achieve its corporate objectives.

For example, consider a company which operates in a highly competitive industry. Product innovation and being first to market are both highly desired by customers. As a result, continual product development is critical to success. Integrating data obtained from the marketing department about customer requirements and demand, with data from the R&D team about the feasibility of new products could help the company to meet customer needs and reduce the time to market.



Context example: Social media analytics

The following worked example explores how organisations are increasingly able to use social media analytics to measure qualitative data.

For large organisations, social media analytics is no longer solely focused on measuring simple quantitative metrics such as the number of times its Facebook account has been 'liked.' Today, there is far greater focus on identifying and measuring qualitative data. The widespread use of social media analytics tools now means that organisations can capture the key words used by customers when posting about the organisation's products and services online. Feedback or comments posted on sites such as X allow organisations to perform 'sentiment analysis' which provides a useful insight into how end users perceive the organisation's offering. Sentiment analysis is helping organisations to co-ordinate operations in response to both positive and negative trends. For example, complaints posted on X about a product can now be directed through to the organisation's customer service teams to help limit the potential negative publicity generated.

2.4.2 Enhanced performance

The availability of vast quantities of transactional data provides organisations with a large amount of detailed, 'real time' performance data. This data can then be analysed to identify fluctuations in performance, and take corrective action in 'real time' where performance deteriorates.



Context example: Data and machine performance

Many supermarkets have taken to installing state-of-the art energy management systems in a bid to improve the efficiency of their in-store refrigeration units. Such systems make use of sensors which track the electricity usage of individual refrigeration units in 'real time.' This data allows the supermarket's own team of data analysts to interpret the results and identify

irregularities in electricity consumption which indicate a fault with the refrigeration unit. In turn, this enables the data analysts to direct the supermarket's in-house maintenance workers to investigate the fault before it becomes more serious.

This example also illustrates the potential for internet of things technology (which we discussed in the chapter, 'Strategic capability') to enable predictive maintenance. Sensors in machines can communicate with each other (as well as producing data for humans to analyse), so there could be scope for 'smart' machines to identify and correct problems, through machine-to-machine communication, without the need for human intervention.

2.4.3 Market segmentation and customisation

Big data should help organisations to further segment their market in 'real time', allowing for the customisation of promotions and advertising to more effectively attract customers to the organisation's product/service offering. The concept of tailored promotions is covered in more detail in the next worked example.

2.4.4 Improved decision making

The use of big data should create value by improving the organisation's approach to decision making. For example, an online retailer could flex the prices it charges for its products using 'real time' sales and inventory data to influence the customer's purchasing behaviour. Such an approach may enable the retailer to entice customers to purchase a slow selling inventory line by reducing the price while the customer is still online.

2.4.5 New products and services

The use of big data obtained from customer shopping habits can also help organisations to develop new and existing products and services to meet customer needs. For example, in response to growing demand from customers wanting to pay lower insurance premiums a number of insurance companies have started to monitor the way in which car policy holders drive their vehicles. Data is collected on how the car is been driven through the use of mobile apps or in some cases through the use of GPS devices. The data obtained then enables the insurance company to monitor the safety of the driving and adjust the car insurance premiums to reflect the level of risk that the insurer is exposed to.



Context example: Netflix

Netflix, the subscription based streaming service, has been at the forefront of using big data to enhance its services and improve the user experience. In fact, its dominance of the streaming industry can in large part be attributed to how effectively it leverages big data across several areas.

Collecting and storing data: *Netflix gathers vast amounts of data from various sources, including personal data when an account is set up and browsing data. Machine algorithms and artificial intelligence (AI) tools analyse this data in real time, to understand user behaviour, preferences and viewing habits and to make decisions quickly and accurately.*

Personalised recommendations: *By analysing user data, viewing history, and preferences, Netflix can provide tailored content suggestions to each user, enhancing their overall streaming experience. The algorithms are constantly learning and adapting, ensuring that viewer recommendations continually improve over time, allowing Netflix to offer a curated selection of content to each user increasing engagement and satisfaction levels.*

Optimising content delivery: *One of the key challenges for streaming platforms is delivering content without interruptions or buffering issues. By analysing data on factors such as user location, network conditions, and device capabilities, Netflix can route content through the best performing networks to guarantee a smooth streaming experience.*

Improved user experience: *Netflix even monitors data on users interaction with the Netflix platform to make changes to the interface that meet the needs of individual users. For example, Netflix conducts frequent A/B testing (split testing) whereby one group of users is shown version A of the user interface and another group is shown version B. By comparing the levels of user interaction with each version, Netflix can determine which one best suits individual customers.*

Original content creation: *Netflix uses its data to identify gaps in the market for new types of programme and can predict what content is likely to be successful amongst different audience segments before a show is released. By analysing data on viewer demographics, genre preferences, and engagement levels, Netflix original programming has a higher chance of success. Once content is released, data analysis of audience reaction is monitored to provide input into what can be changed in the next round of original content creation.*

In summary, analytics and insights derived from big data are vital to Netflix's success in the streaming industry and by constantly making the most of the power of analytics Netflix is able to keep its 270 million customers satisfied with their streaming experience.

(Source: *Collins, N. (2023) How Netflix Uses Big Data. [Online] Available from: https://robots.net/fintech/how-netflix-uses-big-data/ [Accessed on 3 May 2024])*

2.5 Criticisms of big data and data analytics

While big data and data analytics offer a number of potential benefits it is important to appreciate some of the criticisms levied at them.

2.5.1 Latest buzzword

Some critics argue that big data simply represents the latest buzzword to have made it into the business press and suggest that there are insufficient examples to illustrate how analysing larger data sets have revealed anything significant beyond what was already known.

2.5.2 Correlation and causation

It has also been argued that too much of the focus on big data to-date has centred on the identification of correlations in data sets with little consideration given to the causes of such correlation. It is important to remember that a correlation between data sets does not imply causation. For example, in the UK people spend more in shops when it is cold. However, this correlation does not suggest that cold weather causes increased shopping. A better explanation is that cold weather in the UK coincides with Christmas and New Year sales, resulting in higher retail expenditure.

Failing to understand those elements which cause a correlation to exist between data variables ultimately means that analysts have no understanding of those factors or circumstances which could force a correlation to end. The idea of correlation and causation is explored further in the chapter, 'Risk management.'

2.5.3 Data overload

Critics also argue that capturing and analysing all of the available data about a particular subject does not automatically create value for the organisation. For data to be value adding it needs to provide an insight or some meaning that was previously unknown. Failure in this regard simply means the organisation has a lot of data which it cannot do anything with.

2.5.4 Ability to verify

Data captured from external sources increases the difficulty of verifying whether it is suitable as a basis for decision-making. This ultimately reduces any reliance that can be placed on the conclusions drawn from using such data.

2.5.5 Sustainable competitive advantage

Whether or not big data can provide organisations with a sustainable long-term competitive advantage remains to be seen, especially as the use of data analytics gradually becomes common practice in many industries.

2.5.6 Representative data

Finally, it is important to recognise that analysing a lot of data does not necessarily mean that it is fully representative of the whole data population. For example, asking customers via the organisation's X feed which of its products they value most, and analysing the customers tweets would not represent the views of all of the organisation's customers. It is likely that the responses received will largely represent the opinions of younger, urban customers as research suggests that it is this group which tend to use X the most.



Professional skills focus: Applying judgement

When using data from both internal and external sources, apply professional scepticism and critical thinking to identify gaps and inconsistencies in the data. The data may not be fully representative of the whole data population or there may be bias within the data set, affecting the strategic decisions made.

2.6 Barriers to introducing big data and data analytics

An article published by accounting firm EY titled 'Ready for take-off?' focusses on the main strategic challenges organisations face when introducing a big data strategy (some of these points were introduced in the chapter, 'Strategic capability').

The article highlights eight significant barriers:

2.6.1 The unknown destinations

The majority of larger organisations are already aware that they possess useful data, but do not really understand which questions they should be asking of the data they hold to extract meaning from it.

2.6.2 The underlying technology challenges

Many organisations are unable to cope with the amount of data that they have access to. The volume of data flowing into the organisation can be intimidating. Management can be intimidated by the concept of establishing 'real time' data analytics systems, a problem which is further compounded by a lack of knowledge around which technological tools best meet their needs.

2.6.3 The lack of a holistic approach

Particularly in larger organisations the introduction of a big data strategy often occurs on a project- by-project basis, within distinct departments in the organisation. Organisations need to take a more holistic approach when introducing big data. Management need a better understanding of how the value created from big data in one department can be used throughout the organisation. Promoting a holistic approach needs to be driven by senior management.

2.6.4 The shortage of talent

To extract meaning from captured data requires organisations to have employees with specialist skills. Job roles needed are likely to include data scientists, visualisation experts, business intelligence analysts, data warehousing professionals, and data privacy experts. Developing the skills needed in the short term can be particularly difficult, and the cost of buying in such expertise can be prohibitive.

2.6.5 The fear of cyber attack

Later in this chapter we discuss the concept of cybersecurity. The growing dependence on data and data analytics increases the potential exposure to cyber-attack. The reputation damage and potential regulatory risks caused by such a breach may deter the management in some organisations from introducing a big data strategy.

2.6.6 The difficulty of building the business case

While senior management in an organisation may accept the general argument supporting the need for big data initiatives, they often want to understand the potential benefits of specific projects before authorising the related investment needed to improve the IT infrastructure. In cases where a business has undertaken very few big data projects quantifying the benefits can often prove difficult.

2.6.7 The need for legal and regulatory compliance

Organisations need to understand the data privacy issues surrounding the introduction of a big data strategy. Failure to address the legal and regulatory risks could potentially be very costly from both a financial and reputation perspective.

2.6.8 The need for customer data

There is also a growing concern among customers that any data they provide to organisations will be misused or that their privacy will suffer. This presents a challenge for organisations in determining how best to address their customers' concerns, and to comply with the provisions of relevant regulations on data protection such as the UK's Data Protection Act.



Context example: 23andMe data breach

In October 2023, the genetic testing company, 23andMe, admitted to a security breach that had exposed the personal data of more than 7 million users. Of these accounts, hackers actually gained access to 14,000 some of which revealed detailed, sensitive reports on users' health. Following the attack, a hacker offered to sell the names, addresses and genetic details reportedly belonging to a million customers with Ashkenazi Jewish heritage, raising concerns of potential terrorist attacks on this community.

According to 23andMe, the hack was made possible by users failing to update or recycling old passwords which weakened their cybersecurity. However, 23andMe has been criticised by lawyers acting on behalf of affected customers saying that the company should have taken greater measures to protect this highly sensitive data. The company's share price plummeted in the wake of the scandal. As at May 2024, 23andMe's market capitalisation was approximately $244 million, a fall of 93% since it went public in 2021 with a value of $3.5 billion.

*(**Source**: DeGuerin, M. (2024) Hackers for nearly 7 million people's data from 23andMe. The firm blamed users in 'very dumb' move.' The Guardian. [Online] Available from: https://www.theguardian.com/technology/2024/feb/15/23andme-hack-data-genetic-data-selling- response [Accessed on 3 May 2024])*

3 Knowledge management



Section overview

- Knowledge management is a key element in this chapter. Effective use of information requires knowledge. In the chapter, 'Strategic capability,' we explored the increasingly important role that human capital plays in business, where the knowledge of workers is increasingly being viewed as a strategic asset.

- Organisation-specific knowledge, which has been built up over time, is a core competence that cannot easily be imitated.
- Knowledge management (KM) refers to the process of harnessing IT and other systems to develop and disseminate relevant knowledge throughout the organisation.
- KM is a source of competitive advantage because it encourages process improvement and innovation and helps the organisation identify ways to meet customer needs better than the competition. The concept of knowledge management is an area that candidates attempting the Business Strategy and Technology exam have struggled with. It is therefore particularly important that you take your time as you read through the next section.
- Capturing the knowledge is insufficient. Competitive advantage is achieved by acting on the knowledge and using it to make effective decisions.

3.1 Knowledge management programmes



Definitions

Knowledge: Is the potential for action based on data, information, intuition and experience.

Explicit knowledge: Is knowledge that the company knows that it has. This includes facts, transactions and events that can be clearly stated and stored in management information systems.

Tacit knowledge: Is personal knowledge and expertise held by people within the organisation that has not been formally documented.

Knowledge management: Describes a range of strategies and tools that capture all the knowledge that is valuable to an organisation, and deliver it to the people in such a way that it can be acted on quickly, to the competitive advantage of the business.

As we have seen during our discussions of big data, organisations now operate in an age of information. Faced with increasing environmental change, organisations have begun to focus on the importance of exploiting their **intangible assets**, one of which is knowledge. The **effective use of information** requires knowledge.

Knowledge management programmes extend beyond any particular piece of IT/IS and embrace changing the attitudes of management and staff towards sharing information.

They concern:

- Designing and installing techniques and processes to create, protect and use **explicit knowledge**
- Designing and creating environments and activities to discover and release **tacit knowledge**
- **Capturing good quality information** from outside the business as well as within eg, intelligence on competitors

Tacit knowledge is a difficult thing to manage because it is **invisible** and **intangible**. We do not know what knowledge exists within a person's brain, and whether they choose to share knowledge is a matter of choice. Tacit knowledge may be difficult to express or communicate to others but is also invaluable because it is a unique asset that is very hard for other organisations to copy.

Collaboration between employees helps to transform tacit knowledge into explicit knowledge.

The **motivation to share** hard-won experience is sometimes low; the individual is 'giving away' their value and may be very reluctant to lose a position of influence and respect by making it available to everyone.

Organisations should encourage people to share their knowledge. Electronic tools such as online bulletin boards, web logs (blogs) and wikis (editable web pages designed for groups of users) all facilitate knowledge sharing.

However, whilst technology can help unlock knowledge, corporate culture is more important. Collaboration can be encouraged through a culture of openness and rewards for sharing knowledge and information.

3.2 Systems that aid knowledge management

Information systems play an important role in knowledge management, helping with **information flows** and helping formally **capture** the knowledge held within the organisation.

Interactive question 2: Information and knowledge management

Increasingly, the management of information sharing and group working ventures is a fundamental part of business management.

Requirements

2.1 Discuss how the management of information might differ from the management of knowledge.

2.2 How can an organisation develop a knowledge strategy?

See **Answer** at the end of this chapter.

3.3 Knowledge management as a source of competitive advantage

The resource-based view of strategy, discussed in the chapter Strategy and organisations, is one of a successful organisation that acquires and develops resources and competences over time and exploits them to create competitive advantage.

The ability to capture and harness corporate knowledge has become critical for organisations as they seek to adapt to changes in the business environment, particularly those businesses providing financial and professional services.

Knowledge is seen as a strategic asset and organisation-specific knowledge, which has been built up over time, is a core competence that cannot easily be imitated.

Knowledge management can help promote competitive advantage through:

- The fast and efficient exchange of information
- Effective channelling of the information to:
  - Improve processes, productivity and performance
  - Identify opportunities to meet customer needs better than competitors
  - Promote creativity and innovation

In the chapter, 'Strategic capability,' we discussed benchmarking. The process of identifying the outstanding qualities and practices of competitors allows an organisation to capture knowledge that can then be used internally to make improvements that will further its competitive advantage.

If knowledge management is about making sure that information flows efficiently through a company, competitive advantage is achieved by making sure that the knowledge is translated into decisions and actions.

Knowledge must be delivered in a form that can be quickly interpreted and acted on. There must also be a process for learning from past experiences (successes or failures) and identifying opportunities to improve.

Thus competitive advantage is not necessarily achieved by those who have the best knowledge, but by those who use knowledge best.



Interactive question 3: Kid A Ltd

Kid A Ltd (KA) is a company that runs four pre-school nurseries for children aged two to four years old, and is planning to expand the business.

Industry background

Demand for pre-school nurseries has grown substantially in the UK since the start of the 19X0s. Changes in the labour market have resulted in a rise in the number of working women with dependent children. In 20X8, 52% of women whose youngest dependent child was under five years of age were working, compared to only 31% 10 years earlier. During this period unemployment in the UK fell significantly.

This has, however, coincided with a change in demographics in the UK and a drop in the birth rate, resulting in a falling population of children of pre-school age.

There has been an increased emphasis by the Government on the role of childcare and education in raising educational standards and enhancing children's social development. In the UK over the last 10 years, the number of day nursery places more than doubled as more nurseries were opened. Day nurseries are one form of pre-school childcare available in the UK, the other main ones being playgroups, childminders and the use of domestic nannies.

Most day nurseries operate independently from one another and are, in the main, owner managed. Demand for nursery places normally outstrips supply.

Since 20X1, all nurseries have been subject to government registration and inspection, and the inspectors' reports are available on the internet for viewing by interested parties.

Company background

KA was formed in 20X2 by Thom Yorke, a former IT engineer, and Debra Curtis, who previously worked as a manager for another nursery. Both Thom and Debra have a close involvement in the day to day running of the nurseries, which have a maximum of 48 children on each of the premises at any one time.

The company has utilised modern technology to help address concerns of parents. All the children at the nurseries are given GPS (global positioning satellite) wristbands that are linked wirelessly to a series of webcams. Parents can call up the webcams from their workplaces to see the activities of their children.

Parents who have busy work commitments can also utilise a collect and return service whereby KA will pick up children and return them to their homes outside the normal nursery hours of 8am-6pm. This service, although popular with some parents, has attracted criticism from some commentators, who believe it results in disruption to the lives of children.

The KA nursery has a far greater emphasis on learning, especially via computers, than children of the same age-groups at other nurseries, and this has proved to be very popular with some parents.

Parents can review the computer-based games and work undertaken by their children remotely via the internet. These innovations have been praised in the government inspector reports on KA.

KA operates on a staff-to-child ratio that is far higher than the industry average, to ensure the children are given a more structured environment and personal attention. All staff members are NNEB (National Nursery Examining Board) qualified, unlike other nurseries which have a mixture of qualified and unqualified staff. Staff are paid a premium over the salaries of other nurseries. They are expected to have an understanding of the technology that is used within the nurseries and are given a list of tasks each day that they are expected to carry out. Tasks are reviewed by Thom and Joy each week to ensure they have been completed.

KA charge about 40% more than other nurseries, to reflect the extra services provided, but this has not prevented the nurseries being permanently full with a waiting list. Despite extra costs being incurred, KA has a higher profit margin than most other nurseries.

Latest developments

KA is now considering an expansion of the business and opening four more nurseries, with a view to opening additional nurseries if this proves to be a success. Thom and Debra have previously controlled the individual nurseries themselves. They now want to appoint managers to run the nurseries and to set budgets to ensure that they generate a return on the initial investment made, without compromising the standards of care and education offered to children. They are uncertain as to how to assess the managers in terms of how well they are running the individual nurseries.

Thom and Debra will dictate the initial layout of the individual nurseries in terms of building sizes and fixtures and fittings. The nursery managers would be responsible for advertising, bookings, staff, communication with parents and day-to-day running expenses.

Requirements

3.1 Prepare a Porter's Five Forces model for the nursery industry as a whole.

3.2 Identify and explain the critical success factors that operate within the business of KA. Discuss how KA has used information systems and technology to add value and give competitive advantage.

3.3 As an external consultant, prepare a report to Thom and Debra that:

(a) Explains the risks that may arise as a result of expanding the business.

(b) Suggests how they could implement an effective budgeting process for the nurseries.

(c) Explains the strategic, tactical and operational information that Thom and Debra will need to assist in decision making and control.

See **Answer** at the end of this chapter.



Professional skills focus: Structuring problems and solutions

Note that only about 20% of the marks in the Business Strategy and Technology exam are for knowledge, such as Porter's Five Forces, whilst approximately 80% of the marks are for applying this knowledge to the scenario in the question. Models therefore provide a framework for analysis rather than being the answer to the question. In the case of Porter's Five forces you must assess the strength of each force and explain the implications for business competitiveness.

3.4 *The role of cloud computing*

In the chapter, 'Strategic capability,' we introduced the term 'cloud computing,' which plays an increasing role in helping organisations to capture and share corporate knowledge in the pursuit of growth and development. Cloud computing when applied to knowledge management, and business strategy more generally, offers several advantages.

- *Collaboration – cloud computing facilitates the rapid sharing of knowledge internally and also with external partners, regardless of location. In particular, cloud computing has played an important role in the transition to remote working, giving companies and employees flexibility around when and where they work. Cloud-based collaboration tools such as Microsoft Teams and Google Workspace facilitate real-time communication, document sharing and project management.*

- *Scalability – cloud-based systems can scale up or down based on demand, allowing organisations to quickly accommodate growth or changes in workload, avoiding potential performance bottlenecks. Scalability ensures that knowledge creation, storage and sharing remain efficient even during peak times.*
- *Cost efficiency – more generally, cloud-based systems provide cost efficiency since organisations only pay for the resources they use. Upfront investments in hardware and software are avoided and cloud providers manage infrastructure maintenance, allowing companies to focus on strategic initiatives and innovation.*
- *Sustainability – as the importance of sustainability grows, cloud computing can help organisations to reduce their carbon footprint through efficient use of resources, only paying for what is needed. Migrating applications to the cloud and closing data centres can significantly reduce IT carbon emissions, because cloud service providers are able to run highly efficient services, reducing overall resource consumption.*

Definitions

Infrastructure-as-a-service (IaaS): *Provides virtual computing resources over the internet. It allows businesses to rent servers, storage space, and networking capabilities, without needing to own physical infrastructure. Users can manage operating systems, storage, and deployed applications, while the service provider handles the physical hardware.*

Platform-as-a-service (PaaS): *Offers a development and deployment environment delivered over the internet, allowing users to build, test, and deploy applications without the complexity of managing the underlying infrastructure. Providers supply a framework that developers can build upon, including support for software development kits, databases, and development tools. PaaS is designed for developers who want to create applications without worrying about hardware, software updates, or storage.*

Software-as-a-service (SaaS): *Is a delivery model that allows users to access software applications over the internet. These applications are hosted and maintained by external providers, offering convenience and reduced IT overhead for users. SaaS solutions are typically subscription-based, with common applications including email, customer relationship management (CRM) and collaboration tools.*

4 Risks associated with IT/IS

Section overview

- The risks of IT/IS can be summarised as:
  - The **risk of inadequacy**: The failure by the firm to utilise IT/IS as effectively as its rivals will lead to loss of competitive advantage (eg, inferior service, poorer products, excess costs).
  - The **risk of breakdown**: Where the firm depends on IT/IS, a breakdown in its operations threatens the business.
  - The **risk of excess expense**: IT/IS is a significant budget item. Botched projects, expensive contracts, inappropriate systems or non-adoption presents a direct financial risk.
- Risks are present at the system specification stage, and they carry through to implementation. There is also the risk of systems failure.

4.1 Development and implementation – what can go wrong

Problems that occur when implementing a new information system can usually be traced to deficiencies in the development and specification process. Some of these issues involve change management which is covered in the chapter, 'Strategies for change.'
The following table outlines common mistakes that adversely affect the implementation process.

Stage/activity	Problems
Analysis	The problem the system is intended to solve is not fully understood. Investigation of the situation is hindered by insufficient resources. User input is inadequate through either lack of consultation or lack of user interest. The project team is unable to dedicate the time required. Insufficient time is spent planning the project.
Design	Insufficient user input Lack of flexibility. The organisation's future needs are neglected. The system requires unforeseen changes in working patterns. Failure to perform organisation impact analysis. An organisational impact analysis studies the way a proposed system will affect organisation structure, attitudes, decision making and operations. The analysis aims to ensure the system is designed to best ensure integration with the organisation. Organisational factors sometimes overlooked include: • Ergonomics (including equipment, work environment and user interfaces) • Health and safety • Compliance with legislation • Job design Employee involvement
Programming	Insufficient time and money are allocated to programming. Programmers are supplied with incomplete or inaccurate specifications. The logic of the program is misunderstood. Poor programming technique results in programs that are hard to modify. Programs are not adequately documented.
Testing	Insufficient time and money allocated to testing Failure to develop an organised testing plan Insufficient user involvement User management do not review and sign-off the results of testing.
Conversion	Insufficient time and money allocated to data conversion Insufficient checking between old and new files The process is rushed to compensate for time overruns elsewhere.
Implementation	Insufficient time, money and/or appropriate staff mean the process has to be rushed. Lack of user training increases the risk of system under-utilisation and rejection. Poor system and user documentation. Lack of performance standards to assess system performance against. System maintenance provisions are inadequate.

A recurring theme when examining the reasons for information system failure is user resistance. Users may be management and staff, but for outward-facing systems equally could involve customers, suppliers and other partners.

The three types of theories to explain user resistance are explained in the following table.

Theory	Description	Overcoming the resistance
People-oriented	User-resistance is caused by factors internal to users as individuals or as a group. For example, users may not wish to disrupt their current work practices and social groupings.	User training Organisation policies Persuasion User involvement in system development
System-oriented	User-resistance is caused by factors inherent in the new system design relating to ease of use and functionality. For example, a poorly designed user-interface will generate user-resistance.	User training and education Improve user-interface Ensure users contribute to the system design process Ensure the system 'fits' with the organisation
Interaction	User-resistance is caused by the interaction of people and the system. For example, the system may be well-designed but its implementation will cause organisational changes that users resist eg, reduced chance of bonuses, redundancies, monotonous work.	Re-organise the organisation before implementing the system Redesign any affected incentive schemes to incorporate the new system Promote user participation and encourage organisation-wide teamwork Emphasise the benefits the system brings

4.2 Risks from IT systems

There are a range of general risks which IT systems are exposed to. Common risks include:

- **Natural threats** – fire, flood, electrical storms
- **Non-compliance with regulations**:

  The use of IT systems, and the data they contain, is subject to close legal supervision in most countries. In the UK relevant legislation includes the General Data Protection Regulations which can result in large financial penalties where personal data is wrongly held and used.
- **Data systems integrity** – may include incorrect entry of data, use of out-of-date files, loss of data through lack of back-ups
- **Accidents:**

  Staff are a physical threat to IT installations, accidents including spilling a cup of coffee over a desk, or tripping and falling, thereby doing some damage to an item of office equipment.
- **Cybersecurity risks**:

  Hackers and malicious attacks on computer systems represent an increasingly complex threat to organisational IT systems. This is discussed in greater detail in the next section.

4.3 *Managing implementations using agile methodologies*

Agile methodology is a project management framework that can help to manage the risks of project implementation. Agile is an alternative approach to traditional waterfall project management methodologies whereby the project is structured into a series of rigid, sequential phases, with each phase only beginning when the previous phase is completed.

Agile frameworks are often considered to be more appropriate for software implementation. Agile methodologies are iterative, with the project being broken down the project into smaller, manageable phases known as 'sprints.' After each sprint, project teams review their progress and amend their approach ready to complete the next sprint. This enables project teams to ***identify risks*** *early and* ***reduces the chance of major project failures****. In comparison, waterfall approaches follow a rigid linear structure where changes are harder to accommodate once a phase is complete. Agile also focuses on* ***collaboration*** *with stakeholders meaning that feedback is regular and user needs can be met. Agile emphasises* ***continuous testing*** *and* ***regular quality assurance****, mitigating the risk of the problems and defects being encountered only at the end of the project.*

5 Cybersecurity and IT controls



Section overview

- The issue of cybersecurity has become an increasingly important issue for most organisations. Controls are therefore needed to protect data and information.

Cybersecurity is the collection of tools, policies, security concepts, security safeguards, guidelines, risk management approaches, actions, training, best practices, assurance and technologies that can be used to protect the cyber environment and organization and user's assets.

This definition can be extended to include the protection of data from unauthorised modification, disclosure or destruction, and the protection of the information system from the degradation or non- availability of services – in other words, system failure.

In the chapter, 'Strategic capability,' we introduced the concept of cybersecurity in relation to the supply chain. In this section, we consider in more detail the need for organisations to have in place adequate cybersecurity measures.

The widespread use of IT/IS throughout the world presents organisations with the increasingly complex challenge of keeping the systems and data they hold safe from a range of constantly evolving risks. This has made cybersecurity a major issue for most organisations. Cybersecurity is directed towards protecting IT systems from risks which predominantly feature some degree of human involvement.

Although by no means definitive, the list below details some of the most significant cyber risks an organisation's IT systems might encounter:

- **Human threats**:

  Hackers may be able to get into the organisation's internal network, either to steal data or to damage the system. Political terrorism is a major risk in the era of cyber-terrorism.
- **Fraud** – the theft of funds by dishonest use of a computer system
- **Deliberate sabotage** – eg, commercial espionage, malicious damage or industrial action
- **Viruses and other corruptions** can spread through the network to all of the organisation's computers.

- **Denial of Service (DoS) attack**:

  A denial of service attack is characterised by an attempt by attackers to prevent legitimate users of a service from using that service.

In recent years, there has been an alarming rise in the number of so called 'cyber attacks' carried out by hackers and saboteurs, intent on causing maximum disruption to organisational IT systems.



Context example: Risks of cloud computing

In the chapter, 'Strategic capability,' we introduced the term cloud computing and explored some of the advantages that this relatively new approach to data storage may bring to organisations.

The following worked example explores some of the main risks associated with cloud-based services, whether they are public or private clouds.

An article by Wall (2016) highlights comments made by Gavan Egan, the managing director of cloud and IT solutions for global telecoms company, Verizon. 'The biggest risk is giving up control of your data to someone else, using different data centres in remote places. What happens in the event of disaster? You are also putting your data next to someone else's.' Wall (2016) comments that, in such situations, 'data can be lost, wiped, corrupted or stolen.'

This problem is exacerbated by the range of cloud service providers (CSP) that an organisation uses. Many companies sign up to the services of more than one CSP, each with its own configurations and security protocols. Problems arise when the interfaces between each CSP leave security gaps, exposing the organisation to hackers and cybercriminals.

According to a report by IBM (2023), 82% of data breaches involved data stored in the cloud. Cloud services represent a key target for cybercriminals because of the vast amounts of data they hold so organisations must ensure that security is updated frequently. As soon as workloads and assets are moved to the cloud, organisations lose visibility of them so it is essential to work closely with the cloud service provider to protect against cyber threats.

Organisations must also recognise that securing the cloud is not solely the responsibility of the cloud provider with users playing a vital role. Whilst cloud providers offer secure infrastructure, it is the users who manage the controls to protect the data. Leading research and advisory firm, Gartner, expects that 'through 2025, 99% of all cloud security failures will be due to some level of human error' (2019).

***Sources**:*

- *Wall, M. (2016) Can we trust cloud providers to keep our data safe? [Online] Available from: https://www.bbc.co.uk/news/business-36151754 [Accessed 3 May 2024]*
- *IBM (2023) Cost of a Data Breach Report 2023 [Online] Available from: https://www.ibm.com/reports/data-breach [Accessed 3 May 2024]*
- *Gartner (2019) Is the Cloud Secure? [Online] Available from: https://www.gartner.com/smarterwithgartner/is-the-cloud-secure [Accessed 3 May 2024]*

5.1 Cybersecurity and boundary-less organisations

In the chapter, 'Strategy and structure,' we explored the ways in which traditional organisation structures have evolved, including the rise of boundary-less organisations. Boundary-less organisations are structured to make it easier for them to collaborate with external parties, and often involve the removal of barriers (including IT network barriers) that exist between different internal departments, as well as external groups. The creation of these organisations has primarily occurred as a result of outsourcing, and the establishment of virtual communication networks facilitated by internet technologies.

Although, the removal of IT network barriers between organisations has brought significant benefits in terms of data sharing, it has also created significant challenges from a cybersecurity perspective. Organisational data, which traditionally would have been held privately in databases, is now being accessed by employees dispersed all around the world, and by various external groups including customers and suppliers. The interconnectedness of data which can be accessed by various groups has resulted in the creation of a greater number of access points to organisational networks. It is the weaknesses in these access points which criminal groups are keen to exploit.

The removal of such barriers to this data raises some interesting considerations: which party should be deemed ultimately responsible for protecting the data accessed which is in fact owned by another organisation? In the case of data which is accessible from a number of locations around the world, under which countries legal jurisdiction should a data breach be dealt?



Context example: The boundary-less network

The following worked example focusses on the challenges presented by the rise of the boundary- less network.

An article by Watkins (2017) published on the *Computer Weekly* website notes that the interconnected nature of data and IT systems has led to the creation of the 'boundary-less corporate network.' Watkins (2017) suggests that the notion of 'my network and your network' has effectively been made obsolete by the creation of the boundary-less corporate network. Watkins (2017) notes that 'the logical boundary has gone, and along with it the physical – users might be safe and snug within corporate buildings, but are more likely to work at home, on the move, [or work] from someone else's premises entirely. […] Even the information boundary is going – integration and sharing with customers – mash-ups are blending our information with internet sources, with staff pumping corporate information onto the internet.'

These developments have left organisations with increasingly less control over their own data and IT systems. In responding to the challenges that this brings, Watkins (2017) highlights the need for the development of effective policing of the internet, 'an effective international legal framework would be nice, but is probably well beyond the capability of our politicians. Our laws, particularly on personal data, need to catch up with the technology, without killing the creativity and flare the internet has unleashed.'

The article highlights that the loss of traditional boundaries has created uncertainty among organisations as to which data should be made available to share, as well as determining which external data sources they themselves should trust.

(**Source**: Watkins, B. (2017) *The boundary-less network*. [Online] Available from: www.computerweekly.com [Accessed 18 July 2023]).

5.2 The cyber gap

Many organisations suffer from a so-called 'cyber gap,' between the organisations' capabilities to protect their IT infrastructures and the ability of attackers to do damage. To combat these risks effectively, organisations need to make cybersecurity central to everything they do.

5.3 *Building cybersecurity into strategy*

In the chapter, 'Strategy and structure,' we introduced some of the key findings from the report 'Effective Board Governance of Cyber Security' in relation to the role of the board in combating cyber risks. Whilst the board retains overall accountability for cybersecurity, it should be seen as a wider strategic issue which managers must take into account when making strategic decisions. To ensure that cybersecurity is embedded into the decision-making process, organisations can put in place the following measures:

- *Embed a greater awareness of cyber risk throughout the organisation's strategy and operations.*
- *Build a security focused culture based on continuous improvement in terms of cybersecurity.*
- *Articulate the organisation's approach to managing cyber risks through its reporting mechanisms.*
- *Reduce complexity in the organisation's IT environment by implementing effective controls over the use of legacy and modern IT systems.*
- *Develop talent management strategies to attract and retain individuals skilled in managing cybersecurity.*
- *Ensure that suppliers to the organisation comply with relevant cybersecurity laws and standards, as this should help to protect IT assets in the supply chain.*
- *Ensure that checks on the security procedures associated with IT infrastructures are an on-going activity.*

Cybersecurity can significantly impact strategic decisions and should be factored into the risk assessment process. Areas where cybersecurity plays an increasingly important role include:

***Technology investment and innovation:** Organisations must assess the security implications of new technologies, such as cloud-based services, internet of things and AI tools, ensuring security measures are in place to protect data and minimise vulnerability to cyber threats.*

***Supply chain management:** Before working with new suppliers or customers, it is vital to conduct due diligence and ensure alignment with security standards so that hackers cannot use weaknesses in partners' systems to bypass the organisation's security controls. Similarly, tight access controls and security measures should be put in place to allow remote workers to safely access corporate systems. Following the 2020 pandemic, many companies moved to home-based working to allow for business continuity, but suitable security measures were not always put in place to protect systems from hackers.*

***Regulatory compliance and legal frameworks:** All strategic decisions should take into consideration any legislation or compliance requirements, such as data protection laws. Failure to do so exposes the organisation to fines and penalties as well as potential damage to reputation and customer goodwill.*

***Human resources and training:** Organisations need to hire, train and retain skilled cybersecurity professionals with the relevant knowledge to mitigate cybersecurity risks. In addition, cybersecurity training should be integrated into employee training and development programmes in order to embed a security conscious culture across the organisation.*

The increasing use of AI can help in the battle against cybercrime, bringing with it new tools for identifying, analysing and preventing online threats. As discussed in the chapter, 'Strategic capability,' AI has the ability to evaluate enormous data sets allowing it to spot any abnormalities or suspicious patterns, thus preventing potential or early-stage cyber attacks. Some AI tools are able to forecast potential future assaults enabling an organisation to put in place preventative controls. The use of AI also takes the pressure off IT managers and teams since cyber incident planning and response can be automated, with AI automatically assessing event data and taking necessary action before an incident arises.



Context example: Darktrace

Darktrace was set up in 2013 by mathematicians and machine learning specialists from the University of Cambridge, together with input from world-leading intelligence experts from MI5 and GCHQ (the British intelligence services) to bring transformative technology to the challenge of cybersecurity.

Darktrace's enterprise immune system makes use of AI and offers a new approach to cyber defence. A video on the company's website highlights that organisations are gradually moving away from the old model of attempting to define cyber threats and trying to defend the perimeter of the organisation's IT systems, to addressing threats from within.

The enterprise immune system is based on a similar principle to that of the human immune system. Cyber threats are similar in many ways to human DNA, which mutate and evolve over time in order to survive. The human immune system handles undesirable DNA evolutions by learning what is normal and identifies anomalies which it can then address.

In essence, the program is able to learn 'what normal looks like' in terms of an organisation's IT system usage and data flows. It works by automatically modelling the behaviour of every device used throughout an organisation.

The system is then able to identify system anomalies in real time by comparing any deviations detected against normal device usage and prioritise those threats deemed to be most serious. Identification in real time allows threats to be dealt with whilst they are in progress. As a result, the organisation's IT experts can then take action to investigate and stop these anomalies before they cause significant disruption.

*(**Source**: Darktrace (2019) Darktrace. [Online] Available from: www.darktrace.com [Accessed 18 July 2023])*

5.4 Small and medium sized enterprises

It is however important to note that the points outlined above in the ICAEW's report do present some challenges for small and medium sized enterprises. Creating talent management programmes and implementing effective controls over IT systems is very often unviable for smaller entities.

Due to their size, and simpler business models, the associated costs involved in pursuing best practice in the field of cybersecurity often mean that smaller organisations lag behind their larger counterparts.

Smaller organisations are unlikely to have dedicated IT departments or in-house experts monitoring their exposure to cyber risk. It is common for such organisations to rely on external third parties such as outsource firms providing IT services to put in place measures to address the threats posed by cyber risk. As a result it is likely that to some degree a 'cyber gap' will remain.

5.5 Combating IT risks and IT security

There are a number of practical measures that organisations can take in combating IT risks.

- **Business continuity planning:** This means that there should be measures to ensure that if major failures or disasters occur, the business will not be completely unable to function.
- **Systems access control**: This includes protection of information, information systems, networked services, detection of unauthorised activities and security when using the systems.
- **Systems development and maintenance**: This includes security measures and steps to protect data in operational and application systems and also ensuring that IT projects and support are conducted securely.
- **Physical and environmental security**: Measures should be taken to prevent unauthorised access, damage and interference to business premises, assets, information and information facilities and prevention of theft.
- **Compliance** with any relevant legal requirements and also with organisational policies in standards. There is no point in having them if they are not enforced.

- **Personnel security**: This covers issues such as recruitment of trustworthy employees, and also reporting of security-related incidents. Training is particularly important, with the aim that users are aware of information security threats and concerns and are equipped to comply with the organisation's security policy.
- **Security organisation**: It should be clear who has responsibility for the various aspects of information security. Additional considerations will apply if facilities and assets are accessed by third parties or responsibility for information processing has been outsourced.
- **Computer and network management**: This includes ensuring continuity of operations and minimising the risk of systems failures, also protecting the integrity of systems and safeguarding information, particularly when exchanged between organisations. Particularly important is protection from viruses.
- **Asset classification and control**: Information is an asset, just like a machine, building or a vehicle, and security will be improved if information assets have an 'owner,' and are classified according to how much protection they need.
- **Security policy**: A written document setting out the organisation's approach to information security should be available to all staff.

Supply chain management (SCM)

Many businesses succeed or fail depending on the success of their relationship with their suppliers and with those who they supply.

Businesses that rely on other businesses to this extent are in what is called a supply chain – each supplying each other right up to the final link in the chain, the consumer.

Technology can help make this relationship work more effectively and efficiently.

About supply chains

A supply chain encompasses all activities and information flows necessary for the transformation of goods from the origin of the raw material to when the product is finally consumed or discarded.

This typically involves distribution of the product from the supplier to the manufacturer to the wholesaler to the retailer and to the final consumer, otherwise known as nodes in the supply chain.

It is helpful to make a distinction between upstream and downstream supply chain management.

For an internet retailer, for example, upstream SCM would involve transactions between the firm and its suppliers (equivalent to buy-side e-commerce) and downstream, customers (equivalent to sell-side e-commerce).

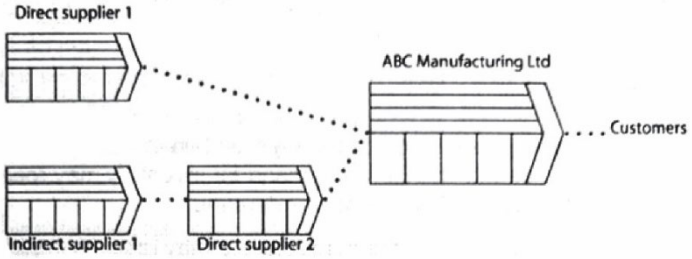
While each firm can be competitive through improvements to its internal practices, ultimately the ability to do business effectively depends on the efficient functioning of the entire supply chain.

Obviously, if e-business capability is present in all members of the supply chain, management of the chain becomes more feasible.

Selling, delivering, ordering, designing and manufacturing can all be linked electronically permitting savings in both cost and time and potentially better innovation.

Illustration 3 – Supply chain management (SCM)

In the supply chain shown, ABC Manufacturing Ltd must be responsive to its customers. Direct supplier 1 and Direct supplier 2 must be responsive to ABC Manufacturing Ltd, and Indirect supplier 1 must be responsive to Direct supplier 2.



Managing the chain

Active management of supply chain activities aims to maximise customer value and achieve a sustainable competitive advantage.

It represents a conscious effort by the supply chain firms to develop and run supply chains in the most effective and efficient ways possible.

Managing the chain primarily therefore concentrates on the management and progression of the following three items:

Inventory control

This will include a consideration of:

- *number, location and size of warehouses and distribution centres*
- *production scheduling (including life cycle management to ensure that new products can be successfully integrated into the chain)*
- *a transportation strategy (in terms of routes, timing etc.)*

Information management

The key elements of information required for successful supply chain management include:

- *potential levels of end-user and customer demand*
- *daily production and distribution plans*
- *resource availability and utilisation.*

Fund management

For the system to work it needs to be sufficiently cash liquid at all levels to ensure that bottlenecks are avoided and supply can be sustained. There also needs to be a strong relationship of trust between each party in the chain.

Importance of Information Technology (IT)

IT plays an obvious role in providing, storing, managing and interrogating the information management part of supply chain management.

However, IT can also provide aid for all the areas through systems such as e-procurement and customer relationship management (covered later in the chapter), and there will also be links to other areas such as project management and organisational structure.

Illustration 4 – Supply chain management (SCM)
A wholesaler's inability to adequately maintain inventory control or respond to sudden changes in demand for inventory may mean that a retailer cannot meet final consumer demand.
Conversely, poor sales data from retailers may result in inadequate forecasting of manufacturing requirements.

Push and pull supply chain models

One key element in supply chain management is choosing between having a 'push' or a 'pull' model.

Push vs pull

Push model

- *Products are built, distributed, and ready for the customer demand.*
- *Product design is led by the manufacturer.*
- *There is little product personalisation to customers.*
- *Low set-up costs and economies of scale are possible.*
- *Inventories are built up waiting for customers to demand them (a push system is sometimes referred to as a Make to Stock (MTS system)).*

Pull model

- *Planning for a product starts when the customer places the order and creates firm demand.*
- *Product design is often customer led (a pull system is sometimes referred to as a Make to Order (MTO system)).*
- *Personalisation of the product by the customer is possible.*
- *Inventory levels are minimised (systems such as JIT and TQM can be used).*
- *Lead times can be much higher.*
- *Set-up costs are higher, and economies of scale are not always possible.*

The pull business model is less product-centric and more directly focused on the individual consumer – a more marketing-oriented approach.

Driven by e-commerce's capabilities to empower clients, many companies are moving from the traditional 'push' business model, where manufacturers, suppliers, distributors and marketers have most of the power, to a customer-driven 'pull' model.

E-commerce creates a much more efficient supply chain that benefits both customers and manufacturers. Companies can better serve customer needs, carry fewer inventories, and send products to market more quickly.

Today, many companies have introduced elements of the pull model so that their systems use a combination of push and pull. Nike By You is a good example of this.

Revisiting the value chain

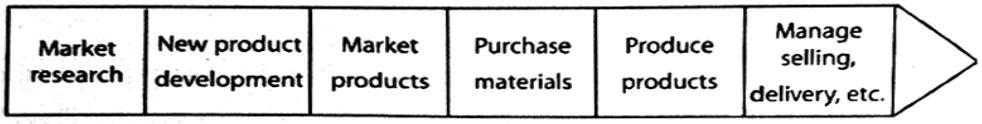
Porter's value chain revisited

The theory behind the value chain is that, somewhere within the organisation, it carries out activities that customers value, are willing to pay for, and that result in profits being made. The activities could be skill and knowhow related, organisational or manufacturing.

The traditional value chain has been criticised because it:

- *is designed primarily to describe manufacturing businesses*
- *does not emphasise the importance of the customer enough. It is product led rather than market led.*

Deise proposed an alternative value chain that places more emphasis on the customer:



In terms of this model, e-business enables companies to:

- *carry out continuous market research as products are ordered and sold*
- *use the results of continuous monitoring to order goods, design new products and design special offers*
- *gain greater flexibility and speed of response, provided that other members of the up-stream supply chain (suppliers) also embrace e-business.*

Upstream SCM

The key activity of upstrea SCM is e-procurement.

What is e-procurement?

The term 'procurement' covers all the activities needed to obtain items from a supplier: the whole purchases cycle.

Electronic Procurement (also known as e-procurement) is the business-to-business purchase and sale of supplies and services over the internet.

It is a way of using the internet to make it easier, faster, and less expensive for businesses to purchase the goods and services they require.

While e-procurement is a general term that covers a wide assortment of techniques, its overall goal is to streamline the purchasing process to reduce costs, increase speed and allow managers to focus on other strategic matters.

E-procurement – more details

E-procurement is the term used to describe the electronic methods used in every stage of the procurement process, from identification of requirement through to payment.

It can be broken down into the stages of e-sourcing, e-purchasing and e-payment:

E-sourcing *covers electronic methods for finding new suppliers and establishing contracts.*

Not only can e-sourcing save administrative time and money, it can enable companies to discover new suppliers and to source more easily from other countries.

E-purchasing *covers product selection and ordering.*

Buying and selling online streamlines procurement and reduces overheads through spending less on administration time and cutting down on bureaucracy.

Features of an e-purchasing system include:

- *electronic catalogues for core/standard items*
- *recurring requisitions/shopping lists for regularly purchased items*
  The standard shopping lists form the basis of regular orders and the lists can have items added or deleted for each specific order
- *electronic purchase orders despatched automatically through an extranet to suppliers*
- *detailed management information reporting capabilities.*

Improvements in customer service can result from being able to place and track orders at any time of day.

User expectations have increased dramatically in recent years as a result of their personal experiences of shopping on the internet. Well-designed websites and mobile apps are essential to offer good functionality so as to maintain user satisfaction.

E-payment *includes tools such as electronic invoicing and electronic funds transfers.*

Again, e-payment can make the payment processes more efficient for both the purchaser and supplier, reducing costs and errors that can occur because of information being transferred manually from and into their respective accounting systems. These efficiency savings can result in cost reductions to be shared by both parties.

Restructuring the supply chain

In an earlier chapter we looked at strategic choices of outsourcing, vertical integration and strategic alliances, where the key issues of cost, quality and control were highlighted.

These are just as relevant for online businesses as they are for conventional 'bricks and mortar' organisations.

All organisations must decide between:

- *vertical integration – manufacturing in-house*
- *virtual integration – the majority of supply chain activities are undertaken by third parties*
- *virtual disintegration (disaggregation) – in between these two extremes.*

However, internet technology allows more efficient and cheaper communications within the chosen structure and may make virtual integration preferable to vertical integration.

5 Downstream SCM

Downstream supply chain management is about managing relationships with both customers and consumers, as well as any other intermediaries along the way.

Examples of downstream supply chain management actions are:

- *providing displays for retailers*
- *creating a website for end users*
- *determining which retailers and distributors to use*
- *use of different logistical methods/providers*
- *changes to finished goods inventory policies*
- *setting recommended retail prices*
- *forward integration*

Advantages	***Disadvantages***
• *can tie in customers/increase switching costs*	• *can be expensive to implement*
• *can improve customer loyalty and retention*	• *as a differentiation strategy it is easily copied*
• *can increase market visibility*	• *it relies on suppliers' willingness to adapt to customer needs (often needs corresponding upstream management)*
• *provides better information on customer needs, tastes etc.*	• *the organisation might become reactive rather than proactive to customer needs*
• *product failure rates can be reduced*	• *forward integration can increase business risk and exit barriers*
• *can facilitate pull supply chain management*	• *must ensure that forum/website users are representative of all users*
• *more regular and better communication with customers (e.g. can provide software/product updates etc.)*	• *requires skills and experience for the benefits to be fully realised*
• *gives users a voice*	• *there is a risk of loss of focus on core competences and activities*

Dealing with intermediaries

A typical downstream for a manufacturer might involve selling to distributors, who then sell on to retailers, who in turn sell on to end users.

Distributors and retailers are therefore intermediaries between the manufacturer and the consumer of the product. One element of supply chain management is to manage intermediaries.

- *E-commerce can lead to **disintermediation**. In this process intermediate organisations (middlemen) can be taken out of the supply chain.*
- *The process of **reintermediation** is also found, i.e. new intermediaries are introduced to the value chain, or at least to some aspects of it.*
- ***Countermediation** is where established firms create their own new intermediaries to compete with established intermediaries.*

Strategic Implications of Technology Adoption

Benefits

*- **Competitive Edge**: Early adoption of AI, IoT, or blockchain can differentiate a company in crowded markets.*

*- **Efficiency Gains**: Automation reduces manual work, improves accuracy, and speeds up processes.*

*- **Data-Driven Decisions**: Advanced analytics provide deeper insights for strategy and operations.*

*- **Scalability**: Cloud platforms and digital tools allow rapid expansion without heavy infrastructure costs.*

*- **Customer Experience**: Personalization and faster service improve satisfaction and loyalty.*

Risks

*- **Cybersecurity Vulnerabilities**: Emerging tech often expands attack surfaces, requiring stronger defenses.*

*- **High Implementation Costs**: Upfront investment in infrastructure, training, and integration can strain budgets.*

*- **Resistance to Change**: Employees may struggle with adoption, lowering productivity during transition.*

*- **Regulatory & Compliance Issues**: Technologies like blockchain or AI may face evolving legal frameworks.*

*- **Obsolescence**: Rapid innovation means today's cutting-edge tools may become outdated quickly.*

Comparison of Key Technologies

***| Technology | Strategic Benefits | Risks & Challenges** |*

*| **Artificial Intelligence (AI)** | Automates tasks, enhances decision-making, enables personalization | Bias in algorithms, ethical concerns, regulatory uncertainty |*

*| **Internet of Things (IoT)** | Real-time monitoring, predictive maintenance, supply chain visibility | Data privacy risks, device security vulnerabilities |*

*| **Blockchain** | Transparency, trust in transactions, reduced fraud | Energy consumption, scalability issues, unclear regulations |*

*| **Cloud Computing** | Flexibility, cost savings, scalability | Dependence on providers, data sovereignty concerns |*

*| **Robotics & Automation** | Increased productivity, reduced human error | Job displacement, high upfront costs |*

Strategic Considerations

*- **Alignment with Business Goals**: Technology should support long-term strategy, not just short-term efficiency.*

*- **Risk Management**: Companies must balance innovation with cybersecurity, compliance, and financial risk.*

*- **Change Management**: Training and cultural adaptation are critical to successful adoption.*

*- **Partnerships & Ecosystems**: Collaborating with tech providers or startups can reduce risk and accelerate integration.*

Artificial Intelligence (AI)

Definition

Artificial Intelligence is an area of computer science that emphasises the creation of intelligent machines that work and react like human beings.

*A common definition from Kaplan and Haenlein describes AI as a "**system's ability to correctly interpret external data, to learn from such data, and to use those learnings to achieve specific goals and tasks through flexible adaptation**".*

This is often considered in the context of human-type robotics (for example health care practices, such as surgery, rehabilitation, therapy, patient companionship) but reaches much further than this, and is set to transform the way we live and work.Some of the more advanced activities and skills artificial intelligence can now master, and therefore present huge opportunity for developers and companies alike, include:

- *Voice recognition*
- *Planning*
- *Learning*
- *Problem solving*

Illustration 8 – Artificial Intelligence
Companies such as Apple and Amazon have developed and marketed voice recognition systems, either to be built into an existing product (such as Apple with its Siri system) or developed new products whose main function is voice recognition (such as Amazon Alexa). A further simple example is that of Netflix, and its process of recommending new programs that subscribers are likely to enjoy.

*There are many, more complex examples of Artificial Intelligence, but a common factor to both the simple and the more involved is **machine learning**.*

Machine learning

Machine learning is a subset of AI where computer code is built to mimic how the human brain works. This enables systems to learn and improve from experience without being explicitly programmed.

It involves algorithms that analyse and interpret data to make decisions or predictions based on past experiences through data, events and connections between events.

The computer then applies this learning to a given situation to give a fact driven plausible outcome.

If the conclusion the computer reaches turns out to be incorrect this will act to add more experience and enhance its understanding further, so in future the same mistake will not be repeated.

Machine learning is used in various applications, such as recommendation systems, fraud detection, and autonomous vehicles, by continuously adapting and optimizing performance based on new data inputs.

Most recent advances in AI have been achieved by applying machine learning to very large data sets.

There are three major types of machine learning:

- *Supervised learning*
- *Unsupervised learning*
- *Reinforcement learning*

Supervised learning

Supervised learning involves training an algorithm with labelled data and human feedback to understand the relationship between inputs and outputs. For example, predicting future house prices might use inputs like "time of year" and "interest rates."

This method is ideal when the user can classify input data and predict desired behaviour but needs the algorithm to apply this to new data.

There are three steps to how supervised learning works:

1. *A human labels every element of the input data (e.g. "time of year", "interest rates" etc.) and defines the output variable (e.g. "house prices").*
2. *The algorithm is trained on the data to find the connection between the input variables and the output.*
3. *Using the trained algorithm to predict outputs from new data.*

Amazon uses supervised learning in its recommendation system, predicting products a customer might like based on past purchases and ratings.

Unsupervised learning

Unsupervised learning involves an algorithm analysing input data without predefined output variables.

For example, it might explore "customer demographic data" to identify patterns.

This method is useful when data lacks known classifications, and the user wants the algorithm to discover patterns and classify the data.

The three steps for using unsupervised learning are:

1. *Receiving unlabelled data, such as customer journeys on a website.*
2. *The algorithm then infers some sort of structure on the data.*
3. *Identifying groups with similar behaviour, like clusters of customers of similar age with similar buying patterns.*

Spotify uses unsupervised learning to create personalised playlists by analysing listening habits and grouping users with similar music preferences.

Reinforcement learning

This is when an algorithm learns to perform a task by trying to maximise the rewards it receives for the actions it takes.

For example, in managing an investment fund, the rewards would be gains in value of the fund based on investment decisions.

Reinforcement learning is useful when there is limited training data, the ideal end state is not clearly defined, or learning about the environment requires interaction.

The three steps for reinforcement learning are:

1. *The algorithm takes an action on the environment; for example, it makes a trade in a financial portfolio.*

2. *It receives a reward if the action taken brings the machine a step closer to its objective. For example, it makes a return on the investment chosen.*
3. *The algorithm optimises for the best series of actions by correcting itself over time. For example, learning from losses or missed opportunities to improve future decisions.*

Rewards for reinforcement learning will depend on the application to which the AI is put.

For example, a courier company that decides to adopt AI in this way might see reductions in fuel consumption, or an increase in the number of deliveries made to customers on time, as a gain.

Artificial Intelligence and finance

Increasingly, we are seeing systems that are producing outputs that far exceed the accuracy and consistency of those produced by humans.

In the short to medium term, AI brings many opportunities for finance professionals to improve their efficiency, provide more insight and deliver more value to businesses. In the longer term, AI brings opportunities for much more radical change, as systems increasingly carry out decision-making tasks currently performed by humans.

AI, no doubt, will contribute to substantial improvements across all areas of accounting and business, equipping those in finance with powerful new capabilities, as well as leading to the automation of many tasks and decisions.

Examples include:

- *using machine learning to code accounting entries and improve on the accuracy of rules-based approaches, enabling greater automation of processes*
- *improving fraud detection through more sophisticated, machine learning models of 'normal' activities and better prediction of fraudulent activities*
- *using machine learning-based predictive models to forecast revenues*
- *improving access to, and analysis of, unstructured data, such as contracts and emails*

Despite the opportunities that AI brings, it must be remembered that it does not replicate human intelligence.

The strengths and limits of this different form of intelligence must be recognised, and we need to build an understanding of the best ways for humans and computers to work together.

Internet of things

The internet of things (IoT) describes the network of smart devices with inbuilt software and connectivity to the internet allowing them to constantly monitor and exchange data.

What devices are connected to the internet of things?
Essentially anything with an on/off switch can become a 'smart' device and be connected over the internet. This allows them to talk to us, applications and each other, which is at the core of their functionality.

Common devices currently connected as part of the internet of things include:

- ***Smart meters** (contain in-home display screens that show how much energy is used in real-time) and **home control thermostat devices** (allow control of heating, electricity and hot water).*
- ***Doorbells and security:** these talk to your smart device. A live connection is established if the doorbell is pressed or the motion sensors activated, allowing immediate interaction.*
- ***Wearable tech** such as smart watches and fitness trackers capture and record an array of data to monitor and record your fitness.*
- ***Home appliances** such as smart lights, fridges, washing machines, ovens etc… the connectivity built in allows remote access and control of these devices, for instance turning your lights on if you're out at night can be done using a smart phone.*

- ***Cars:*** *the computer systems used to control cars are increasingly sophisticated. They track and monitor thousands of parameters on every journey. This capability is central in the continued pursuit of autonomous vehicles.*
- ***Transport and infrastructure:*** *smart motorways are a common feature in many countries with traffic sensors monitoring the flow and build-up of traffic and responding to provide extra lanes or activate temporary speed limits.*
- ***Manufacturing equipment and plant:*** *monitoring of business assets facilitates efficient utilisation, it allows continual live feedback to track performance and flag maintenance requirements earlier.*

*The growth in the **internet of things** often termed 'smart technology' is fuelled by improvements in broadband connectivity and the development of 4G and 5G communication networks.*

*Coupled with the fact that people and businesses are increasingly comfortable with the idea and operation of this **smart technology**, it is anticipated that the **internet of things** will continue to grow, becoming increasingly central to how we live and work as new and innovative applications for the technology emerge all the time.*

SMART technology

*The word "SMART" refers to **"self-monitoring, analysis, and reporting technology."***

It is a technology that uses artificial intelligence, machine learning, and Big Data analysis (further discussed above) to provide cognitive awareness to objects that were in the past considered inanimate.

Internet of Things (IoT) devices

A network of devices that make use of sensors, chips, software, online connectivity, analytics, and applications to bring static physical objects to life. These devices create substantial value, and they are futuristic, scalable, and automated.

Some prominent examples include smart watches, smart cities, smart homes, and smart factories.

Smart Connected Devices

Controlled via a remote and connected via the internet or Bluetooth, smart connected devices can offer a customised experience but have to be handled given that they don't adapt to the extent IoT devices do.

Examples: smart security cameras, smart bulbs, and smartphones.

Smart Devices

With limited automation, no need for internet connectivity, and programmable nature, smart devices, such as smart coffeemakers, provide certain personalised services at a particular time.

Examples of the benefits and risks of SMART technology

Benefits	***Risks***
• *Convenience – provides the ability to do many tasks simultaneously with minimal effort.*	• *Privacy intrusion – for example, having house appliances and devices so connected gives service providers a great chance to collect vital information about you.*
• *Sustainability – smart technology can play a pivotal role in helping conserve energy, regulating and automating the use of energy, for*	• *Cyber-attacks – aimed at extorting money from unsuspecting users, interrupting the normal operation of connected appliances, accessing, altering or destroying sensitive*

example by switching off or adjusting lights, heating, and cooling appliances when they are not in use, or when the required conditions have been achieved.	*information and personal data, and interfering with normal business operations and processes.*
• *Security – offers more reliable security than traditional, manually operated systems. Smart digital smoke, gas, water, and sewage leakage detection enables real-time preventive action.*	• *Power outages and technology failure – systems run through the internet are vulnerable during power outages.*
• *Efficiency – uses data to understand and track improvements, delivering better information for future results.*	• *Identity theft – cybercriminals can hack devices and steal sensitive information such as bank accounts, social security numbers, and passwords.*

Potential Risks of AI

Artificial Intelligence has many risks associated with its use and application.

Elon Musk and Professor Stephen Hawking have warned about AI's exponential growth and potential dangers.

- ***Automation-spurred job loss*** *– job automation is viewed as an immediate concern. AI will replace certain types of jobs, especially those involving predictable and repetitive tasks.A 2019 study stated that* ***36 million people work in jobs with "high exposure" to automation****, predicting that roles in retail sales, market analysis, hospitality, and warehouse labor will soon be done using AI.*
- ***Privacy violations*** *– malicious use of AI could threaten digital security (e.g. through criminals training machines to hack or socially engineer victims at human or superhuman levels of performance), physical security (e.g. weaponising consumer drones), and political security (e.g. through privacy-eliminating surveillance, profiling, and repression, or through automated and targeted disinformation campaigns).*
- ***'Deepfakes'*** *– created by manipulating voices and likenesses. Using AI that's involved in natural language processing, an audio clip of any given politician could be manipulated to make it seem as if that person voiced racist or sexist views when in fact, they said nothing of the sort. If the clip's quality is high enough to fool the general public and avoid detection, it could completely derail a political campaign and career.*
- ***Socioeconomic inequality*** *– Widening socioeconomic inequality sparked by AI-driven job loss is another cause for concern. Along with education, work has long been a driver of social mobility. However, when it's a certain kind of work, the predictable, repetitive kind that's prone to AI takeover, research has shown that those who find themselves without work are much less able to obtain alternative work or seek retraining compared to those in higher-level positions who have more money and resources.*
- ***Market volatility*** *– algorithmic trading occurs when a computer, unencumbered by the instincts or emotions that could cloud a human's judgement, executes trades based off of pre-programmed instructions. These computers can make extremely high-volume, high-frequency and high-value trades that can lead to big losses and extreme market volatility.*

Mitigating the risks of AI

One way to prevent or at least temper the most malicious AI from wreaking havoc is some sort of regulation.

Elon Musk suggested: "It needs to be a public body that has insight and then oversight to confirm that everyone is developing AI safely. This is extremely important."

His comments were regarding the regulation of AI implementation, he said, but not of the research itself which needs to be less regulated in his opinion.

Any country that lags in AI development is at a distinct disadvantage militarily, socially and economically. The solution is therefore selective application.

Information Technology risks

Computer systems have unique risk and control issues that need to be addressed by business.

As with any risk factor the company needs to assess the risks and decide on the appropriate level of control to reduce the risks to an acceptable level.

Risks to a computer system

A risk to a computer system could be anything that prevents the managers getting the information they need from the system at the time that they need it.

Risks to information processing facilities may arise from:

- *Dissatisfied employees or hackers might deliberately modify or destroy information in the system.*
- *Viruses or malicious software could be introduced.*
- *Accidental mistakes could be made on input to the system.*
- *Inadequate security of the hardware or data.*
- *Faults in the hardware system.*

Such risks result in the loss of information (or the loss of its integrity or confidentiality), business disruption and a loss of time and money.

Further detail on risks

Information security

Risks to information security can be categorised as follows:

Risks	***Description***
Risk of hardware theft	*The theft of computer hardware is common.*
Physical damage to hardware and computer equipment (e.g. servers)	*Physical damage can be caused by:* • *malicious damage or poor operating conditions causing damage to equipment and files* • *natural disasters, such as fire and flooding.*
Damage to data	*Data can be damaged by hackers (or unhappy employees) attacking the system, viruses, program faults in the software and faults in the hardware or data storage media.* *Software, particularly purpose-written software, can become corrupted. Programs might be altered by a hacker or computer fraudster. Alternatively, a new version of a program might be written and introduced but contain a serious error that results in the corruption or loss of data on file.*
Operational mistakes	*Unintentional mistakes can cause damage to data or loss of data; for example, using the wrong version of a computer program, or the*

Risks	Description
	wrong version of a data file, or deleting data that is still of value.
Fraud and industrial espionage	*This can lead to the loss of confidentiality of sensitive information, or the criminal creation of false data and false transactions, or the manipulation of data for personal gain.*

Data protection legislation

Some countries give individuals the right to seek compensation against an organisation that holds personal data about them, if they suffer loss through the improper use of that data. In the UK, for example, rights are given to 'data subjects' by the Data Protection Act.

There could be a risk that an organisation will improperly use or communicate personal data about individuals, in breach of the legislation.

Hacking

Hacking is the gaining of unauthorised access to a computer system. It might be a deliberate attempt to gain access to an organisation's systems and files, to obtain information or to alter data (perhaps fraudulently).

Once hackers have gained access to the system, there are several damaging options available to them. For example, they may:

- *gain access to the file that holds all the user ID codes, passwords and authorisations*
- *discover the method used for generating/authorising passwords*
- *interfere with the access control system, to provide the hacker with open access to the system*
- *obtain information which is of potential use to a competitor organisation*
- *obtain, enter, alter or delete data for a fraudulent purpose*
- *cause data corruption by the introduction of unauthorised computer programs and processing on to the system (computer viruses).*

Viruses

A virus is a piece of software that seeks to infest a computer system, hiding and automatically spreading to other systems if given the opportunity. Most computer viruses have three functions – avoiding detection, reproducing themselves and causing damage. Viruses might be introduced into a computer system directly, or by disk or email attachment.

Viruses include:

- ***trojans*** *– whilst carrying on one program, secretly carry on another*
- ***worms*** *– these replicate themselves within the systems*
- ***trap doors*** *– undocumented entry points to systems allowing normal controls to be by-passed*
- ***logic bombs*** *– triggered on the occurrence of a certain event*

Controls in an information systems environment

To combat the types of risks discussed above companies will put in place control procedures.

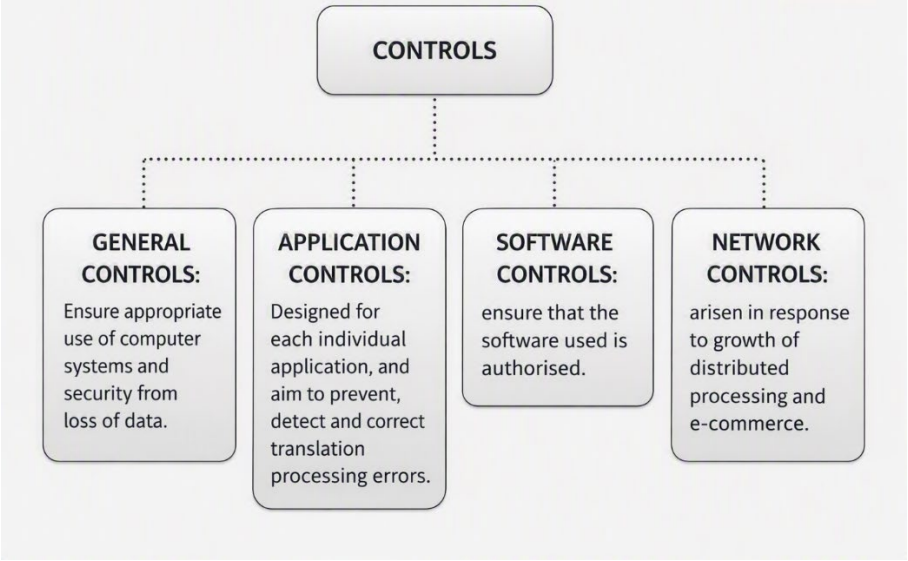
The rationale for strong information system controls

From a strategic perspective, effective information system controls are vital for a business.

The realisation of IT risks can have many consequences for a business such as:

- *Incorrect decisions based on incorrect information. If, say, standard cost systems are recording standard costs incorrectly then strategic decisions on areas such as pricing may be incorrect.*
- *Poor performance measurement and appraisal. For example, if sales totals have been input incorrectly then this will impact on the appraisal of sales targets for staff and may, in turn, impact on bonus and rewards payments. Knock-on effects on motivation levels could have serious consequences for the business.*
- *Unhappy customers and suppliers. If customers are receiving incorrect information, such as totals on sales invoices or payments then they can lose trust in the business. This may mean that they take their business elsewhere.*
- *Reputational damage may arise from deliberate hacks and viruses that corrupt, steal or destroy data in the system. Potential customers may not believe that their data is safe and may not want the business to hold their data. A loss of a sense of security can drive customers elsewhere – especially if the business is transacting online.*
- *Correcting problems and creating adequate new controls can lead to significant business disruption. This can result in a loss of business as well as significant costs in solving the problems.*
- *Hackers or malicious employees may gain access to crucial business information. This may be very valuable to competitors or may prematurely disclose business plans. It may even be that the information is of a negative nature such as problems with proposed new developments that could adversely impact on the business' share price and ability to raise finance, for example.*
- *Protection of data is highly regulated. Failure to do so may result in fines being imposed upon and restrictions being placed on the business.*

Examples of information system controls

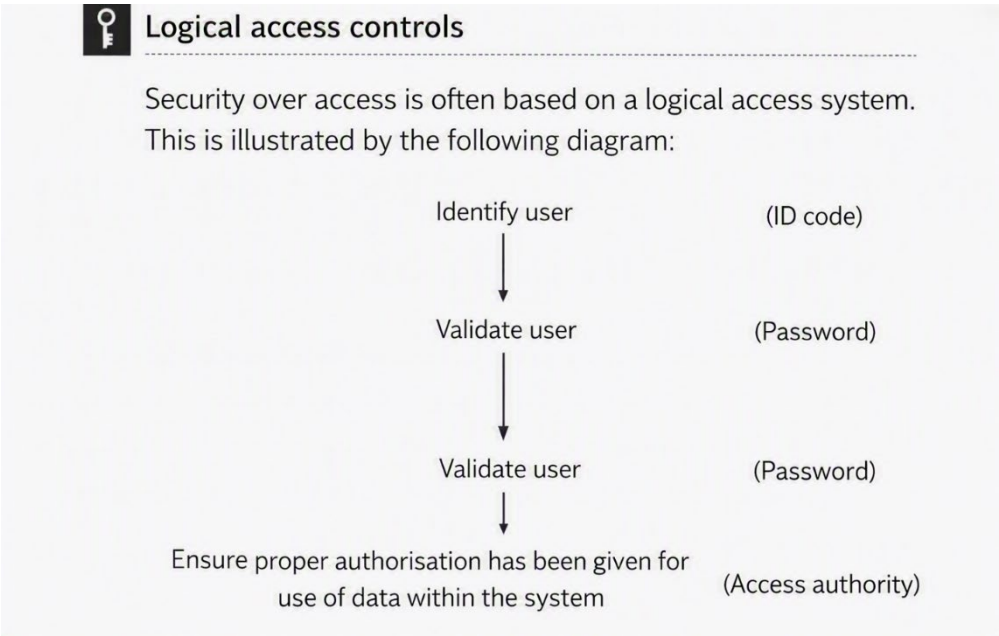


General controls

Personnel controls

Recruitment, training and supervision needs to be in place to ensure the competency of those responsible for IT and data entry.

Logical access controls



Passwords and user names are a common way of identifying who is authorised to access the system, and granting access to the system, or to specific programs or files, only if an authorised password is entered.

There may be several levels of password, with particularly sensitive applications protected by multiple passwords and two factor authentication (2FA) via, say, an app or text message to a different device.

Problems with passwords

Password systems can only be effective if users use them conscientiously. There are several inherent problems with a password system:

- *Authorised users may divulge their password to a colleague.*
- *Many passwords may have associations with the user so that a hacker can discover them by experimentation.*
- *Passwords are often written down close to the computer (e.g. pinned to the notice board inside the office) and so easily discovered.*

To protect passwords and user numbers against discovery, a number of precautions should be adopted:

- *Users should be required to change their passwords regularly.*
- *Passwords should be memorable but not obviously related to a user's private life.*
- *Users should be encouraged never to write down their passwords.*
- *There should be strict controls over passwords – they should never be 'lent' or written down where they can be easily seen.*
- *There should be automatic sentinel or watchdog programs to identify when a password has been keyed incorrectly.*

Access logging

The system will produce regular reports including a system access report and various pre-determined exception reports. The effectiveness of these reports is determined by:

- *The frequency of report production.*
- *The follow up of detected breaches in security.*

Audit trail

An audit trail consists of a record or series of records that allows the processing of a transaction or an amendment by a computer or clerical system to be identified accurately, as well as verifying the authenticity of the transaction or amendment, including the identity of the individuals who initiated and authorised it.

Audit trails are also used to record customer activity in e-commerce on a company's website. The audit trail records the customer's initial access to the website, and then each subsequent activity (purchasing and payment, confirmation of order and delivery of the product).

The audit trail can be used to deal with any subsequent enquiry or complaint from the customer. In some cases, 'audit trails' can be used to track down hackers into a system. A hacker might sometimes unknowingly leave a trail of where he/she/they came from, for example through records in the activity log of the hacker's internet service provider.

Facility controls

Physical access

There are various basic categories of controlling access to sensitive areas. These include:

- *security guards in buildings*
- *working areas to which access is through a locked door or a door with an ID card entry system or entry system requiring the user to enter a personal identification code (PIN number)*
- *using safes and lockable filing cabinets*
- *closed circuit TV used to monitor what is happening in a particular part of a building – this may be backed up by security video cameras*
- *doors automatically locked in the event of a security alarm.*

Location of IT facilities

It is imperative that the location of the system is considered, and hence all equipment is located so as to protect against:

- *Floods, fires or smoke*
- *Food and drink spillages*
- *Power failure*
- *Environment (e.g. extreme temperatures, dust, rain)*

*Business continuity planning (**disaster recovery planning**) takes place in order to recover information systems from business critical events after they have happened. It involves:*

- *Making a risk assessment*
- *Developing a contingency plan to address those risks.*

More on disaster recovery plans

An unexpected disaster can put an entire computer system out of action. For large organisations, a disaster might involve damage from a terrorist attack. There could also be threats from fire and flood damage. A disaster might simply be a software or hardware breakdown within a system.

Disaster recovery planning involves assessing what disasters might occur that would pose a serious threat to the organisation and trying to ensure that alternative arrangements are available in the event that a disaster occurs.

In the case of a computer system for a clearing bank, this would mean having an entire back-up computer system in place that could be brought into operation if a disaster puts the main system out of action.

Not all organisations have extensive disaster recovery plans. Certainly, however, back-up copies of major data files should be kept, so that if the main files are destroyed, the data can be re-created with the back-up files.

System back-ups

All files containing important information should be backed up on a regular basis. Backing up provides protection against the loss or corruption of data due to:

- *faults in the hardware (e.g. hard disk)*
- *the accidental deletion of a file by a computer operator*
- *damage to a data file by a hacker.*

A back-up is simply a copy of the file. If the original file is lost or becomes corrupt, the back-up can be substituted in its place, and the master file can be re-created.

There will be some loss of data if the input to the system since the most recent back-up copy of the file was made.

Back-up copies might be stored on the same physical computer file as the original file, but this is risky, since damage to the physical file will result in the loss of the back-up as well as the main file.

To counter the risk of damage to a file due to a fire or similar disaster at the premises where the IT system is located, a back-up copy might be taken off-site and held somewhere else.

Application controls

These are controls to ensure that data are correctly input, processed and maintained, and only distributed to authorised personnel.

Application controls are specific to each application, but these can include controls such as spell checks and format checks that ensure that inputted data is accurate and complete.

The accounting system may have additional controls such as credit limits for customers which cannot be altered.

Software controls

Software control prevents making or installing unauthorised copies of software. Illegal software is more likely to fail, comes without warranties or support, can place systems at risk of viruses and the use of illegal software can result in significant financial penalties.

Software can be controlled by:

- *Buying only from reputable dealers.*
- *Ensuring that the original disks and/or licences come with the software.*
- *Retaining all original disks/licences and documentation.*

Network controls

Risks on networks

The increase in popularity of the LAN (local area network) has brought concerns in relation to system security.

An LAN allows for many more breaches of security than does a single computer.

The main areas of concern are:

- *Tapping into cables*
- *Unauthorised log in*
- *Computer viruses*
- *File copying*
- *File server security.*

Controls
Controls must exist to prevent unauthorised access to data transmitted over networks and to secure the integrity of data.

Methods include:

- *Firewalls*
- *Flow*
- *Data encryption*
- *Virus protection.*

More on network controls

Firewalls:
A firewall will consist of a combination of hardware and software located between the company's intranet (private network) and the public network (internet). A set of control procedures will be established to allow public access to some parts of the organisation's computer system (outside the firewall) whilst restricting access to other parts (inside the firewall).

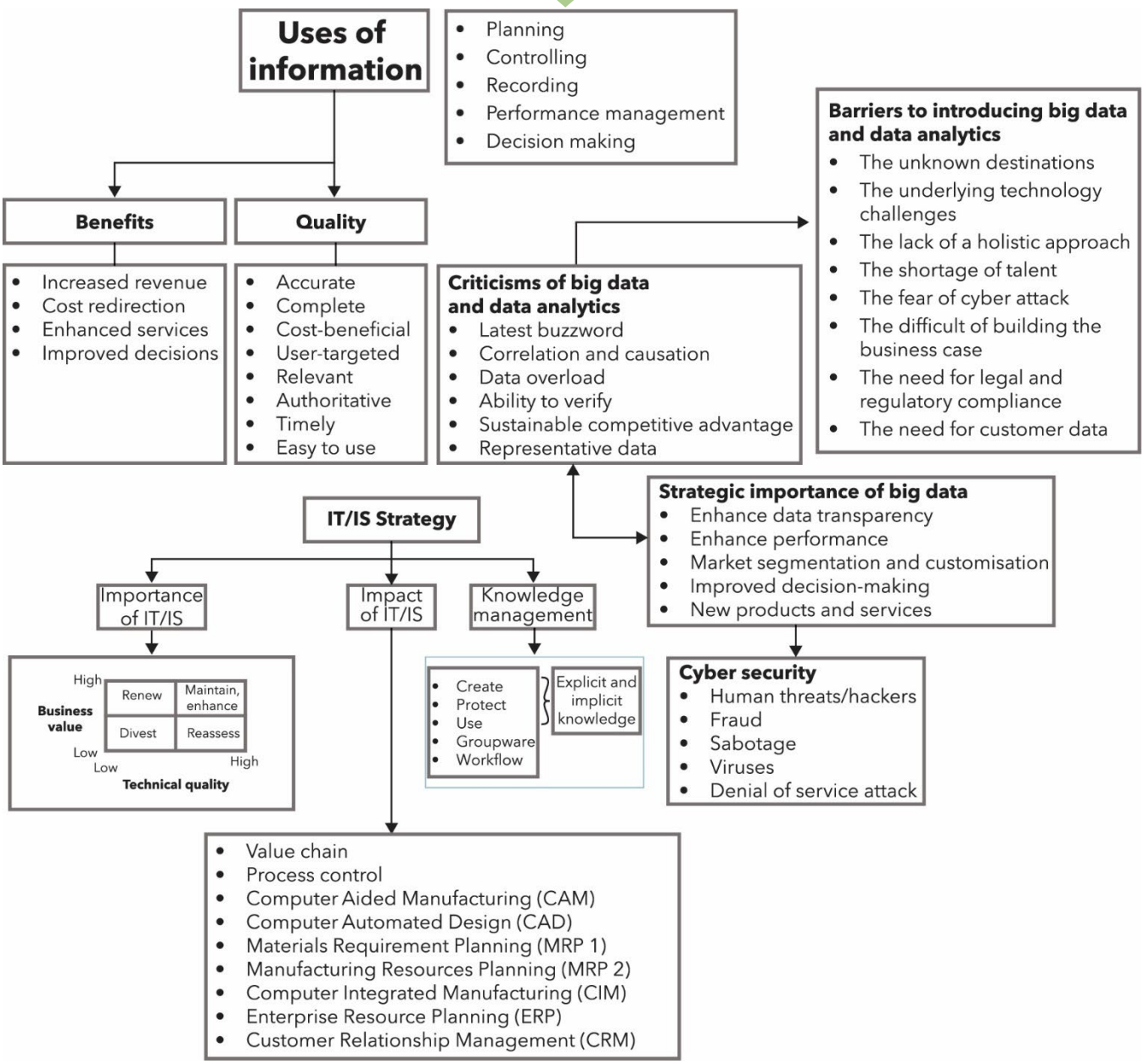
Flow:
This regulates movement of data from one file to another. Channels are specified along which information is allowed to flow, i.e. confidential/non-confidential, and these are linked by authority levels.

Data encryption:
Encryption is a technique of disguising information to preserve its confidentiality. It is disguised during processing/storage. In essence it is a method of scrambling the data in a message or file so that it is unintelligible unless it is unscrambled (or decrypted).

Virus protection:
It is extremely difficult to protect systems against the introduction of computer viruses. Preventative steps may include:

- *control on the use of external software (e.g. checked for viruses before using new software)*
- *using anti-virus software, regularly updated, to detect and deal with viruses*
- *educating employees to be watchful for viruses being imported as attachment files to email messages.*

Summary


Uses of information
Planning
Controlling
Recording
Performance management
Decision making
Benefits
Quality
Increased revenue
Cost redirection
Enhanced services
Improved decisions
Accurate
Complete
Cost-beneficial
User-targeted
Relevant
Authoritative
Timely
Easy to use
Criticisms of big data and data analytics
Latest buzzword
Correlation and causation
Data overload
Ability to verify
Sustainable competitive advantage
Representative data
Barriers to introducing big data and data analytics
The unknown destinations
The underlying technology challenges
The lack of a holistic approach
The shortage of talent
The fear of cyber attack
The difficult of building the business case
The need for legal and regulatory compliance
The need for customer data
IT/IS Strategy
Strategic importance of big data
Enhance data transparency
Enhance performance
Market segmentation and customisation
Improved decision-making
New products and services
Importance of IT/IS
Impact of IT/IS
Knowledge management
High
Business value
Renew
Maintain, enhance
Divest
Reassess
Low
Low
High
Technical quality
Create
Protect
Use
Groupware
Workflow
Explicit and implicit knowledge
Cyber security
Human threats/hackers
Fraud
Sabotage
Viruses
Denial of service attack
Value chain
Process control
Computer Aided Manufacturing (CAM)
Computer Automated Design (CAD)
Materials Requirement Planning (MRP 1)
Manufacturing Resources Planning (MRP 2)
Computer Integrated Manufacturing (CIM)
Enterprise Resource Planning (ERP)
Customer Relationship Management (CRM)

Further question practice

1 Knowledge diagnostic

Once you have worked through this guidance, you are ready to attempt the further question practice included at the end of this chapter.

Confirm your learning	
(1)	What are the features of strategic information? (Topic 1)
(2)	What is the strategic importance of big data? (Topic 2)
(3)	How is knowledge management a source of competitive advantage? (Topic 3)
(4)	What can go wrong during IT system development and implementation? (Topic 4)
(5)	How can IT risks be mitigated? (Topic 5)

2 Question practice

Aim to complete all self-test questions at the end of this chapter. The following self-test questions are particularly helpful to further topic understanding and guide skills application before you proceed to the next chapter.

Question	Learning benefit from attempting this question
6	This short, knowledge-based question assesses whether you can list four practical controls for assuring IT security.
7	Are you able to identify computer system risks and describe relevant controls to manage the risk?

Once you have completed these self-test questions, it is beneficial to attempt the following questions from the Question Bank for this module. These questions have been selected to introduce exam style scenarios that will help you improve your knowledge application and professional skills development before you start the next chapter.

Question	Learning benefit from attempting this question
Forsi Ltd (Sept 2014)	Part 3 of the question requires you to explain how knowledge management is strategically important to the company. This requires application to the scenario. The remainder of the question tests your commercial awareness and common sense in asking how to implement a knowledge management strategy. There is no technical knowledge to guide you, so adopt a logical and practical approach here.
Air Services UK Ltd (Sept 2017)	Part 3 of the question asks you to think practically to explain the cybersecurity risks that the organisation faces and how the risks can be managed. Avoid the temptation to regurgitate knowledge that you have learnt and instead apply your answers to the scenario. You can also use your own experience and wider business reading to trigger ideas and thoughts.

Self-test questions

Answer the following questions.

1 List five uses of information.

2 List five characteristics of strategic information.

3 List five characteristics of tactical information.

4 List five characteristics of operational information.

5 Distinguish between explicit knowledge and tacit knowledge.

6 Identify four controls for assuring IT security.

7 Computer security

Computer security is of vital importance to all organisations. Security is the means by which losses are controlled and therefore involves the identification of risks and the institution of measures to either prevent such risks entirely or to reduce their impact.

Requirements

7.1 Identify the main areas of risk which may arise in relation to a computer system.

7.2 Describe the different forms of control which should be instituted to safeguard against computer security risks.

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

1.1 **Long-term outlookxxxx**

Information flows at a strategic level will be geared towards information expected to impact on the long-term future of the business. There is a need for high quality information to enable sound long-term decisions to be made.

Flexible

Organisations are complex systems.

To maintain control in a constantly changing environment the information flows and systems at strategic level need to be flexible and able to respond quickly to new demands.

For example, if a new competitor enters the marketplace, or environmental legislation is introduced affecting production processes or a trading opportunity arises in an overseas territory, then strategic decisions will be required to formulate the most appropriate response.

Good decisions can only be made when all the implications can be quantified with an acceptable degree of certainty. Management information systems must be flexible enough to provide concise, accurate and timely information relevant to the new environment.

Multi-directional

In a business organisation, information flows both vertically and horizontally. The familiar pyramid hierarchy of an organisation sets out three levels of control and information.

Information flows at the strategic level facilitates decision-making that will affect the whole organisation and provide the framework for long term strategic plans. Internal information will flow from middle management up to senior management and *vice versa*.

Information also flows horizontally between different activities in a business. For instance, overtime hours provided by the payroll section may be used by the production department, delivery lead times from warehouses may be used by the sales team and the level of future orders from the sales department will be used to forecast turnover and cash flows by the accounts department. The quality of these information flows will dictate both the efficiency of the business operations and will impact on the type and quality of information received by senior management.

The complex nature of strategic decisions makes information sharing vital.

An external component

Strategic level information flows will include information from external sources (eg, government, suppliers, media etc). Strategic decisions are generally non-routine and require a high degree of judgement. The quality of information is critical at the strategic management level.

1.2 Information used at a strategic level is often ad hoc – strategic decision making is non-routine and potentially risky.

Strategic information comes from both internal and external sources.

Internal sources

Most management information systems are now computer-based because of processing speed, accuracy and the ability to process large volumes of data. Internal data needs to be

captured from day-to-day operations, processed into relevant information and made available in a suitable form at a strategic level.

Senior management information requirements should play a part in the development of information systems to ensure that the information required for strategic decision making is able to be produced.

Centralised systems are relatively powerful and usually controlled at senior level. However, a flexible response to non-routine problems may be lacking.

Decentralised systems may provide more flexibility, however central control and standard formats are often lacking.

Currently, popular solutions revolve around **networked and distributed** processing systems. These can provide information to different levels of management – often from the same database. They combine the advantages of local control, speed and ease of use with flexibility and the potential for standardised presentation of information.

Executive support systems (ESS) can be particularly useful in this area, providing summarised high-level information with the ability to view the underlying data if required.

The introduction of an ESS will also encourage senior management to consider which type of information is really relevant to the business.

An **intranet** may also be appropriate to encourage the sharing of knowledge and opinions relevant to strategic decisions (eg, bulletin boards).

External sources

External information may be in the form of official reports, tax leaflets, technical updates, press updates and often just word of mouth.

Much of this information is now available via the **internet**. **Intelligent agents** and **newsclipping services** can also be utilised on the internet, to find user-defined information and forward it – usually via email.

Some organisations are able to access external information through an **extranet** – allowing them to enter certain parts of another organisation's intranet.

Answer to Interactive question 2

2.1 **Information management** entails identifying the current and future information needs, identifying information sources, collecting and storing the information and facilitating existing methods of using information and identifying new ways of using it. It should also ensure that the information is communicated to those who need it and not to those who are not entitled to see it.

Mayo defines **knowledge management** (KM) as 'the management of the information, knowledge and experience available to an organisation – its creation, capture, storage, availability and utilisation – so that organisational activities build on what is already known and extend it further.'

More specifically, **knowledge** is interpreted in terms of **potential for action** and distinguished from **information** in terms of its more immediate link with performance. This interpretation is consistent with what the information systems philosopher and professor Charles West Churchman observed three decades ago in his pioneering work, *The Design of Inquiring Systems*: 'knowledge resides in the user and not in the collection of information […] [I]t is how the user reacts to a collection of information that matters.'

Databases, or more correctly **knowledge bases**, need to be designed and developed to store the organisation's knowledge. This will be particularly difficult because of the **tacit** nature of much of the knowledge and also because it is largely inside the heads of individuals. Recording this knowledge will be very different to recording the fields and records of a traditional database. An **expert system** seems one way forward as this has a knowledge base made up of facts, rules and conditions. But much of this knowledge may

have to be represented pictorially as images and 'knowledge maps.' An intranet that lends itself to full multimedia representation and intelligent searching may therefore be the way forward.

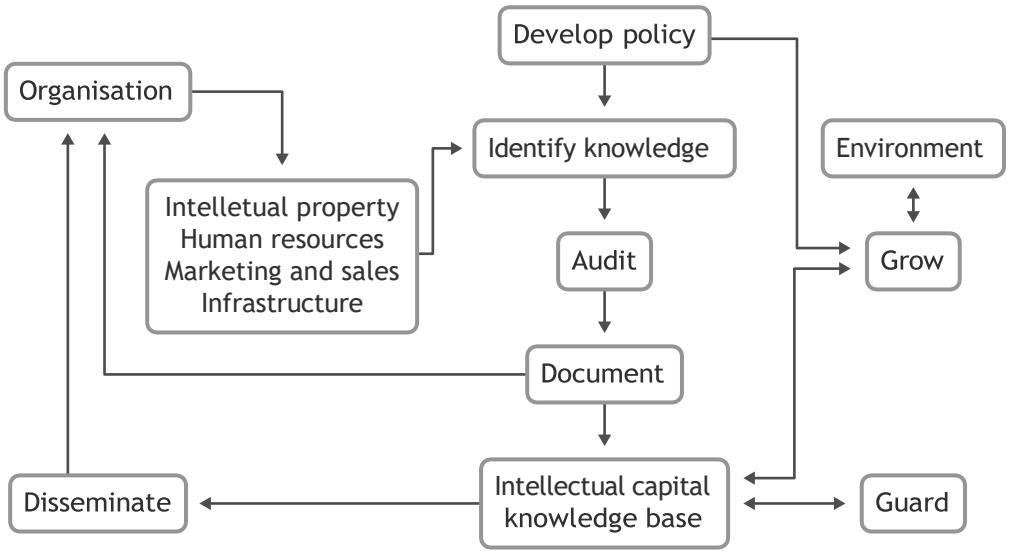
Knowledge-based companies will vary from industry to industry but there are some broad common principles about where knowledge resides and how to capture its value. The intellectual capital can be divided between human capital (the bodies that go home at night), and structural capital. Structural capital includes innovation capital (intellectual property), customer capital (address lists and client records), and organisational capital (systems for processing policies and claims). A number of organisations are creating knowledge management programmes for **protecting and distributing knowledge** resources that they have identified and for discovering new sources of knowledge.

- One such programme is the identification and development of informal networks and communities of practice within organisations. These self-organising groups share common work interests, usually cutting across a company's functions and processes.
- Another means of establishing the occurrence of knowledge is to look at knowledge-related business outcomes eg, product development and service innovation. While the knowledge embedded within these innovations is invisible, the products themselves are tangible.
- Every day companies make substantial investments in improving their employees' knowledge and enabling them to use it more effectively. Analysis of these investments is a third way of making KM activities visible. For example, how much technical and non-technical training are individuals consuming? How much is invested in competitive and environmental scanning, and in other forms of strategic research?

The process by which an organisation develops its store of knowledge is sometimes called **organisational learning**. A **learning organisation** is centred on the people that make up the organisation and the knowledge they hold. The organisation and employees feed off and into the central pool of knowledge. The organisation uses the knowledge pool as a tool to teach itself and its employees.

There are dozens of different approaches to KM, including document management, information management, business intelligence, competence management, information systems management, intellectual asset management, innovation, business process design, and so on. Many KM projects have a significant element of information management. After all, people need information about where knowledge resides, and to share knowledge they need to transform it into more or less transient forms of information.

2.2 A **knowledge management strategy** might take the form shown in the diagram below.



The stages may include the following:

- Develop and determine a policy for owning, growing and sharing knowledge within the organisation.
- Identify critical knowledge functions within each department.
- Audit knowledge to determine its type, location, longevity and whether it is explicit or tacit.
- Document knowledge in a medium that best suits the purpose for which it will be used.
- Store it in a repository where it can be easily updated and retrieved.
- Determine ways in which it can be grown and tracked.
- Decide how the knowledge will be disseminated inside the organisation and possibly outside.
- Ensure this valuable organisational asset is kept secure.

Answer to Interactive question 3

3.1 **Porter's five forces Customers**

- Parents individually have very little power/influence over the industry.
- This is because they have low buying power in relation to the size of the industry.
- Because demand exceeds supply of childcare places even if parents take their children elsewhere, it is unlikely to cause any problems for the industry as others will take their places.
- However, if there were a series of damaging reports into the industry, then parents may decide to use alternative forms of childcare. Their children are very precious to them and they would not be prepared to accept substandard levels of service.
- Switching costs are low in terms of money; however parents would have to consider issues such as distress to the children involved and the problem of finding a better alternative in the local vicinity due to the lack of supply.
- Overall influence: Low

Suppliers

- Supply of products to nurseries is likely to be quite diverse as individual nurseries will have different demands. The main issues would be the provision of food, clothing, nappies and toys for the children in the nurseries. The market for these products is likely to be competitive and therefore the bargaining power of individual suppliers is weak.
- However, given that the majority of nurseries themselves are relatively small, their bargaining power is also weak which will neutralise the above position.
- The other main supplier group will be skilled staff. Their power is likely to be low on an individual basis and they are unlikely to be unionised.
- Only if nurseries form groups or chains, to place volume orders, is the negotiating position likely to improve.
- Overall influence: Low

Substitutes

- Demand currently exceeds supply within the childcare industry. Substitutes exist in the form of domestic nannies, playgroups and childminders.
- These competitors individually have little power, although they may offer a service which is more convenient and local to that of nurseries, especially in the form of domestic nannies and childminders.

- Considerations such as the level of service offered by the competition would have an influence on whether they could take market share away from nurseries. Parents would be concerned about the level of regulation over the substitutes, due to the importance of high quality childcare to them.
- Overall influence: Medium

Competitors

- The majority of entities in the industry are small, owner managed businesses. As such competitors' power is low, as one nursery will not be able to significantly influence the behaviour of others.
- Even if individual nurseries set up close to one another, because demands outstrips supply there is potentially room for both to survive.
- If there were a number of chains of nurseries then they might be able to demonstrate some power, but there seems to be little evidence of this.
- Overall influence: Low/medium

New entrants

- Set up costs are relatively low, as demonstrated by the large number of owner managed nurseries in existence. The main costs to be incurred would be premises and equipment, along with ensuring that staff were appropriately qualified and trained.
- The excess demand for childcare is likely to attract new entrants to the market. This may come from existing nursery staff who decide to set up businesses themselves.
- Government regulation and red tape may however act as a deterrent to new entrants.
- The other main restraint on new entrants would be the ability to find suitable premises from which to operate.
- Overall influence is high.

3.2 Critical success factors

Innovation

KA has a range of services and tools that distinguish it from competitors due to the company embracing modern technologies and applying them to the nursery model.

This will help to attract new children as parents will be keen to take advantage of being able to see the activities their children undertake during the day.

This is coupled with flexibility in terms of offering services such as the collect and return service, which puts KA in a position of offering unique services that competitors are either unable or unwilling to provide.

This is likely to result in fewer children being removed from the nursery, and customers being willing to accept the higher pricing structure charged by KA.

Compliance with regulations

KA has a good reputation in the form of the positive feedback from the government reports. Independent approval from a third party is one of the considerations that parents would take when deciding where to send their children.

This will keep parents satisfied that the level of care given to their children is appropriate.

Quality of staff

KA offers a more personal level of service to children, through having a higher staff-to-child ratio, with staff being qualified to a degree in excess of competitors.

In addition, one of the owners of the company has extensive experience of working in the industry and knows the concerns of customers. This will be a comfort factor to parents.

Use of information systems and technology

In relation to Porter's generic strategies Kid A is operating a differentiation strategy.

Parents are prepared to pay a premium for the factors noted above and as a result KA can charge 40% higher fees and still have full nurseries with waiting lists.

Porter's value chain can be used to identify areas where IT could be used to add value. KA makes use of IT in the following areas:

- The use of GPS wristbands linked to webcams allays parents' concerns regarding the safety and welfare of their offspring. This may make them more likely to choose KA over other nurseries.
- Computers are used within the nursery for learning purposes. In addition parents are able to get feedback via the internet on the computer-based games and work that their children have been involved in. These have helped KA gain a good reputation in government reports which is likely to increase demand and retention rates.
- Staff are expected to have knowledge of the systems used and are paid more highly as a result. This may encourage better quality nursery nurses who are more able to communicate with parents.

3.3 **Report**

To:	The directors of Kid A Ltd
From:	XYZ Consultants
Date:	21 June 20X8
Subject:	Expansion plans

(a) **Risks caused by expansion Loss of control**

As the number of nurseries increases you will have to take a more strategic and hands off approach to running the business. This will require developing a new range of skills and provide new problems.

The managers recruited must be competent at the tasks that are being delegated to them. If not, the business could suffer damage to its reputation that would be difficult to repair.

Given the sensitive nature of the business, one poorly performing nursery could have a detrimental impact on the reputation of the whole business. There are also the potential conflicts that would arise between yourselves and new managers.

Financial/economic risks

Fresh finance will be required to expand the business. Because you are running a small business this will have to come from your own savings or new borrowings. If you borrow money then the lenders will require security, either over the nursery properties themselves (if purchased as opposed to rented) or in the form of personal guarantees.

Interest on borrowings will eat into profit and is a fixed cost. If demand for nursery spaces falls then there is a risk that the interest payments may not be fulfilled.

To attract good nursery managers you may have to increase salaries, which will also have an impact on profitability.

A downturn in the economy could have a significant impact on the business. Higher unemployment could result in more families where both parents are not working, and therefore demand for nursery spaces may fall. Financial insecurity amongst the working population may mean that fewer families are willing to pay the premium prices charged by KA.

(b) **Budget issues**

There are a variety of budget elements that could be adopted to make the process more effective.

Profit centres

Because you want to transfer responsibility for each individual nursery to the managers, consider making each nursery a profit centre. Here the managers are responsible for both revenues and costs within the nursery.

Controllable costs

Budgets and income statements should be produced for you to review on a regular (say monthly) basis for each nursery.

These should focus on controllable issues over which managers have influence (local advertising, recruitment, pay and day-to-day expenses). It is not appropriate to assess them on costs over which they have no control.

Delegation/participation

Managers should be involved in the budget setting process, rather than it being imposed by you. This is because managers will be motivated by the increased responsibility and their knowledge of local conditions may improve the budget.

Authorisation

Care must be taken however that managers do not build in slack into their budgets to make them too easy. It may therefore be useful to compare the different budgets prepared by individual managers and adopt one of them as the approved model.

Once the budget has been approved this will act as authorisation to the manager to incur expenditure. Budgets should be created by computer, as this will allow for greater sensitivity analysis.

Evaluation of performance

Actual performance should be compared to the budget plan, feedback provided to the managers and any necessary action taken. The budgets should be flexed to allow you to compare the performance of different sized nurseries operated by KA.

Rewards

Managers should be given rewards for achieving and exceeding budgets. This could be extended to their staff too.

(c) **Information for control and decision making**

Strategic information

Thom and Debra will require strategic information to help with long term planning and assess whether objectives are being met. This information will be both external and internal:

- Information about the external environment useful to assess the future political, economic, social and technological changes that may impact on KA (eg, proposed changes to health and safety regulations, the government's inspection process)
- Market information to estimate future demand (eg, demographic information regarding the local population for each nursery – number of working parents with pre-school age children living or employed locally who might choose to use the nursery)
- Data about competitors to provide a benchmark against which the performance of individual nurseries and the business as a whole might be assessed

  This should consider financial and non-financial factors (eg, the fees charged, government ratings, range of services offered, opening hours, etc).

- Financial information to assess the overall profitability of the business and each nursery
- Information regarding fixed asset investments (eg, premises and capital structure/financing)

Tactical information

This might focus on key performance indicators such as the utilisation of resources by each nursery, particularly in relation to staff requirements eg, ratio of children to staff, fees earned per staff member, staff turnover rates, expenditure as a percentage of fees earned.

Thom and Debra should monitor customer satisfaction, eg, by asking parents to complete a survey online. They should monitor the number of parents who switch from them to other nurseries, the number of parents they attract from competitors.

They could monitor utilisation of capacity by considering nursery places filled as a percentage of places available and the waiting lists at each nursery. They can also monitor the ability of individual managers to convert enquiries into bookings.

It would be useful to monitor the take up of some of the additional services offered eg, the number of parents who actually use the webcam, the number of times work is reviewed by parents on the internet. This will help assess the demand for these services and the value being ascribed to them by parents. KA should ensure that they generate and monitor performance indicators for the areas that the government inspection focuses on.

Operational information

This should be prepared very frequently to help track the specific day-to-day activities of the nursery:

- Staff timesheets to monitor hours worked and sickness
- Weekly task completion sheets
- Weekly running costs in comparison to budget; fees invoiced/collected

Answers to Self-test questions

1 Planning, controlling, recording transactions, measuring performance and making decisions

2 Five of:
- Derived from both internal and external sources
- Summarised at a high level
- Relevant to the long term
- Concerned with the whole organisation
- Often prepared on an *ad hoc* basis
- Both quantitative and qualitative
- Uncertain, as the future cannot be predicted accurately

3 Five of:
- Primarily generated internally (but may have a limited external component)
- Summarised at a lower level
- Relevant to the short and medium term
- Concerned with activities or departments
- Prepared routinely and regularly
- Based on quantitative measures

4 Five of:
- Derived from internal sources
- Detailed, being the processing of raw data
- Relevant to the immediate term
- Task-specific
- Prepared very frequently
- Largely quantitative

5 Explicit knowledge is knowledge that an organisation already stores in formal systems. It includes facts, transactions and events that can be clearly stated and stored in information systems.

Tacit knowledge is expertise held by people within the organisation that has not been formally documented.

6 Four from:
- Business continuity planning
- Systems access control
- Systems development and maintenance
- Physical and environmental security
- Compliance
- Personnel security

- Security organisation
- Computer and network management.
- Asset classification and control
- Security policy

7 Computer security

7.1 The main areas of risk to which a computer system is exposed, and some of the factors which may lead to the exposure are as follows.

Accidental destruction or **loss of data** by operator or software error. The auditors should pay particular attention during their audit to recovery procedures. In addition the possibility of accidental destruction of programs or hardware, particularly the dropping of a disk pack, by an operator, and the consequences thereof, should not be overlooked.

The **acceptance of inaccurate input data** due to inadequate edit or other checks is another frequent cause of loss of data.

A **complete systems failure** can lead to loss of data and may be caused by a failure in the power supply or possibly a failure of the air conditioning or other environmental controls.

Theft of data from a **batch processing system** by an operator copying or removing data files, particularly where these are on easily transportable media such as magnetic tapes.

Theft of data from an **online or real time system** by a person obtaining unauthorised access to the system via a remote terminal and either using passwords illegally or alternatively using a 'piggyback system' (in which a valid transmission is intercepted and the final 'logging off' operation stopped in transmission to permit the illegal operator to continue in operation masquerading as the authorised user).

Theft of software either by operators copying or removing the program file, and in the latter case possibly demanding a ransom from the rightful owner, or alternatively by programming staff copying and attempting to sell the source documentation, with or without the object program.

Deliberate destruction of the hardware has been known to occur, and where adequate protection has not been provided, such acts have also led to the simultaneous destruction of software and data. Similar results may occur as a result of fire or explosion either in the computer room or adjoining premises.

7.2 The different forms of control which should be instituted may be sub-divided into three main headings.

Physical security

Strict control of access to the computer area, using such devices as magnetic keys and alarm systems.

Effective precautions against **fire** or other **natural disruption** including alarm systems, automatic extinguishing systems and regular inspections.

Established and well-practised **emergency procedures** in the event of fire or other disorder and alternative power supply.

Location of the computer so that it is difficult for unauthorised personnel to have access with the minimum of entrances and exits.

Possibility of **remote storage of security copies of data**.

Location of the computer room so that it is, if possible, **situated away from known** hazards such as flooding, radiation from X-ray equipment and radio systems and fire/explosion risks in adjoining premises.

Software security

Effective control over the **preservation of information** contained on files by ensuring that before a file is to be overwritten a check is made on the file version.

Prevention of unauthorised access by the use of devices such as passwords.

Systems security

Strict control and verification of all **input data**, where possible with control totals prepared outside the computer department.

All input should pass through an **'edit' program** as the first stage in being entered on to the computer files. This program clearly indicates all items accepted and rejected, the latter to be investigated by the user department.

Adequate controls should be in force to ensure that **amendments to programs** are properly **authorised, checked out** and **validated** before use.

There should be **adequate recovery, restart and standby procedures** in the event of power failure or machine breakdowns, which can be facilitated by a 'log' of all work performed and by frequent dumping of files.

Controls should be instituted to ensure that **computer output** is **properly distributed**, especially confidential print-outs, payments and so on.

Proper control over **storage and issue of electronic media** with manual records being kept of physical maintenance performed. Such records frequently also record current status of the media and the details of the file(s) currently stored upon it.

Chapter 15

Strategies for change

Introduction

Learning outcomes

Syllabus links

Examination context

Chapter study guidance

Learning topics

1 The need for change

2 Strategic change in organisations

3 Change and the individual

Summary

Further question practice

Self-test questions

Answers to Interactive questions

Answers to Self-test questions



Introduction

Learning outcomes

- Explain and justify the key issues which should be addressed by the management of an organisation during the planning and implementation of change.

Syllabus links

This chapter involves the implementation of the strategies discussed in the preceding chapters. It has not been covered in the syllabus of previous examinations.

Examination context

Previous chapters concentrate on identifying the need for, and the direction of change if an organisation is to remain successful. This chapter looks at the issues which need to be addressed by an organisation during the planning and implementation of such changes.

In exams so far, most candidates have demonstrated a reasonable knowledge of change management but a common mistake has been to provide a generic answer, repeating elements by rote from the learning materials. As we have mentioned throughout this Workbook, your ability to apply your knowledge to the question scenario will be vital in earning you marks.

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this chapter.

Topic	Practical significance	Study approach	Exam approach	Interactive questions
1	**The need for change** Strategy changes organisations and this needs management. The sources of change are both internal and external and must be handled carefully if the organisation wishes to remain broadly on track, delivering a return to shareholders.	**Approach** Read through the triggers of change and the types of change outlined by Johnson, Scholes and Whittington. The worked example gives you an understanding of change driven by the gig economy; short term contracts and flexible working replacing full employment of staff. **Stop and think** Technological developments present organisations with new opportunities	When using the Johnson, Scholes and Whittington model of change ensure you justify the company's position in the matrix by explaining each axis. For example, to justify the type of change you identify, explain both the nature of change and the management role in dealing with the change. Your explanation must refer to detail in the scenario in order to be well applied.	**IQ1: Types of change** This question encourages you to use the Johnson, Scholes and Whittington model to explain change in the car and supermarket industries.

Topic	Practical significance	Study approach	Exam approach	Interactive questions
		and improved ways of working. However, they also present significant challenges. Think of two organisations that you are familiar with and consider the impact that technological change has had on their business models. Has technological change been beneficial or not?		
2	**Strategic change in organisations** Change affects people inside and outside the organisation and it is essential to avoid resistance from significant groups who might derail the changes. These could include major investors, key groups of organised labour or significant clients. Most change models describe the stages of psychological transition to manage people's expectations and attitudes throughout the change process.	**Approach** Read through the different frameworks for managing change and take a look at the worked example discussing change at the wearable fitness technology company, Fitbit. **Stop and think** The first step in change management, according to Lewin's iceberg model, is to unfreeze existing behaviour. Think about any changes that have occurred in your life, either in a work or personal context. What behaviours and attitudes needed 'unfreezing?' How was this done?	Most well-prepared students demonstrate a reasonable knowledge of change management but, according to the examiners, a common mistake has been to provide a generic answer repeating elements by rote from the learning materials. Only refer to a change model if it provides structure to your answer or enhances it in any way. Theory and models earn very few marks and instead must be applied to the scenario.	**IQ2: Transforming the National Mail** This scenario- based question requires you to apply the Gemini 4Rs approach to understand how successfully change has been managed.
3	**Change and the individual** Organisations are made up of collections of individuals whose needs must be considered when managing change. Individual and collective resistance can result in the	**Approach** Spend time understanding individual barriers to change and the approaches management can take to communicate with staff and breakdown barriers. **Stop and think** It is a natural human	It is important to provide practical suggestions and recommendations to managing individual resistance to change. Use your own work experience to help generate sensible suggestions that can be applied to the scenario provided.	**IQ3: Gerrard, Dudek and Smicer** Apply Lewin's forcefield model to this short scenario-based question, in order to recommend how change can be facilitated. **IQ4: Burgermania** This exam standard

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	failure of change programmes so efforts must be made to win individuals over.	tendency to resist change. How can organisations develop a culture whereby staff are more change adept?		question requires you to identify the type of change under consideration and the barriers to change. You are then asked to draft a report on how to manage and communicate the change.

Once you have worked through this guidance, you are ready to attempt the further question practice included at the end of this chapter.

1 The need for change

Section overview

- The sources of change are both internal and environmental.
- The range of impacts of a 'change trigger' will be felt as shock waves through the organisation and will affect staff and management at all levels.
- Change must be handled carefully to avoid the organisation being overwhelmed.

1.1 The need for an organisation to change or develop

The need for an organisation to change or develop could be caused (or 'triggered') by a number of factors:

- **Changes in the environment**: These could be changes in what competitors are doing, what customers are buying, how they spend their money, changes in the law, changes in social behaviour and attitudes, economic changes, and so on.
- **Changes in the products the organisation makes, or the services it provides**: These are made in response to changes in customer demands, competitors' actions, new technology, and so on.
- **Changes in technology and changes in working methods**: These changes are also in response to environmental change such as the advent of new technology and new laws on safety at work.
- **Changes in management and working relationships**: For example, changes in leadership style, and in the way that employees are encouraged to work together. Changes could also be triggered by the growing flexibility in workplace arrangements. Also changes in training and development.
- **Changes in organisation structure or size**: These might involve creating new departments and divisions, greater delegation of authority or more centralisation, changes in the way that plans are made, management information is provided and control is exercised, and so on. **Organisation re- structuring** will be made in response to changes of the types discussed above.
- **Post–acquisition**: Incoming management will wish to improve and integrate the firm into the new parent's structure and systems. This will involve visible change to names and signage but also deeper changes to organisational structures, culture, job roles, staff numbers and management systems.

Context example: The gig economy

In the chapter, 'Strategic capability,' we explored the concept of workforce flexibility and considered some of the impacts that it is having on the relationships which exist between organisations and workers.

Advances in technology have been key facilitators of the so-called gig economy in the UK. However, as the following worked example illustrates a number of legal challenges are likely to lead to significant changes in how this 'economy' operates in the future.

Wilson (2017) highlights that the gig economy is a term used to describe a labour market in which work is undertaken by freelancers and people on short-term contracts, as opposed fully employed workers. Proponents argue that the gig economy provides workers with flexibility around the hours they work, whilst critics suggest that it is fundamentally exploitative by nature, as individuals are not eligible to receive standard employment benefits or trade union

representation. In recent times a number of firms operating in the gig economy have been accused of deliberately attempting to misclassify workers as independent subcontractors in order to get around legal requirements governed by employment law.

Companies including Uber and Deliveroo, have become synonymous with the gig economy. Both companies operate through the use of internet technologies including apps. On its website, Uber (2022) claims that it is an 'app enabling transportation requests, whose mission is to reimagine the way the world moves for better. Our technology helps us develop and maintain multi-sided platforms that match consumers looking for trips with drivers.' Drivers are effectively freelancers who use the Uber app as a means of picking up fares. Uber then collects a percentage of the fare charged, as commission. Deliveroo operates in a similar manner, whereby customers use the company's homepage to order food from a selection of local restaurants. As Deliveroo (2018) highlights, food is then couriered by its team of independent contractors by bicycle from the selected eatery to the end customer. Deliveroo riders are paid a standard rate per hour with an additional amount per delivery made.

In recent times, a number of high-profile cases have made the news as groups working for the likes of Deliveroo and Uber have claimed that the nature of their relationships entitles them to certain employment rights including: sick and holiday pay and the right to receive the minimum wage.

Sources:

- Deliveroo (2018) *About Deliveroo*. [Online] Available from: https://deliveroo.co.uk/faq
- Uber (2018) *What is Uber?* [Online] Available from: https://help.uber.com
- Wilson, B. (2017) *What is the 'gig' economy?* [Online] Available from: www.bbc.co.uk



Professional skills focus: Assimilating and using information

Change within an organisation can often be controversial with some stakeholder needs being ignored in order to improve shareholder return. When analysing change it is important to identify and assess different stakeholder perspectives and to consider the ethical implications of the proposed changes.

1.2 Types of change

Change itself may be divided into two types, **incremental** and **transformational**, and so may the management approach to change be divided into **reactive** and **proactive**.



Definitions

Incremental change: Is characterised by a series of small steps. It is a gradual process.

Transformational change: Is characterised by major, significant change being introduced relatively quickly.

Step change: Describes an unexpected jump (upwards) or drop (downwards) in the pace of change. The step is caused by an unexpected event (eg, environmental disaster, unexpected change in government etc).

Planned change: Involves following a series of pre-planned steps.

Emergent change: Views change as a series of continuous open-ended adjustments to the environment.

Johnson, Scholes and Whittington suggest the model of change shown below: Figure 15.1: Types of change

Management role	Nature of change: Incremental	Nature of change: Transformational
Pro-active	Tuning	Planned
Reactive	Adaption	Forced

The importance of **proactive management** is that it implies that organisational change may be undertaken **before** it is imposed by events. It may, in fact, result from the process of forecasting and be a response to expected developments. **Forced change** is likely to be both painful and risky.

Interactive question 1: Types of change

Classify and explain the following changes using Johnson, Scholes and Whittington model.

Requirement

Classify and explain the following changes using Johnson, Scholes and Whittington model.

(a) An international car maker undertaking a turnaround strategy in response to external pressures following years of underperformance.

(b) A nationwide UK supermarket retailer announcing its entry into a large and profitable foreign market. (It is anticipated that the change will have a far-reaching impact on the retailer's current business model.)

See **Answer** at the end of this chapter.

1.3 Levels at which change efforts may focus

There are three main levels to which organisational development and change efforts may be directed:

- **Individual level** where the focus is on improving individual skill levels, attitudes and motivation. Techniques employed could include education and training, management development, coaching and counselling, team building activities, inter-group activities, role analysis, job re- design, planning and objective setting activities and process consultation.
- **Organisation structure and systems level**: The characteristics of the organisational situation in which people work (eg, job redesign, reward systems, setting clear objectives) that help achieve organisational goals.
- **Organisational climate and interpersonal style levels**: The improvement of social and other informal processes among organisation members by creating a system with a wide climate of high interpersonal trust and openness and a reduction in the negative consequences of excessive social conflict and competitiveness.

2 Strategic change in organisations

Section overview

- Change management involves management of people's expectations and attitudes. This can range from the willingness of an operative to learn a skill or change a shift up to the willingness of the Board to authorise the necessary resources.
- Therefore most change models seek to describe stages of psychological transition.
- These stages may be the focus of planned change management initiatives.

2.1 Change processes

For an organisation to be innovative, and continually responsive to the need for change, a systematic approach should be established, for planning and implementing changes.

Although each situation should be considered individually, the following general steps can be identified in a major change initiative:

Step 1 Determine need or desire for change in a particular area.

Step 2 Prepare a tentative plan. Brainstorming sessions are a good idea, since alternatives for change should be considered.

Step 3 Analyse probable reactions to the change.

Step 4 Make a final decision from the choice of alternative options. The decision may be taken either by group problem-solving (participative) or by a manager on their own (coercive).

Step 5 Establish a timetable for change.

- 'Coerced' changes can probably be implemented faster, without time for discussions.
- Speed of implementation that is achievable will depend on the likely reactions of the people affected.
- Identify those in favour of the change, and perhaps set up a pilot programme involving them. Talk with any others likely to resist the change.

Step 6 Communicate the plan for change. This is really a continuous process, beginning at Step 1 and going through to Step 7.

Step 7 Implement the change.

Step 8 Review the change. This requires continuous evaluation.

2.2 The three-stage approach (iceberg model)

The Lewin/Schein three-stage **model of change** identifies key steps as **unfreeze**, **move** and **refreeze**.

Unfreeze		Move		Refreeze
Existing behaviour	→	Attitudinal/behavioural change	→	New behaviour

Step 1 Unfreeze is the most difficult (and in many cases neglected) stage of the process, concerned mainly with selling the change, with giving individuals or groups a motive for changing their attitudes, values, behaviour, systems or structures.

Unfreezing processes require four things:

- A trigger
- Someone to challenge and expose the existing behaviour pattern
- The involvement of outsiders
- Alterations to power structure

Step 2 Move is the second stage, mainly concerned with identifying what the new, desirable behaviour or norm should be, communicating it and encouraging individuals and groups to 'own' the new attitude or behaviour. This might involve the adoption of a new culture. To be successful, the new ideas must be shown to work.

Step 3 Refreeze is the final stage, implying consolidation or reinforcement of the new behaviour. Positive reinforcement (praise and reward) or negative reinforcement (sanctions applied to those who deviate from the new behaviour) may be used.

Overcoming resistance

Lewin

Kurt Lewin developed a simple model for achieving successful change and overcoming resistance in three steps:

- ***Unfreezing** – create the initial motivation to change by convincing staff of the undesirability of the present situation.*
- ***The change process itself** – mainly concerned with identifying what the new behaviour or norm should be. This stage will often involve new information being communicated and new attitudes, culture and concepts being adopte*

Further details

Unfreezing

The first stage of the model is to break down and remove the current resistance to change. Staff need to be convinced that the current system is not successful and needs to change. Failures in the current culture, whether that be in its structure, people's routines or organisational controls etc. need to be examined and explained. The benefits of a proposed new system need to be clearly communicated to staff. This may require strong leadership styles, which will be discussed later in this chapter.
The goal is to motivate employees to embrace the change and reduce their resistance. Their attitudes are 'unfrozen', meaning they are no longer 'stuck in their ways' and are now willing to accept the change.

Changing

With the groundwork laid, the new system and associated changes can be implemented. This may require hands-on leadership, and it will most certainly require good communication. Some change will have to be supported by additional training and resources. For example, if a company is transitioning to a new accounting software, staff will need training to become proficient in its use.

Refreezing

The new system now needs to become part of the organisational culture. It needs to feel natural to staff. This may involve creating a new reward structure linked to the new system, establishing new rules and controls, and setting new targets.

Organisations should create new stories around the success of the change, to celebrate successes and communicate to staff how the expected benefits are being achieved. This helps embed the change as part of their new culture, ensuring staff do not fall back on old habits and want to return to the old way of doing things.

Some argue that the refreezing step is outdated in contemporary business. Due to the continuous need for change it is unnecessary to stabilise a new state when it may need re-evaluation and further change.
However, without the refreezing step, there is a high chance that people will revert to the old way of doing things, leading to a cycle of taking one step forward and two steps back. Therefore, it is crucial for organisations to solidify changes to prevent regression.

There are several problems with a coercive approach:

- Underestimation of the forces of resistance
- Failure to muster forces in favour
- Failure to attack root causes of resistance
- Management shift their attention too quickly elsewhere
- Failure to ensure implementation

This approach is necessary in situations of **crisis** where there simply is no time to consult, or where decisions need to be taken quickly. Crisis has the effect of inducing panic – which managers must do what they can to minimise – but **it can also promote an immediate willingness to change**: it can be the necessary 'unfreeze' process before an organisational change. An example is a sudden environmental shock that endangers the company's survival. Most of the time, a **mixed approach between coercive change and adaptive change** is suitable. Adaptive change may be too slow, whereas coercive change is often resented and therefore not accepted.

2.3 Change agents



Definition

Change agent: Is an individual (sometimes called a champion of change), a group or external consultancy with the responsibility for driving and 'selling' change.

The role of the change agent varies depending on the brief they are given. It may include:

- Defining the problem
- Suggesting possible solutions
- Selecting and implementing a solution
- Gaining support from all involved

To be effective a change agent should have the following skills and attributes:

- Communication skills
- Negotiation and 'selling' skills
- An awareness of organisational 'politics'
- An understanding of the relevant processes

The **champion of change model** recognises the importance of change being led by a **change agent**, who may be an individual or occasionally a group.

Step 1 Senior management are the change strategists and decide in broad terms what is to be done. There is a need for powerful advocacy for change at the strategic apex. This will only occur if senior management are themselves agreed on the need for change. This is a role requiring a clear vision of what the change is to achieve.

Step 2 They appoint a change agent to drive it through. Senior management has three roles:

- Supporting the change agent, if the change provokes conflict between the agent and interest groups in the organisation
- Reviewing and monitoring the progress of the change
- Endorsing and approving the changes, and ensuring that they are publicised

Step 3 The change agent has to win the support of functional and operational managers, who have to introduce and enforce the changes in their own departments. The champion of change has to provide advice and information, as well as evidence that the old ways are no longer acceptable.

Step 4 The change agent galvanises managers into action and gives them any necessary support. The managers ensure that the changes are implemented operationally, in the field. Where changes involve, say, a new approach to customer care, it is the workers who are responsible for ensuring the effectiveness of the change process.

It is important to realise that successful change is not something exclusively imposed from above. There is a sense in which middle and junior managers are **change recipients** in that they are required to implement new approaches and methods. However, they are themselves also **change agents** within their own spheres of responsibility. They must be committed parts of the change process if it is to succeed.



Context example: Fitbit

The following worked example focuses on the changes being introduced at wearable technology company, Fitbit.

In March 2017, wearable technology firm Fitbit announced changes to its leadership team. A press release issued by the company announced that Jeff Devine, an expert with 25 years' experience in 'scaling global technology brands including Cisco, Nokia, and Hewlett-Packard' had been appointed as executive vice president of operations (Fitbit, 2017).

An article by Pressman (2017) highlighted that Devine's appointment was part of a 'multi-pronged strategy' by Fitbit's CEO, James Park to help turn the business around. The announcement followed tough trading conditions in 2016 in which profits fell and sales flat-lined as the company suffered from excess inventory holdings. In January 2017, Fitbit announced that it was laying off 6% of its workforce in response to the company's deteriorating performance.

Pressman (2017) noted that 'in Fitbit's view, the sales slump was due to the fact that the market for trackers among early adopters had been saturated' with mainstream buyers, the so-called late adopters, failing to show sufficient interest in the company's offering. Pressman (2017) highlighted 'with over 50 million devices sold in the past three years, and over 23 million active users, the company must now reach more casual, less tech savvy buyers' (Pressman, 2017).

Pressman (2017) likened the appointment of Jeff Devine to Steve Jobs hiring of Tim Cook (the current CEO of Apple), who was brought in 'from Compaq Computer to overhaul Apple's operations in 1998.' Central to Devine's role will be improving Fitbit's gross margins and improving the quality of the company's products. The changes at Fitbit would also appear to be part of CEO James Park's desire to enter the smartwatch category, with a view to taking on Apple and Samsung.

Pressman (2018) highlighted that, in April 2018, Fitbit agreed a deal with Google to 'use Google's recently announced health data standards for apps […] to connect its wearable devices to the electronic medical records systems used by doctors and hospitals.' It is Fitbit's intention that, in time, doctors will be able to access patient health data from their Fitbit device.

Sources:

- Pressman, A. (2018) *Fitbit Strikes Deal With Google That Could Lead to Wearables Collaboration.*

[Online] Available from: http://fortune.com

- Pressman, A. (2017) *Fitbit CEO Offers Turnaround Strategy After a Tough Year*. [Online] Available from: http://fortune.com; Fitbit (2017)
- Google (n.d.) *Fitbit Announces Changes to Leadership Team*. [Online] Available from: https://investor.fitbit.com

2.4 The Gemini 4Rs framework for planned strategic change

Management consultants Gouillart and Kelly describe a four-dimensional process for business transformation in their book *Transforming the Organisation*. Known as the Gemini 4Rs framework, this approach aims to cover all the important components of the organisation's identity.

Reframing involves fundamental questions about what the organisation is and what it is for:

- **Achieve mobilisation** – create the will and desire to change.
- **Create the vision** of where the organisation is going.
- **Build a measurement system** that will set targets and measure progress.

Restructuring is about the organisation's structure, but is also likely to involve cultural changes:

- **Construct an economic model** to show in detail how value is created and where resources should be deployed
- Align the physical infrastructure with the overall plan
- Redesign the work architecture so that processes interact to create value.

Revitalising is the process of securing a good fit with the environment:

- Achieve market focus
- Invent new businesses
- Change the rules of competition by exploiting technology

Renewal ensures that the people in the organisation support the change process and acquire the necessary skills to contribute to it:

- **Create a reward system** to motivate
- **Build individual learning**
- **Develop the organisation** and its adaptive capability

Professional skills focus: Structuring problems and solutions

When discussing change management it can be helpful to use a model to structure your answer and ensure breadth of analysis. The Gemini 4R framework allow a comprehensive assessment of the stages of change management and can be used to provide focus to your answer.

Context example: National Mail Group

The following worked example sets out the difficulties faced by the National Mail group (National Mail) at two points in its history. National Mail was a previously-state owned postal service provider in a developed European country. You should assume that the current date is June 20X9.

May 20X0

In May 20X0, the government announced that from 1 January 20X1 any licensed operator would be allowed to deliver mail to business and residential customers, effectively ending the state-owned business's monopoly:

'National Mail [17 May 20X0] today announced a record €537 million profit on its operations for the last 12 months with quality of service to customers now hitting the highest levels in a decade.

National Mail's then chair, Jennifer Smith, said: 'Postmen and women have achieved a fantastic turnaround. They will now deservedly each get a 'Share in Success' payment of €1,074, amounting to €218 million of the company's profit. It's one of the biggest profit shares with employees in European corporate history.'

But Jennifer Smith warned: 'There is still a huge amount to do. Transforming our operations, cutting our costs and, above all, winning the support of our people for the modernisation plan with its top priority being to improve customer service, has been National Mail's greatest achievement in decades. But competing successfully in an open mail market is going to be even more difficult.
We've a mountain to climb and we've only reached the base camp.

The greatest challenge now is to bring about a complete culture change in National Mail. We need everyone in the company focused on ensuring that we consistently deliver high quality, value-for- money services that customers need and want. Competition has arrived and customers have a choice; so we need to prove that National Mail is the best and our people are key to that.'

The company, Smith said, was also facing other daunting hurdles:

National Mail will have to generate sufficient profit to pay millions of pounds into its pension fund to tackle the €2.5 billion deficit, the 14,609-strong network of Letter and Parcel Office (LPO) branches made a loss on its operations last year of €110 million. Its rural network of 8,037 branches is fundamentally uneconomic and needs an injection of €3 million a week to survive. However, with the current annual Government funding of €150 million due to end in 20X3, Letter and Parcel Office Ltd cannot be expected to absorb extra costs at this level.

National Mail lags behind its major rivals in automated sorting technology. It needs to make a several billion euro investment if it is to compete successfully and that means being more profitable to invest. However, the 8.6% return on its domestic letters business last year compares with the 16.4% Fast Post (a major postal service in another European country) makes in its home market, and 22.2% made by RNT Post Group (a commercial postal service).

National Mail's chief executive at the time, Carl Ling, also stressed the challenges ahead.

'The huge task now facing National Mail is to make the cultural change needed to succeed as a commercial business and to become the postal operator of choice for customers in an open competitive market.

Our vision remains to be demonstrably the best and most trusted mail company in the world. With the dedication and commitment of our people, we are confident we can achieve our goal.'

October 20X8

'Since 20X0 the National Mail has undergone some significant changes. In October 20X8, the government controversially sold off the National Mail for €3.3 billion. The flotation saw a significant number of National Mail employees (15,000) purchase shares in the company. In May 20X9, it was reported in the press that National Mail was embroiled in a row with the postal regulator over the company's ability to meet its obligations to provide a 'universal postal-delivery service.' National Mail's commitment to the universal postal-delivery service requires it to deliver post to all addresses in its country six days a week.

It was also reported that at the same time as she unveiled a 12% rise in full-year operating profits to €671 million, National Mail's chief executive, Sandra Khan, warned that there is a prospect that National Mail will not be able to sustain a commercial rate of return if it maintains its commitment to deliver mail to all of the country's homes six days a week.

A spokesman for the regulator commented: 'we do not believe there is presently a threat to the financial sustainability of the universal postal service. National Mail should take appropriate steps to respond to the challenge of competition, including improving efficiency.'

When the government sold off the National Mail in 20X8 it guaranteed that it would continue with its universal six-day service.

A press release concerning the National Mail's 20X8/X9 financial results noted the following comments made by Sandra Khan.

'Our performance was in line with our expectations. We delivered 2% revenue growth, controlled operating costs and drove strong free cash flow. We are facing a couple of key issues. The competitive environment on the parcels side is more intense. We are taking steps to remain the leader in this growing market. On the letters side, the main issue is direct delivery and we have strategies in place to counter its adverse financial impact. However, without timely regulatory action, direct delivery could undermine the economics of the universal service and our ability to generate sustainably a 5–10% EBIT margin in our reported business.

Our key value drivers of single digit revenue growth, margin expansion and underlying free cash flow growth remain the objectives for the Group for the 20X8–X9 financial year.'

Below is a selection of key highlights from the National Mail's press release:

- 'Group revenue increased by 2%, due to parcel revenue growth. Parcels are the largest contributor to Group revenue, accounting for 51%.
- Group operating profit before transformation costs grew to €671 million.
- Transformation costs of €241 million for the year include a provision of €104 million in relation to the management reorganisation programme, announced on 25 March 20X9, which will be implemented post 20X9.
- The Board has recommended a final dividend of 13.3 cents per share, subject to shareholder approval at the annual general meeting, to be held on 24 July 20X9.
- Based on our estimates of the impact of RNT Post's publicly-stated plans, direct delivery could reduce National Mail revenue by over £200 million in the next 10 years.

Summary outlook

We are facing increasing challenges in the parcels and letters markets.

National Mail's annual report 20X8/X9 highlighted a number of initiatives the company has undertaken to boost its performance. It has acknowledged the decline in the number of letters now being posted, and has moved to streamline its operations, with plans to reduce the number of people it employs by 1,300 over the coming year.

In May 20X8, the company unveiled an agreement it has entered into with the Letter and Parcel Office (LPO) to offer retailers and customers the option of having parcels delivered to a local LPO branch ready for collection. The 'local click and collect' service has led National Mail to modernise a number of its operations.

National Mail also invested €70 million in the year on improving its business mail operations. And it intends to introduce barcode technology to allow business customers to track their parcels prior to delivery.

Interactive question 2: Transforming the National Mail

Consider the cultural change described in the final paragraphs of the National Mail press release from 20X0 and the steps taken to respond to the deregulation of the postal service.

Requirement

Apply the activities of the Gemini 4Rs approach to this change. Refer to the information provided in respect of the period 20X8/X9, where relevant, for evidence of how successfully the various elements of the four Rs framework have been applied.

See **Answer** at the end of this chapter.

3 Change and the individual

Section overview

- Understanding the implications of change for the individual is essential to a change agent or other would-be 'midwife' of change.
- The way the individual views the change will be influenced by the manner of its introduction and the forces driving it and restraining it.

3.1 How change affects individuals

Change may affect individuals in several areas.

- **Physiological** changes – pattern of shift-working, location of place of work
- **Circumstantial** changes – unlearning previous knowledge and learning new ways of doing things, new work-mates, using new IT/IS
- **Psychological** changes – feelings of disorientation, insecurity, changing relationships, etc

3.2 Reactions to proposed change

The importance of staff support during a change programme should not be underestimated. We look here in more detail at the barriers to change they may cause to arise.

3.2.1 Cultural barriers

Structural inertia is the cumulative effect of all the systems and procedures the organisation has installed over the years to ensure consistency and quality. These act as barriers to change. For example, selection processes systematically select certain kinds of people; promotion processes regularly reward certain kinds of people.

Group inertia may block change where changes are inconsistent with team/departmental norms, or where they threaten to make skills and expertise of a particular professional or skill group less important or even redundant.

Power structures may be threatened by the redistribution of decision-making authority or resources, or the re-routing of lines of communication.

3.2.2 Personnel barriers

In addition to cultural barriers, there are also barriers which affect individuals and result in them seeing the change as a threat.

- Habit, because habitual ways of work are hard to change, and the new and unknown is often uncomfortable.

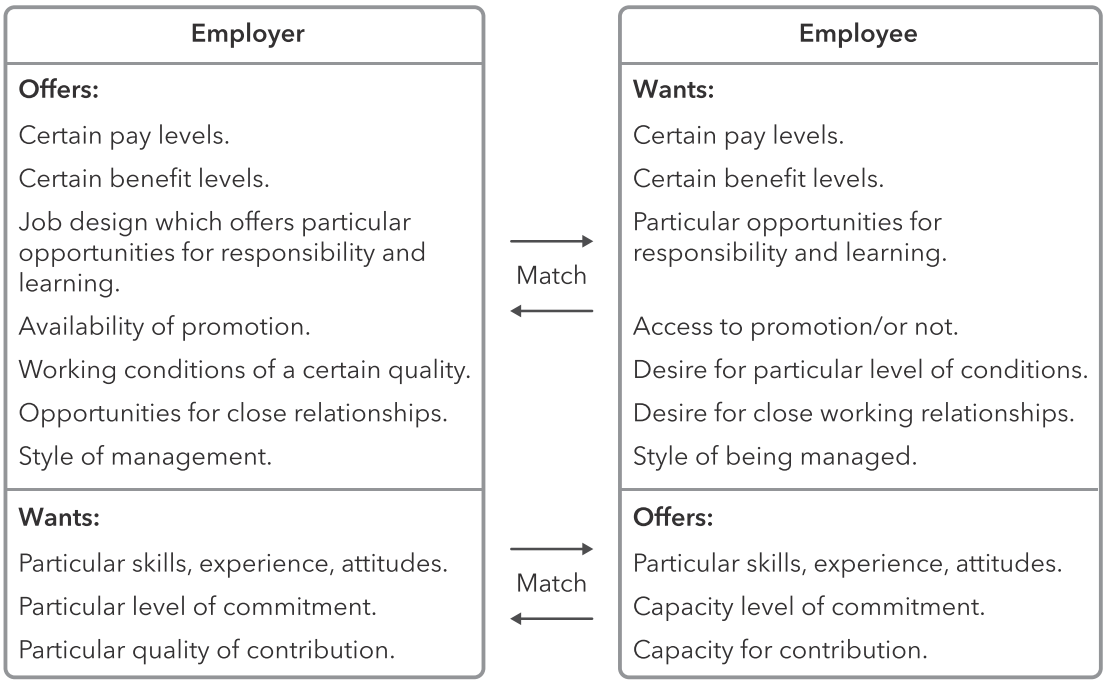
- Security is almost inevitably threatened – job security and the security of familiarity.
- Effect on earnings may be considerable: impact on promotion and income.
- Fear of the unknown reduces people's willingness and interest in learning new skills and processes; they may lack the confidence to take on a new challenge.
- Selective information processing results in employees choosing what to hear and what to ignore; they can then use their selected information to justify their position and ignore management argument for change.

3.2.3 The psychological contract

The psychological contract is the 'deal' between employer and employee. It covers the full set of expectations each party has of the other.

On either side of the contract, there are offers and wants.

Figure 15.2: The psychological contract



The contract works if offers and wants match. The selection process is the means both parties use to assess the initial match. The promotion process is the means used to check the ongoing match. In addition, both parties continuously monitor the match. The model is dynamic, any change in one part triggers changes in others. If a change programme threatens any item listed under employer offers, the employee will reconsider what they offer as their part of the deal. If the employee believes that their offers will be made wholly or partially redundant by the changes, they will anticipate that their wants will no longer be satisfied.

3.2.4 Identification of pressure points

Two important pressure points can develop in the psychological contract during times of change.

- **Lack of appropriate skill levels**: This can happen if the employer fails to reconsider how employee wants will alter as a result of changes, or if the employer fails to redesign their offers to ensure these attract and encourage the right kind of people needed in the new set-up.

- **Declining staff morale**: The driver of employee performance is the human emotions which underlie employee wants. These wants express personal motives, desires, ambitions and needs and determine people's commitment to the changes.

A change programme must be preceded by a review of:

- New competences and qualities required
- Selection processes to ensure they identify the above
- Promotion processes to ensure they identify the above

A change plan must include a staff development plan covering:

- Communication of implications for jobs
- Communication of required skills
- Discussions about individual development needs/options
- Learning and training opportunities
- Opportunities for making a contribution to changes



Context example: Teachers' performance-related pay

It is interesting to set side by side the comments made by a Department of Education spokesperson and the General Secretary of the National Union of Teachers in a developed country regarding the introduction of performance-related pay. It isn't difficult to tell which is which!

- 'There is strong support from teachers across the country for our plans to let schools pay good teachers more. This year schools will be able to reward the most effective teachers who get the best out of their pupils with higher pay. We already have the best generation of teachers ever, and the changes that we are making to teachers' pay arrangements will further raise the status of the profession and encourage more great people into teaching.'
- 'Despite the spin, this research proves again that teachers oppose the government's new performance-related pay measures. Historically movement up the pay scale was linked to seniority and increasing professional skills and competence. It is hardly surprising, therefore, that teachers continue to favour the previous system, which is transparently fairer and less open to biased judgements. While effective appraisal linked to professional development is useful, provided it encourages collaboration, this will not be achieved by the new pay system.'

3.3 Introducing the change

Three factors for managers to consider when dealing with resistance to change.

- The **pace** of change
- The **manner** of change
- The **scope** of change

3.3.1 Pace

Given time, people can get used to the idea of new methods – can get acclimatised at each stage, with a consequent confidence in the likely success of the change programme, and in the individual's own ability to cope.

- Presenting the individuals concerned with a *fait accompli* may short-circuit resistance at the planning and immediate implementation stages. But it may cause a withdrawal reaction (akin to 'shock'), if the change is radical and perceived as threatening, and this is likely to surface later, as the change is consolidated – probably strengthened by resentment.
- **Timing** will also be crucial: those responsible for change should be sensitive to incidents and attitudes that might indicate that 'now is not the time.'

3.3.2 Manner

The manner in which a change is put across is very important: the climate must be prepared, the need made clear, fears soothed, and if possible the individuals concerned positively motivated to embrace the changes as their own.

- **Resistance should be welcomed and confronted**, not swept under the carpet. Talking through areas of conflict may lead to useful insights and the adapting of the programme of change to advantage. Repressing resistance will only send it underground, into the realm of rumour and covert hostility.
- **There should be free circulation of information** about the reasons for the change, its expected results and likely consequences. That information should appear sensible, clear, consistent and realistic: there is no point issuing information which will be seen as a blatant misrepresentation of the situation.
- **The change must be sold to the people concerned**: People must be convinced that their attitudes and behaviours need changing. Objections must be overcome, but it is also possible to get people behind the change in a positive way. If those involved understand that there is a real problem, which poses a threat to the organisation and themselves, and that the solution is a sensible one and will solve that problem, there will be a firm rational basis, for implementing change. The people should also be reassured that they have the learning capacity, the ability and the resources to implement the plan. It may even be possible to get them really excited about it by emphasising the challenge and opportunity by injecting an element of competition or simply offering rewards and incentives.
- **Individuals must be helped to learn**, that is, to change their attitudes and behaviours. Few individuals will really be able to see the big picture in a proposed programmed of change. To put across the overall objective, the organisation should use **visual aids** to help conceptualise. Learning programmes for any new skills or systems necessary will have to be designed according to the abilities of the individuals concerned.
- The effects of **insecurity**, perceived **helplessness**, and therefore **resentment**, may be lessened if the people can be **involved** in the planning and implementation of the change, that is, if it is not perceived to have been imposed from above.

3.3.3 Scope

The scope of change should be carefully reviewed. Total transformation will create greater insecurity – but also greater excitement, if the organisation has the kind of innovative culture that can stand it – than moderate innovation. There may be hidden changes to take into account: a change in technology may necessitate changes in work methods, which may in turn result in the breaking up of work groups. Management should be aware of how many various aspects of their employees' lives they are proposing to alter – and therefore on how many fronts they are likely to encounter resistance.

3.4 Force field analysis

There is a technique developed by Lewin to visualise the change process called **force field analysis** It is based on the idea that in any group or organisational situation there is an **interplay of restraining and driving forces that keeps things in equilibrium**.

Lewin's **force field analysis** maps the forces that are pushing **toward the preferred state** and the restraining forces, which are **pushing back to the current state**. They can then be presented in a chart.

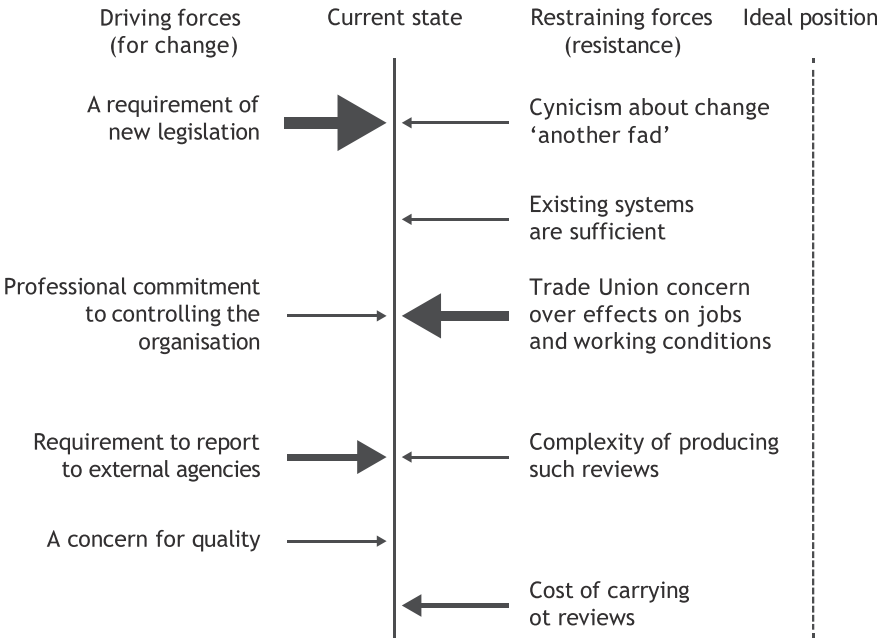
The example below describes a public sector organisation whose management are introducing a performance review system. A group of workers are producing at 70% of the efficiency that might be expected on purely technical grounds. Their output can be visualised as a **balance** between two opposing sets of forces, ie, **driving forces** which are propelling their output upwards and **restraining forces** which are preventing it from going beyond the 70% level.

The role of change management is to help

- Weaken the resisting forces
- Strengthen the driving forces

The Lewin/Schein three-step (unfreeze-move-refreeze) approach is one way to do this.

Figure 15.3: Illustration of the force field analysis model



Arrow sizes denote the relative strength of the forces.

Professional skills focus: Applying judgement

One of the skills tested in the Business Strategy and Technology exam is your ability to assess different stakeholder perspectives when evaluating options. When recommending change management strategies, use your judgement to identify the main sources of stakeholder resistance and the likely reactions of different stakeholder groups in response to any strategies you propose.

Interactive question 3: Gerrard, Dudek & Smicer

Gerrard, Dudek & Smicer are a small, long-established firm of solicitors. They have recently appointed Eva, a new business development manager. They have suggested that the firm introduce tablet computers with email and diary features as staff often don't know where to find each other. Secretaries also struggle to book meetings as the partners often keep their agendas with them, which has annoyed some key clients. However, the partners are extremely reluctant to consider such a departure from their current methods.

Requirement

Using Lewin's force field model, analyse the above situation and suggest how the new manager could bring about the change she wants.

See **Answer** at the end of this chapter.

When to use each style

Strategies: addressing resistance to change

Approach/style	*Situations used*	*Advantages*	*Disadvantages*
Education + Communication	*Where there is a lack of or inaccurate information and analysis.*	*If persuaded, people will help with the implementation of change.*	*Can be time consuming if many people are involved.*
Participation + Involvement	*Where initiators do not have all information to design change, and where others have power to resist.*	*Participants are committed to implementing change including their relevant contribution.*	*Can be very time consuming with possibly inappropriate changes made.*
Facilitation + Support	*Where resistance stems from adjustment problems.*	*Best approach for adjustment issues.*	*Can be time consuming, expensive and still fail.*
Negotiation + Agreement	*Where one group will lose out and has power to resist.*	*Can be an easy way to avoid major resistance.*	*Can be too expensive if it leads to general compliance.*
Manipulation + Co-optation	*Where other tactics won't work or are too costly.*	*Can be a relatively quick and inexpensive solution to resistance.*	*Can lead to future problems if people feel they have been manipulated.*
Explicit + Implicit Coercion	*Where speed is essential, and the change initiators possess considerable power.*	*It is speedy and can overcome any kind of resistance.*	*Can be risky if it leaves people angry at the initiators.*

◆ *Blue Ocean Strategy (BOS)*

- ***Definition***: *Simultaneous pursuit of differentiation and low cost to open new market space and make competition irrelevant.*
- ***Origin***: *Developed by W. Chan Kim and Renée Mauborgne (2005).*
- ***Core Idea***: *Instead of competing in "red oceans" (crowded markets), firms create "blue oceans" (new demand, uncontested space).*
- ***Tools & Frameworks***:
  - ***Strategy Canvas***: *Visualizes current competition and identifies gaps.*
  - ***Four Actions Framework***: *Eliminate, Reduce, Raise, Create.*
  - ***Value Innovation***: *Align innovation with utility, price, and cost positions.*

Examples:

- *Cirque du Soleil (reinvented circus by blending theater and circus arts).*
- *Nintendo Wii (focused on casual gamers instead of competing with high-performance consoles).*

Adaptive Strategies

- ***Definition**: Strategic initiatives that prioritize flexibility, responsiveness, and resilience in uncertain or rapidly changing environments.*
- ***Core Principles**:*
  - ***Continuous Learning**: Monitor market signals and adjust quickly.*
  - ***Experimentation**: Test small-scale initiatives before scaling.*
  - ***Resilience Building**: Diversify revenue streams and supply chains.*
  - ***Agility**: Empower teams to make fast decisions.*

***Examples**:*

- *Tech firms shifting to cloud-based services during digital transformation.*
- *Retailers adopting omnichannel strategies (online + offline integration).*
- *Businesses pivoting during COVID-19 (restaurants moving to delivery-first models).*

Comparison Table

Aspect	***Blue Ocean Strategy***	***Adaptive Strategy***
Goal	*Create new demand, uncontested markets*	*Survive & thrive in uncertainty*
Approach	*Innovation + cost efficiency*	*Flexibility + resilience*
Focus	*Long-term growth via differentiation*	*Short-term responsiveness & evolution*
Risk	*High (new market creation may fail)*	*Moderate (constant adjustment may dilute focus)*
Best Use Case	*Stable industries needing disruption*	*Volatile industries facing rapid change*

Strategic Fit

- ***Blue Ocean Strategy** is ideal when a company seeks breakthrough growth and wants to escape saturated competition.*
- ***Adaptive Strategy** is essential when markets are volatile, uncertain, complex, and ambiguous (VUCA).*
- ***Combined Approach**: Firms can use BOS to innovate, then apply adaptive strategies to sustain and evolve once the new market space matures.*

3.5 Communicating change

Stakeholder	Their needs	What they want to know	How to communicate
Shareholders	Reassurance	That there is well thought-through strategy How the strategy will benefit them	The press Financial statements AGM Website

The press	A good story	What is happening, the rationale, and whether the changes are under control	Briefings
Suppliers	Information	How the changes will affect their working relationship	Meetings face-to-face with major suppliers Letters/email to small suppliers
Customers	Motivation	That the service will continue uninterrupted	The press Advertisements
Senior managers	Acknowledgement and involvement	How they will be involved and opportunities in the new structure Reassurance over employment positions	One-to-one meetings
Staff	Help to adapt	Training and support Job security	Briefings One-to-one with line manager
Line managers	Involvement	Opportunities to be involved and opportunities to learn	Briefings One-to-one with senior manager/HR



Professional skills focus: Concluding, recommending and communicating

Change in an organisation needs to be carefully communicated since it can be an uncertain and potentially worrying time for stakeholders. Consider the style and tone of your communication as well as the format used to ensure the message is conveyed in the most appropriate manner.



Interactive question 4: Burgermania

Company background

Burgermania plc (hereafter Burgermania) is the world's largest fast-food chain. The company has 30,000 restaurants in over 100 countries, and has been committed to opening over 700 new restaurants each year in recent times. The company has benefited from consumers wanting quick and filling foodstuffs, and has met this need with its style of product for both adults and children. The company has also had many tie-in arrangements with sports and films, to encourage consumers to buy its products. The range is very limited, focussing mainly on burgers, fries and soft drinks. There are occasional variations in some countries. This approach has proved popular with consumers who have conservative tastes and are happy to know the food will be consistent wherever it is sold.

The company operates mainly via a combination of owned and franchised restaurants. Franchisees sign a 15-year deal in which they are given the right to use Burgermania trademarks, restaurant decor designs, formulae and specifications for menu items, use of Burgermania's inventory control systems, bookkeeping, accounting, marketing and the right to occupy the restaurant premises.

In return, the franchisee agrees to operate the business in accordance with Burgermania's standards of quality, service, cleanliness and value. The company is anxious to avoid any

negative publicity, so it employs 'secret shoppers' who randomly visit and assess restaurants on the above criteria. The company pays relatively low wages and has high staff turnover; this has prevented Burgermania being able to recruit high calibre staff, as the image of the jobs offered is poor.

Recent events

The company has had stagnant profits and sales growth in the last two years. The main reasons for this are increased health concerns in its major geographical markets. The company's products are high in both fat and carbohydrate content, and stories in the media about preventable obesity, heart disease, diabetes and cancer have caused turnover to falter. Furthermore, new style diets, whilst promoting meat consumption, have discouraged many consumers from eating bread or fries due to their high carbohydrate content. Initially Burgermania claimed that its products were nutritious and healthy, but this policy was deemed to be a failure as it did not reverse the stagnation of sales. Some high-profile legal claims against the company from consumers claiming that Burgermania products caused health problems have also had a negative impact on the company's image.

The company has decided at board level to respond to this deterioration in its financial performance by introducing and promoting a healthier menu, with options that include burgers but no burger buns, chicken and fish-based products, grilled rather than fried meats, salads, fruit and healthier drinks, as well as phasing out extra-large portions of food on its menu. The directors want to give more autonomy to individual restaurant managers to offer a wider range of products that will vary between different locations.

In addition, the board intend to change the company name to 'Eatwise' to move away from the association with burgers and fries. The cost of changing all the company, logos, livery, advertising campaigns and cooking equipment is expected to be massive. Staff will also have to be trained to use the new equipment and promote healthy eating to customers. The company is uncertain how the public and franchisees will respond to these changes in the business. Moreover, Burgermania is under pressure to deliver improved financial performance as its share price has underperformed that of the overall stock market by 20% in the last 18 months.

Requirements

4.1 Identify the type of change that Burgermania is considering.

4.2 Outline the potential barriers to the proposed changes from:

(a) Franchisees

(b) Restaurant staff

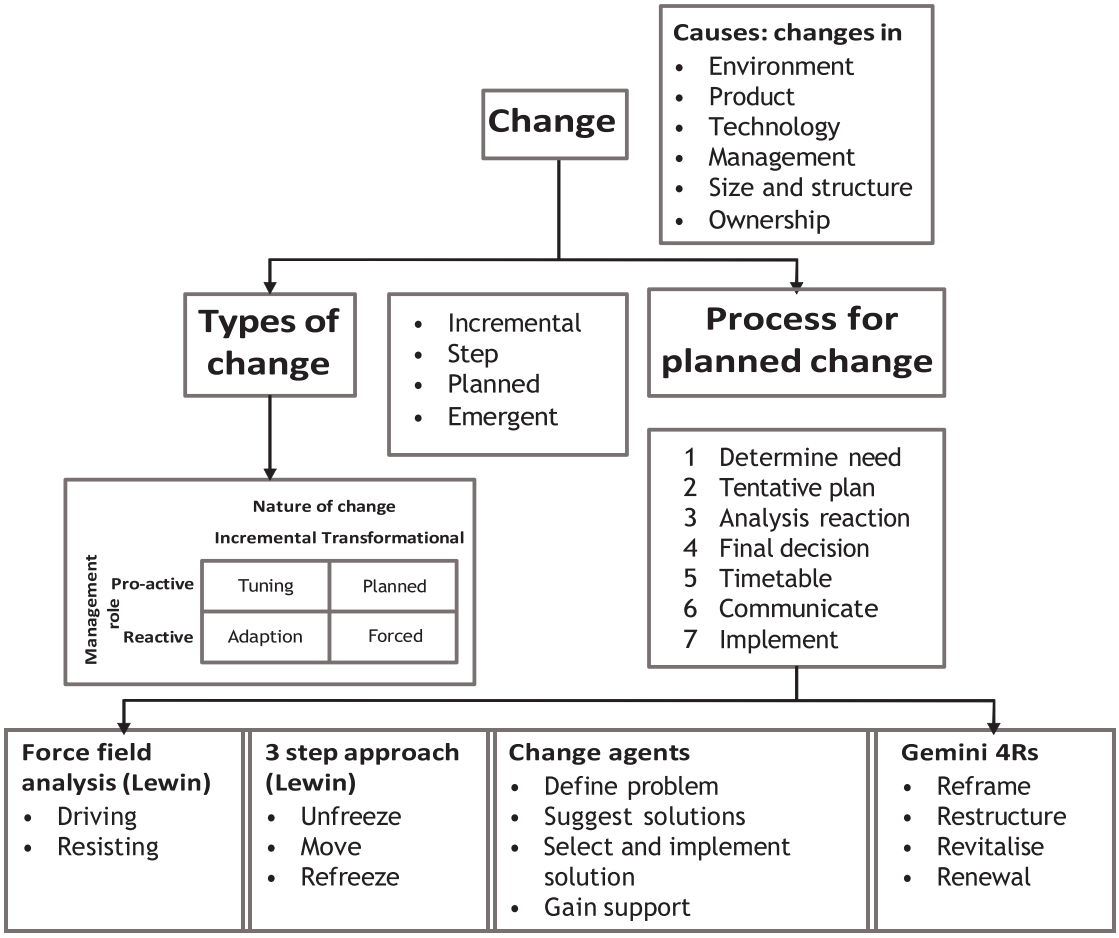
(c) Customers

4.3 Prepare a draft blueprint for submission to the board of directors of Burgermania that details how the company will deal with the changes of menus and name.

4.4 Suggest ways in which the company can communicate successfully the changes to customers, franchisees and shareholders.

See **Answer** at the end of this chapter.

Summary


Change
Causes: changes in
• Environment
• Product
• Technology
• Management
• Size and structure
• Ownership
Types of change
• Incremental
• Step
• Planned
• Emergent
Process for planned change
1 Determine need
2 Tentative plan
3 Analysis reaction
4 Final decision
5 Timetable
6 Communicate
7 Implement
Nature of change
Incremental Transformational
Management role
Pro-active
Reactive
Tuning
Planned
Adaption
Forced
Force field analysis (Lewin)
• Driving
• Resisting
3 step approach (Lewin)
• Unfreeze
• Move
• Refreeze
Change agents
• Define problem
• Suggest solutions
• Select and implement solution
• Gain support
Gemini 4Rs
• Reframe
• Restructure
• Revitalise
• Renewal

Further question practice

1 Knowledge diagnostic

Before you move on to question practice, confirm you are able to answer the following questions having studied this chapter. If not, you are advised to revisit the relevant learning from the topic indicated.

Confirm your learning	
(1)	Can you name four internal and four environmental triggers for change? (Topic 1)
(2)	What is Lewin's iceberg model and do you know how to apply it? (Topic 2)
(3)	What are the four stages in the Gemini 4R framework? (Topic 2)
(4)	Why do individuals resist change? (Topic 3)
(5)	What should managers consider when dealing with resistance to change? (Topic 3)
(6)	How can Lewin's Force Field analysis be used in managing change? (Topic 3)

2 Question practice

Aim to complete all self-test questions at the end of this chapter. The following self-test questions are helpful to further topic understanding and guide skills application before you proceed.

Question	Learning benefit from attempting this question
Herring plc	Using the Johnson, Scholes and Whittington model identify the type of change and the best way to manage it. Remember to explain and justify the type of change you select, with reference to both axes on the matrix.
Freddis plc	Consider the cultural barriers to change that are likely to exist. This is a short scenario, so use your common sense and experiences to generate ideas.

Once you have completed these self-test questions, it is beneficial to attempt the following questions from the Question Bank for this module. These questions have been selected to introduce exam style scenarios that will help you improve your knowledge application and professional skills development before you start the next chapter.

Question	Learning benefit from attempting this question
Vision Bank (Sept 2018)	Part 2 requires you to identify and explain change management issues. Ensure you apply your answer to the scenario and consider the practicality of your suggestions. Incorporate learned technical knowledge, but generate ideas using your experience and wider business reading.

Refer back to the learning in this chapter for any questions which you did not answer correctly or where the suggested solution has not provided sufficient explanation to answer all your queries. Once you have attempted these questions, you can continue your studies by moving on to the next chapter.

Self-test question

Answer the following questions.

1 List the three levels at which organisational development/change efforts may focus.

2 'An organisation enjoying good profitability should not consider change.' True or false?

3 List the three stages of Lewin's model for changing the behaviour of individuals.

4 What is meant by 'refreezing'?

5 List three possible problems with coercive change.

6 What causes resistance to change?

7 What is force field analysis?

8 Fill in the following blanks.

(a) C _____ , (b) c _____ and (c) c _____ play important roles when introducing change to an organisation.

9 In the Lewin/Schein three-stage change management model, what general principle does management use to embed the desirable new behaviour?

10 List four areas that a project manager should be skilled in.

11 Herring plc

Herring plc is in crisis. Major changes in its culture, structure and working practices are needed if it is to survive. The effects of any changes made are highly uncertain. Identify the type of change involved for Herring plc and the best approach to managing such change.

12 Freddis plc

Freddis plc has sold its single product in a single market segment for many years. It's only competitor has just brought out a technically superior and far cheaper version of the product. Outline the cultural barriers to change which are likely to be present in Freddis plc.

13 Plant Ltd

Plant Ltd has just finished implementing a major cost cutting project which involved automating a number of previously paper processes across the whole company. The project did not go as well as had been hoped and at one point the telephone ordering system was out of action for over a week as a result of unexpected problems.

Requirement

Explain how this type of change could be categorised. What kind of investigation is likely to be instigated as a result of the problems that occurred?

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

(a) **International car maker**

Forced: The car maker is reacting to external competitive and environmental changes. Such a change may result in significant changes in the way the company is structured.

(b) **Supermarket retailer**

Planned: It is proactive, aimed at seizing an opportunity. Furthermore, the change is anticipated to have a far reaching impact on the existing business model. The large size of the foreign market will change the character of the retailer in terms of size but also its global profile.

Answer to Interactive question 2

Reframing

The major reframing is the term 'commercial business' which implies making profits as the primary goal, or at least providing as good a service as competitors in a de-regulated environment.

The ensuing strong emphasis on profits and termination of loss-making activities underlines this.

The challenge for management would be to make this a goal that staff wished to support. The government's decision in 20X8 to sell off National Mail appears to have been well received by employees, with 15,000 choosing to purchase shares. This change in the ownership structure of the National Mail is likely to be positive in aligning the aims of the company with its employees.

Restructuring

Although not obviously stated in the 20X0 press release, references to automated sorting suggest changes were required here. The points raised from the 20X8/X9 annual report indicate that National Mail is attempting to automate and enhance its current operations in response to changes in the demand for its services.

Revitalising

In 20X0, Jennifer Smith recognised the challenge of adapting to the new competitive environment: 'Competing successfully in an open mail market is going to be even more difficult. We've a mountain to climb and we've only reached the base camp.' The 20X8/X9 annual report explains how National Mail is looking to increase its use of technology to respond to the change in the nature of the marketplace and the increasing competition.

Renewal

The statements from 20X0 are clearly inclusive and emphasise the importance of gaining staff commitment: 'with the dedication and support of our people.' The offer of shares to employees as part of the 20X8 flotation might be seen as a further attempt to gain support from staff.

Answer to Interactive question 3

Driving forces for change: Technological developments raising customer expectation, irritated customers, new innovative staff member

Restraining forces: Partners' habits and possible fear of the unknown, possible concerns over expense

Eva should strengthen the driving forces by stressing to the partners the level of clients' concerns and gain the support of the secretaries in encouraging the partners to consider the use of the tablet computers.

The restraining forces should be weakened by producing costings, and explaining clearly how the tablets would work and what benefits would arise.

Answer to Interactive question 4

4.1 Type of change

Burgermania is undertaking a change in the business that would be deemed revolutionary in terms of impact on the business. Both the process and the underlying paradigm of the company will have to change. The retraining and rebranding of the business suggests a low degree of predictability, as it will be difficult to anticipate the reaction of stakeholders, especially customers, to the proposed changes.

4.2 Barriers to change

(a) **Franchises**

- Franchisees will fear that new system is too radical and may have detrimental impact on revenues of restaurants.
- They may fear that costs of changing name and livery of the restaurants will be recovered by Burgermania in the form of higher franchise fees.
- Franchisees will be reluctant to change to a new system of delivering food if they are happy with current products and procedures.
- They may lack initiative when it comes to introducing new products that they can add to the menu.

(b) **Restaurant staff**

- Staff in fast food restaurants tend to be poorly paid and have short periods of employment. Therefore, they may be reluctant to be retrained or adopt a pro-active policy with customers in terms of promoting the new healthier food lines.
- Staff may feel that if Burgermania is suffering then jobs may be lost – likely to cause demotivation.
- Staff may fear that management will use the change as a means of introducing new working practices that could reduce income, overtime, bonuses etc.
- Staff may lack the skills to adapt to the new nature of the business.
- The decision is likely to be enforced by coercion if necessary due to Burgermania's poor recent results, potential for conflict and resentment by staff.

(c) **Customers**

- Customers are more concerned with being fed quickly than with the quality of what they eat, so may choose to go elsewhere. Many people have conservative tastes and so will be reluctant to try the new products on offer.
- They may feel that the new products are being forced on them and again vote with their feet by eating at other fast food restaurants.

4.3 **Draft blueprint**

To: Board of directors, Burgermania plc

From: B. Cowie, consultant

Date: Today

Subject: Blueprint for menu and name changes

Action required

- Introduction of healthier menus and local autonomy
- Launch of new name/brand for the company

Purpose

The objective of the changes is as follows

- To reverse the decline in the company's performance.
- To adapt to changes in consumer demands and tastes.
- To promote the company as a family-friendly and healthy eating place.

Timing of changes

- Given the nature of the company's products and positioning in the market, a national change will have to be launched if it wants to maximise publicity for the changes. The company could try introducing the new menus on a trial basis in selected restaurants.
- The transformation will be very time-consuming given the global nature of the company and its products.

Responsibilities

A committee needs to be formed to co-ordinate the changes, headed by a senior manager/director.

Evaluation

- Market research needs to be undertaken to establish public reaction to both the menu and name changes.
- Performance data can be produced to determine the popularity of the new food products on a trial basis in selected stores.

4.4 **Communication of changes**

- Franchisees could be invited to a conference to see the new 'Eatwise' logos and product range.
- A major PR campaign, with TV, radio and press adverts should take place to coincide with the changes to advise customers. Perhaps special promotions relating to the new products could be used to encourage customers to try them out.
- Shareholders could be briefed via email and the company's website, together with press releases in the financial media.

Answers to Self-test questions

1 Individual; organisation structure and systems; organisation climate and interpersonal style

2 False. Organisations should embrace change, and not wait until the obsolescence of their current practices leads to crisis.

3 Unfreeze, change (or move) and refreeze

4 The final stage of the Lewin/Schein three-stage approach in which the desired new behaviour is consolidated.

5 Underestimating resistance; not coercing hard enough or long enough; failure to ensure implementation

6 Attitudes, beliefs, loyalties, habits and norms, politics

7 Lewin's analysis of the driving and restraining forces which underlie any group or organisational equilibrium.

8 (a) C ommitment , (b) c o-ordination , (c) c ommunication play important roles when introducing change to an organisation.

9 Reinforcement – both positive and negative

10 Four of:
- Leadership
- Team building
- Organisation
- Communication
- Technical expertise
- Personal qualities

11 Herring plc

Revolutionary change

Approach – emergent (ie, flexibility is essential)

12 Freddis plc

Structural inertia – systems and procedures are likely to have been in place for many years.

Group inertia – production and marketing departments, for example, may resist change (probably used to doing things in the same way for many years).

Power structures – lack of pressure to change in the past may have resulted in clear lines of decision- making authority and/or control over resources.

13 Plant Ltd

The change could be categorised as reconstruction – it affects the entire organisation, but does not appear to require a paradigm shift.

A post-project audit would be required. It would be carried out by an independent team who would review all elements of the project process in detail to identify why the problems arose.

Chapter 16
Ethics, sustainability and ESG

Introduction

Learning outcomes

Syllabus links

Examination context

Chapter study guidance

Learning topics

1 Strategy and ethics

2 Marketing and ethics

3 Ethical issues in manufacturing and procurement

4 Ethical issues in the use of data

5 The concepts of sustainability and ESG

6 Sustainable development

7 Social responsibility

8 Monitoring and measuring sustainability

9 Developing an approach to ethics requirements in the exam

Summary

Further question practice

Self-test questions

Answers to Interactive questions

Answers to Self-test questions



Introduction

Learning outcomes

- Analyse the external factors which may impact upon an organisation's performance and position, evaluating significant issues in areas such as:
  - Natural capital and climate change
  - Macroeconomic forces
  - International trade, financial systems and global economic factors
  - Government policies
  - Its industry and markets, including competition
  - Cultural environment
  - Stakeholder, ecosystem and participant impact
  - Markets for finance, labour and other resources
  - Supply chain factors
  - Technology developments, including those relating to automation and intelligent systems
- Explain and evaluate the ethical and business trust factors to be considered in determining the scope and nature of an organisation's objectives and its strategic analysis, giving appropriate consideration to the legitimate interests of all stakeholder groups and the public interest.
- Explain and demonstrate how an organisation compares and selects from competing strategies in order to maximise the achievement of its key objectives.
- Evaluate the ethical implications of an organisation's strategies and operations for the organisation and for individuals including the accountant in business and others.
- Explain and evaluate the ethical, sustainability, ESG and public interest implications of how an organisation chooses to implement and modify its strategies, suggesting appropriate courses of action to resolve ethical dilemmas that may arise.

Syllabus links

Some coverage was given in your Business Technology and Finance studies to the policies that can be used to help ensure an organisation acts ethically, in addition to issues of sustainability and environmental, social and governance (ESG) issues. The knowledge from Business Technology and Finance is developed further in Business Strategy and Technology and applied in the context of different business scenarios. At Advanced Stage, sustainability and ESG are integrated into the technical areas of the syllabus.

This chapter also considers how best to apply knowledge of ethics, sustainability and ESG in the exam, in the context of a specific scenario. This is an approach which will help you in developing the necessary skills to make ethical judgements at Advanced Stage.

Examination context

Ethics

At least one of the exam scenarios is likely to include an ethical issue or dilemma, possibly in the context of the ethics of an organisation or in relation to a professional accountant working in business. Between 7 and 10 marks of the exam will be allocated to the appropriate discussion of such ethical issues, but they are often not clear cut. The important skills here are for you to develop a balanced argument, use appropriate ethical language and have regard to any relevant professional principles.

Scenarios might include topics such as:

- Conflict between the accountant's professional obligations and their responsibilities to the organisation

- Lack of professional independence (eg, personal financial interest in business proposals)
- Conflicts of interest among stakeholders
- Attempts to intentionally mislead key stakeholders (by disclosure or non-disclosure of information)
- Doubtful accounting or commercial business practice
- Facilitation of unethical strategies
- Inappropriate pressure to achieve a result
- Actions contrary to law, regulation and/or technical and professional standards
- The ethical usage of data or technology

ESG and sustainability

Scenarios may also consider how organisations create, embed and monitor a sustainability strategy, an environmental, social and governance (ESG) strategy or a corporate social responsibility (CSR) strategy. CSR is a management concept whereby companies integrate social and environmental concerns in their business operations and interactions with their stakeholders. The term CSR is used less often in recent times as 'sustainability' and 'ESG' tend to be more favoured and form the central pillar of the ACA framework for sustainability.

Areas that might be examined in the context of ESG and sustainability include:

- The involvement of key stakeholders in identifying objectives, conflicts of interest, and communication with such stakeholders
- The development of appropriate strategy and governance structures to implement ESG and sustainability
- Determination of appropriate KPIs to measure performance of the ESG strategy

The examiners have noted that a significant number of candidates attempting the Business Strategy and Technology examination in recent sittings appear to specifically lack knowledge surrounding the concept of sustainability and sustainable enterprise.

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this chapter.

Topic	Practical significance	Study approach	Exam approach	Interactive questions
1	**Strategy and ethics** The practice of accountancy grew from the need for the users of capital to be accountable to the owners of the capital. Today's organisations need to be accountable, in varying degrees, to a wider range of stakeholders. In deciding on strategies an organisation must take account of ethical issues such as its impact on the natural environment; its	**Approach** You are required to know the five fundamental principles outlined in the ICAEW *Code of Ethics* (2020). Also ensure you are familiar with the threats to the fundamental principles and examples of safeguards. **Stop and think** What considerations of ESG or ethical issues might get in	At least one exam scenario will include some kind of ethical dilemma. Ensure you develop a balanced argument making reference to any relevant professional principles.	**IQ1: Why have ethical standards?** This question asks why professional bodies, such as the ICAEW and the FRC have ethical codes. **IQ2: Professional judgement and scepticism** This question requires you to use professional judgement in considering the ethical threats present in the scenario

CHAPTER 16

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	human resource management policies; the ethical use of data; the extent of its charitable support; and how far it goes beyond the minimum standards required by laws and regulations.	the way of organisations doing just what they like?		information. You are also required to suggest practical solutions to the threats. **IQ3: Ethical dilemmas** Consider the ethical threats present in the question and suggest how they can be reduced. **IQ4: Pay to stay** Evaluate the ethics of a decision proposed by your line manager.
2	**Marketing and ethics** Marketing is an attempt to sell products to customers, irrespective of need. It therefore raises many ethical issues around whether it wastes the world's resources and whether it pressurises consumers into making unnecessary purchases.	**Approach** Read through the topic and assess the ethical arguments in favour of and against marketing. **Stop and think** Is it ethical to market new drugs and medical equipment to doctors and hospitals? Assess both sides of the argument.	When answering ethics questions ensure you consider both sides of the argument and do not simply state your personal opinion. Organisations have to meet the demands of many stakeholders, and therefore have a duty to shareholders to deliver a return, whilst acting ethically.	**IQ5: Socially responsible marketing** This question considers the ethical and social responsibility issues facing marketers in food manufacturing.
3	**Ethical issues in manufacturing and procurement** Production of goods should not cause harm to workers, customers, the environment or society. Procurement should be sustainable, and suppliers should be treated fairly.	**Approach** Read quickly through this short section to understand the ethical issues surrounding manufacturing and procurement. **Stop and think** How does a weapons manufacturer reconcile ethics and the production of armaments?	An exam question could look at the ethical issues present in manufacturing organisations. Whether or not you personally agree with the product or production methods, strong answers will explore both sides of the ethical argument.	
4	**Ethical issues in the use of data**	**Approach** The use, storage	Whenever data is being used by an	

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	More than ever data is being used by organisations to understand their customers and the impact of their strategies. However, the use and storage of personal data gives rise to a number of ethical issues.	and protection of data has been discussed in detail in earlier chapters. Read quickly through this short reminder topic. **Stop and think** Many websites now use analytics cookies to anonymously review and analyse how individuals use websites. Are you happy to accept analytics cookies? What ethical issues does this practice raise?	organisation consider whether the company is accessing, using and storing it ethically.	
5	**The concepts of sustainability and ESG** Sustainability concerns maintaining, rather than depleting, the world's resources. Environmental, social and governance (ESG) examines how organisations can approach sustainability-related issues.	**Approach** Read through the topic to understand the difference between sustainability and ESG. Look at the examples of social, environmental and economic sustainability. **Stop and think** Would you consider the organisation you work for to be sustainable?	Exam scenarios may consider how organisations create, embed and monitor an ESG strategy. Areas that might be examined include: the involvement of key stakeholders in identifying objectives and conflicts of interest; the development of appropriate strategy and governance structures to implement an ESG strategy; determination of appropriate KPIs to measure performance of the ESG strategy.	
6	**Sustainable development** Governments, businesses and other global organisations are attempting to promote sustainable development through, amongst others, corporate policies,	**Approach** This short section outlines the different approaches being used to encourage sustainable development. **Stop and think** What are the	When recommending a strategy in the exam, consider the implications for sustainability.	

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	voluntary codes and taxes or subsidies.	strategic benefits of an organisation developing a sustainable business model?		
7	**Social responsibility** The social element of ESG describes the actions an organisation may take to develop a culture of ethical, responsible and sustainable behaviour with regard to all stakeholders.	**Approach** Understand the benefits of adopting the social element of ESG as well as the scope of social factors to consider in a business. **Stop and think** Consider an organisation you are familiar with. Can you identify any socially responsible actions they are taking? How is the ESG strategy congruent with delivering value to shareholders?	Consider the stakeholders present in the exam scenario and assess their needs. The scope of social factors in the ESG strategy varies depending on which stakeholders are considered to be most strategically relevant.	**IQ6: Being socially responsible** This question examines the short- and long-term implications of an organisation acting in a socially responsible manner. **IQ7: Skin Care company** This scenario-based question looks at ethical and ESG issues for a skin care company.
8	**Monitoring and measuring sustainability** Once a sustainable development strategy has been instigated, it needs to be monitored regularly to assess progress against initial objectives.	**Approach** The worked examples show how real companies are monitoring and measuring sustainability. Understand the GRI and A4S approaches to sustainability reporting. **Stop and think** If an organisation is acting in a sustainable manner why do they need to report on this?	A scenario may consider how an organisation monitors an ESG strategy so ensure you are familiar with the various sustainability reporting approaches.	**IQ8: Ashdene homes** This exam standard question looks at the implications of poor ethical principles as well as assessing why a company may wish to act in a socially responsive way.
9	**Developing an approach to ethics requirements in the exam** Ethics, by its nature, is	**Approach** Read through the recommended approach. Apply it in the exam by	Ethical issues in the exam may not be directly highlighted with a separate question requirement	

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	a grey area and there is no one right answer. It is therefore important to analyse all arguments carefully whilst retaining a logical structure and flow to your response.	reading through the worked example. **Stop and think** What is the problem in answering all ethics- based questions using a standardised approach?	but instead ethical issues may be part of a broader requirement. Ethics and ethical issues are ever present in business and the same applies to the exam.	

Once you have worked through this guidance, you are ready to attempt the further question practice included at the end of this chapter.

1 Strategy and ethics

Section overview

- Morals and ethics involve doing the 'right' thing. This may not always be the same as 'best for the individual manager' or 'best for the organisation as a whole.'
- Some ethical imperatives may be enshrined in laws but ethics and law are not the same thing.
- Ethical issues exist at the level of the individual, the business and, at its widest, ESG strategy.
- A professional accountant is required to comply with the five fundamental principles of the ICAB *Code of Ethics* (2020).
- The Code provides guidance for the professional accountant working in business on how to identify, evaluate and address any threats to those fundamental principles and the safeguards to implement.
- The desire by management to act ethically affects the scope of strategies adopted but also requires that management keep an eye on the ethical consequences of its operations.
- Section 8 of this chapter gives guidance on how to approach ethics in the context of an examination question.

1.1 Ethics and morals

The meanings of the words 'ethics' and 'morals' are intermingled and difficult to distinguish. For example, the *Concise Oxford Dictionary* offers the following two definitions:

- **Morals** are 'standards of behaviour or principles of right and wrong.'
- **Ethics** are 'the moral principles governing or influencing conduct.'

Morals or ethics will also differ depending on the beliefs, value systems and norms of a particular society or culture. So, ethics might be considered as:

Definition

Ethics: A system of behaviour which is deemed acceptable in the society or context under consideration.

Professional skills focus: Applying judgement

Ethical issues are often not clear cut and there is unlikely to be a 'correct' answer to dealing with the issue presented in the scenario. Instead, you should present a balanced argument to demonstrate that you understand the concerns held by various stakeholder groups.

Another area in which ethics can be invoked is environmental, social and governance (ESG) which is discussed in more detail later in this chapter. For present purposes, the field can be simplified by suggesting that business ethics exist at three levels.

(a) **Personal ethical behaviour:** This relates to the way you as an individual conduct yourself. Bad behaviour would include bullying, stealing, discrimination against a colleague and giving away business secrets to a rival.

(b) **Business ethics:** This is the way a firm as a whole behaves. Bad conduct here would include offering bribes to win contracts, distorting the accounts, victimisation or discrimination against certain workers and telling lies to regulators.

(c) **ESG:** This is the belief that a firm owes a responsibility to society and its wider stakeholders, as well as to shareholders. Examples of bad behaviour would include pollution, mass redundancies and dangerous products.

1.2 Ethical stance of corporations

An organisation can adopt a range of ethical stances:

- Meet minimum legal obligations and concentrate on short-term shareholder interests.
- Recognise that long-term shareholder wealth may be increased by well-managed relationships with other stakeholders (corporate governance approach which was discussed in the chapter, 'Strategy and structure').
- Go beyond minimum legal and corporate governance obligations to include explicitly the interests of other stakeholders in setting mission, objectives and strategy. In this context, issues such as environmental protection, sustainability of resources, selling arms to tyrannical regimes, paying bribes to secure contracts, using child labour etc, would be considered.
- Public sector organisations, charities, etc, where the interests of shareholders are not relevant.

The ethical stance taken by an organisation is often reflected in its mission statement.

Context example: Starbucks

Coffee chain, Starbucks, places a strong emphasis on its ethical credentials. This is best illustrated by the company's mission statement which was posted on its website in 2019:

Our mission

'To inspire and nurture the human spirit – one person, one cup and one neighbourhood at a time.'

Starbucks underpins its mission by stating its corporate values which place an emphasis on its customers and alludes to the important role that its employees play in its operations. According to the company's website, employees are referred to as 'partners.' Starbucks values are as follows:

Our values

'With our partners, our coffee and our customers at our core, we live these values:

- Creating a culture of warmth and belonging, where everyone is welcome
- Acting with courage, challenging the status quo and finding new ways to grow our company and each other
- Being present, connecting with transparency, dignity and respect
- Delivering our very best in all we do, holding ourselves accountable for results

We are performance driven, through the lens of humanity.'

(**Source**: Starbucks (2019) *Our mission.* [Online] Available from: www.starbucks.co.uk/about-us/company-information/mission-statement [Accessed 18 July 2023])

1.3 Business trust

Definition

Business trust: Exists when stakeholders act as willing and positive participants in their dealings with an organisation because their experience of the organisation, and its reputation, support open and mutually beneficial relations. Loss by an organisation of business trust can have multiple and serious consequences.

Context example: The importance of building trust

Business success is rooted in how stakeholders view an organisation. If the company is trusted, suppliers are more willing to deal with the organisation since they can be more certain of a positive trading experience and of getting paid on time. Building trust with customers leads to loyalty and brand strength.

According to the 2019 Edelman Trust Barometer special report, 'In Brands we trust', 67% of respondents agreed with the statement, 'A good reputation may get me to try a product but unless I come to trust the company behind the product, I will soon stop buying it.' Trust in a company can also have a hugely positive impact on the workforce. According to a study by the Harvard Business review, employees at 'high-trust companies report: 74% less stress, 106% more energy at work, 50% higher productivity, 13% fewer sick days and 76% more engagement.' It is therefore clear that business trust is an important component in operating as a successful business.

Sources:

- Edelman (2019) *In Brands We Trust*. [Online] Available from: https://www.edelman.com/sites/g/files/aatuss191/files/2019-07/2019_edelman_trust_barometer_special_report_in_brands_we_trust.pdf
- HBR (2017) *The Neuroscience of Trust*. [Online] Available from: https://hbr.org/2017/01/the- neuroscience-of-trust

1.4 Regulating ethical behaviour

Ethical business regulation operates in two ways:

(a) **Forbidding or constraining certain types of conduct or decisions** – eg, most organisations have regulations forbidding ethically inappropriate use of their IT systems. Similarly many will forbid the offering or taking of inducements to secure contracts.

(b) **Disclosure of certain facts or decisions** – eg, because the board sets its own pay they disclose it, and sometimes the reasons behind the awards, to shareholders in final accounts.

In other Professional Stage exams, you will have been introduced to the following codes potentially binding on you as a trainee Chartered Accountant.

The Financial Reporting Council's (2019) Ethical Standard

You should recall from your Audit and Assurance studies that auditors are expected to comply with the Financial Reporting Council's (2019) Ethical Standard, which covers:

- Introduction and scope
- Part A: *Overarching Principles and Supporting Ethical Provisions*
- Part B:
  - Section 1: *General Requirements and Guidance*

- Section 2: *Financial, Business, Employment and Personal Relationships*
- Section 3: *Long Association with Engagements and with Entities Relevant to Engagements*
- Section 4: *Fees, Remuneration and Evaluation Policies, Gifts and Hospitality, Litigation*
- Section 5*: Non-Audit / Additional Services*
- Section 6: *Provisions Available for Audits of Small Entities*

The ICAB Code of Ethics for members



Definition

Code of ethics: A codification of principles or rules to help prevent wrong-doing and to avoid conflicts of interest and to promote honest and ethical conduct among staff in an organisation.

The International Ethics Standards Board (IESBA) is one of the standard setting boards of the International Federation of Accountants (IFAC). The IESBA issues a *Code of Ethics for Professional Accountants*, with which you should be familiar from your Audit and Assurance studies. As a member of IFAC the ICAEW amends the Code of Ethics to align with IESBA code. Additional material is only included where the IESBA code is considered to offer insufficient discussion or detail for the purposes of ICAEW members. To avoid the disruption that annual updates bring, ICAEW accumulates any changes made to the IESBA code and only re-issues the ICAEW code when changes are substantive or significant improvements have been made. Consequently, the ICAEW *Code of Ethics* approved from January 2020 is the first revision to the code since 2011. Compliance with the ICAEW code ensures compliance with the IESBA Code.

The following sections give specific consideration to some of the key provisions of the ICAEW *Code of Ethics*.

The ICAEW *Code of Ethics* (2020) s. 110.1, sets out the following five fundamental principles:

(a) **Integrity** – to be straightforward and honest in business and professional relationships

(b) **Objectivity** – to not allow bias, conflict of interest or influence of others to override professional or business judgement

(c) **Professional competence and due care** – to maintain professional knowledge and skill at the level required to ensure that a client or employer receives competent professional services, based on current developments in practice, legislation and techniques, and to act diligently and in accordance with applicable technical and professional standards

(d) **Confidentiality** – to respect the confidentiality of information acquired as a result of professional and business relationships and, therefore, not disclose any such information to third parties without proper and specific authority, unless there is a legal or professional right or duty to disclose, nor use the information for the personal advantage of the professional accountant or third parties

(e) **Professional behaviour** – to comply with laws and regulations and not to discredit the profession



Interactive question 1: Why have ethical principles?

The Codes of the ICAEW and of the FRC above seek to regulate the behaviour of accountants. Why do the bodies have these?

See **Answer** at the end of this chapter.

In the next section, we explore in more detail how the ICAEW Code applies to both professional accountants working in business and those working in public practice.

1.5 ICAB Code of Ethics for Professional accountants working in business

Investors, creditors, employers, the business community, government and the public may **rely** on the work of professional accountants in business in the context of:

- Preparation and reporting of financial and other information
- Providing effective financial management
- Competent advice on a variety of business-related matters

It is important that professional accountants do not allow pressure from others, nor place pressure on others, to breach compliance with the fundamental principles (s270). If an accountant feels under any such pressure, they are advised to escalate the issue to higher levels of management including their superior, the auditors or even governance committees, depending on the severity.

1.5.1 Threats

Section 120.6 of the Code (2020) outlines areas where there may be **conflict** for the professional accountant between furthering the legitimate aims of their organisation and their absolute duty to comply with the fundamental principles. According to s. 120.6 A3, threats to compliance with the fundamental principles fall into one or more of the following categories:

(a) **Self-interest** – financial or other interests will inappropriately influence a professional accountant's judgement or behaviour.

(b) **Self-review** – a professional accountant will not appropriately evaluate the results of a previous judgment made, or an activity performed by the accountant, or by another individual within the accountant's firm or employing organisation, on which the accountant will rely when forming a judgment as part of performing a current activity.

(c) **Advocacy** – a professional accountant will promote a client's or employing organisation's position to the point that the accountant's objectivity is compromised.

(d) **Familiarity** – due to a long or close relationship with a client, or employing organisation, a professional accountant will be too sympathetic to their interests or too accepting of their work.

(e) **Intimidation** – a professional accountant will be deterred from acting objectively because of actual or perceived pressures, including attempts to exercise undue influence over the accountant.

The ICAEW *Code of Ethics* 2020 requires professional accountants to apply the Conceptual Framework to identify, evaluate and address threats to compliance with the fundamental principles. In other words, the accountant should exercise professional judgement when making decisions, ensuring that they remain alert for new information and changes to facts as well as using the reasonable and informed third party test. This test considers whether another party, using the same information that the accountant is privy to, would come to the same conclusion regarding a potential breach of fundamental principles (s. 120.5 A4).

1.5.2 Safeguards

To comply with the Code, professional accountants are required to consider whether their actions or relationships might constitute **threats** to their adherence to the fundamental principles and where these are significant, to implement **safeguards.**

These safeguards might be generic, created by the profession or regulation or developed in the working environment by the individual or their organisation.

If effective safeguards are not possible, they are required to **refrain** from the action or relationship in question.

Examples of safeguards in the **work environment** which might be applied to overcome these threats include:

- The employing organisation's systems of corporate oversight or other oversight structures
- The employing organisation's ethics and conduct programs
- Recruitment procedures in the employing organisation emphasising the importance of employing high calibre competent staff
- Strong internal controls
- Appropriate disciplinary processes
- Leadership that stresses the importance of ethical behaviour and the expectation that employees will act in an ethical manner
- Policies and procedures to implement and monitor the quality of employee performance
- Timely communication of the employing organisation's policies and procedures, including any changes to them, to all employees, and appropriate training and education on such policies and procedures
- Policies and procedures to empower and encourage employees to communicate to senior levels within the employing organisation any ethical issues that concern them without fear of retribution
- Consultation with another appropriate professional accountant

In addition to formal safeguards, professional accountants are expected to encourage and promote an ethics-based culture (s. 200.5 A3), although the ability to do this is heavily influenced by the accountant's position and seniority in the organisation. Examples of possible actions to introduce, implement and oversee an ethics-based culture include:

- Ethics education and training programs
- Ethics and whistle-blowing policies
- Policies and procedures designed to prevent non-compliance with laws and regulations

Ethical threats and safeguards

Ethical threat	*Safeguard*
Conflict between requirements of the employer and the fundamental principles. *For example, acting contrary to laws or regulations or against professional or technical standards.* *(Intimidation threat)*	• *Obtaining advice from the employer, professional organisation or professional advisor.* • *The employer providing a formal dispute resolution process.* • *Legal advice.*
Preparation and reporting on information *Accountants need to prepare/report on information fairly, objectively and honestly. However, the accountant may be pressurised to provide misleading information. (Intimidation threat)*	• *Consultation with superiors in the employing company.* • *Consultation with those charged with governance.* • *Consultation with the relevant professional body.*
Having sufficient expertise *Accountants need to be honest in stating their level of expertise – and not mislead employers by implying they have more expertise than they actually possess.*	• *Obtaining additional advice/training.* • *Negotiating more time for duties.* • *Obtaining assistance from someone with relevant expertise.*

Ethical threat	*Safeguard*
Threats that may result in lack of expertise include time pressure to carry out duties, being provided with inadequate information or having insufficient experience.	
Financial interests *Situations where an accountant or close family member has financial interests in the employing company. Examples include the accountant being paid a bonus based on the financial statement results which they are preparing, or holding share options in the company. (Self-interest threat)*	• *Remuneration being determined by other members of management.* • *Disclosure of relevant interests to those charged with governance.* • *Consultation with superiors or relevant professional body.*
Inducements – receiving offers * Refers to incentives being offered to encourage unethical behaviour. Inducements may include gifts, hospitality, preferential treatment, or inappropriate appeals to loyalty. Objectivity and/or confidentiality may be threatened by such inducements. (Self-interest threat)*	- *Do not accept the inducement! - Inform relevant third parties such as senior management and professional association (normally after taking legal advice).*
Inducements – giving offers * Refers to accountants being pressurised to provide inducements to junior members of staff to influence a decision or obtain confidential information. (Intimidation threat)*	- *Do not offer the inducement! If necessary, follow the conflict resolution process outlined in the next section.*
Confidential information * Accountants should keep information about their employing company confidential unless there is a right or obligation to disclose, or they have received authorisation from the client. However, the accountant may be under pressure to disclose this information as a result of compliance with legal processes such as anti-money laundering/terrorism – in this situation there is a conflict between confidentiality and the need for disclosure.*	- *Disclose information in compliance with relevant statutory requirements, e.g. money laundering regulations.*
Whistleblowing * Situations where the accountant needs to consider disclosing information, where ethical rules have been broken by the client.*	- *Follow the disclosure provisions of the employer, e.g. report to those responsible for governance. Otherwise disclosure should be based on assessment of: legal obligations, whether members of the public will be adversely affected, gravity of the matter, likelihood of repetition, reliability of the information, reasons why employer does not want to disclose.*

1.5.3 Professional judgement

All ethical dilemmas must be dealt with using professional judgement (s. 120.5 A3). This involves the application of relevant training, skills, professional knowledge and experience when applying the Conceptual Framework to make decisions about potential courses of action and their suitability in the specific circumstances. In exercising professional judgement an accountant might consider, amongst other matters, whether:

- relevant information might be missing from the facts known to the accountant;
- there is inconsistency between the known facts and the accountant's expectations;
- the accountant's expertise and experience are sufficient to reach a conclusion (is there a need to consult experts on the issues involved?);
- the information provides a reasonable basis on which to reach a conclusion; and
- the accountant's own bias might be affecting the exercise of professional judgement

1.5.4 Action required in unethical circumstances

In circumstances where a professional accountant in business believes that unethical behaviour or actions by others will continue to occur within the employing organisation, they should consider seeking **legal advice**.
In extreme situations where all available safeguards have been exhausted and it is not possible to reduce the threat to an acceptable level, a professional accountant in business may conclude that it is appropriate to **disassociate** from the task and/or **resign** from the employing organisation.



Interactive question 2: Professional judgement and scepticism

GSA is a listed pharmaceutical company that operates across different countries but has its headquarters in a European country. You are a qualified accountant and now work as a consultant for a small professional firm which has been asked to provide tax advice to GSA.

The finance director of GSA studied with the engagement partner so is happy to give them the work, especially as they believe that the firm's experience in tax advice can help identify any future tax loopholes.

The engagement is very high-profile for your firm, so it has been offered at a discount to the firm's normal fees, although you understand that the GSA board has indicated that it expects fees to stay low as it wants your firm to act as both internal and external auditors in order to improve efficiencies. Tax is not your area of expertise, but you have worked with GSA on many other projects so have developed a good relationship with the management team. You often attend GSA social events and would consider some of the team to be good friends.

Before you go into a planning meeting you overhear the end of a conversation between the engagement partner and the GSA finance director in the meeting room: '[…] obviously, if the tax savings identified are sufficiently imaginative and creative, there will be a special bonus for you, although I would rather the savings are kept quiet because it is likely to generate a lot of negative press. It seems we should all be acting in the public interest these days.'

Requirement

Identify the ethical threats present in the scenario. Are you in a position to exercise professional judgement in dealing with the ethical dilemmas? What action should you take?

See **Answer** at the end of this chapter.

1.6 Other ethical codes

Other ethical codes impacting on businesses include the *Sarbanes-Oxley Act* (2002) requirement, policed by US Securities and Exchanges Commission (SEC), that public

corporations filing returns in the US have a Code of Ethics and that this should be detailed in their annual filings.

It is described by SEC as **a codification of standards that is reasonably necessary to deter wrongdoing and to promote:**

(a) Honest and ethical conduct, including the ethical handling of actual or apparent conflicts of interest between personal and professional relationships

(b) Full, fair, accurate, timely, and understandable disclosure in reports and documents that a company files with, or submits to, the Commission and in other public communications made by the company

(c) An environment that encourages disclosure of ethical and compliance related concerns

(d) Compliance with applicable governmental laws, rules and regulations

(e) The prompt internal reporting of code violations to an appropriate person or persons identified in the code; and accountability for adherence to the code

Many firms use these headings to develop their own internal code of ethics for management.

1.7 ICAB *Code of Ethics* for professional accountants working in public practice

The following paragraphs are based on s. 300.6 A1 of the ICAEW's *Code of Ethics* (2020) for **accountants working in public practice**. They outline some of the threats that professional accountants experience. Note that the following section is not exhaustive but gives an indication of the types of issue that may arise in a Business Strategy and Technology exam question scenario.

1.7.1 Threats

(a) **Self-interest** – a firm or professional accountant having a financial interest in an assurance client; a firm which has undue dependence on fees from a client or fears losing a significant client; a firm which enters into a contingent fee arrangement with a client; a member of the firm enters into employment negotiations with a client; a firm which develops a significantly close relationship with a client; a professional accountant discovers a significant error when evaluating the results of professional services carried out in the past by a member of the firm.

(b) **Self-review** – a firm designs a financial reporting system and is then asked to issue an assurance report on the effectiveness of the same system, a firm which has prepared the original data used to produce records that are reviewed as part of an assurance engagement; a member of the assurance team has been a director of the client; a member of the assurance team has been employed by the client and is now able to significantly influence the subject matter of the engagement.

(c) **Advocacy** – the firm promotes shares in an audit client; the professional accountant acts as an advocate on behalf of a client in litigation.

(d) **Familiarity** – a member of the engagement team having an immediate or close family member who is a director or officer of the assurance client; a member of the engagement team having a close or immediate family member who is employed by the client who is in a position to significantly influence the subject matter of the engagement; a director or officer of the client or employee is in a position to exert significant influence over the subject matter having recently been the engagement partner at the firm providing assurance services; the acceptance by a member of the engagement team of gifts or preferential treatment from a client, unless the value is inconsequential; members having a long association with the assurance client.

(e) **Intimidation** – a firm which is threatened with dismissal by a client; an audit client indicating that it will not award non-assurance related work to the firm if the firm disagrees with the client's accounting treatment; a firm being threatened with legal action by a client; a firm

being pressured to reduce fees; a professional accountant being pressured to agree with a member of the client's staff due to the employee's level of expertise on a particular matter.

1.7.2 Safeguards

Safeguards vary depending on the facts and circumstances. Section 300.8 A2 of the Code (2020) details the following **safeguards** relating to the professional accountant's work environment to overcome the threats explored above, these include:

- Assigning additional time and qualified personnel to required tasks when an engagement has been accepted might address a self-interest threat.
- Having an appropriate reviewer who was not a member of the team review the work performed or advise as necessary might address a self-review threat.
- Using different partners and engagement teams with separate reporting lines for the provision of non-assurance services might address self-review, advocacy or familiarity threats.
- Involving another firm to perform or re-perform part of the engagement might address self- interest, self-review, advocacy, familiarity or intimidation threats.
- Disclosing to clients any referral fees or commission arrangements received for recommending services or products might address a self-interest threat.
- Separating teams on matters of a confidential nature might address a self-interest threat



Interactive question 3: Ethical dilemmas

This next question should serve as a refresher from your earlier studies as to how ethics and professional standards impact the work of accountants in different settings.

Russell Ltd

Your company has been acting as consultants to Russell Ltd providing advice about the need to cut costs and restructure the company's manufacturing operations. In a recent meeting with the production director of Russell Ltd you are made aware that the entire consulting team will be invited to the company's Christmas celebrations which run over the course of a weekend and are being held at an expensive Castle-Keep Hotel.

Ravencroft White plc

You are a manager in the audit firm of Swain Brothers. Ms Singh has been the engagement partner for a client Ravencroft White plc for the last 11 years. They have excellent knowledge of the client and know all of the directors of Ravencroft White exceptionally well.

Sole Records Ltd

Mr Minus, the principal consultant in the advisory firm for which you work, has been assisting a client Sole Records Ltd to help improve the company's performance. Mr Minus's daughter, Jemima, recently joined the finance department of Sole Record's largest competitor.

Requirement

Explain the ethical threats which may affect the independence of the accountants/consultants featured in the three scenarios.

For each threat explain how it might be reduced.

See **Answer** at the end of this chapter.

1.8 Simple ethical tests for a business decision

Compliance with codes of conduct will be mandatory for members and employees subject to the codes.

Ethical tests enable managers to consider ethical consequences of decisions where:

- The wording of codes may be imprecise
- Situations arise that are not covered in the codes

Initially at least the situation should be evaluated as to:

(a) the facts of the situation and whether all relevant information is available for making the decision; and

(b) whether there is any possibility of illegality – if there is then seeking legal advice should be considered.

Consider whether the situation is characterised by integrity, transparency and honesty. Would a third party consider that there had been transparency about what was being done or planned?

Also, think about the impact the situation is likely to have on stakeholders. Who is affected and how? Have all the interests of the stakeholders been considered? Do the involved parties consider that they have been fairly treated and would third party observers think that the situation has been handled in a fair and just manner?

To understand how ethical values could be examined in a broader business setting, you are advised to review the following interactive question.

Interactive question 4: Pay to stay

You are a supply chain manager working for Happy Foods, a leading food production company renowned for making a range of well-known household food brands including cakes, bread and custard. A significant part of your day-to-day work involves dealing with the company's many suppliers to ensure Happy Foods pays the best possible price for the ingredients needed in its operations. Earlier in the year, tough trading conditions required the company to approach shareholders and request a fresh injection of funds to support the business. Shortly after this the Happy Foods board issued a profit warning.
A week ago, your line manager called you into a confidential meeting. During the meeting, they explained that the board had recently decided to introduce a scheme which would require each of the company's suppliers to make an 'annual voluntary investment' to remain on its books as an approved supplier. They added 'given your many years' experience of dealing with Happy Foods suppliers the board feel that you are ideally placed to inform our current suppliers of the new scheme. The directors would, however, prefer if you could avoid unnecessarily publicising this beyond the discussions with the suppliers themselves.

Requirement

Evaluate whether your line manager's suggestion, that you should commence the launch of the 'annual voluntary investment', is ethical.

See **Answer** at the end of this chapter.

Professional skills focus: Structuring problems and solutions

Ethics requirements can be difficult to answer due to the lack of clarity inherent in ethical problems. Consider a wide range of issues but ensure that your response has a clear structure, using headings to introduce individual points. Use the information in the scenario to guide your response and to ensure that you address the key issues.

1.8.1 Philosophical model of ethical decision making

The philosophical model of ethical decision making offers an alternative approach to evaluating ethical issues, in particular by focusing on whether a suggested decision or course of action in the face of an ethical issue is the right one.

By considering each of the four questions the user can determine whether the suggested course of action is appropriate.

(a) Do the benefits outweigh the harm to oneself?

(b) Do the benefits outweigh the harm to others?

(c) Are the rights of individual stakeholders considered and respected?

(d) Are the benefits and burdens justly distributed?

Considering a wider range of ethical issues should help to ensure that better business decisions are reached.

1.9 Impact of ethics on strategy

In the chapter, 'Strategy and organisations,' we considered the points in the strategy process where ethics may have an impact:

- In the formulation of strategic objectives, some firms will not consider lines of business for ethical reasons.
- External appraisal will need to consider the ethical climate in which the firm operates. This will raise expectations of its behaviour.
- Internal appraisal: Management should consider whether present operations are 'sustainable' (ie, consistent with present and future ethical expectations).
- Strategy selection: Management should consider the ethical implication of proposed strategies before selecting and implementing them.

1.9.1 Conflict between ethics and business

Potential areas for conflict between ethics and business strategy include:

- **Cultivating and benefiting from relationships with legislators and governments**: Such relationships may lead politicians to ignore the national interest (eg, of the people who elected them) to further their own personal interests.
- **Fairness of labour contracts**: Firms can use their power to exploit workers, including child labour, and subject them to unethical treatment in areas where jobs are scarce.
- **Privacy of customers and employees**: Modern databases enable tracking of spending for marketing purposes or to discriminate between customers on basis of their value. Staff can be subject to background checks and monitored through their use of email and the location of their mobile phones. This could be particularly pertinent given the rise in the number of staff now using wearable technology while at work.
- **Terms of trade with suppliers**: Large firms may pay poor prices or demand long credit periods and other payments from weak suppliers. This has been a particular criticism of large retail food stores in North America and Europe who are blamed for the impoverishment of farmers at home and in developing countries.
- **Prices to customers**: Powerful suppliers of scarce products such as energy, life-saving drugs or petrol, are able to charge high prices that exclude poorer individuals or nations. Examples here include anti-aids drugs to Africa or purified water to developing countries.

- **Managing cross cultural businesses**: Different countries of operation or different ethnic groups within the domestic environment can present ethical issues affecting what products are made, how staff are treated, dress conventions, observance of religion and promotional methods.

2 Marketing and ethics



Section overview

- Ethical issues begin with questioning whether marketing exists to sell people things they don't need and so wastes resources and cause envy and dissatisfaction.
- The nature of products, the means by which they are promoted, the level of prices and the selective way they are made available are also ethical issues.
- Marketers may defend themselves to some extent by noting that ethics are culturally relative and therefore, given that marketing seeks to identify and satisfy needs, it will follow and not lead ethical consciousness.

2.1 Ethical issues in the marketing concept

The most fundamental ethical issue facing marketing is the potential accusation that marketing wastes the world's resources by making things that people do not really need and then using promotion to convince people that they are not satisfied without them.

It is hard to accept this argument in the areas of essential products, but the destruction of clothes, electrical appliances etc, on the basis that they are outmoded or have obsolete technologies, and the huge amount of paper and electricity used to advertise products and to run pop-up and social media advertisements does lend weight to the argument.

The envy created by development and presentation of products that only some can have, such as personal digital players, mobile phones, expensive trainers etc, also contributes to street crime. Sports shoe manufacturer Nike was caught-up in accusations that its slogan 'just do it' was encouraging street thefts of its shoes.

To tilt the argument back in favour of marketing the following points can be made:

- **Value judgements**: Phrases like 'things that people don't really need' are not helpful because if people will spend money and effort to get them, who is to say they don't need them?
- **Employment effects**: The production of goods and services creates jobs. This is an ethical good to come from marketing.
- **Proper target marketing may reduce waste:** Marketing tries to ensure that unwanted products and marketing effort is reduced. This saves resources.
- **Ethical marketing**: Marketing has been used to promote alternative and more ecologically responsible products and ways of life (eg, to quit smoking). This suggests that it is not the practice of marketing that is unethical but rather some of the ends to which it has been put.

2.2 Ethical issues and the marketing mix

The way that marketing is carried out may raise ethical concerns:

Product issues

Some products are dangerous but are sold without this being considered in some parts of the world. Cigarettes and alcohol are obvious examples, but increasingly processed foods are being criticised as leading to illness due to high fat, sugar and salt content.

Some products are wasteful of resources, or lead to environmental pollution. The sports utility vehicle (SUV) has been a criticised example of this.

Decisions to cease provision of products, or never to offer them at all, can reduce the welfare of society and possibly inflict hardship. Pharmaceutical firms have been accused of suppressing drugs that could cure a condition in favour of continued sales of palliatives.

Price issues

Pricing products to maximise returns may mean pricing them out of the reach of many who want them. Where products are essential, such as life savings drugs or affordable homes, this has clear ethical implications.

Price discrimination remains discrimination. To charge higher prices to people in rural areas compared to towns because competition is less, or to charge people differently because of their gender or age seems unfair.

Pricing low to encourage usage may have the effect of increasing consumption to unhealthy levels. Two-for-one offers on alcohol, or multi-buy offers on confectionary are examples of this.

Promotion issues

The public has expressed concern that they are being 'brainwashed' by clever advertising techniques such as hidden messages in subliminal advertising (flashes of images too brief to be registered and filtered by the conscious mind). It is unlikely that these techniques work as feared but the broader principle that promotion manipulates behaviour remains an ethical issue.

Some images used in advertising and promotion may be offensive to some in society. Promotion has been blamed for causing anti-social behaviour. A soft drink advertisement was withdrawn following protests from schools that children were emulating a character in the advertisement by ambushing and slapping schoolmates so they could experience the 'burst of taste.'

Some advertising is upsetting to casual observers. Road safety advertisements showing slow motion footage of children being propelled over the top of speeding cars have shock value but are potentially very effective.

Place issues

The principal ethical issues here revolve around encouraging or denying access.

Stores have been accused of encouraging consumption of confectionery by mounting displays at children's eye levels at the check-out leading to parents being pestered.

The use of premium rate telephone lines for enquiries and service calls, often with substantial waiting times, has been criticised as a hidden charge. These are also common on children's phone-in quizzes and television shows.

Closures of branches to save cost leaves some customers with a lack of service and poorer quality of life. This has been a common criticism of banks.

Migration of customer service from the high street to call centres and websites excludes those without IT access, credit cards or those put off by the impersonal contact.

2.3 Ethical marketing

Marketers defend their profession against the issues above in a number of ways.

- **Personal choice**: Providing the information given is clear about the content and effects of products, it is for the buyer to decide for themselves their ethical priorities, rather than for the firm to decide it for them.
- **Codes of practice**: Most marketers abide by voluntary codes of practice in how they promote products and deal with customers. These are developed after taking a balanced view of the issues rather than accepting the assurances of the industry or the accounts of their critics.

- **Societal marketing concept**: Marketing is about meeting people's needs and expectations and not about leading them. If issues such as junk food are bothering society, then effective marketing should reduce reliance on the product and as demand falls refinements or new products can be found. Ethical investment funds, cosmetics and energy are all examples of marketing following a genuinely widespread ethical issue and responding appropriately with new market offerings to meet the need.

Interactive question 5: Socially responsible marketing

Your senior partner has requested a memo for developing into a report to a global food manufacturer regarded as being a leading exponent of modern marketing practice.

Requirements

5.1 Explain what you understand to be the advantages of a marketing orientation to food consumers and food manufacturers.

5.2 Identify some of the ethical and social responsibility issues that face marketers in the food manufacturing industry.

See **Answer** at the end of this chapter.

3 Ethical issues in manufacturing and procurement

Section overview

- Organisations have a duty to ensure that production processes and products do not cause harm to workers, customers, the environment and wider society.
- Many firms seek to fulfil their own CSR commitments by demanding similar commitments from their suppliers.

3.1 Ethical issues in manufacturing

The ethical issues in manufacturing cover the duties of an organisation to ensure that products and production processes do not cause harm either to workers, customers, wider society or the environment.

Businesses involved in manufacturing face both opportunities and risks arising from environmental and social issues. Stakeholders are becoming increasingly aware of ESG issues, so a company that factors social and environmental concerns into its manufacturing strategy can benefit from improved brand strength and customer loyalty. However if ESG issues are ignored, companies risk a negative reaction from stakeholders including damaged reputation, loss of investor confidence and possible fines or penalties for breaking laws. Potential areas of risk where businesses need to respond to ESG issues, and to consider whether their manufacturing strategy and processes are ethically sound include:

Social issues

- The manufacture, marketing and distribution of defective, addictive and inherently dangerous products and services (eg, tobacco, alcohol, weapons, motor vehicles and some chemicals) that companies knew were harmful to the buyers of those products.
- The use of child labour often in hazardous jobs such as the manufacture of matches, fireworks and carpets.
- Ethical problems arising out of new technologies such as genetically modified food, mobile phone radiation and health.

- Product testing ethics concerning animal rights and animal testing and the use of economically disadvantaged groups (such as students) as test objects.

Environmental issues

- Ethical relations between the organisation and the environment in terms of pollution, carbon emissions trading, and environmental ethics.
- The environmental impact of products at the end of their lives, such as discarded electric equipment, including computers and smart phones, or scrapped cars.
- The environmental effect of using natural resources, including the effects of deforestation and mining, to manufacture consumer products.

Consumers have moved towards products that are considered more environmentally and ecologically friendly and where people have not been exploited in their manufacture.

Hence, we now see a greater range of products labelled. For example:

- 'Organically grown'
- 'Not tested on animals'
- 'The manufacturer is a carbon neutral firm'
- 'This paper is from recycled sources'
- 'This wood is from sustainable forests'
- 'The chemicals contained in this product are biodegradable'

Consumer awareness and a move towards sustainability represents both an opportunity and a risk for firms. If companies act ethically and integrate environment, social and governance issues into their products and processes this may be used as a means of differentiation, to increase competitive advantage. However, if companies fail to take action on ESG and sustainability issues this could negatively impact stakeholder relationships and business reputation.

3.2 Ethical aspects of purchasing and procurement

Definition

Modern slavery: Is the severe exploitation of people for commercial gain. Modern slavery covers a wide range of abuse and exploitation including sexual exploitation, domestic servitude, forced labour, criminal exploitation and organ harvesting.

Many firms seek to fulfil their own ESG commitments by demanding similar commitments from their suppliers. This gives rise to **ethical procurement.** For example, in the UK it is mandatory for companies with a turnover of more than £36 million to address and report on modern slavery in their own operations and in those of their supply chain. There are two main parts to the Modern Slavery Act (2015) requirements:

- Take action to identify, prevent and mitigate modern slavery in your operations and supply chain.
- Publish an annual statement to report on those actions.

There is increasing pressure on companies to take action on ESG and sustainability issues in their supply chains. Ethical procurement requires a company to understand its societal obligations and expectations so that it can build trust amongst its stakeholders.

Adopting an ethical procurement strategy presents an organisation with the opportunity to differentiate from competitors, allowing it to attract and retain customers. However, unethical procurement practices, such as the exploitation of workers, can give rise to both damaged reputation and stakeholder relations, harming profits through negative press. There are many

challenges to adopting an ethical procurement strategy due to the complex nature of modern supply chains.

Examples include:

(a) Illegal supply chain practices – organisations are at risk of purchasing products that have been produced by illegal or immoral means, for example stolen or counterfeit goods, use of child labour or other abuses of human rights in production, environmental damage caused by the production or consumption of goods.

(b) Bribery and coercion – special consideration is made for candidates during contract negotiations or supplier selection and evaluation, in return for illicit payments or to avoid threats of harm or loss.

(c) Nepotism – the practice of giving preference to suppliers due to family connections.

(**Source**: Oxford College of Procurement and Supply (n.d) *What is Ethical Procurement and why is it important?* [Online] Available from: https://www.oxfordcollegeofprocurementandsupply.com/what- is-ethical-procurement-and-why-is-it-important/ [Accessed 25 January 2024])

To counteract some of these problems many UK firms require that suppliers sign up to the Code of Practice of the Ethical Trading Initiative, a charity with representation from industry, government, non- governmental and voluntary organisations. The ETI base code presents nine conditions that companies should work towards as part of their ethical procurement practice, as follows:

- *Employment is freely chosen (ie, no forced, slave or prison labour).*
- *Freedom of workforce association and collective bargaining are respected – workers have the right to join trade unions.*
- *Working conditions are safe and hygienic.*
- *Child labour shall not be used.*
- *Living wages are paid.*
- *Working hours are not excessive.*
- *No discrimination is practised.*
- *Regular employment is provided.*
- *No harsh or inhumane treatment is allowed.*

***(Source**: Ethical Trading Initiative (n.d) ETI Base Code. [Online] Available from: https://www.ethicaltrade.org/eti-base-code [Accessed 24 January 2024])*

*Promoting an **ethical corporate culture** requires embedding integrity into the organization's DNA through **whistleblower protections, legal compliance, and strong governance frameworks**. Here's how these elements interconnect strategically:*

◆ *Whistleblower Protections*

- ***Purpose**: Encourage employees to report misconduct without fear of retaliation.*
- ***Key Practices**:*
  - *Anonymous reporting channels (hotlines, digital platforms).*
  - *Clear non-retaliation policies backed by enforcement.*
  - *Independent investigation mechanisms.*
- ***Strategic Advantage**: Builds trust, surfaces risks early, and prevents reputational damage.*
- ***Professional Skepticism**: Ensure reports are investigated objectively, avoiding bias or suppression by management.*

🔹 *Legal Compliance*

- ***Purpose**: Align operations with laws, regulations, and industry standards.*
- ***Key Practices**:*
  - *Compliance audits and risk assessments.*
  - *Training programs on anti-corruption, labor laws, and data protection.*
  - *Integration of compliance into performance metrics.*
- ***Strategic Advantage**: Reduces legal exposure, enhances credibility with regulators and investors.*
- ***Professional Skepticism**: Question whether compliance is formalistic (box-ticking) or genuinely embedded in decision-making.*

🔹 *Governance Frameworks*

- ***Purpose**: Provide oversight, accountability, and ethical direction.*
- ***Key Practices**:*
  - *Independent board committees (audit, ethics, risk).*
  - *Transparent reporting (financial and non-financial disclosures).*
  - *Clear delegation of authority and accountability structures.*
- ***Strategic Advantage**: Strengthens investor confidence, ensures long-term sustainability, and aligns with ESG expectations.*
- ***Professional Skepticism**: Scrutinize whether governance structures are effective in practice or symbolic.*

🔹 *Integrated Ethical Culture Model*

Pillar	*Mechanism*	*Strategic Outcome*
Whistleblower Protections	*Safe reporting channels, non-retaliation*	*Early risk detection, trust*
Legal Compliance	*Audits, training, monitoring*	*Reduced liability, credibility*
Governance Frameworks	*Independent oversight, transparency*	*Sustainability, investor confidence*

🔹 *Practical Example (Bangladesh Context)*

- *A **listed company under BSEC** could:*
  - *Establish a whistleblower hotline monitored by an external auditor.*
  - *Conduct quarterly compliance reviews aligned with NBR tax and labor laws.*
  - *Strengthen governance by appointing independent directors with audit expertise.*
- *This approach not only ensures **regulatory compliance** but also positions the firm as a **trusted market leader**, attracting both local and foreign investors.*

✅ ***Recommendation**:*

Treat ethics as a strategic asset, not just a compliance requirement. By embedding whistleblower protections, compliance, and governance into the corporate culture, firms create resilience, credibility, and long-term value.

4 Ethical issues in the use of data

Section overview

- The increasing use of data in business has given rise to a number of ethical issues in terms of how it is used and stored.

In the chapters, 'Strategic capability,' 'Business planning and functional strategies' and 'Strategies for information,' we explored some of the key considerations concerning the increased use of data in business. As organisations strive to collect greater quantities of data about customers, this has raised concerns that such information will be used to target customers with increasingly personalised marketing messages, to facilitate sales which may not otherwise have been made.

Concerns have also been raised about the ethical and legal issues in holding vast quantities of customer data in corporate databases, solely for commercial gain. This issue is further compounded, as it is unclear as to who actually owns the customer data held by organisations and whether or not commercial entities have the right to sell on customer data to third parties.

In 2022, the International Ethics Standards Board for Accountants (IESBA) updated the *Handbook for Professional Accountants* to spell out the ethical threats relating to the use of technology and, in particular, the use of AI and generative AI.

A summary of the recommendations is as follows:

- ***Integrity*** *– it is essential to be transparent and honest in the use of generative AI, especially if it has been used to support research or analysis on behalf of a client.*
- ***Objectivity*** *– it is important to exercise professional judgement when using the outputs of generative AI, recognising that there may be bias in the data depending on the source. When using generative AI, accountants must have comfort over the sources of data used to train models and how considerations of bias have been managed.*
- ***Professional competence and due care*** *– generative AI tools lack the skills, training and judgement of a professionally qualified accountant and also may not be aware of the latest legislation that needs to be taken into consideration. Accountants should therefore exercise professional scepticism when reviewing the output of generative AI. Accountants should also continue to develop their knowledge and skills in the use of generative AI.*
- ***Confidentiality*** *– accountants should continue to respect the confidentiality of client information by not loading confidential information into public generative AI tools, even if such information has become publicly available. This is also important for compliance with data protection laws.*
- ***Professional behaviour*** *– action that discredits the profession around the use of generative AI could include being overly reliant on the output of generative AI tools and a failure to understand any bias or inaccuracies in AI output. It is also important to be aware of legal frameworks governing the use of AI.*

*(**Source**: ICAEW (n.d) Generative AI and ethics. [Online] Available from: https://www.icaew.com/technical/technology/artificial-intelligence/generative-ai-guide/ethics [Accessed on 13 May 2024])*

In the chapter, 'Strategies for information,' consideration was given to the growing need for organisations to have in place robust security measures over personal data, held in commercial databases, to prevent hackers from gaining access.

Ethics of technology usage

Legal considerations of technology use

As discussed above, data is a central aspect of the technology developments driving the 4th industrial revolution.
The amount of personal data available to and used by organisations means that the privacy, sensitivity and security of this data are very significant issues in modern business.
A business must ensure it is compliant with all legislation but there are also considerations from an ethical and social responsibility point of view in terms of what is right and wrong in the eyes of the public.

Data protection

General Data Protection Regulation (GDPR) *was introduced throughout the UK and EU in 2018. It is legislation which details the following principles about data:*

- *Used fairly, lawfully and transparently*
- *Used for specified, explicit purposes*
- *Used in a way that is adequate, relevant and limited to only what is necessary*
- *Accurate and, where required, kept up to date*
- *Kept for no longer than is necessary*
- *Handled in a way that ensures appropriate security. Including protection against unlawful or unauthorised processing, access, loss, destruction or damage.*

Data controllers *within organisations have to ensure adequate safeguards and controls to implement these data principles.*

Ethical and social considerations of technology use

Ethics *is a system of moral principles that affects how people and organisations make decisions. It involves acting in a way deemed acceptable by society.*

Ethical and social considerations

Compliance with legislation such as the ***GDPR laws*** *noted above is something that all companies should achieve.*

However, the question of whether this is sufficient opens up ethical considerations as to what companies should do and how they should act.

When considering the way in which an organisation handles technology and data, ***ethical and social considerations are important*** *for the following reasons:*

- *A company that handles technology and data in an ethical way will give investors confidence*
- *Customer confidence in data security is vital in the digital age and is an important element of brand confidence*
- *Consumers feel safer dealing with companies that are proactive and responsible in their use of data*
- *Employees are attracted to companies that exhibit a strong ethical stance when it comes to how they use technology and data*
- *Ethical handling of technology and data helps ensure long-term sustainability of an organisation through stakeholder confidence and trust*
- *It demonstrates a proactive awareness of the risks associated with data and technology*

Illustration 10 – Microsoft AI ethical principles
Microsoft is one of a number of high profile technology companies to lay down a voluntary set of ethical principles surrounding its use of ***artificial intelligence (AI)****.*
AI is an area of technology that presents some of the most challenging ethical questions.

Microsoft's AI principles:

- ***Fairness*** *– AI systems should treat all people fairly*
- ***Reliability & safety*** *– AI systems should perform reliably and safely*
- ***Privacy & security*** *– AI systems should be secure and respect privacy*
- ***Inclusiveness*** *– AI systems should empower everyone and engage people*
- ***Transparency*** *– AI systems should be understandable*
- ***Accountability*** *– AI systems should have algorithmic accountability*

Corporate digital responsibility

Corporate Digital Responsibility (CDR) extends the idea and ethos of Corporate Social Responsibility (CSR) to the digital world.

It is a voluntary commitment by organisations to go beyond legal compliance in handling technology and data ethically, aiming for a positive technological future.

A CDR strategy can include the following areas:

- ***Digital stewardship:*** *Using data responsibly and securely in line with customers' and employees' expectations.*
- ***Customer expectations:*** *Increasing transparency and rewarding customers for sharing data.*
- ***Giving back:*** *Sharing data benevolently to help society, such as banks using financial data to assist customers.*
- ***Data value:*** *Recognising the importance of data and incentivising customers to share it.*
- ***Digital inclusion:*** *Ensuring all members of society have the skills and tools to access the digital world.*

5 The concepts of sustainability and ESG



Section overview

- Sustainability is about maintaining the world's resources rather than depleting or destroying them. This will ensure they support human activity now and in the future. Sustainability is closely linked to the concept of natural, social and human capital which were introduced in the chapter The macro environment.
- Environmental, social and governance (ESG) examines how organisations can approach sustainability.
- A commonly employed and useful way of thinking about sustainability issues is under three key headings: social, environmental and economic.

In the chapter, 'The purpose of an organisation,' we discussed the fact that although the primary stakeholders of a company are its shareholders, increasingly there is a need for an organisation to consider other stakeholders and wider areas of social responsibility.
This is enshrined in the concept of corporate governance, covered in the chapter 'Strategy and structure.'

5.1 What are sustainability and ESG ?

Definitions

Sustainability: The ability to 'meet the needs of the present without compromising the ability of future generations to meet their own needs' (Brundtland Report 1987).

Sustainable development: Is the process by which we achieve sustainability.

Sustainable enterprise: A company, institution or entity that generates continuously increasing stakeholder value through the application of sustainable practices through the entire base activity – products and services, workforce, workplace, functions/processes, and management/governance (Deloitte Creating the Wholly Sustainable Enterprise).

ESG: Environmental, social and governance (ESG) examines how organisations can approach sustainability-related issues. Therefore, ESG considers the impact of these risks on business and enterprise value (rather than on society more broadly). ESG is frequently used terminology in corporate and investment communities (see double materiality).

ESG is another characterisation of sustainability, providing the three lenses of environment, social and corporate governance to use when exploring relevant sustainability factors.

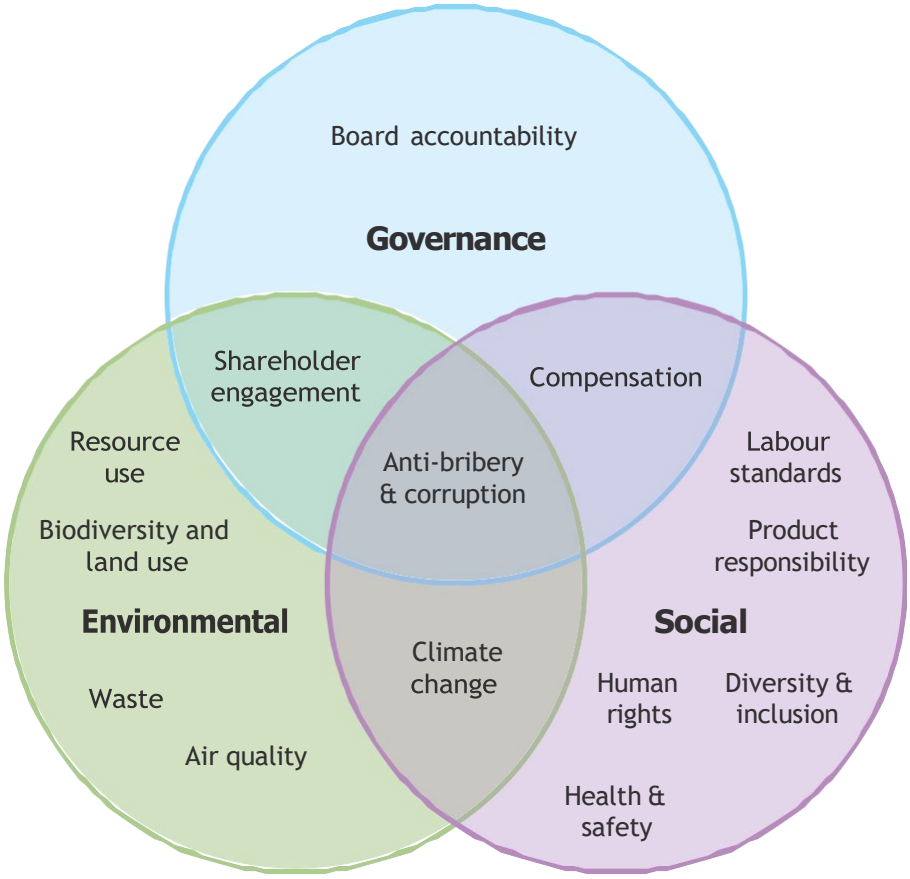


Figure 16.1: Environmental, social and governance

Sustainability is a broad term that encompasses an environmental and social focus to improve people's lives and improve the planet for the long-term. It is about maintaining the world's resources rather than depleting or destroying them. This will ensure they support human

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activity now and in the future. In the chapter, 'The macro environment,' we introduced **natural capital, social capital** and **human capital,** which are key parts of the concept of sustainability.

Sustainability is not limited to the environment. Interpretations of the scope of sustainable development have developed from a narrow interpretation which focuses on 'green issues' to broader interpretations which include concerns such as:

1. Increasing extremes of poverty and wealth
2. Population growth
3. Biodiversity loss
4. Deteriorating air and water quality
5. Climate change
6. Human rights
7. Bribery and corruption

A commonly employed and useful way of thinking about these issues is under three key headings:

social, environmental and **economic** (or **financial).**

Issues	Examples
Social	Health and safety, workers' rights (in the business itself and its supply chain), pay and benefits, diversity and equal opportunities, impacts of product use, responsible marketing, data protection and privacy, community investment, and bribery/corruption.
Environmental	Climate change, pollution, emissions levels, waste, use of natural resources, impacts of product use, compliance with environmental legislation, air quality.
Economic	Economic stability and growth, job provision, local economic development, healthy competition, compliance with governance structures , transparency, long-term viability of businesses, investment in innovation/new product development.

To achieve sustainable success, these three issues must be addressed and balanced; not just the concrete issues such as raw materials or energy use, but also less tangible ones such as relationships with wider stakeholders and employees, whose goodwill is necessary to the long-term success of the business.

Natural capital

During our discussion of natural capital, social capital and human capital in the chapter The macro environment we highlighted that a growing number of large organisations are now starting to attach monetary values to the natural, social and human capital assets that they impact or depend on when undertaking their business activities.
By attaching monetary values in this way, it is believed that the senior management of participating organisations will be better placed to understand the implications that the organisation's activities are having on the planet, which should in turn enable better-informed business decisions to be made.



Context example: National Grid

National Grid owns approximately 1,800 hectares of non-operational land, including habitats ranging from ancient woodland to peatbogs. To protect and improve the nature of these landholdings, National Grid has created a bespoke Natural Capital Tool. The tool outlines 12 ecosystem services, or benefits, that the land provides and, using third-party data, National Grid has been able to assign an estimated financial value to each of these services.
Recognising the value of the natural assets, including assets such as water and air quality, has

helped National Grid to focus on managing them effectively, in order to deliver the greatest value to stakeholders.
(**Source**: https://www.nationalgrid.com/electricity-transmission/pathway-to-net-zero/caring-natural- environment [Accessed 24 January 2024])

Sustainable development recognises the interdependence between business, society and the environment, since without the environment neither business nor society could exist.

Thus, for businesses to deliver value to their shareholders, they must respond to the needs and priorities of their stakeholders and not exhaust the world's capital.

ESG is a strategic business issue, since by addressing their social and environmental impacts, organisations can have a positive effect on their economic performance.

An organisation can adopt different **strategic approaches** to corporate responsibility:

Proactive strategy	A business is prepared to take full responsibility for its actions. For example, a company which discovers a fault in a product and recalls the product without being forced to, before any injury or damage is caused, acts in a proactive way.
Reactive strategy	This involves allowing a situation to continue unresolved until the public, government or consumer groups find out about it.
Defence strategy	This involves minimising or attempting to avoid additional obligations arising from a particular problem.
Accommodation strategy	This approach involves taking responsibility for actions, probably when one of the following happens: • Encouragement from special interest groups • Perception that a failure to act will result in government intervention

5.2 Differences between sustainability and ESG

Sustainability describes a world of thriving economies and just societies based on what nature can afford. It incorporates consideration of both the impacts and dependencies of an organisation and so includes those factors that are material both to the organisation but also to society.

Environmental, social and governance (ESG) approaches these issues from the perspective of an organisation and considers only how these risks and opportunities affect a business and its enterprise value.

Whilst sustainability includes the concept of environmental and social limits (planetary limits and a social foundation) within which there is a safe operating space, ESG focusses on enterprise value and so does not incorporate considerations of a safe operating space.

5.3 Sustainability impacts and dependencies

Impacts consider how the decisions an organisation makes, or its actions, either positively or negatively affect environmental, societal and governance issues.
Dependencies consider how the current and future environmental, social and governance issues can affect the organisation's ability to create and maintain value.
Impacts and dependencies are fundamental aspects of sustainability.
Information on **impacts** is generally useful for broader stakeholders, including consumers, civil society and employees. The decisions they make are more attuned to information on how an organisation is impacting on livelihoods, for example: deforestation; equality; and health.

Examples of impacts include:

- Worker rights
- Human rights
- Health and safety policy
- Waste
- Greenhouse gas emissions
- Water and land usage
- Biodiversity

Information on **dependencies** is generally more useful for investors, who want to assess how well a company is managing it is exposure to long-term ESG risks, and hence assess the value of the company to inform their investment decisions.

Examples of dependencies which may affect an organisation are:

- Worker health and workplace diversity
- Climatic conditions
- Resource availability
- Regulation
- Consumer and other stakeholder expectations
- Risks to organisational reputation

6 Sustainable development



Section overview

- This section reviews the mechanisms by which governments, businesses and other global organisations support and promote sustainable development.

The term 'sustainable development' originally appeared in literature on development economics, often as a contrast to the 'unsustainable development' of some countries that were receiving large inward investment from multinational corporations. This was believed to be destroying the social and ecological infrastructure and hence hampering the future development of the countries.

Following the Rio 'Earth Summit' of 1992 and ensuing global political support the need for 'sustainable development' has, amongst political initiatives, led to a focus on firms developing appropriate strategies and thereby becoming '**sustainable enterprises**.' The focus has moved from just developing countries to include impacts on mature home economies too.

6.1 Global initiatives to promote sustainable development

There are a wide variety of UK, European and global initiatives to foster sustainable development, including steps taken by governments, business and other organisations. Such actions include:

- **Corporate policies** whereby the perceived expectations of society convince organisations of the merits of adopting policies on sustainability and publishing information about the policies and their impact.
- **Supply chain pressure** by which the expectations of society drive purchasers to promote a desired standard of sustainable performance and reporting amongst suppliers and others in the supply chain.

- **Stakeholder engagement** enabling those with a particular interest to influence the decisions and behaviour of an organisation, to engage an organisation in ongoing dialogue and a process of feedback to and from stakeholders, supported by information flows about sustainable performance.
- **Voluntary codes** through which society encourages organisations to improve particular aspects of their sustainability performance, often requiring a statement for stakeholders regarding compliance or an explanation of non-compliance.
- **Rating and benchmarking** by which investors and others, or agencies working on their behalf, grade organisations through the use of benchmarks or ratings on the basis of information on sustainability policies and performance and thus influence the behaviour of organisations and stakeholders.
- **Taxes and subsidies** to incentivise organisations to operate in ways that contribute to sustainability, requiring information in the form of tax returns and grant claims.
- **Tradable permits** whereby governments ration allocations of scarce resources or undesirable impacts so as to improve sustainability, requiring information about quota utilisation and prices to support the operation of fair markets.
- **Voluntary and mandatory ESG reporting** following recommendations made by the Task Force on Climate-related Financial Disclosures (TCFD). The TCFD disbanded in October 2023 having fulfilled its remit, with the IFRS foundation taking over the monitoring of progress on companies' climate-related disclosures.

In 2015, the United Nations (UN) introduced its Sustainable Development Goals (SDGs). The UN's Sustainable Development Goals are aimed at tackling a number of global sustainability challenges. The 17 goals focus on a broad range of issues ranging from ending poverty to protecting the planet and ensuring global peace.

The goals have been configured so that they are interconnected, meaning that efforts to address one will also involve addressing those issues associated with a different goal. Each goal is supported by a series of targets which are intended for use in measuring its achievement. The UN has set a deadline of 2030 for the achievement of these goals.

The UN's Sustainable Development Goals are aimed at getting people, organisations and governments around the World to engage in their achievement by integrating them into their everyday thinking, corporate strategies and national legislation.

7 Social responsibility



Section overview

- *The 'S' in ESG stands for 'social' and addresses human rights and equity with regard to a company's stakeholders, policies and behaviours.*
- *The social element of ESG can be a source of both risk and opportunity.*
- *This section considers the reasons why organisations might choose to demonstrate the social element of ESG, and the extent to which addressing social factors is congruent with delivering value to shareholders.*
- *The scope of the social factors to be managed varies with the nature of the organisation.*
- *Embedding social responsibility and ESG within an organisation requires the formulation, implementation and control of a strategic plan and the involvement of senior management.*

7.1 *Why consider social factors?*

Behaving in a socially responsible manner is no longer just a public relations exercise but is a strategically fundamental driver for business.

*The social element of ESG is both a source of **risk** – the reputational damage that can be done by 'bad' behaviour – and of **opportunity** – companies that take care of their employees will experience a more motivated workforce, reducing staff turnover and improving employee efficiency and output.*

*Attention to **social, environmental** and **economic** aspects of business activity and performance is increasing from customers, suppliers, investors, pressure groups and the community. In a sense each group's support gives a company a licence to operate. Without the support of these stakeholders a business will find its ability to operate effectively is impaired, thus damaging its performance over the long-term.*

Firms that misbehave will find fewer and fewer trading partners, consumers will stop buying their products, host nations will withdraw licences or impose fines, local communities will withdraw their support and possibly become hostile, and workers will become demotivated and unproductive. In addition, investors will stop buying their stock, and existing shareholders will sell, banks will increase their lending rates and the cost of capital will increase.

***Justifications** offered for management seeking to demonstrate social responsibility outside a business's normal operations are:*

- *Self-regulation by the firm or industry now is likely to be more flexible and less costly than ignoring the social element of ESG and facing statutory regulation later.*
- *It improves relations with key external stakeholders such as regulators, government and legislators.*
- *Donations, sponsorship and community involvement are a useful medium of public relations and can reflect well on the business and its brands.*
- *Involving managers and staff in community activities develops them more fully.*
- *It helps create a value culture in the organisation and a sense of mission, which is good for motivation.*
- *In the long-term, upholding the community's values, responding constructively to criticism and contributing towards community well-being might be good for business, as it promotes the wider environment in which businesses flourish.*

***Tangible benefits** of addressing the social element of ESG include:*

- *Firms may achieve a lower cost base through the efficient use of resources, including people, which would help improve competitive advantage.*
- *Socially responsible behaviour may provide opportunities to enter new markets or attract new ethical customers.*
- *A policy that is socially responsible may help protect the company's licence to operate and its reputation (eg, Anglo American), the global mining company is committed to minimising any negative impacts of its operations and works with local communities to ensure that they benefit from its activities. The focus on safe and responsible extraction enhances its image and helps it retain a social licence to operate.*
- *Behaving in a socially responsible manner offers an opportunity to attract top talent to your organisation since prospective employees increasingly seek out companies that align with their own personal values.*
- *Social responsibility may enhance a firm's reputation and hence attract more finance from ethical investors – eg, by gaining a listing on the FTSE4Good index of companies that meet globally recognised ESG standards.*

Interactive question 6: Being socially responsible

What are the implications of a company acting in a socially responsible manner in the:
(a) *Short term*
(b) *Long term*

*See **Answer** at the end of this chapter.*

7.2 Scope of Social in ESG

The scope of the social element of ESG varies from business to business. Factors frequently included are:

- ***Health and safety** – includes workplace injury, customer and supplier injury and harm to third parties*
- ***Environmental protection** – energy use, emissions (notably carbon dioxide), water use and pollution, impact of product on environment, recycling of materials and heat*
- ***Staff welfare** – issues such as stress at work, personal development, achieving work/life balance through flexibility, equal opportunities for disadvantaged or minority groups*
- ***Customer welfare** – through content and description of products, non-exclusion of customer groups, fair dealing and treatment*
- ***Supply-chain management** – insisting that providers of bought-in supplies also have appropriate ESG policies, ethical trading, elimination of pollution and unrecycled packaging, eliminating exploitative labour practices amongst contractors*
- ***Ethical conduct** – staff codes for interpersonal behaviour, prohibitions on uses of data and IT, management forbidden from offering bribes to win contracts, ensuring non-exploitation of staff*
- ***Engagement with social causes** – includes secondment of management and staff, charitable donations, provision of free products to the needy, involvement in the local community, support for outreach projects such as cultural improvement or education*

*Successful ESG programmes focus on issues of most strategic **relevance** to an organisation which offer an opportunity to meet both shareholder and wider stakeholder expectations.*

Examples include:

- *A service organisation might focus on a staff welfare programme to encourage retention.*
- *A supermarket could stock ethically sourced food and attract new customers as a result.*
- *A software company might introduce a new programme designed for the elderly to tackle digital exclusion and create an additional revenue stream in the process.*
- *A bank may allocate funds to invest in technology to combat climate change.*

Interactive question 7: Skin Care Company

You have recently accepted a position as a Chartered Accountant working for the Skin Care Company.

The Skin Care Company (SCC) specialises in making beauty products, predominantly for women. All products are made at the company's UK factory and are sold around the world. The company's founder was keen to introduce a range of ethically produced beauty products made from sustainable ingredients. The founder's vision was to source ingredients from farmers and growers in some of the world's most impoverished regions through establishing long term relationships with suppliers. As a result, SCC pays its growers higher prices for ingredients than competing beauty care companies.

SCC's ethical stance helps the company command premium prices for its products and it remains well known among 'ethically minded' customers.

In the years since the founder's retirement, demand for SCC's products has continued to increase and the company has recently finalised plans to list on the London Stock Exchange. One week before the official listing, you are accidentally copied in on an email from the company's MD to the head of production.

The email reveals that one of the ingredients used in one of SCC's most popular skin care products has been found to contain a toxin which can cause nasty rashes on certain skin types. Although not illegal to use the ingredient in such products, several neighbouring countries have recently commenced an investigation into its safety. The UK government has yet to announce plans to conduct its own investigation.

The MD suggests in the email that because only they and the head of production are aware of the exact recipe for the product, they should not reveal this potential problem to the rest of the directors for fear of lowering the company's share price when it floats on the exchange the following week.

Requirement

Describe the ethical and ESG related issues that should be considered upon discovery of the contents of this email and suggest any appropriate actions that should be taken.

*See **Answer** at the end of this chapter.*

7.2.1 *Achieving ESG*

The management of the business must put ESG at the centre of its activities by:

(a) *identifying the sustainability and ESG issues that are relevant to the organisation; and*

(b) *developing an organisational strategy to embed ESG into the organisation's corporate policies.*

As with all strategic management, this involves the formulation, implementation and control of a strategic plan and the involvement of senior management.

*There are a number of key factors required for an organisation to **embed sustainability in its day-to- day operations**:*

(a) *Board and senior management commitment*

(b) *Understanding and analysing the key sustainability drivers for the organisation*

(c) *Integrating the key sustainability drivers into the organisation's strategy*

(d) *Ensuring that sustainability is the responsibility of everyone in the organisation*

(e) *Breaking-down sustainability targets and objectives for the organisation as a whole into targets and objectives which are meaningful for individual subsidiaries, divisions and departments*

(f) *Processes that enable sustainability issues to be taken into account clearly and consistently in day-to-day decision-making*

(g) *Extensive and effective sustainability training*

(h) *Including sustainability targets and objectives in performance appraisal*

(i) *Champions to promote sustainability and celebrate success*

(j) *Monitoring and reporting sustainability performance*

Later in the chapter, we look at how to ensure that organisations and their stakeholders have the information available to support the mechanisms that will enhance sustainability.

8 Monitoring and measuring sustainability

Section overview

- A sustainable development strategy needs to be implemented and regularly monitored.
- Monitoring involves measuring progress in terms of both processes and outcomes.
- The GRI reporting standards are a global, voluntary code for sustainability reporting. It provides a series of standards that organisations can use to measure and report their economic, environmental, and social performance.

8.1 Monitoring process

It is not enough for an organisation to develop a sustainable development strategy. Delivery of the strategy needs to be implemented and regularly monitored:
As with any strategy the implementation and monitoring process involves:

- Identifying the key sustainability issues for the business and the factors that drive them – the social, environmental and economic framework discussed earlier can be used.
- Sustainability issues are not static. Changes to the activities and operations of the organisation may give rise to new issues, so one aspect of monitoring involves giving consideration to any new developments.
- Set targets and standards for sustainability objectives. The standards might originate from government guidance and policy, legislation, public commitments or commonly accepted industry or sector standards.
- Monitor progress towards sustainability objectives in terms of both processes and outcomes.

  Process – is the organisation doing the things that it said that it would in order to implement sustainability initiatives?

  Outcome – having implemented initiatives, is the organisation achieving the outcomes towards which it is striving?
- Report progress and evaluate the implications for future decision making and performance (feedback).

An organisation will need to develop its management information and data analytics systems to meet the requirements of sustainable development monitoring, reporting and evaluation. This may involve making better use of existing financial and non-financial information and data.

Context example: SSE – developing a net zero transition plan

SSE is a UK-listed energy company operating mainly across the UK and Ireland. It is a leading generator of renewable electricity and has a commitment to achieve net zero greenhouse gas (GHG) emissions by 2050 at the latest.

To achieve this, SSE has developed a transition plan which could be monitored and measured to track progress. The plan includes a range of short- (2025), medium- (2025–2035) and long-term (2035–2050) emissions targets. SSE has processes in place to gather the relevant data to monitor these targets and is working to automate this by putting in place a platform for data collection.

As a result of the data collected, SSE is able to report on its performance against targets in its bi-annual sustainability statement.

*(**Source**: A4S (n.d.) SSE: Developing a net zero transition plan – Setting, tracking and reporting targets. [Online] Available from: https://www.accountingforsustainability.org/en/knowledge-hub/case- studies/sse-developing-net-zero-transition-plan-setting-tracking-reporting-targets.html [Accessed on 24 January 2024])*

8.2 Reporting requirements

Effective reporting involves ensuring that the organisation is transparent in communicating with internal and external stakeholders. In the public sector, there are a number of organisations that have a role in evaluating the outcomes of sustainability initiatives. These include independent advisory bodies, such as the National Audit Office.

Definition

The Taskforce on Climate-related Financial Disclosures (TCFD): Was formed in 2015 by the Financial Stability Board (FSB) with a goal of developing a set of voluntary climate-related financial risk disclosures. The disclosure recommendations are structured around four core elements: governance; strategy; risk management; and metrics and targets.

Following recommendations from the Task Force on Climate-related Financial Disclosures (TCFD), climate-related financial disclosures are mandatory for publicly quoted companies and all private and listed liability companies with more than 500 employees and turnover greater than £500 million. This recognises the increasing importance of addressing ESG issues in creating long-term shareholder value.

In particular, the TCFD focus on forward-looking financial disclosure. TCFD recommendations ask organisations to identify climate-related risks and opportunities, to consider the financial implications, and to assess the resilience of the business strategy to future climate outcomes.

TCFD also recommends organisations use scenarios to explore future business risks arising from climate change and present the findings in the narrative report.

The TCFD was disbanded in 2023 after it had fulfilled its remit, and the FSB has asked the IFRS foundation to take over the monitoring of companies' climate-related disclosures.

Definition

Corporate Sustainability Reporting Directive (CSRD): *In tandem with the development of standards by the International Sustainability Standards Board (ISSB), the European Union (EU) has adopted the Corporate Sustainability Reporting Directive (CSRD). The CSRD introduces detailed sustainability reporting requirements for large and listed companies and paves the way for mandatory sustainability assurance.*

The CSRD incorporates the concept of double materiality. Companies have to report on:

(a) ***Financial materiality*** *– sustainability issues which might create financial risks for the company*

(b) ***Impact materiality*** *– how an organisation's activities impact on people and the environment*

Definition

Double materiality: *Means considering the sustainability issues that might create financial risks for the company (financial materiality) and also sustainability issues where an organisation's activities materially impact on people and the environment (impact materiality). This concept is fundamental to the Corporate Sustainability Reporting Directive (CSRD).*

8.3 Performance measurement

8.3.1 Global Reporting Initiative

The Global Reporting Initiative (GRI) is the world standard setter in sustainability reporting, agreed through a multi-stakeholder approach. In October 2016 the GRI launched the first global standards for sustainability reporting. The GRI standards replaced the G4 guidelines. The standards help organisations report on the economic, environmental and social impacts of their activities:

- **Economic** – concerns the organisation's impacts on the economic conditions of its stakeholders and on economic systems at local, national, and global levels
- **Environmental** – concerns an organisation's impacts on living and non-living natural systems, including ecosystems, land, air, and water
- **Social** – concerns the impacts an organisation has on the social systems within which it operates

Performance indicators to measure the three impacts mentioned above might include:

Economic	Environmental	Social
Revenues and costs	Materials used	Employee turnover and absenteeism
Wages, pensions, other employee benefits	Energy consumption	Diversity of workforce, incidents of discrimination
Retained earnings and payments to providers	Water use	Employee health and safety
Taxes paid, subsidies and grants received	Greenhouse gas emissions	Child labour
Geographic analysis of key markets	Effluents and waste produced	Training undertaken
Return on capital employed	Significant spillages	Bribery and corruption
	Fines and penalties	Community relations
	Impact of activities on biodiversity	Complaints re breaches of customer privacy
		Standard of Product labelling

(Source: GRI, 2024)

The GRI standards are intended to show how organisations contribute to sustainable development.

The reporting standards issued by the GRI are the best-known example of a global, voluntary approach for sustainability reporting. The introduction of the GRI standards is intended to make sustainability reporting relevant and more responsive to new developments. Each specific standard outlines the reporting requirements, reporting recommendations and provides guidance to the user organisation.

The standards are designed as an easy-to-use modular set incorporating:

- Universal standards – apply to all organisations. There are three universal standards. GR1 explains the purpose and system of GRI Standards as well as explaining the key concepts for sustainability reporting. GR2 contains disclosures to provide information on an organisation's activities, governance and policies. GR3 provides guidance on how to report information on the organisation's most important, 'material' topics.
- Sector standards – standards are available for 40 different sectors and indicate the disclosures that are likely to be relevant for each sector. For example, GRI 11 provides information for organisations in the Oil and Gas sectors. The aim is to increase the quality, completeness and consistency of reporting across sectors.
- Topic standards – contain disclosures for providing information on specific topics. For example GRI 405 sets out the reporting requirements on the topic of diversity and equal opportunity whereas GRI 416 addresses Customer Health and Safety.

(**Source**: GRI (2024) *GRI Standards by language*. [Online] Available from: https://www.globalreporting.org/standards/download-the-standards/ [Accessed on 25 January 2024])

To keep the GRI Standards relevant and up to date, the Global Sustainability Standards Board (GSSB) sets out a new work program every three years, which aims to review existing standards as well as introducing new standards.

In March 2022, the International Sustainability Standards Board (ISSB) and the GSSB announced their intention to coordinate work programs and standard setting activities. The agreement to align their work will help to reduce the reporting burden for companies and harmonise sustainability reporting more generally. According to the chair of the GRI GSSB, Judy Kuszewski: 'The collaboration between the GSSB and the ISSB demonstrates our shared commitment to the global alignment of disclosure requirements. This is crucial if we are to enable consistent reporting by companies, which increases accountability and drives responsible business practices' (IFRS, 2022).

(**Source**: IFRS (2022) *IFRS Foundation and GRI to align capital market and multi-stakeholder standards to create an interconnected approach for sustainability disclosures*. [Online] Available from: https://www.ifrs.org/news-and-events/news/2022/03/ifrs-foundation-signs-agreement-with-gri/ [Accessed 25 January 2024])

8.3.2 Accounting for Sustainability

The **Accounting for Sustainability (A4S)** project was established in 2004 to provide organisations with methodologies and tools to enable them to embed sustainability into day-to-day processes and to report more effectively on their sustainability performance.



Context example: A4S

The aim of the A4S project is to 'inspire action by finance leaders to drive a fundamental shift towards resilient business models and a sustainable economy. To do this, A4S has three core aims:

- Inspire finance leaders to adopt sustainable and resilient business models.
- Transform financial decision-making to enable an integrated approach, reflective of the opportunities and risks posed by environmental and social issues.
- Scale up action across the global finance and accounting community.'

The A4S website sets out the importance of making sustainability an embedded issue in the strategy, operations and reporting of all types of organisations.

The work of A4S highlights the roles that chief financial officers (CFOs) and the accounting profession play in ensuring that organisations 'create long term sustainable value by

integrating economic, environmental and social matters into their decision-making.' A4S engages CFOs and their finance teams, equipping them with practical tools to build sustainability into everything they do. A4S adheres to the view that the transformation of finance is essential to achieving a sustainable future and their research demonstrates that there are many financial benefits to considering sustainability in decision making, including:

Cost cutting and efficiency

Sustainable business can deliver cost savings and efficiencies (eg, lower resource usage).

Risk mitigation

Environmental risks can result in material financial risks so must be managed as much as possible. For example, insurance claims are growing because of damage caused by extreme weather.

New competitive and revenue opportunities

There is increasing evidence of the link between sustainable business models and improved commercial returns, with sustainability and profit being interconnected.

Innovation

Long-term integrated thinking helps finance teams to foster innovation by updating their business models, technology and processes.

Improved employee development and retention

Valuing social and human capital and investing in people enhances business resilience and increases competitive advantage.

(**Source**: Accounting for sustainability (2024) *A4S Aims*. [Online] Available from: https://www.accountingforsustainability.org [Accessed 25 January 2024])



Context example: Marks and Spencer

Retailer Marks and Spencer takes a proactive approach to sustainability having launched its 'Plan A' sustainability scheme in 2007. The original 'Plan A' set out 100 commitments on the most important social, environmental and ethical challenges facing Marks and Spencer's business. Since this time Marks and Spencer has continually updated its 'Plan A' scheme with a current goal to be net zero across its entire supply chain and products by 2040, ten years ahead of the UK government's strategy.

Marks and Spencer is focusing on ten priority areas to achieve the Plan A commitment:

- Zero deforestation – 100% of soy to be sourced from verified deforestation-free sources by 2026
- Sustainable sourcing – 100% verified recycled polyester by 2026
- Low-impact farming – supporting farmers to grow low carbon food, using fewer pesticides
- Increasing the range of plant-based protein – double the sales of vegan and vegetarian products by 2025
- Suppliers and business partners on net zero journey – supporting decarbonisation across the supply chain
- Circular economy – incentivising clothes recycling scheme
- Reduce food waste by 50% by 2030
- Reduce and recycle packaging – remove one billion units of plastic packaging by 2028
- Zero emissions property with a more efficient store estate
- Zero emissions transport with reduced dependency on diesel and increased use of new technologies and cleaner fuels (Marks and Spencer, 2021).

(Source: Marks & Spencer (2021) *Plan A Our Planet* [Online] Available from: https://corporate.marksandspencer.com/sustainability/plan-a-our-planet [Accessed 25 January 2024])

Interactive question 8: Ashdene Homes

Ashdene Homes is a regional house builder, having considerable knowledge and experience in the South of England where the current UK housing shortage is centred. The company caters for the mid to lower end of the market, with prices normally below £500,000, on relatively small and individual sites which tend to be too large for the resources of local builders but too small for the high-volume national house builders. Any mass release of land for development in the south-east due to government initiatives is likely to be centred in one area. The development of any such land would take many years given delays within the planning process.

The company, worth £67 million has looked like a takeover target for a while but unfortunately, the company's reputation for internal control has been damaged somewhat by a qualified audit statement last year (over issues of compliance with financial standards) and an unfortunate internal incident which concerned an employee expressing concern about the compliance of one of the company's products with an international standard on fire safety. They raised the issue with their immediate manager but when they failed to obtain a response, they decided to report the lack of compliance to the press. This significantly embarrassed the company and led to a substantial deterioration in their reputation, especially as there have been more press releases about the company's failure to adhere to the high welfare, health and safety, financial, marketing and ethical principles that the founder practised when they started Ashdene Homes.

Requirements

8.1 Outline the implications of poor ethical principles and damaged reputation on the relationship between the affected stakeholder groups and Ashdene Homes.

8.2 What are the main issues concerned with sustainability and why might Ashdene Homes choose to act, or at least claim to act, in a socially responsive way?

8.3 Explain, with reference to Ashdene Homes as appropriate, the ethical responsibilities of an accountant both as an employee and as a professional.

See **Answer** at the end of this chapter.

9 Developing an approach to ethics requirements in the exam

Section overview

- The intention of this section is to demonstrate the thought processes to be adopted when tackling a question involving ethics and the attributes of an answer that will score well in the exam.
- You will find further examples of requirements covering ethical issues in the Business Strategy and Technology Question Bank.

9.1 What will the ethics requirement involve?

Between 7 and 10 marks in the Business Strategy and Technology exam will be allocated to the appropriate discussion of ethical issues, possibly in the context of the ethics of an organisation or an individual working in business who may be a professional accountant.

As the Business Strategy and Technology exam continues to evolve, it is important to note that ethical issues appearing in question requirements may appear in more subtle ways in future sittings. This means that ethical issues in question scenarios may not have an 'ethical issue' heading to grab your attention, it also means that there may not be a separate question requirement which is solely dedicated to a discussion of ethical issues.

Instead, question requirements may ask you to consider ethical issues as part of a broader requirement relating to other business matters raised in the scenario, for example, a requirement might ask you to discuss the commercial and ethical considerations relating to the featured entity's data strategy.

You need to be alert to the fact that it may also be appropriate to highlight ethical considerations when attempting other requirements in the exam. For example, when answering a requirement which asks for an evaluation of a proposed strategy, you should also consider whether the proposal will give rise to any ethical or sustainability issues.

Scenarios might include topics such as:

- The impact of ethics on strategies and strategic choice
- Conflicts of interest among stakeholders
- Attempts to intentionally mislead key stakeholders (by disclosure or non-disclosure of information)
- Doubtful accounting or commercial business practice
- Facilitation of unethical strategies
- Inappropriate pressure to achieve a result
- Conflict between the accountant's professional obligations and responsibilities to the organisation
- Lack of professional independence – eg, personal financial interest in business proposals
- Actions contrary to law, regulation and/or technical and professional standards

Such issues are often not clear-cut, and it is important that you can develop a balanced argument, use appropriate ethical language and have regard to any relevant professional principles.



Professional skills focus: Assimilating and using information

One of the skills tested in the ACA exams is the ability to identify ethical issues, including public interest and sustainability issues, in a range of business situations. Read the scenario detail carefully and use ethical language and relevant professional principles to help frame your response. Wider business reading can help you to understand the many ethical dilemmas that real businesses face and the actions they take to manage them.

9.2 Recommended approach

The following is a suggestion for the approach to adopt when tackling questions involving ethical issues:

(a) Do I have all the facts and/or information? For instance, if the information is from a newspaper report, is it credible?

(b) Is there a legal or regulatory issue (criminal or civil law)? For instance, are the persons in question directors who owe statutory duties to their companies? When attempting exam questions in some cases it may be unclear whether or not there is a legal issue, if you are unsure you are encouraged to consider whether legal advice should be sought.

(c) Do any other codes or professional principles apply? For instance, is the individual with the ethical dilemma a professional accountant, or is the organisation with an ethical issue subject to its industry's particular code of conduct?

(d) Which stakeholders does the decision/action impact?

(e) Which ethical concepts/principles/threats are relevant?

(f) Ask the key questions of a suggested course of action:

  (1) Do the benefits outweigh the harm to oneself?

  (2) Do the benefits outweigh the harm to others?

  (3) Are the rights of individual stakeholders considered and respected?

  (4) Are the benefits and burdens justly distributed?

(g) If this action/decision is NOT taken, what are the issues?

(h) Are there any sustainability or ESG issues?

(i) What actions should be taken next?

These steps will be illustrated using the worked example that follows.

Key weaknesses in answering ethical questions have been:

(a) Failure to identify the ethical issue (eg, dishonesty, illegality, lack of transparency or unfairness)

(b) Failure to use ethical language

(c) Answering simply by asserting an opinion rather than applying balanced reasoning using ethical principles

(d) Failure to identify appropriate safeguards

(e) Application of professional accountants' ethical codes to individuals in the scenario who are not accountants

(f) Failure to distinguish between the ethical responsibilities of the individual and those of the organisation

(g) Confusing commercial and ethical implications or solely focusing on commercial considerations and ignoring ethical factors

To produce a good answer when attempting an ethical question it is critical that you adopt a structured approach. In some cases, it may be worth considering ethics at a personal, business and corporate level.



Professional skills focus: Concluding, recommending and communicating

It is always good practice to end your discussion of ethical issues with a conclusion.
Do not use the concluding paragraph to repeat the content of your earlier discussion but instead use it to pull together the strands of your argument and to state your final position on how the ethical dilemma is best resolved. Conclusions in ethics questions should always provide a clear course of action for the relevant affected parties discussed in the scenario.

Worked example: Pharmaceutical dilemma

Pharma Co is a highly successful global pharmaceutical company.
Pharma's board has recently received advance warning of the likely results of a recent study which is not due to be completed for another 12 months. The study is being conducted by Pharma as part of on-going product testing for its drugs that are already available in the marketplace.
Although the study is only 30% complete, the initial indications are that one of Pharma's leading drugs, which significantly improves the quality of life for sufferers of a long-term chronic illness, may cause harmful side-effects if taken at the same time as a newly available over-the-counter medicine that can be self-prescribed for pain relief.
The board, mindful of the company's social responsibilities and the potential for bad publicity, is currently discussing what action to take in response to this discovery but is concerned about the harmful impact of any premature announcement on its share price.

Requirement

Discuss the ethical implications of this situation for Pharma and any potential for stakeholder conflict that arises as a result.

Solution

(The following solution uses the questions in the recommended approach as a prompt to generate a balanced answer.)

Ethical implications

(1) **Do we have all the facts and/or information?**

The study is only 30% complete, so the possibility that in the end the situation may be better than it appears currently should be considered. It is also sensible to double check that correct information gathering procedures are being implemented in the study, and that the preliminary conclusion has been soundly arrived at.

(2) **Is there a legal issue (criminal or civil law)?**

There is likely to be considerable UK and international legislation governing the pharmaceutical industry and the safety of drugs and medicines. This may dictate the responsibilities of Pharma's directors in relation to transparency and communication of any possible safety issues and the timing of any announcements.

(3) **Do any other codes or professional principles apply?**

The pharmaceutical industry will almost certainly have its own code of conduct and guidelines to ensure members behave in an ethical and professional manner. As a highly successful player, Pharma is likely to have a high profile in the industry and will be expected to set high standards of responsible behaviour. Most drugs on the market carry warnings of possible side-effects. A key issue here is the timing of any announcement and the point at which the legislation and/or any professional code of conduct specifies that an announcement and warning of possible side- effects is necessary.

(4) **Which stakeholders does the decision/action impact?**

Pharmaceutical companies are commercial organisations and it is reasonable to assume Pharma's long term objective is to maximise shareholder wealth. If the board announces the research findings and they subsequently prove to be unfounded, then Pharma's share price and shareholders are likely to have suffered unnecessarily. Also sufferers of the chronic illness who stop taking the drug immediately may have a lower quality of life as a result. However were the board not to announce preliminary findings and the tests went on to be conclusive, patients taking the drug may experience adverse side-effects in the meantime.

(5) **Which ethical concepts/principles are relevant?**

Relevant concepts and principles here include issues of integrity and honesty. By announcing the information from the study at this stage, it could be argued that the directors at Pharma would be acting with integrity and honesty, as this would allow users of the drug to make better informed decisions about the combination of medications that they take. This approach would, however, create a conflict of interest between the directors' duties to shareholders and Pharma's responsibility to potential users.

(6) **If this action/decision is NOT taken, what are the issues?**

Here, it is appropriate to consider the implications in terms of transparency, effect and fairness.

(7) **Are there any sustainability or ESG issues?**

The sustainability of Pharma's reputation and its long-term profitability could be significantly affected if it comes to light that the drug is harmful, and that Pharma's board were aware of this in advance but delayed giving warnings.

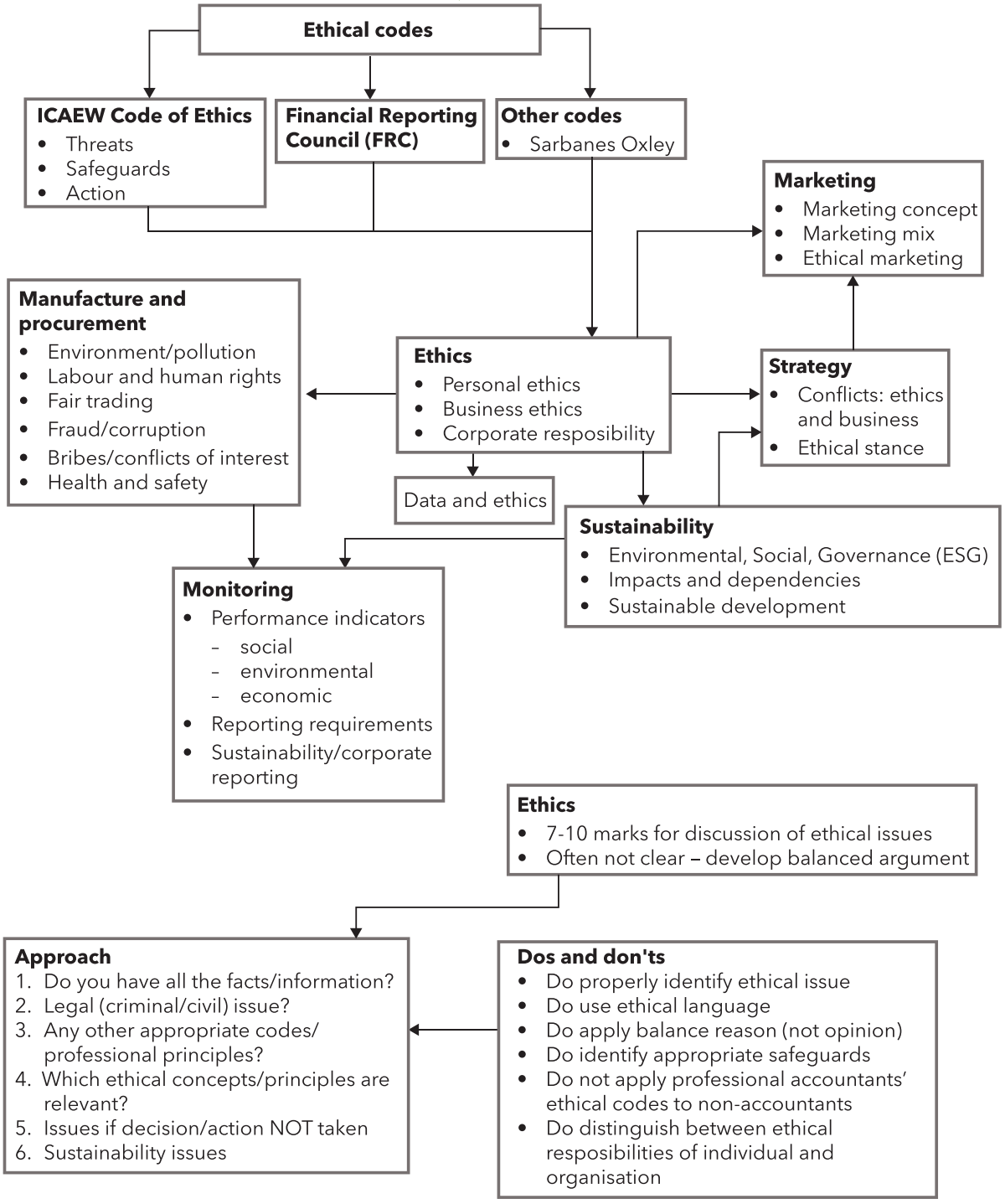
Stakeholder conflict

Stakeholders are groups of people who are interested in what Pharma does. In the case of Pharma's approach to the safety of its products this would include internal stakeholders (employees, management), connected stakeholders (shareholders, customers) and external stakeholders (government/regulatory bodies).
Here, the issue highlights the potential for conflict within and between various stakeholder groups. Shareholders want profitability so some may prefer Pharma to limit the damage and keep the information confidential until it is forced by legislation and/or regulation to announce it. Users of the drug and regulatory bodies may value safety more highly however and prefer to be in possession of the relevant information to make an informed choice.

Conclusion

Which stakeholders' interests determine Pharma's actions will depend to an extent on their relative power. Clearly Pharma's primary focus is to maximise shareholder wealth. However, the users of the drugs are very important to success and may choose to stop buying the product or look to other companies if they think their safety is being compromised. Also, shareholders may acknowledge that going beyond the basic requirements and demonstrating a responsibility to wider society will enhance Pharma's reputation and ensure government and public support, giving it a competitive advantage and higher profits in the long term.

Summary



CHAPTER 16

Further question practice

1 Knowledge diagnostic

Before you move on to question practice, confirm you are able to answer the following questions having studied this chapter. If not, you are advised to revisit the relevant learning from the topic indicated.

Confirm your learning	
(1)	Do you know the ICAEW Code of Ethics' five fundamental principles and the threats to these? (Topic 1)
(2)	Do you know the difference between sustainability and ESG? (Topic 5)
(3)	What are the benefits of being socially responsible? (Topic 7)
(4)	How can sustainability be measured and reported on? (Topic 8)
(5)	Do you know the recommended approach to dealing with ethics requirements in the exam? (Topic 9)

2 Question practice

Aim to complete all self-test questions at the end of this chapter. The following self-test questions are particularly helpful to further topic understanding and guide skills application before you proceed to the next chapter.

Question	Learning benefit from attempting this question
1	This is a knowledge-based question asking you to explain the three levels at which ethics can exist.
3	This question tests your knowledge of appropriate safeguards for a professional accountant working in business.
8	This short questions tests whether you understand the Global Reporting Initiative framework.

Once you have completed these self-test questions, it is beneficial to attempt the following questions from the Question Bank for this module. These questions have been selected to introduce exam style scenarios that will help you improve your knowledge application and professional skills development before you start the next chapter.

Question	Learning benefit from attempting this question
Zeter Domestic Chemicals plc (Dec 2016)	Part 3 of the question asks you to assess threats to the fundamental principles including a familiarity threat and an intimidation threat.
Muple plc (June 2018)	Part 2 of the question requires you to address the ethical issues of lack of integrity and intimidation.

Refer back to the learning in this chapter for any questions which you did not answer correctly or where the suggested solution has not provided sufficient explanation to answer all your queries. Once you have attempted these questions, you can continue your studies by moving on to the next chapter.

Self-test question

Answer the following questions.

1 Explain the three levels at which ethics can exist.

2 Give some examples of threats to self-interest that might arise for a professional accountant in business.

3 What are the sources of possible safeguards for a professional accountant working in business?

4 What action should a professional accountant take if threats cannot be reduced and safeguards have been exhausted?

5 What areas of a supplier contract may be affected by an organisation's ethical procurement policy?

6 What is meant by a sustainable enterprise?

7 Explain how questions of sustainability tie in with the short/long term debate?

8 What three areas of performance should be measured according to the Global Reporting Initiative?

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

The work of the accountancy profession is crucial to the effective working of the capital markets (that is the mechanism for the provision of finance to business and the protection of those who supply it). Put very simply, investors may lose financially if their investment decisions are based on inadequate information, and this would deter further investment. A loss of confidence in financial reporting could therefore undermine the economy.

Working to the highest principles of ethics and professionalism allows the public, investors and regulators to have confidence in the profession. Ethical behaviour of Chartered Accountants not only helps to protect their own, individual reputation but that of the profession as a whole. Chartered Accountants have a collective responsibility to apply principles, such as integrity and objectivity, that will enable high quality financial reporting, and effective financial management and business practices; to protect investors, businesses and the wider public interest.

Answer to Interactive question 2

Ethical threats

Advocacy – acting as tax advisers at the same time as being independent external auditors (there is some self-review threat here too)

Self-interest – low fees; compromised ethics in order to retain the engagement

Intimidation – pressure to reduce fees; could a small firm stand up to a large multinational if they disagreed on a proposed accounting treatment or modification to the auditor's opinion?

Familiarity – the consultant is friends with many senior managers at GSA which could lead to a lack of objectivity and independence in certain judgements during the audit

Self-review – internal audit services will need to be relied upon for external audit opinion

Professional judgement

You do not have all the relevant information to make a judgement. You overheard the end of a conversation and have not seen any written evidence of the proposed tax scheme. Without all the facts and evidence, it is possible that the situation has been misinterpreted. Despite this as a professional firm you would not expect the engagement partner to be willing to accept inducements to act unethically and against the public interest.

Furthermore, you lack experience in providing tax advice so the proposed scheme may be within the confines of tax law and deemed to be ethically acceptable, despite the comments made by the FD to keep the tax savings quiet.

Because of the close relationship with GSA and many of the staff working there it will be difficult for you to act impartially and to exercise professional judgement without any bias, when assessing the ethical implications of the engagement.

Action

There is clear evidence of breaches to the fundamental principles and suggestions that the engagement partner is acting unethically by keeping audit fees low and potentially accepting inducements. However, you may struggle to exercise professional judgement for the aforementioned reasons. The ethical issues should therefore be escalated to more senior management. In this instance, it would appear that the engagement partner is unlikely to be sufficiently independent of the client in order to exercise appropriate professional judgement.

You should therefore speak to a superior or partner who is removed from the GSA relationship.

Answer to Interactive question 3

Threat	Safeguards
Russell Ltd This involves a **self-interest threat** as the consultants' acceptance of the invitation to enjoy the lavish hospitality on offer increases the danger that the consulting team may be reluctant to raise any contentious matters as part of the cost cutting review. There is also a **familiarity threat** because involvement in social events with the client is likely to increase the consulting team's familiarity with client staff which may impair their ability to give impartial advice.	Gifts and hospitality should not be accepted unless the value is trivial and inconsequential. In this case it, would be appropriate to decline the weekend away, so as not to impair the consulting firm's independence.
Ravencroft White plc This involves a **familiarity threat** because Ms Singh has been the engagement partner for 11 years and their long association with the client could mean that they do not question judgements made by the client and do not exercise sufficient professional scepticism.	Ms Singh should be rotated off the audit and another partner assigned to the client. Given that Ravencroft White is a public listed entity, the engagement partner should serve ideally for no more than five years before being rotated off. The five-year term for a listed client engagement partner may be extended for an additional period of up to two years if the client's audit committee considers that this extension is necessary to safeguard the quality of the audit.
Sole Records Ltd Given that the consultant's daughter works for Sole Records largest competitor, there is a danger that a **confidentiality threat** could arise. Mr Minus may be tempted to pass on confidential information about Sole Records' operations to his son. An **intimidation threat** could also exist if the daughter tries to pressure their father into providing Sole Records with poor consulting advice to give their own employer a competitive advantage. It should, however, be noted that Mr. Minus's daughter has only taken up a very junior position with the competitor so in reality their degree of influence over their father's work is likely to be insignificant.	Mr Minus should be removed from the consulting team and replaced by an independent consultant. If it is felt that Jemima does not have any influence over their father's work at Sole Records, it may be appropriate for Mr Minus to remain as the principal consultant. This illustrates the need for discretion and tact when assessing ethical issues as a perceived threat may not actually represent a threat at all. Mr Minus may wish to disclose their connection with the competitor to avoid any perceived conflict of interest in his dealings with Sole Records Ltd.

Answer to Interactive question 4

The board's decision to request an 'annual voluntary investment' from the company's suppliers seems ethically dubious at best. It could be argued that the scheme represents a form of 'blackmail' as failure to comply will result in a supplier's removal as an approved supplier for reasons that are divorced from the quality of the supplies they make.

Honesty, integrity and transparency

As the board of directors have requested that you avoid 'unnecessarily publicising' the scheme this clearly indicates that they as a collective do not wish to be open about their suggestion and see it publicised at all. Consumer groups, industry bodies and government watchdogs would most likely perceive this move as dishonest, lacking in integrity and going against general wisdom on how business should be conducted. Furthermore, calling the scheme an 'annual voluntary investment' could be regarded as being misleading, as the suppliers are not able to opt out from making the payment without losing a major customer.

Impact on stakeholders

If you were to proceed and introduce the scheme, then it is likely that a number of groups could be affected by your decision. Most obviously, the decision to charge suppliers a fee to continue supplying Happy Foods may put them under financial pressures, especially smaller suppliers.

The degree of financial distress this may cause is difficult to measure as no details are given as to the size of payments the board at Happy Foods expect the suppliers to make. Clearly for some suppliers the ability to make such payments every 12 months may prove to be unsustainable, thereby forcing them out of business altogether.

Given that Happy Foods has experienced tough trading conditions it seems highly plausible that suppliers will have also been encountering similar financial problems which will only be made worse by the introduction of the scheme. This increases the danger that certain suppliers may need to rationalise their own operations to make the annual payments and maintain supply to Happy Foods. Such moves are likely to impact on the suppliers' own employees (through pay cuts and redundancies).

Consumers are also likely to be affected this decision, as suppliers of ingredients may increase the prices that they charge other food producers that they supply, who in turn increase the prices of the food products they sell. As a stakeholder in this situation you too are clearly going to be affected.

Careful consideration needs to be given to the motives of the board in asking you alone to contact the suppliers about the charge. Could this represent an attempt by the board of directors to distance themselves from your actions? In the event that the scheme became publicised in the press it seems plausible that the board could attempt to place the accountability for the scheme at your feet, by downplaying your actions as that of an individual 'rogue' employee acting on their own initiative.

Such an occurrence would leave your own career in a potentially precarious position.

Equity and fairness

Although unknown at this point, it seems reasonable to suggest that the affected groups outlined above would most likely consider the introduction of the scheme and your support in facilitating it to be unfair. On the face of it the scheme seems to prop up Happy Foods' own operations at the expense of suppliers who up until now had received fair payment for the goods supplied.

This point is exacerbated as the affected groups do not stand to gain anything from the payment beyond what they already had in the first place ie, a deal to supply ingredients to Happy Foods. However, it is worth noting that the notion of what is fair is likely to vary from stakeholder group to stakeholder group.

The perception that your own actions (and those of Happy Foods) could be regarded as being fair is further undermined by the board's attitude to the company's stakeholders, as no consideration appears to have been given to the impact of this decision on them.

Answer to Interactive question 5

5.1 Memorandum

To: Senior partner

From: Accountant

Date: Today

Subject: Marketing and responsibility issues

I shall outline the broader perspectives of marketing to food consumers and food manufacturers and consider the implications of these issues.

Benefits to business organisations, consumers and society

Marketing touches everyone's life. It is the means by which a standard of living is developed and delivered to people and is a human activity directed at satisfying needs and wants through exchange processes. Marketing oriented companies combine many activities – marketing research, product development, distribution, pricing, advertising, personal selling and others – designed to sense, serve and satisfy consumer needs whilst meeting the organisation's goals. The core concepts of marketing are needs, wants, demands, products, exchange, transactions and markets that will benefit the individual, consumers and society at large, organisations and national and international governments. Marketers must be able to manage the level, timing and composition of demand from these different beneficiaries to satisfy their needs and wants. For instance, McDonald's have adopted the broader marketing concept on a global scale through understanding and responding to the changing needs of their customers.

Modern marketing is guided by a number of converging philosophies. The production concept holds that the consumer favours products which are available at low cost and that marketing's task is to improve production efficiency and bring down prices. The marketing concept holds that a company should research the needs and wants of a well-defined target market and deliver the desired satisfactions, which is accompanied by long-run societal wellbeing. In marketing-led organisations the entire workforce share, the belief that the customer is all important and that building lasting relationships is key to customer retention. A company's sales are derived from satisfying existing customers and attracting new customers. This approach benefits the livelihoods of the employees and suppliers and their staff. Successfully adopting a marketing approach improves customer retention and minimises additional costs. A satisfied customer buys more, stays loyal longer, talks favourably to others, pays less attention to competing brands and is less price sensitive. These benefits are transferred into gains for consumers and ultimately society at large as success breeds success. McDonald's divert much of their energies to ensuring that customers repeatedly return to them satisfied and content with their offering.

5.2 Responsibilities of the modern marketer

Marketing is seen as a social force which not only conveys a standard of living but also serves as a force that reflects and influences cultural values and norms. The boundaries of marketing extend beyond economic considerations. Marketing concepts and techniques are used to promote the welfare of society as a whole instead of the traditional approach of providing

products that satisfy consumers' needs efficiently and profitably. They could encompass reduction in poverty, improved education and improved healthcare. For instance, marketing tools are used in promoting healthier lifestyles through better diets, encouraging leisure activities and pursuits and social behaviour. Social marketing suggests that a more ethical and moral orientation be incorporated into companies' marketing strategies: marketers should consider and incorporate the wider social implications of their products and services, such as natural conservation or labour exploitation in emerging countries.

Social marketing does not imply a replacement of the traditional marketing concept, but it is an extension so as to recognise and encompass the wider needs of society at large. The criticisms of marketing generally focus on ethical issues and the extent to which marketing is responsible for a variety of social and environmental problems. Whatever the reasons, voluntarily or otherwise, marketers have to consider ecological, environmental and consumer welfare issues together with their wider social role more frequently in their marketing plans and activities.

Effective and aware marketers have responded to these developments in a number of ways, for instance, by producing recyclable products and packaging, reducing pollution generated by toxic products or from contamination and protecting consumers against harmful or hazardous products by modifying them or withdrawing them from sale.

Answer to Interactive question 6

(a) **Short term**
- Additional costs – using greener fuels, buying fairly traded products, rebuilding green areas, paying higher salaries
- Reduced income – no sales to unethical purchasers
- Distribution of wealth to non-shareholders – giving to charity
- Distorts focus – one more target for managers

(b) **Long term**
- Avoids legislative imperative – will otherwise fall to Government to enforce and measures may be more draconian
- May avoid financial penalties where behaviour is covered by legislation
- Improves PR
- Attracts ethical investors
- Attracts ethical consumers
- Improves staff morale

Answer to Interactive question 7

Ethical issues

Personal ethical issues

As an ICAEW chartered accountant you have a responsibility to follow the ethical code of ICAEW.

Just because a product contains an ingredient which is not illegal for use at the current time does not mean that it is ethical for consumers to use it if concerns over the products safety come to light. The fact that you became aware of the issue with the ingredient via an email which discusses non- disclosure to the other directors supports the need to act.

The first thing you should do is discuss the matter with your manager in the finance department, and then contact the ICAEW ethical department for advice.

Directors' ethical responsibilities

The fact that both the MD and head of production are aware of the potential problem is a governance issue, and the fact that they are keeping it secret from the remainder of the board

is a breach of their duties to the company as a whole and potential shareholders who may invest in SSC when the company floats.

The MD and head of production have not however, done anything that appears to be illegal. There is a difference between breaches of legislation and activities that appear to be unethical.

Corporate ethical responsibilities:

The Skin Care Company has a responsibility towards its customers and their wellbeing. Many corporate entities do have a code of ethics which covers all stakeholders, given SSC's historic approach to business it is highly likely that the company does have a code of ethics. Although the other directors are unaware of the issues relating to the ingredients in the product, it could be argued that this is due to poor internal controls, which should identify issues of this nature. The responsibility for maintaining a system of internal controls lies with the board as a whole, and so they could be collectively responsible for the failings to identify any health-related concerns.

Answer to Interactive question 8

8.1 When more than one stakeholder group has reason to question the otherwise good reputation of an organisation, the effect can be a downward spiral leading to a general lack of confidence which, in turn, can have unfortunate financial effects. In particular, however, poor ethical principles are likely to affect one or more of the organisation's interactions with:

- **Customers:** Customers will expect certain standards of health and safety and ethical behaviour from Ashdene Homes, especially regarding its treatment of employees. The recent damage to their reputation may reduce confidence among customers leading to reduced sales – with a subsequent impact on corporate profits.
- **Shareholders:** Investor confidence is important in public companies and any reputation risk is likely to be reflected in market value. Shareholders may invest in buying shares, or their wealth, tied up in pension funds, may be invested for them by investment firms. The growth in ethical funds management where investment firms guarantee their customers not to invest in ethically-unsound organisations has led to company directors addressing the issue in earnest rather than giving it cursory attention.
- **Senior management:** Poor ethical behaviour from them creates a poor perception of the organisation in the market. However, poor ethical behaviour from those below can also have a negative impact on such executives and make them wish to disassociate themselves from a failing enterprise; the loss of key talent may be sorely felt by those who remain.
- **Employees:** Although not directly affected, poor ethical principles may leave the employee feeling that they no longer have a worthy association with the firm, which may cause them to leave or be de-motivated as a result. Also, if the organisation exhibits poor ethical principles, employees may feel that they either can or even should follow suit, and a general decline in standards will follow.
- **Suppliers:** Also not greatly affected, but it may be the case that suppliers decide not to deal with Ashdene Homes because they feel that the poor ethical principles will in some way implicate themselves.

8.2 Acting in a socially responsive way generally refers to business decision-making linked to ethical values, compliance with legal requirements, and respect for people, communities and the environment.

There is a growing view that the best-managed companies are those that are aware of their wider responsibilities to the social community and to the environment. To ensure that a company honours those responsibilities and protects its reputation, it is necessary to embed these core values into the policies, practices and programmes of the company's systems and decision-making processes.

The sustainability issues that affect companies vary according to the nature of the company but there are five areas where sustainability might be relevant:

- To treat employees fairly and with respect
- To operate in an ethical way and with integrity
- To respect human rights
- To sustain the environment for future generations
- To be a responsible neighbour in the community

There are several reasons why Ashdene Homes might choose to act in a socially responsible manner:

- They might want to act voluntarily to avoid legislation. For example, to avoid excessive pollution of the environment in their methods of working and by buying materials locally to reduce transport use and avoid allegations of their suppliers adding to deforestation.
- They might want to act in an ethical and socially responsible way by making the houses eco-friendly, reducing carbon emissions, having rigorous health and safety checks on their building sites and incorporating recycled materials where possible into the buildings.
- They might want to respond to pressure from shareholders. Some institutional shareholders have a policy of investing only in socially responsible and ethical companies.
- To protect their reputation.

The risk to the company's reputation from adverse publicity about social and environmental factors is always difficult to assess. Ashdene Homes will be aware that adverse publicity can have a damaging effect on customer goodwill – and sales and profits. Management might need to consider sustainability as a strategic issue when evaluating their strategic options.

8.3 Ethical responsibilities of a professional accountant

A professional accountant has two 'directions' of responsibility: one to their employer and another to the highest standards of professionalism.

Companies provide a Code of Ethics that all employees are expected to follow to maintain a culture of corporate ethics. Issues to be included in such a Code of Ethics are:

- Avoiding conflicts of interest
- Compliance with laws and regulations
- Rules about disclosure or avoidance of opportunities for personal gain through use of company property or their position in the company
- Confidentiality – extending to absolute discretion of all sensitive matters both during and after the period of employment
- Fair dealing with customers, suppliers, employees and competitors
- Encouragement to report illegal and unethical behaviour

The responsibilities also include the expectation that the accountant will act in shareholders' interests as far as possible and that they will show loyalty within the bounds of legal and ethical good practice.

In addition to an accountant's responsibilities to their employer, there is a further set of expectations arising from their membership of the accounting profession. In the first instance, professional accountants are expected to observe the letter and spirit of the law in detail and of professional ethical codes where applicable (depending on country of residence, qualifying body, etc).

In any professional or ethical situation where codes do not clearly apply, a professional accountant should apply 'principles-based' ethical standards (such as integrity and probity) such that they would be happy to account for their behaviour if so required. Finally, and in common with members of other professions, accountants are required to act in the public interest that may involve reporting an errant employer to the relevant authorities. This may be the situation that an accountant may find themselves in at Ashdene Homes. It would clearly be unacceptable to be involved in any form of deceit and it would be the accountant's duty to help to correct such malpractice if at all possible.

Answers to Self-test questions

1 **Personal ethical behaviour**

This relates to the way you as an individual conduct yourself.

Business ethics

This is the way a firm as a whole behaves.

ESG

This is the belief that a firm owes a responsibility to society and its wider stakeholders, as well as to shareholders.

2 Threats to self-interest include:

- Financial interests
- Loans or guarantees
- Incentive compensation arrangements
- Inappropriate personal use of corporate assets
- Concern over employment security
- Commercial pressure from outside the employing organisation

3 Safeguards might be generic, created by the profession or regulation or developed in the working environment by the individual or their organisation.

4 In circumstances where a professional accountant in business believes that unethical behaviour or actions by others will continue to occur within the employing organisation, they should consider seeking **legal advice**.

In extreme situations where all available safeguards have been exhausted and it is not possible to reduce the threat to an acceptable level, a professional accountant in business may conclude that it is appropriate to **disassociate** from the task and/or **resign** from the employing organisation.

5 Contract terms might cover:

- The use of child labour
- Human rights
- Health and safety standards
- Working conditions
- Minimum wages
- Environmental protection
- Terms for contract negotiation and renewal
- Basis of contract price

6 Sustainable enterprise: A company, institution or entity that generates continuously increasing stakeholder value through the application of sustainable practices through the entire base activity – products and services, workforce, workplace, functions/processes, and management/governance (Deloitte: Creating the Wholly Sustainable Enterprise).

7 The short-term/long-term debate refers to the trade-off management must make between decisions with short-term impacts on the business and those with impacts on its longer-term success. Here, the assumption is that sustainability will have an adverse short-term impact on the business, for example due to the enhanced costs of compliance, but that it is essential to its long-term success in the face of mounting social and legal pressure to improve ecological performance.

Some writers suggest that there may be short-term benefits from sustainability, such as reduced costs from using less energy and other resources or attracting customers who will place contracts or buy the offerings of firms with better sustainability postures (eg, 'carbon neutral').

8 **Economic** – the organisation's impacts on the economic conditions of its stakeholders

Environmental – an organisation's impacts on living and non-living natural systems, including ecosystems, land, air, and water

Social – the impacts an organisation has on the social systems within which it operates.

Chapter 17
Spreadsheet functionality for Business Strategy and Technology

Introduction

Examination context

Chapter study guidance

Learning topics

1 Spreadsheet formulae in Business Strategy and Technology

2 Further examples of using spreadsheets in Business Strategy and Technology



Introduction

Examination context

You will be expected to be able to use a number of spreadsheet functions and functionality in your BST exam in order to assimilate and analyse data for use in decision making.

This chapter summarises each of the functions that are likely to be examinable.

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this Chapter.

Topic	Practical significance	Study approach	Exam approach
1	**Spreadsheet formulae in Business Strategy and Technology**		In the exam, you may be asked to perform calculations to support your analysis of a company's performance. It is much easier to do this using the spreadsheet functions covered in this chapter.
2	**Further examples of using spreadsheets in Business Strategy and Technology**	In this section, we provide some comprehensive worked examples, which demonstrate the formulae	It is advised that you work through each question prior to the exam.

1 Spreadsheet formulae in Business Strategy and Technology

Section overview

- There are a number of calculations in the BST exam that can be performed more efficiently using spreadsheet functions. This chapter covers the spreadsheet functions that are examinable in BST which are:
  - AVERAGE
  - CORREL
  - COUNTIF, COUNTIFS
  - IF
  - IFS (Including nested IF statements)
  - RANK
  - STDEV
  - SUM, SUMIF
  - SUMIFS
  - TREND
- This chapter also introduces you to relative and absolute cell referencing which are important when analysing large data sets with the spreadsheet functions listed above. You will also need to be able to format cells correctly (percentages, monetary values, decimal places) ensuring consistency throughout your answer.
- Several questions have been amended in the ICAEW BST Question Bank so that you can practise using spreadsheet functionality. These are:

Question name	Spreadsheet functionality
Wheely Bus	AVERAGE
Reyel	CORREL
ARU	COUNTIF
Just Houses	IF
Zed Museum	IFS
Ignite plc	RANK
Kentish Fruit Farms	STDEV
Forsi	SUM
Zeter Domestic	SUMIF
Taxi Tracker	SUMIFS
The Healthy Vegetarian	TREND
Thistle Engineering	What If analysis

1.1 AVERAGE

The AVERAGE function is used to calculate the arithmetic mean for a set of values. This function allows an average to be calculated quickly. Because it can apply to a range of cells, the user does not have to specify every cell to be included (although they can do so if they wish). The AVERAGE formula format was introduced in Business, Technology and Finance (BTF).

1.2 CORRELATION

The CORREL spreadsheet function calculates the correlation coefficient between two data sets. For further information on how to calculate and use CORRELATION see the chapter, 'Risk management.'

1.2.1 CORREL formula format

=CORREL (cell range for array 1, cell range for array 2))

1.2.2 Using the CORREL formula

Type **=CORREL (**to begin the function entry. Drag the cursor over the first range of cells then insert a comma followed by the second range of cells. Finally insert a closed bracket).



Worked example: CORREL formula

The results from a school survey examining the hours that students spent studying at home and their subsequent exam grades are shown below:

	A	B
1	Hours studied	Exam grade
2	8	82
3	11	94
4	3	70
5	5	69
6	15	96
7	8	74
8	2	66
9	0	43
10	7	78

Requirement

Using the CORREL functionality calculate the correlation between exam grades and time spent studying and comment on the implications of your results.

Solution

C2=CORREL(A2:A10,B2:B10)

	A	B	C
1	Hours studied	Exam grade	Correlation coefficient
2	8	82	0.93
3	11	94	
4	3	70	
5	5	69	
6	15	96	
7	8	74	
8	2	66	
9	0	43	
10	7	78	

The correlation coefficient is 0.93 which is a strong positive correlation. A correlation coefficient of 1 would indicate perfect correlation, such that each extra hour spent studying would be reflected in a proportionally higher exam grade. The results can be used by the school to demonstrate to students the importance of home study in order to achieve the highest exam grades.

1.3 COUNTIF

The COUNTIF function counts the number of cells in a range that contain specific content. The COUNTIF function can be used, for example, to count the number of cells in a spreadsheet with a value over a certain amount, or those containing a specific word (eg 'Yes'; or 'No'). The COUNTIF formula format was introduced in Business, Technology and Finance (BTF).

1.4 COUNTIFS

The COUNTIFS function counts the number of cells in a range that meet one or more criteria.

1.4.1 COUNTIFS formula format

=COUNTIFS(range1, criteria1, range2, criteria2,...) where the range is the range of cells being examined and criteria being the content being looked for.

1.4.2 Using the COUNTIFS formula

Type=COUNTIFS(to begin the function entry. Drag the cursor over the range of cells to be analysed, add a comma then type the criteria. The criteria can be a number, a cell reference containing a number or word, or a formula, but it should be enclosed within " ". Repeat this process for all ranges and criteria you are wishing to examine.

Worked example: COUNTIF formula

A company's profit projections and forecast return on capital employed (ROCE) for the next six years are as follows:

	A	B	C
1	Year	Profit (£)	ROCE (%)
2	1	15,000	14
3	2	20,000	15.5
4	3	25,000	15
5	4	40,000	15.8
6	5	60,000	16.2
7	6	90,000	15.9

Requirement

Calculate how many years the company is expected to earn over £22,500 profit and also achieve a target ROCE of at least 15.5%.

Solution

To calculate how many years the company is expected to earn over £22,500 profit and achieve a target ROCE of at least 15.5% the instruction in Cell A8 would be:

=COUNTIFS(B2:B7,">22500",C2:C7,">=15.5")

There are three years out of six where profit exceeds £22,500 and ROCE is equal to or greater than 15.5%.

	A	B	C
1	Year	Profit (£)	ROCE (%)
2	1	15,000	14
3	2	20,000	15.5
4	3	25,000	15
5	4	40,000	15.8
6	5	60,000	16.2
7	6	90,000	15.9
8	31		

1 =COUNTIFS(B2:B7,">22500",C2:C7,">=15.5")

1.5 IF

The IF function makes a logical comparison between two values and returns one of two results depending on the outcome of the comparison.

The IF function enables the user to analyse the meaning of cells quickly. For example, the function can compare an 'actual' number with a 'budget' number, and return the text "Under budget" or "Over budget" depending on the result.

1.5.1 IF formula format

=IF(logical test, value if true, value if false] where the logical test performs a calculation and the value if true and value if false is what is displayed depending on the result of the logical test. If the cell is to be left empty when the value is false, a comma is required eg =IF(B1>20000,B1*C1,)

1.5.2 Using the IF formula

Type =IF(to begin the function entry. Then type the logical test (for example A3>1), then type a comma followed by what should be returned if the test is true (for example "Yes"), then type a comma followed by what should be returned if the test is false (for example "No").



Worked example: IF formula

The following report shows a company's actual and budgeted profits from the last six years:

	A	B	C	D
1	Year	Actual profit	Budgeted profit	
2	1	10,000	15,000	
3	2	15,000	20,000	
4	3	30,000	25,000	
5	4	35,000	40,000	
6	5	70,000	60,000	
7	6	95,000	90,000	

Requirement

Indicate in column D whether the actual profit for each year was "Under budget" or "Over budget".

Solution

To indicate in column D whether the actual profit for the year was "Under budget" or "Over budget", the instruction would be (for cell D2) **=IF(B2>C2, "Over budget", "Under budget")**

Once entered into cell D2 the formula should be dragged down to cell D7.

D2=IF(B2>C2, "Over budget", "Under budget")

	A	B	C	D
1	Year	Actual profit	Budgeted profit	
2	1	10,000	15,000	**Under budget**(1)
3	2	15,000	20,000	Under budget
4	3	30,000	25,000	Over budget
5	4	35,000	40,000	Under budget
6	5	70,000	60,000	Over budget

7	6	95,000	90,000	Over budget
8				
9				

Spreadsheet notes

(1) Cell D2 could be copied to cells D3, D4, D5, D6 and D7. This would mean, for example, that cell D3 would contain the formula. **=IF(B3>C3, "Over budget", "Under budget")**

This is an example of relative cell referencing whereby the cell reference within the formula changes when the formula is dragged into new cells.

1.6 IFS

IFS statements make a logical comparison between two or more values and return a value that corresponds to the first TRUE value. Nested IF statements are an alternative to IFS when you are looking to test for more than one condition.

1.6.1 IFS formula format

=IFS(logical test1, value if true1,logical test2, value if true…) Note that the IFS function only returns a result that corresponds to the condition being TRUE.

1.6.2 Using the IFS formula

Type =IFS(to begin the function entry. Then type the logical test (for example A2>100), then type a comma followed by what should be returned if the test is true (for example "Higher").

Type another comma before entering the second logical test (for example A2<100) followed by a comma. Now type what should be returned if the second test is true (for example "Lower").

The IFS statement will now look like this:

=IFS(A2>100,"Higher", A2<100, "Lower")

You can continue to add logical tests and values until you have tested all conditions.

Tutorial Note

If no TRUE conditions are found (for example if A2=100), IFS returns #N/A. To return a value when no TRUE conditions are found, type TRUE, followed by the value you wish to return as a default (for example "Same").

The IFS statement will now look like this:

=IFS(A2>100,"Higher", A2<100, "Lower",TRUE,"Same")

This can be seen in the worked example below.

Worked example: IFS formula

The HR director is going through a salary benchmarking process to ensure that employees are fairly remunerated. One particular role has a recommended salary of £35,000 and the HR director would like to compare the salaries of employees currently in the role against this salary figure.

	A	B
1	Recommended salary:	£35,000
2		
3	Employee number	Salary (£'000)
4	235	28,960
5	478	37,065
6	569	38,410
7	301	35,000
8	104	36,500
9	783	34,385
10	250	35,000
11	456	29,540
12	767	38,401

Requirement

Using the IFS formula, identify which employees are paid more than, less than or the same as the recommended salary of £35,000.

Solution

The IFS formula has been used to identify which employees are paid more than, less than or the same as the recommended salary.

The formula is entered into Cell C4 and can then be copied into Cells C5:C12.

Before copying the formula, it is important to fix the reference to Cell C1 (£35,000) by inserting $ before the column and row reference.

C4=IFS(B4>$C$1,"More",B4<$C$1,"Less",TRUE,"Same")

	A	B	C
1	Recommended salary		£35,000
2			
3	Employee number	Salary (£'000)	More/Less/Same
4	235	28,960	Less[1]
5	478	37,065	More
6	569	38,410	More
7	301	35,000	Same
8	104	36,500	More

9	783	34,385	Less
10	250	35,000	Same
11	456	29,540	Less
12	767	38,401	More

1 Cell B4 must be a relative cell reference so that the formula can be copied in C5:C12

1.7 Nested IF statements

Nested IF statements are an extension of IF statements, grouping several IF statements together. Like IFS, nested IF statements can be used when you are looking to test for more than one condition and return different results depending on those tests. IFS and nested IF statements can be used interchangeably so you should choose the approach you prefer.

1.7.1 Nested IF formula format

=IF(logical test, value if true, value if false,IF(logical test, value if true, value if false, IF(logical test, value if true, value if false))) Every function requires opening and closing brackets and the closing brackets go at the end of the statement. For example, three IFs mean three closing brackets.

Tutorial Note

It is not always necessary to have 'value if true' AND 'value if false' for each IF component. In the below example, students can either get an A grade, B grade or C grade. If they get none of these grades, they will get an F grade, which represents the 'value if false'. "F" is placed at the end of the statement rather than being repeated for each IF statement.

If we used the Nested IF formula approach to identify which employees are paid more than, less than or the same as the recommended salary (see IFS example above), the formula entered into Cell C4 would be: =IF(C4>$C$1,"More",IF(C4<$C$1,"Less","Same"))

	A	B	C
1	Recommended salary		£35,000
2			
3	Employee number	Salary (£'000)	More/Less/Same
4	235	28,960	Less1

1 Cell B4 must be a relative cell reference so that the formula can be copied in C5:C12



Worked example: Nested IF formula

The following report shows a list of student exam results. If they score 85 or more, they get an A grade. If they score 75 or more, they get a B grade. If they score 65 or more, they get a C grade. For any other score the student fails, denoted by F.

	A	B	C
1	Student	Exam results	
2	Angel	85	
3	Bradley	77	
4	Caroline	68	
5	Dan	73	
6	Elsie	78	
7	Fred	91	
8	Georgie	66	
9	Hugh	65	
10	Ida	63	
11	Jack	81	

Requirement

Indicate what exam grade Angel and Fred received.

Solution

	A	B	C
1	Student	Exam results	Exam grade
2	Angel	85	A
3	Bradley	77	B
4	Caroline	68	C
5	Dan	73	C
6	Elsie	78	B
7	Fred	91	A
8	Georgie	66	C
9	Hugh	65	C
10	Ida	63	F
11	Jack	81	B

To identify the grades achieved by the students a nested IF statement is used. Within the statement there are three 'value if true' instructions grouped together. To test whether Angel's exam results (cell B2) should be awarded an A grade, the instruction in C2 is =IF>=85"A"

To test whether Angel should be awarded a B grade, the instruction in C2 is =IF>=75"B"

To test whether Angel should be awarded a C grade, the instruction in C2 is =IF>=65"C"

If Angel does not get an A, B or C grade they will automatically get an F grade. The instruction placed at the end of the nested IF statement is ,"F". When combined, the instruction in C2 is:

=IF(B2>=85,"A",IF(B2>=75,"B",IF(B2>=65,"C","F")))

There are three IFs in the statement so there are three closed brackets at the end of the statement. Once entered into cell C2 the formula can be dragged down to cell C11.

Tutorial Note

The nested IF statement in this example could be replaced with an IFS statement as follows:

=IFS(B2>=85,"A",B2>=75,"B",B2>=65,"C",TRUE,"F")

IFS delivers the same output as Nested IF statements, but can be simpler to input and read when many conditions are tested.

1.8 RANK

The RANK function provides the rank of a number from within a series of numbers.

The rank of a number is determined by its size in comparison to the other numbers in a series. In other words, if the series of numbers was displayed as a list, the number's rank is the position of the number within the list.

1.8.1 RANK formula format

=RANK(number, ref, order) where number is the number which is being ranked, ref is an array, or a list, or a sequence of numbers and order refers to whether this list is to be sorted in ascending or descending order. If order is '0' or if no number is given for order, the list will be sorted in descending order (highest to lowest). If the value is any other value, the list will be sorted in ascending order (lowest to highest).

1.8.2 Using the RANK formula

Type =RANK(to begin the function entry. Then either type the number you are looking to rank (for example 56) or enter a cell reference containing that number, then type a comma and drag the cursor over the cells containing the numbers you wish to analyse. Then type another comma and the optional order value (0 or blank for descending order, any non-zero number for ascending order).



Worked example: RANK formula

The following is a list of ages of students attending a university course:

	A	B	C
1	Student	Age	
2	Angel	18	
3	Bradley	22	
4	Caroline	20	
5	Dan	30	
6	Elsie	26	
7	Fred	40	

8	Georgie	19	
9	Hugh	21	
10	Ida	24	
11	Jack	28	

Requirement

Calculate where Elsie's age ranks in the group of students, ranking in ascending order.

Solution

To calculate where Elsie's age ranks in the group of students, ranking in ascending order, the instruction would be =RANK(B6, B2:B11,1)

C6 = RANK(B6, B2:B11,1)

	A	B	C
1	Student	Age	
2	Angel	18	1
3	Bradley	22	5
4	Caroline	20	3
5	Dan	30	9
6	Elsie	26	**7**(1)
7	Fred	40	10
8	Georgie	19	2
9	Hugh	21	4
10	Ida	24	6
11	Jack	28	8

Spreadsheet notes

(1) This is the RANK for Elsie's age (26). **C6 = RANK(B6,B2:B11,1)**

Note, any non-zero number could have been entered instead of 1 as the optional order value. If a zero had been entered, or if the optional order number was omitted, the ranking would have been in descending order and Elsie's age would have ranked as 4.

1.9 STDEV

The STDEV function calculates the standard deviation based on a series of values which are the sample from the population. Standard deviation measures how wide the sample values are dispersed from the average value of the sample. To learn more about standard deviation look at the chapter, 'Risk Management.'

1.9.1 STDEV formula format

=STDEV(number 1, number 2…) where number 1 contains a range of cells or numbers containing the sample. Number 2 and onwards are optional ranges of cells or numbers also containing samples to be analysed.

1.9.2 Using the STDEV formula

Type =STDEV(to begin the function entry. Then drag the cursor over the cells containing the sample of numbers you wish to analyse. If you wish to add further ranges of cells for analysis, add a comma and then drag the cursor over the next group of cells and repeat as necessary.

Only cells containing directly typed numbers are counted. Empty cells, text cells and cells containing logical values (formulae) are ignored.

Instead of dragging the cursor over a range of cells you can also directly type the numbers you wish to analyse, each separated by a comma or include a logical value (formula) to calculate each number in the sample.



Worked example: STDEV formula

The following is a list of ages of students attending a university course.

	A	B	C	D
1	Student	Age	Student	Age
2	Angel	18	Kelly	23
3	Bradley	22	Luke	19
4	Caroline	20	Molly	32
5	Dan	30	Neil	28
6	Elsie	26	Olive	27
7	Fred	40	Peter	18
8	Georgie	19	Queenie	20
9	Hugh	21	Roger	21
10	Ida	24	Stephanie	20
11	Jack	28	Thomas	33

Requirement

Calculate the standard deviation of the two groups of student ages.

Solution

To calculate the standard deviation, the instruction would be **=STDEV(B2:B11, D12:D11)**

A12=STDEV(B2:B11, D12:D11)

	A	B	C	D
1	Student	Age	Student	Age
2	Angel	18	Kelly	23
3	Bradley	22	Luke	19
4	Caroline	20	Molly	32
5	Dan	30	Neil	28
6	Elsie	26	Olive	27
7	Fred	40	Peter	18
8	Georgie	19	Queenie	20
9	Hugh	21	Roger	21
10	Ida	24	Stephanie	20
11	Jack	28	Thomas	33
12	**5.9**			

1.10 SUM

The SUM spreadsheet function adds the values in a range of cells.

1.11 SUMIF

The SUMIF Function adds the values in a range of cells that meet certain criteria.

1.11.1 SUMIF formula format

=SUMIF(range, "criteria", sum range) where range is a range of cells to be added and criteria is a number, expression, cell reference or text that defines the cells to be added. The sum range is an optional argument which contains a range of cells that you actually wish to add, other than those specified in the range argument.

1.11.2 Using the SUMIF formula

Type **=SUMIF(** to begin the function entry. Drag the cursor over the range of cells containing the numbers that you wish to add as part of the range, then add a comma and enter the criteria for adding the numbers. The criteria should be expressed within " ". If you wish to add a sum range, add a comma after the criteria argument and enter the cells you actually wish to add. The sum range cell range should be the same size and shape as the range, for this to work properly.

Worked example: SUMIF formula

The following is a list of products and associated profits that were recently sold by a furniture store.

	A	B
1	Type of product	Profit
2	Bed	100
3	Cupboard	50
4	Table	150
5	Cupboard	75
6	Sofa	300
7	Bed	120
8	Bed	150
9	Sofa	250
10	Table	200
11	Cupboard	600

Requirement

Calculate the total profit earned from products that earned a profit of over £180.

Solution

To calculate the total profit earned from products that earned a profit of over £180, the instruction would be **=SUMIF (B2:B11,">180")**

	A	B	C
1	Type of product	Profit	
2	Bed	100	
3	Cupboard	50	**1,350**
4	Table	150	
5	Cupboard	75	
6	Sofa	300	
7	Bed	120	
8	Bed	150	
9	Sofa	250	
10	Table	200	
11	Cupboard	600	

Worked example: SUMIF formula (2)

The following is a list of products and associated profits that were recently sold by a furniture store.

	A	B
1	Type of product	Profit
2	Bed	100
3	Cupboard	50
4	Table	150
5	Cupboard	75
6	Sofa	300
7	Bed	120
8	Bed	150
9	Sofa	250
10	Table	200
11	Cupboard	600

Requirement

Calculate the total profit earned from sales of beds.

Solution

To calculate the total profit earned from sales of beds, the instruction would be **=SUMIF(A2:A11,"Bed",B2:B11)**

C3=SUMIF(A2:A11,"Bed",B2:B11)C3=SUMIF(A2:A11,"Bed",B2:B11)

	A	B	C
1	Type of product	Profit	
2	Bed	100	
3	Cupboard	50	**370**
4	Table	150	
5	Cupboard	75	
6	Sofa	300	
7	Bed	120	
8	Bed	150	
9	Sofa	250	
10	Table	200	
11	Cupboard	600	

1.12 SUMIFS

SUMIFS adds the values in a range of cells that meet multiple criteria, in comparison to SUMIF which only looks at one criterion in a range.

1.12.1 SUMIFS formula format

=SUMIFS(sum range, criteria range 1,"criteria1", criteria range 2, "criteria2") where sum range is the range of cells to be added, criteria range is the range of cells to be tested using the criteria and criteria is a number, expression, cell reference or text that defines the cells to be added.

1.12.2 Using the SUMIFS formula

Type **=SUMIFS(** to begin the function entry. Drag the cursor over the range of cells containing the numbers that you wish to add then add a comma. Now drag the cursor over the range of cells containing the first criteria you wish to test. Add a comma and enter the criteria you are testing in " ".

To test a second criteria add a comma after "criteria 1" and drag the cursor over the range of cells containing the second criteria you wish to test. Add a comma and enter the second criteria you are testing in " ". Repeat this process for as many criteria as you want.



Worked example: SUMIFS formula

The following data shows the volume of products sold by month and by country.

Requirement

Using SUMIFS functionality calculate:

- The volume of Product B sold in February
- The volume of Product B sold in February in France
- The volume of Product B sold in February that exceeds 80 units

	A	B	C	D	E	F	G	H	I
1	Month	Product	Country	Sales volume (units)		**Criteria**			
2	Jan	C	UK	88		Feb	Product B		
3	Jan	A	France	66		Feb	Product B	France	
4	Feb	D	France	74		Feb	Product B	>80 units	
5	Feb	B	Ireland	96					
6	Feb	A	Ireland	87					
7	Feb	B	France	90					
8	Mar	D	UK	43					
9	Mar	C	UK	52					

Solution

The formulae used are:

- The volume of Product B sold in February:
  =SUMIFS(D2:D9,A2:A9,"Feb",B2:B9,"B")
- The volume of Product B sold in February in France:
  =SUMIFS(D2:D9,A2:A9,"Feb",B2:B9,"B",C2:C9,"France")
- The volume of Product B sold in February, that exceeds 80 units:
  =SUMIFS(D2:D9,A2:A9,"Feb",B2:B9,"B",D2:D9,">80")

	A	B	C	D	E	F	G	H	I
1	Month	Product	Country	Sales volume (units)		**Criteria**			
2	Jan	C	UK	88		Feb	Product B		186
3	Jan	A	France	66		Feb	Product B	France	90
4	Feb	D	France	74		Feb	Product B	>80 units	186
5	Feb	B	Ireland	96					
6	Feb	A	Ireland	87					
7	Feb	B	France	90					
8	Mar	D	UK	43					
9	Mar	C	UK	52					

1.13 TREND

The TREND function returns values along a linear trend. TREND uses regression analysis to calculate the predictive values of Y for a given range of X values.

1.13.1 TREND formula format

=TREND(known y values, known x values, new x values) where known y values are the data you will use to predict new data, for example revenues. The known x values are a set of (optional) values that you may use to help predict new data, for example months.

The new x values are data for which you want the TREND to return corresponding Y values, for example future months.

1.13.2 Using TREND formula

Highlight ALL cells where you wish trend values to be returned. It is important to do this before you input the TREND formula.

Type =TREND(to begin the function entry. Drag the cursor over the range of cells containing the Y values (values you wish to find a trend or prediction for).

Type a comma and drag the cursor over the range of cells containing the X values. Type a comma and drag the cursor over the new X values for which you want to return a TREND value.

Notes

1. The values of X and Y must be the same length, ie both ranges should be, for example, 10 cells.
2. All cells that require a TREND value to be returned must be highlighted **before** typing =TREND(
3. X and Y values must be numeric, for example instead of typing January use 1 to indicate the first month.



Worked example: TREND formula

A school teacher wishes to predict the grades of three students, based on their recent exam scores. They have accessed past exam data from a previous class (Column A) and the corresponding exam grades (Column B).

The three students' exam scores are provided in Column C.

	A	B	C
1	Exam score (x values)	Exam grade (y values)	Student score
2	56	5	78
3	75	6	98
4	86	8	60
5	66	6	
6	96	9	
7	95	8	
8	85	7	
9	65	5	
10	100	9	
11	76	7	

Requirement

Calculate the predicted grades of the three students using the data in Columns A–C.

Solution

=TREND(B2:B11,A2:A11,C2:C4)

	A	B	C	D
1	Exam score (x values)	Exam grade (y values)	Student score (new x values)	Predicted grade
2	56	5	78	7

3	75	6	98	9
4	86	8	60	5
5	66	6		
6	96	9		
7	95	8		
8	85	7		
9	65	5		
10	100	9		
11	76	7		

1.14 Absolute cell referencing

When entering a formula that uses values from specific cells, a relative cell reference is usually used. This means that, if the formula is dragged vertically, the number part of the cell reference will change accordingly. If the formula is dragged horizontally, the letter part of the cell reference will change accordingly.

For example, the formula below (=B2-C2) is being entered into cell D2 to calculate the profit for Year
1. If the formula is dragged downwards to calculate the profit for Years 2–5, it will change to "=B3-C3" for Year 2 and "B4-C4" for Year 3 and so on.

D2=B2-C2

	A	B	C	D
1	Year	Revenue	Costs	Profit
2	1	150,000	100,000	**50,000**
3	2	275,000	195,000	
4	3	250,000	225,000	
5	4	425,000	365,000	
6		655,000	400,000	

There are times when a formula needs to use the contents of a particular cell or cells as part of the calculation. If the formula is dragged to other cells, the references to these particular cells will change and the calculation will be incorrect. However, cell references can be 'fixed' so they do not change as the formula is dragged. These are known as absolute cell references.

To create an absolute cell reference, a "$" is inserted before the letter or number part (or both parts) of the cell reference that needs to be fixed.

If the "$" is inserted before both parts of the reference, the cell reference will not change regardless of the direction the formula is dragged. For example, $A$2 ensures that neither the column nor row change when copied.

If the "$" is inserted only before the letter part of the cell reference, for example $A2, then the column does not change when copied.

If the "$" is inserted only before the number part of the cell reference, for example A$2, then the row does not change when copied.

The formula below is being entered into cell C2 to calculate the variable cost of product A (sales price multiplied by 60%). By including the "$", cell D2 is fixed and the formula can be dragged down to calculate the variable cost of the other products.

C2=B2*$D$2

	A	B	C	D
1	Product	Sales price	Variable cost	Variable cost percentage
2	A	150,000	**90,000**	60%
3	B	275,000		
4	C	250,000		
5	D	425,000		
6	E	500,000		

1.15 Formatting

It is important to ensure that values in cells are formatted correctly and consistently so that it is clear to the marker what they are looking at.

The key rules to remember are:

- Large monetary values (eg, a company's quarterly revenue) should be formatted as Accounting £ to 0 decimal places.
- Small monetary values (eg, the sales price of a product) should be formatted as Accounting £ to 0 or 2 decimal places.
- Any calculation resulting in a percentage should be formatted as a percentage cell to 0,1 or 2 decimal places.
- Pre-populated spreadsheets in the exam may provide dates. When a date is presented in long form (eg 1-June-21) it can be used to perform calculations so the format should not be changed. An example of this can be seen in the question HGF Ltd (see the ICAEW practice software). Using the pre-populated order and delivery dates, it is possible to calculate how many days early or late the delivery arrives.

Other quantitative analysis techniques

Businesses may also want to quantify forecasts for budgeting, planning and evaluation purposes. There are several quantitative techniques students should be familiar with, such as correlation, linear regression and time series analysis.

In exam scenarios, students must be able to explain and apply those techniques. They also must be able to discuss the suitability, principles, uses and limitations of these techniques.

Linear regression

The high-low method only takes account of two observations – the highest and the lowest. To take account of all observations, a more advanced calculation is needed. It is known as ***linear regression*** *and uses a formula to estimate the linear relationship between the variables.*

Linear regression is a simple statistical tool used to model the dependence of a variable (say, costs) on one or more variables (say, volume). This functional relationship may then be formally stated as an equation, with associated statistical values that describe how well this equation fits the data.

Linear regression equation

The equation of a straight line is:

$$y = a + bx$$

where y = dependent variable
a = intercept (on y-axis)
b = gradient
x = independent variable

and $$b = \frac{n\Sigma xy - \Sigma x \Sigma y}{n\Sigma x^2 - (\Sigma x)^2}$$

where n = number of pairs of data

and $$a = \bar{y} - b\bar{x}$$

Regression analysis is a technique for estimating the line of best fit, given a series of data. It is essentially a statistical technique, based on the concept of 'drawing the line that minimises the sum of the squares of the deviations of the line from the observed data' (so it is sometimes referred to as the 'least squares' method).

The regression line of y on x is used when an estimate of y (the dependent variable) is required for a given value of x (the independent variable).

Linear regression analysis can be used to make forecasts or estimates whenever a linear relationship is assumed between two variables, and historical data is available for analysis.

Two such relationships are:

- ***A time series and trend line****. Linear regression analysis is an alternative to calculating moving averages to establish a trend line from a time series. (Time series is explained later in this chapter.) The independent variable (x) in a time series is time. The dependent variable (y) is sales, production volume or cost.*

- ***Total costs, where costs consist of a combination of fixed costs and variable costs*** *(for example, total overheads, or a semi-variable cost item). Linear regression analysis is an alternative to using the high-low method of cost behaviour analysis. It should be more accurate than the high-low method, because it is based on more items of historical data, not just a 'high' and a 'low' value. In this case:*

- *The independent variable (x) in total cost analysis is the volume of activity.*
- *The dependent variable (y) is total cost.*
- *The value of a is the amount of fixed costs.*
- *The value of b is the variable cost per unit of activity.*

When a linear relationship is identified and quantified using linear regression analysis, values for 'a' and 'b' are obtained, and these can be used to make a forecast for the budget.

For example, a sales budget or forecast can be prepared, or total costs (or total overhead costs) can be estimated, for the budgeted level of activity.

Illustration 4 – Linear regression

X Co is forecasting its sales for the four quarters of 20X5.

It has carried out a linear regression exercise on its past sales data and established the following:

a = 20

b = 0.7

The equation of the regression line is therefore:

y = 20 + 0.7x

When x is number of the quarter and y is the sales value in $000s.

Calculate the sales for each of the quarters in 20X5.

Solution

Quarter	*Equation*	*$000*
Quarter 1	*(y = 20 + (0.7 × 1))*	*20.7*
Quarter 2	*(y = 20 + (0.7 × 2))*	*21.4*
Quarter 3	*(y = 20 + (0.7 × 3))*	*22.1*
Quarter 4	*(y = 20 + (0.7 × 4))*	*22.8*

Regression analysis is based on sample data. If we selected a different sample, it is probable that a different regression line would be constructed.

For this reason, regression analysis is most suited to conditions where there is a relatively stable relationship between the variables.

Illustration 5 – Marcus Aurelius

Marcus Aurelius is a small supermarket chain that has 6 shops. Each shop advertises in their local newspapers and the marketing director is interested in the relationship between the amount that they spend on advertising and the sales revenue that they achieve.

She has collated the following information for the 6 shops for the previous year:

Shop	*Advertising expenditure ($000)*	*Sales revenue ($000)*
1	*80*	*730*
2	*120*	*610*
3	*160*	*880*
4	*90*	*750*
5	*70*	*650*
6	*30*	*430*

She has further performed some calculations for a linear regression calculation as follows:

- *the sum of the advertising expenditure (x) column is 450*
- *the sum of the sales revenue (y) column is 4,050*

- *when the two columns are multiplied together and summed (xy) the total is 326,500*
- *when the advertising expenditure is squared (x^2) and summed, the total is 38,300, and*
- *when the sales revenue is squared (y^2) and summed, the total is 2,849,300.*

Calculate the line of best fit using regression analysis:

Advertising expenditure ($000) (x)	***Sales ($000) (y)***	***xy***	***x^2***	***y^2***
80	*730*	*58,400*	*6,400*	*532,900*
60	*610*	*36,600*	*3,600*	*372,100*
120	*880*	*105,600*	*14,400*	*774,400*
100	*750*	*75,000*	*10,000*	*562,500*
70	*650*	*45,500*	*4,900*	*422,500*
30	*430*	*12,900*	*900*	*184,900*
450	***4,050***	***326,500***	***38,300***	***2,849,300***

$$b = \frac{n\Sigma xy - \Sigma x \Sigma y}{n\Sigma x^2 - (\Sigma x)^2}$$

$$= \frac{6 \times 326{,}500 - 450 \times 4{,}050}{6 \times 38{,}300 - 450^2}$$

$$= \frac{136{,}500}{27{,}300} = 5$$

$$a = \bar{y} - b\bar{x}$$

$$a = \frac{4{,}050}{6} - 5 \times \frac{450}{6} = 300$$

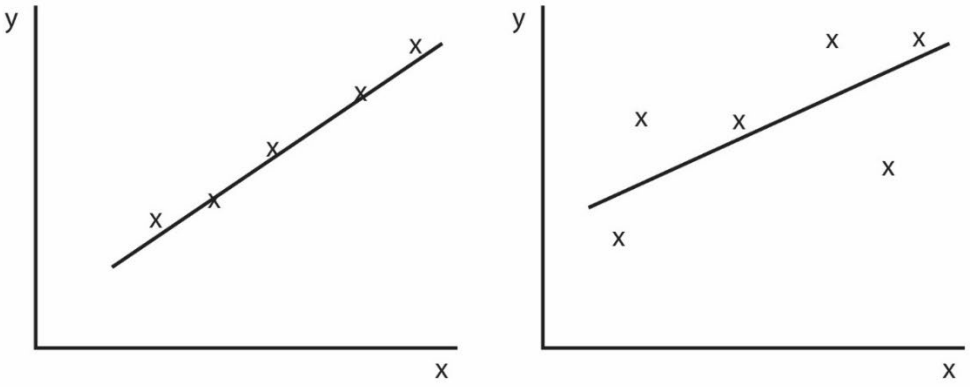
The regression equation is $y = 300 + 5x$

Mathematical interpretation:
If (x = 0), then (y = 300). Each time (x) increases by 1, (y) increases by 5.

Business interpretation:
If no money is spent on advertising then sales would still be $300,000. For every additional $1 increase in advertising, sales revenue would increase by $5.

Correlation

Regression analysis attempts to find the relationship between a number of variables. Correlation is concerned with establishing how strong the relationship is.

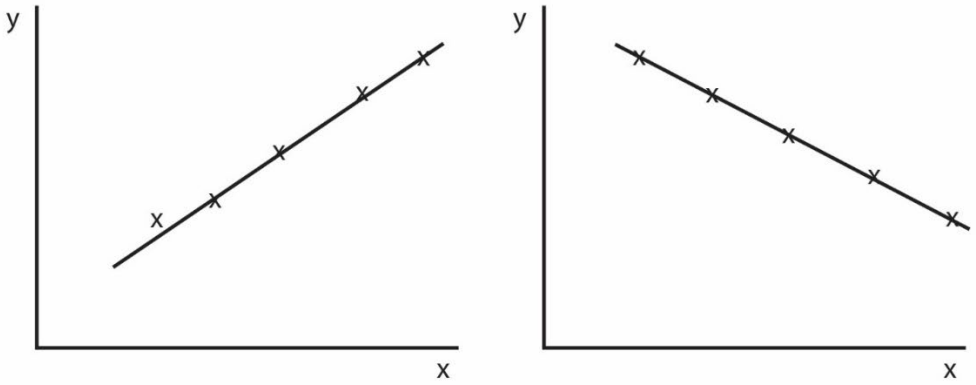


Clearly in the first diagram, the regression line would be a much more useful predictor than the regression line in the second diagram.

In terms of degrees of correlation, two variables might be perfectly correlated, partly correlated, or uncorrelated.

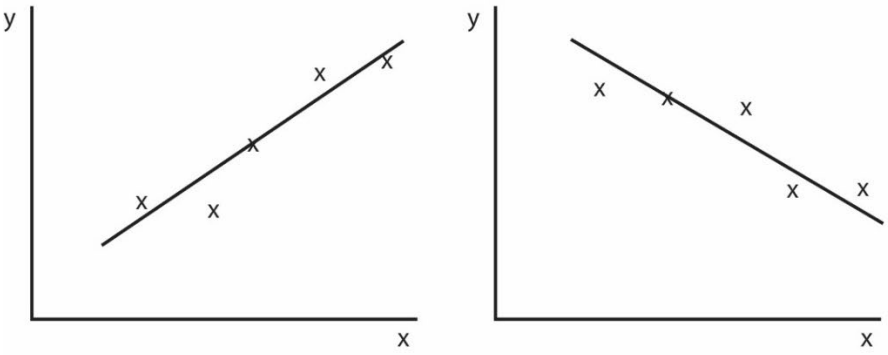
Different types of correlation explained

Perfect correlation



All the pairs of values lie on a straight line. There is an exact relationship between the two variables.

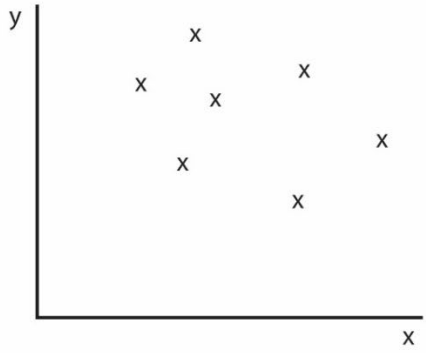
Partial correlation



In the first diagram there is not an exact relationship, but low values of x tend to be associated with low values of y, and high values of x tend to be associated with high values of y.

In the second diagram again there is not an exact relationship, but low values of x tend to be associated with high values of y and vice versa.

No correlation



The values of the two variables seem to be completely unconnected.

Positive and negative correlation

Correlation can be positive or negative.

Positive correlation means that high values of one variable are

associated with high values of the other and that low values of one are associated with low values of the other.

Negative correlation means that low values of one variable are associated with high values of the other and vice versa.

The correlation coefficient

The degree of correlation can be measured by the Pearsonian correlation coefficient, r (also known as the product moment correlation coefficient).

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{\left(n\sum x^2 - \left(\sum x\right)^2\right)\left(n\sum y^2 - \left(\sum y\right)^2\right)}}$$

The calculation of correlation coefficients can be useful in telling us how strongly two sets of data are related to each other – for instance, factory costs in relation to production levels or sales volumes in relation to advertising spend – so that we can decide whether the data can be used to help us forecast.

r must always be between –1 and +1.

If r = 1, there is perfect positive correlation. The data points lie exactly on a straight line of positive gradient.

If r = 0, there is no linear correlation.

If r = –1, there is perfect negative correlation (again the data points lie on a straight line, but with a negative gradient).

For other values of r, the meaning is not so clear. It is generally taken that if r > 0.8, then there is a strong positive correlation and if r < –0.8, there is a strong negative correlation.

However more meaningful information can be gathered from calculating the coefficient of determination, r^2.

2 Further examples of using spreadsheets in Business Strategy and Technology



Section overview

- In this section, we provide five comprehensive worked examples, which demonstrate three or four formulae used in the same worksheet. In the exam some questions will provide pre-populated spreadsheet data whereby the spreadsheet area of the exam screen already contains some data. Be careful not to alter this data when using it for your analysis. Examples of pre-populated data in this section are Pico Fogo Ltd, By Camplight Ltd, Chrono Ltd and HGF. Other questions will not have pre-populated data and you can choose which data you wish to copy from the question into the spreadsheet area of the exam in order to perform your analysis (see the question Zinnia below).

2.1 Comprehensive worked example



Worked example: Pico Fogo Ltd

Pico Fogo Ltd (PF) is a European speciality organic coffee company. PF sources beans from West Africa and ships them to its two European sites to be roasted and packaged.

PF prides itself on sourcing the highest quality beans and paying its suppliers a fair price. Its packaging carries both fair trade and organic certifications. PF is known for its high-quality speciality organic coffee beans and charges premium prices.

The speciality coffee beans are sold via PF's website to: hospitality customers (cafes and restaurants), speciality coffee retailer customers, and individual consumers. PF sells its coffee beans in three different sizes of bag: small, medium and large. The small bags are predominantly purchased by consumers for personal use while the medium and large bags are bought by its hospitality and retailer customers. As the majority of its income comes from hospitality and retailer customers, in the past year PF has targeted its marketing efforts towards them.

Financial information for the year ended 31 December 2021 is in **the pre-populated spreadsheet.**

Coffee market

Europe is a major market for speciality organic coffee. Average margins for selling these coffee beans are 31%.

Coffee beans are sold all year around but purchases by most retailers and hospitality venues follow a seasonal pattern with the largest peak in Q4 when shops and restaurants are at their busiest.

A large majority of the sales of speciality coffees by hospitality venues are as hot drinks, but some are sold as cold drinks.

Recent board meeting

At a recent board meeting, the directors discussed reducing the number of bag sizes in the PF range.

The CEO opened the meeting: 'I think that we should reduce our bag sizes to just two: medium and large. This will be better for the environment as the smaller bags have a lot of packaging for such a small amount of coffee beans. It will also reduce the number of our deliveries. I'm sure most people who order small bags would swap to buying the medium bag, so we would retain some of their revenue, but with fewer bags and fewer deliveries.'

The marketing director disagreed: 'I think it is important we still have the choice of three bag sizes. We have a lot of customers who buy small bags as they want fresh coffee beans, and a medium bag is too big for them. In the long term, we might lose revenue from these customers altogether.'

The finance director presented the best-case and worst-case scenarios for discontinuing the small bag in 2022.

Requirements

Using the data in the Exhibit, the prepopulated spreadsheet (Exhibit 2) and other information provided:

(a) Evaluate PF's quarterly performance for the year ended 31 December 2021.

(b) Calculate the impact of the CEO's suggestion on profit in 2022, under the best-case and worst- case scenarios. Briefly discuss any other factors PF should consider when making this decision.

Exhibit 1: Assumptions for best-case and worst-case scenarios

	A	B	C	D	E
80	Scenario	Best-case		Worst-case	
81	% of revenue retained from customers of small bags	75%		25%	
82	Retained revenue from small bags split between medium and large bags	Medium	Large	Medium	Large
83		80%	20%	80%	20%

Other assumptions:

- Gross profit margins will be the same as in 2021.

Exhibit 2: Pre-populated spreadsheet data

	A	B	C	D	E	F	G
1							
2			**Q1**	**Q2**	**Q3**	**Q4**	**Total**
3		**Small bag**					
4		Quantity sold	6785	7080	6490	9145	29,500
5		No. of customer	587	691	630	857	2765
6		Sales	£	£	£	£	£
7		Individual	49,666	53,985	51,826	60,463	215,940
8		Retailer	18,992	24,638	20,019	39,010	102,659
9		Hospitality	6726	8142	7080	13,452	35,400
10		Total sales	75,384	86,765	78,925	112,925	353,999
11		Cost of sales	47,115	53,361	48,703	68,885	218,064
12		Gross profit	28,269	33,404	30,222	44,040	135,935
13		**Medium**					
14		Quantity sold	45,225	56,250	51,525	72,000	225,000
15		No. of customers	876	1089	1002	1390	4357
16		Sales	£	£	£	£	£
17		Individual	188,528	225,355	223,540	260,348	879,751
18		Retailer	533,520	684,000	615,600	902,880	2,736,000
19		Hospitality	118,631	141,075	150,694	230,850	641,250
20		Total sales	840,679	1,050,410	989,834	1,394,078	4,275,001
21		Cost of sales	562,414	695,372	660,723	920,091	2,838,600
22		Gross profit	278,265	355,038	329,111	473,987	1,436,401
23		**Large**					
24		Quantity sold	14,640	18,720	16,320	30,320	80,000
25		No. of customers	191	248	215	394	1,048
26		Sales	£	£	£	£	£
27		Individual	21,760	28,160	26,880	51,200	128,000
28		Retailer	161,280	206,080	188,160	340,480	896,000
29		Hospitality	276,480	368,640	307,200	583,680	1,536,000
30		Total sales	459,520	602,880	522,240	975,360	2,560,000
31		Cost of sales	323,962	423,222	366,332	675,924	1,789,440
32		Gross profit	135,558	179,658	155,908	299,436	770,560

Solution

Note: The answers in the spreadsheet are examples of what can be achieved with the spreadsheet functionality in the exam software, not what a candidate would be expected to produce in the exam.

	B	C	D	E	F	G
1	**Total**	**Q1**	**Q2**	**Q3**	**Q4**	**Total**
2	Total quarterly revenue	£1,375,583 (1)	£1,740,055 (3)	£1,590,999	£2,482,363	£7,189,000
3	Total quarterly gross profit	£442,092 (2)	£568,100	£515,241	£817,463	£2,342,896
4	GPM	32.1%	32.6%	32.4%	32.9%	32.6%
5						
6	Small GPM	37.5%	38.5%	38.3%	39.0%	38.4%
7	Medium GPM	33.1%	33.8%	33.2%	34.0%	33.6%
8	Large GPM	29.5%	29.8%	29.9%	30.7%	30.1%
9						
10	**Spread of quarterly revenue**					
11	Individual	20.9%	24.8%	24.3%	30.0%	100.0%
12	Retailer	19.1%	24.5%	22.1%	34.3%	100.0%
13	Hospitality	18.2%	23.4%	21.0%	37.4%	100.0%
14	% of total revenue per quarter	19.1% (4)	24.2%	22.1%	34.5%	100.0%
15						
16						
17	Individual	18.9%	17.7%	19.0%	15.0%	17.3%
18	Retailer	51.9%	52.6%	51.8%	51.7%	51.9%
19	Hospitality	29.2%	29.8%	29.2%	33.4%	30.8%
20						
21	**Revenue mix (by product size)**					
22	Small	5.5%	5.0%	5.0%	4.5%	4.9%
23	Medium	61.1%	60.4%	62.2%	56.2%	59.5%
24	Large	33.4%	34.6%	32.8%	39.3%	35.6%
25						
26	**Average revenue per transaction**					

27	Small	£11.11 (5)	£12.25	£12.16	£12.35	£12.00
28	Medium	£18.59	£18.67	£19.21	£19.36	£19.00
29	Large	£31.39	£32.21	£32.00	£32.17	£32.00
30						
31	**Revenue per customer**					
32	Small	£128.42	£125.56	£125.28	£131.77	£128.03
33	Medium	£959.68	£964.56	£987.86	£1,002.93	£981.18
34	Large	£2,405.86	£2,430.97	£2,429.02	£2,475.53	£2,442.75
35						
36	**Quarterly sales mix by volume**					
37	Small	23% (6)	24%	22%	31%	100%
38	Medium	20%	25%	23%	32%	100%
39	Large	18%	23%	20%	38%	100%
40	Total	20%	25%	22%	33%	100%
41						

Spreadsheet notes – functions and functionality used

(1) **SUM - individual cell references**

The SUM function adds values together. For example, individual cells can be added: Total quarterly revenue Q1 =SUM(C10+C20+C30)

(2) **Relative cell referencing**

When we SUM the quarterly gross profit to get a total for Q1 (in C36), because the method is the same for each quarter we can copy and paste the formula in C36 into D36, E36 and F36 to give us the quarterly totals for Q2, Q3 and Q4 respectively. This is possible because the same relative cells are used for each quarter i.e the cells are in the same row but are in each particular quarter's column in the spreadsheet. So the formula for Q1 =SUM(C12+C22+C32) is copied across and becomes, for Q2, =SUM(D12+D22+D32) etc.

(3) **Formatting monetary cells – large numbers**

Larger monetary amounts (eg total quarterly revenue Cell D35) should be formatted as Accounting £ to 0 decimal places.

(4) **Absolute cell referencing**

When looking at a calculation like the spread of quarterly revenue in row 47, each quarter's revenue in columns C to F are divided by the same number (the total revenue for the year in column G). Therefore using a relative cell reference will not work. In this case an absolute cell reference can be used by inserting a $ before each of the two elements of the cell reference which is fixed. This means that the formula can still be copied and pasted across but instead of changing to the relative column it will keep the fixed cell reference.

For example:

% of total revenue for Q1 in cell C47 =(C10+C20+C30)/($G$10+$G$20+$G$30) When this is copied and pasted across the quarters, Q2 for instance becomes:

=(D10+D20+D30)/($G$10+$G$20+$G$30)

The quarterly revenue changes as these are relative cell references but the total revenue does not change as it is an absolute cell reference.

(5) **Formatting monetary cells – small numbers**

Smaller monetary amounts (eg average revenue per transaction in cell C60) should be formatted as Accounting £ to 0 or 2 decimal places.

(6) **Formatting percentages**

A calculation resulting in a percentage (eg quarterly sales by volume C70) should be formatted as a percentage cell to 0,1 or 2 decimal places.

Solutions to (a) and (b):

(a) **Sales**

It can be clearly seen from the data that the majority (34.5%) of PF's revenue is made in Q4. This is a trend replicated across all sizes of bags and types of customer, although it is not as prominent with individual customers (which has a more even spread over the quarters but still has the largest proportion of revenue in Q4 at 30.0%). Individuals presumably buy for personal use at home all year around and do not follow the same seasonal trends of retail and hospitality. The trend can also be seen in quarterly sales by volume, with the majority made in Q4 and a distinct dip in Q1.

Q1 is the quietest quarter with just 19.1% of revenue earned here, perhaps after the seasonal peak in sales there is a dip in hospitality and retailer sales which is mirrored in coffee consumption. There is also a dip in revenue in Q3 which could be due to the warmer weather at this time of year in Europe and possibly less focus on hot drinks.

Revenue per customer is fairly similar across the quarters with a slight peak in Q4 which indicates potentially higher prices in Q4 or that customers buy more per transaction in this period as they stock up for the busy period for the retailer and hospitality sector.

Sales mix

By customer

Individuals account for 17.3% of the revenue, meaning they are the smallest customer type. This is because individuals buy mostly small bags and perhaps this explains why PF focussed their marketing effort on their business customers. However, it could be argued that individuals need more marketing attention to improve their sales value and volume on the small bags as these have the highest margin (see below).

The majority of sales (51.9%) are made to retailers. These are speciality coffee retailers who buy a mix of all types of bags but mainly medium bags, whereas the hospitality customers buy the majority of the large bags. This shows that having a range of different sizes helps attract many different customers, which is considered below with the CEO's suggestion.

By size of bag

Overall, the small bag accounts for just 4.9% of the total sales, with the majority (59.5%) of sales coming from the medium size and 35.6% from the large. This is interesting with regards to the CEO's suggestion which will be dealt with below. It is clear that hospitality customers buy both medium and large sizes and efforts could be made to try and get customers to 'size up' for future sales and move towards the large size, especially as most speciality coffee sales are made by hospitality venues. It could be that PF needs better relationships within the hospitality sector eg by becoming the coffee bean of choice in an upmarket chain.

Gross profit

The gross profit margin remains consistent across the quarters at between 32.1% and 32.9%. This is higher than the average margin in the industry which is 31.0%, which perhaps reflects the premium price charged by PF and that its customers value the quality of the product. The highest margins are gained in Q4 when the majority of the sales are made, indicating possibly higher prices charged in this quarter and/or economies of scale in its cost of sales. Partly this is borne out by the slightly higher average revenue per transaction in Q4, with all sizes of product having a higher price. For example, the small bags have a per transaction value of £12.35 in Q4 versus £11.11 in Q1. It could be that PF charges higher prices around the busiest part of the year or offers discounts to stimulate sales in the quieter quarters.

The gross profit margin varies across the sizes of bag, with the small size getting the highest margin of 38.4% overall and the large bags the lowest at 30.1% overall. There is likely to be a larger mark-up on cost with the smaller bags, plus business customers who buy the larger bags are likely to be trying to obtain bulk discounts for their purchases. As with the overall margin, quarterly margins are highest in Q4 and lower in Q1 for all sizes.

(b) Impact of the CEO's suggestion on profit:

	A	B	C	D	E
80	Scenario	Best-case		Worst-case	
81	% of revenue retained from customers of small bags	75%		25%	
82	Retained revenue from small bags split between	Medium	Large	Medium	Large
83		80%	20%	80%	20%

	A	B	C	D	E	F	G	H
85								**Comparison to existing**
86		**Best case**	**Medium**	**Large**	**Total**	**Medium**	**Large**	
87		Existing 2021 revenue	£4,275,001[1]	£2,560,000[2]				
88		Additional revenue	£212,399[3]	£53,100[4]	£265,499[5]			
89		New revenue	£4,487,400[6]	£2,613,100	£7,100,500	5.0%	2.1%	-1.2%
90		Profit @ 2021 GPM	£1,507,767	£786,543	**£2,294,310***	5.0%	2.1%	-2.1%
91		New profit margin			**32.3%**			
92								
93		**Worst case**						

94		Existing 2021 revenue	£4,275,001	£2,560,000				
95		Additional revenue	£70,800[7]	£17,700[8]	£88,500[9]			
96		New revenue	£4,345,801[10]	£2,577,700	£6,923,501	1.7%	0.7%	**-3.7%**
97	Profit @ 2021 GPM	£1,460,190		£775,888	**£2,236,077**	1.7%	0.7%	**-4.6%**
98	New profit margin				**32.3%**			

[1] =G20

[2] =G30

[3] =B81*G10*B83

[4] =B81*G10*C83

[5] =SUM(D88:E88) The formula can be copied in F89:F90

[6] =SUM(D87:D88) The formula can be copied in E89

[7] =D81*G10*D83

[8] =D81*G10*E83

[9] =SUM(D95:F95) The formula can be copied in F96:F97

[10] =SUM(D94:95) The formula can be copied in E96

Spreadsheet notes - functions and functionality used

*** SUM - ranges of cells**

The SUM function adds values together. For example, a range of cells can be added:

Total profit for the best-case scenario =SUM(D90:E90)
Under the best-case scenario overall profit falls by 2.1% and under the worst-case scenario it falls 4.6%. Under both scenarios the overall margin falls from 32.6% to 32.3%. Given that the small size has the highest margins a fall in margin would be expected.

The small size represents 4.9% of total revenue, and in the best-case scenario 75% of these sales are retained but redistributed. However, under the worst-case PF only redistributes 25% of the small bag sales. The CEO appears confident that the best-case scenario is likely, and if this is the case the change would not be too significant for the business. Indeed, if PF is doing this for environmental reasons this may help attract customers. Additionally, it might reduce the cost base as it saves on making multiple deliveries of small items.

However, if the marketing director is correct PF will lose 75% of its customers for small bags. Despite it being a small number of customers, losing 75% of 4.9% of overall revenue is not insignificant. Furthermore, in the long term all small customers may be lost because, after buying the bigger sizes, the 25% may decide they prefer the small bags after all and go elsewhere.

The small bag is not just bought by individuals, as the hospitality and retailer customers also buy the small size possibly to sell onto their customers. They may be upset at this change and decide they want to change to a company that can offer them a full range of sizes.

PF should not make this change without discussing this with their biggest customers first.

Worked example: Zinnia

Zinnia Ltd (Zinnia) is an online florist operating only in the north of England. Zinnia was established five years ago by two brothers, Gerard and Joseph. Zinnia currently delivers arrangements of flowers (an arrangement) to customers' individual orders, so it has peaks across the year related to special occasions such as Mother's Day. The average price for an arrangement delivered by Zinnia is £35.

In a recent board meeting, the brothers considered launching a subscription service for customers, under which a very distinctive and long-lasting arrangement would be delivered each month to a customer who maintains a regular subscription. This service would provide Zinnia with more consistent revenue across the year. Some larger nationwide florists already offer a subscription service.

Details of the subscription service

Customers will subscribe for flowers to be delivered each month in a box. The customer will then arrange the flowers in a suitable vase or jug following detailed instructions, so it resembles the distinctive Zinnia style of flower arrangement. The minimum subscription period will be three months, after which a customer can cancel their subscription.

In order to generate interest for the service, Joseph wants to give customers the first month's box free if they enter a code, disclosed in marketing material. Each customer can only obtain a free box once, when they initially subscribe to the service.

Based on market research, Joseph used the following original assumptions when appraising the first six months for the subscription service.

Working assumptions for the subscription service:

Monthly subscription fee	£30
Variable costs per box	£5
Additional fixed costs for six months	£100,000
% of new subscribers each month expected to use a code to obtain the first box free	60%
Total new subscribers each month	300

A working assumption is that no customers will cancel their subscription in the first six months.

Alternative scenario for the subscription service:

At the board meeting Gerard identified an alternative scenario. They feel that many customers will take advantage by subscribing with the code then cancelling after three months. Instead, they suggest retaining the code for a free box on initial subscription but reducing the subscription fee per box from £30 to £26. They believe this will increase the number of new subscribers per month by 10%.

Requirement

- Calculate the contribution for the first six months based on the figures provided and the working assumptions.
- Calculate the impact on the contribution figure of the alternative scenario.
- Calculate the break-even number of subscribers and the margin of safety for the working assumptions and for the alternative scenario.
- Discuss the financial and non-financial factors to be considered in deciding whether to launch the subscription service.

Total: 18 marks

Solution

Calculations:

	A	B			C		D		
4		**Original assumptions**(1)					**Alternative scenario**		
5		Monthly subscription fee			£30		£26		
6		Variable costs per box			£5		£5		
7		Additional annual fixed costs			£100,000		£100,000		
8		% of new monthly subscribers expected to use a code to obtain the first box free			60%		60%		
9		Average new monthly subscribers			300		330		
12			**M1**	**M2**	**M3**	**M4**	**M5**	**M6**	**Total**
13		Paying subscribers1	120	420	720	1020	1320	1620	
14		New, non-paying subscribers	180	180	180	180	180	180	
15		Total subscribers	300 (2)	600	900	1,200	1,500	1,800	**6300**
16		Revenue	£3,600 (4)	£12,600 (4)	£21,600	£30,600	£39,600	£48,600	£156,600 (5)
17		Variable costs	£1,500	£3,000	£4,500	£6,000	£7,500	£9,000	£31,500
18		Contribution	£2,100	£9,600	£17,100	£24,600	£32,100	£39,600	**£125,100**
19						**Contribution per box**			£19.86 (3)
20									
21						**Boxes to break even**			5036
22									
23						**Margin of safety**			25.10% (6)
24									
25			**M1**	**M2**	**M3**	**M4**	**M5**	**M6**	**Total**
26		Paying subscribers2	132	462	792	1122	1452	1782	
27		New, non-paying subscribers	198	198	198	198	198	198	
28		Total subscribers	330	660	990	1,320	1,650	1,980	**6,930**
29		Revenue	£3432	£12,012	£20,592	£29,172	£37,752	£46,332	£149,292
30		Variable costs	£1,650	£3,300	£4,950	£6,600	£8,250	£9,900	£34,650

	A	B			C		D		
31		Contribution	£1,782	£8,712	£15,642	£22,572	£29,502	£36,432	**£114,642**
32						**Contribution per box**			**£16.54**
33									
34						**Boxes to break even**			**6045**
35						Increase in boxes		1009	**20%**
36						**Margin of safety**			**14.64%**
37						**Fall in contribution from original**			**(£10,548)**
38						**% fall in contribution**			**(8.4%)**

1 These calculations are based on the original assumptions in C5:C9 of the question information. These can be copied from the question into the spreadsheet area of the exam.
2 These calculations are based on the alternative scenario information in D5:D9.

Cells B12:I23 can be copied and absolute cell references changed to the assumptions in D5:D9. This will allow the Whatif analysis to be performed quickly in the exam.

Spreadsheet notes

(1) The assumptions data from the exam question should be copied into the spreadsheet so the cells can be referenced in the calculation.

(2) **SUM**

The Sum function adds values together: This can be individual cell references or a range of cells. For example: Total number of subscribers for Month 1 =SUM(C13:C14)

(3) **Relative cell references**

When we calculate the contribution per box, we divide the total contribution by the total number of subscribers =I18/I15. The method for this is the same for the alternative scenario so we can then copy and paste that calculation into cell J32 for the alternative scenario. This will give us the contribution per box for the alternative scenario because it references the same relative cells i.e the cells in the same row but in that particular column in the spreadsheet. So the formula changes from =I18/I15 to =I31/I28.

(4) **Absolute cell references**

When looking at a calculation like the monthly revenue, the number of paying subscribers are multiplied by the same number (the £30 monthly subscription fee) so a relative cell reference will not work. In this case an absolute cell reference can be used, which means that the formula can still be copied and pasted but instead of changing to that relative column it will keep the fixed cell reference.

For example, the monthly revenue for month 1 in the original scenario =C13*$C$5 and when this is pasted into the cells for month 2 in the next column it is now =D13*$C$5. The subscriber number changes because it is a relative cell reference, but the monthly subscription fee does not change as it is an absolute cell reference.

(5) **Formatting monetary cells**

Monetary cells (eg, total revenue) should be formatted as Accounting £ to 0 decimal places.

(6) **Formatting percentage cells**

A calculation resulting in a percentage (eg, margin of safety) should be formatted as a Percentage cell to 0, 1 or 2 decimal places.

Contribution

The original assumptions generate a contribution of £125,100 compared with £114,642 if the price is lowered to £26, so the alternative would result in £10,458 or 8.4% less contribution.

Breakeven and margin of safety

Under the original assumptions Zinnia will need to deliver a total of 5,036 boxes over the six months to break even based on the average contribution per box. Under the alternative scenario this increases by 1,009 to 6,045 boxes which is a 20.0% increase.

The margin of safety is 25.10% under the original assumptions and decreases under the alternative assumption to 14.64% because the contribution per box falls from £19.86 to £16.54.

Discuss the financial and non-financial factors to be considered in deciding whether to commence the subscription service

Financial

- Under either set of assumptions the subscription will increase Zinnia's contribution, there are however £100,000 additional fixed costs associated with the introduction of the service so profits are low at £25,100 under the original assumptions and £14,642 under the alternative assumption.
- The calculations do not take into account any cannibalisation of Zinnia's existing arrangement sales in respect of customers who bought on an ad hoc basis but are now subscribing instead. This would reduce the net additional contribution from the subscription service.
- The calculations assume that 60% use the code to receive a free box, so if this figure was higher it would reduce the contribution. However, it may be that customers use the code at less than the rate of 60% of new subscribers which would increase the contribution, though this might depend on Zinnia's marketing efforts. If, after a month or two, the subscription service is going well Zinnia could stop the free box promotion for new subscribers.
- The original price of £30 is cheaper than the average price of an arrangement usually sold by Zinnia of £35 so this may encourage regular subscribers to sign up. However, some may see the lower price as an indication that the flowers are of lower quality or a smaller arrangement.
- There is a large contribution on a paid-for box (£25 per unit or 83%). Whilst buying in bulk is likely to give Zinnia a good margin it would be useful to have more information about the usual margins on flowers for both Zinnia and the industry overall.
- Under the alternative scenario the reduction in price to £26 is a 26% discount on the usual £35 price for Zinnia. Again, this may indicate a lack of quality to customers but if demand for the product is price-elastic then it is likely that a reduction of this kind could drive up the quantity of subscribers, as suggested by Gerard.
- There is no further information about the fixed costs, again it would be useful to have a breakdown of these to see if any fixed costs have been omitted e.g. additional website costs for setting up the subscription platform or additional staff costs.
- Both options provide additional contribution for Zinnia at a steady rate across this year which may be helpful from a cashflow perspective.

Non-financial

- Some larger florists offer a similar subscription service for flowers which may mean that they would be able to offer these at a cheaper rate. However, they may not be as long-lasting and distinctive as Zinnia's arrangements.
- Florists appear to operate on a seasonal basis with peaks based around special occasions. It might be that customers do not want flowers all year around or they might have peaks of sign-ups around these events which are then cancelled.
- There is nothing in the scenario to indicate the levels of spare capacity for Zinnia. At times of high demand, it may not have enough capacity to service a monthly subscription. On the other hand, at times of high volumes the variable costs per arrangement may be reduced and the profit on the subscription service might increase.
- The use of a code for a free box might be a useful strategy to pull in customers and make them sign up for the service but they then may not stay beyond the initial three-month sign-up period. The calculations above assume that there are no cancellations which seems highly unlikely.
- Even using the instructions, customers may be unable to recreate the distinctive flower arrangement from their box of flowers. This could lead to dissatisfaction with the service which may also result in loss of existing customers for their usual flower arrangements in the long term.



Worked example: By Camplight Ltd

By Camplight Ltd (BC) operates four luxury camping sites across Europe with 27 permanently situated tents in total. Each site is attached to a working farm and has between four and eight tents.

Each tent is fully equipped with most of the amenities found in a luxury hotel. However, BC markets the sites as an escape from the stresses of daily life so, unlike a hotel, guests are not provided with internet access. As the sites are attached to working farms, BC offers a range of activities for guests, including animal feeding, horse riding and cooking classes. Guests rent a tent for a week at a time, but any activities on the farm are an additional charge per guest.

Rental prices and activity charges vary according to location, but each site has peak and off-peak rates reflecting the seasonality of the business. The peak season runs from May to October. Each tent can accommodate up to five guests, but the rental price is the same regardless of how many guests stay in the tent. The popularity of each site during the year is impacted by both seasonal weather and the availability of year-round activities.

The site location, number of tents and weekly rental prices are as follows:

Location	Bavaria	Cornwall	Dyfed	Zeeland	Total
Number of tents	8	7	8	4	27
Rental price per tent per week (£)					
Off-peak	250	400	400	300	
Peak	300	450	450	325	

BC markets itself as an environmentally friendly holiday option and records an estimate of the CO_2 emissions for each site. The industry benchmark for holiday businesses is 31.1kg CO_2 per

room (or tent per night. For BC this is calculated as the total CO_2 emissions divided by the number of tents and the number of days a year the tents are available. BC operates 365 days a year and aims to score well below the benchmark of 31.1kg as it strives to become carbon neutral in the long term.

Annual review

Monthly data is collated by each site and submitted to BC's head office team.

This data is given in the **pre-populated spreadsheet**. The board wants to understand which are the best and worst performing sites this year.

BC measures performance according to the following three KPIs:

	A	B	C
1			
2		**KPI**	**Target**
3		Guest satisfaction	>80%
4		Contribution margin	>70%
5		Environmental impact	<15.55kg CO2 per tent per night

Requirement

Prepare a report for the board which analyses the performance of the sites against BC's KPIs and shows how the sites rank against each other. Provide supporting data and calculations.

Pre-populated data

	A	B	C	D	E	F	G	H	I	J	K	L	M
61		**Jan**	**Feb**	**Mar**	**Apr**	**May**	**Jun**	**Jul**	**Aug**	**Sep**	**Oct**	**Nov**	**Dec**
62	**Bavaria**												
63	No of employees	2	2	2	3	4	5	5	5	5	4	2	4
64	% of tents occupied	38	13	38	44	63	88	100	100	63	50	25	38
65	% guest satisfaction	82	82	84	78	79	73	73	74	76	81	88	83
66	Accom. revenue (£)	3,250	1,050	3,210	3,240	6,080	8,555	9,610	9,720	6,090	4,800	2,100	3,040
67	Activity revenue (£)	352	118	417	389	729	856	1,057	875	670	533	2,058	316
68	Total revenue (£)	3,602	1,168	3,627	3,629	6,809	9,411	10,667	10,595	6,760	5,333	4,158	3,356
69	Contribution (£)	2,311	796	2,611	2,649	4,766	7,434	8,107	8,052	4,934	3,627	2,827	2,148
70	CO2 emissions estimate (kg)	2,315	792	2,315	2,680	3,837	5,360	6,091	6,091	3,837	3,045	1,523	2,315
71													
72	**Cornwall**												
73	No of employees	2	2	2	3	4	5	5	5	5	4	2	4
74	% of tents occupied	50	43	57	57	71	86	100	93	86	79	64	57
75	% guest satisfaction score	87	89	84	87	84	79	72	74	79	79	84	81

	A	B	C	D	E	F	G	H	I	J	K	L	M
76	Accom. revenue (£)	5,800	4,850	6,600	6,620	9,000	10,800	12,600	11,950	10,800	9,950	7,400	6,580
77	Activity revenue (£)	580	534	660	596	990	1,296	1,386	1,315	1,296	1,095	740	790
78	Total revenue (£)	6,380	5,384	7,260	7,216	9,990	12,096	13,986	13,265	12,096	11,045	8,140	7370
79	Contribution (£)	3,988	3,553	4,792	4,762	6,693	8,225	9,930	9,418	8,588	7,510	5,535	4,717
80	CO2 emissions estimate (kg)	2,665	2,284	3,045	3,045	3,807	4,568	5,330	4,949	4,568	4,187	3,426	3,045
81													
82	**Dyfed**												
83	No of employees	2	2	3	3	5	5	6	6	5	4	3	4
84	% of tents occupied	44	38	50	50	63	75	88	81	75	69	56	50
85	% guest satisfaction score	87	89	84	87	84	7	72	74	79	79	84	81
86	Accom. revenue (£)	5,700	4,950	6,800	6,650	9,200	10,800	12,700	11,760	10,800	9,920	7,200	6,750
87	Activity revenue (£)	570	545	680	599	1,012	1,296	1,397	1,294	1,296	1,091	720	810
88	Total revenue (£)	6,270	5,495	7,480	7,249	10,212	12,096	14,097	13,054	12,096	11,011	7,920	7,560
89	Contribution (£)	4,264	3,670	5,139	4,922	7,148	8,467	10,291	9,529	8,830	8,093	5,544	5,141
90	CO2 emissions estimated (kg)	2,665	2,284	3,045	3,045	3,807	4,568	5,330	4,949	4,568	4,187	3,426	3,045
91													
92	**Zeeland**												
93	No of employees	2	2	2	2	3	3	4	4	3	3	2	3
94	% of tents occupied	75	63	75	75	81	94	100	94	88	88	75	75
95	% guest satisfaction score	93	91	93	92	93	91	89	92	89	91	91	90
96	Accom. revenue (£)	3,750	3,150	3,750	3,600	4,225	4,875	5,325	4,875	4,720	4,550	3,600	3,750
97	Activity revenue (£)	975	788	975	900	1,014	1,268	1,331	1,268	1,133	1,138	828	975
98	Total revenue (£)	4,725	3,938	4,725	4500	5,239	6,143	6,656	6,143	5,853	5,688	4,428	4,725
99	Contribution (£)	2,977	2,520	2,953	2,835	3,353	3,870	4,127	3,931	3,687	3,526	2,701	3,024
100	CO2 emissions (kg)	2,284	1,903	2,284	2,284	2,474	2,855	3,045	2,855	2,665	2,665	2,284	2,284

Solution

Note: Answers in the spreadsheet below are examples of what can be achieved with the spreadsheet functionality in the spreadsheet software, not what a candidate would be expected to produce in the exam. The yellow shaded rows are the minimum calculations required to see if sites have met KPIs.

To: BC board

From: An advisor

Date: Today

Subject: Report on KPIs

(1) **Guest satisfaction**

In terms of guest satisfaction, Zeeland ranks the highest with a score of 91% and Bavaria ranks in fourth place with 79% (see Appendix). The target is over 80% so Bavaria has only just missed out on this target, but it is still far behind the 91% achieved by Zeeland.

The highest satisfaction score is Zeeland which is the smallest site, it is possible that guests looking for a stress-free holiday value the smaller size of site that is offered in this location.

It also appears there may be a correlation with the number of staff a site has per tent. Zeeland has, on average, 2.8 employees across the year for four tents giving a ratio of 0.7, whereas Bavaria has a ratio of 0.4 employees per tent. It is likely that guests receive a lower level of customer service from sites with a lower ratio of staff to tents. Given the sites are marketed as luxury camping sites it is likely that a high level of service is expected.

Looking at the monthly data for all locations, there appears to be a dip in satisfaction scores in the peak summer months. This may be due to the sites being busier and staff being unable to give the desired high level of service.

It also appears from reviewing the data that whether guests take part in activities might have an impact on the satisfaction scores as Zeeland has a much higher proportion of revenue from activities (20% of total revenue). The three other sites have between 10-12% of revenue from activities. Again, it is possible that the lower level of staff available at sites such as Bavaria have impacted on the availability of staff to organise the activities. The popularity of these sites amongst guests may also be due to the fact that there are a lot of activities to do regardless of the weather conditions/time of year.

(2) **Contribution margin**

The KPI target was a contribution margin above 70% (see Appendix). Overall this target is not met. When looking at individual sites only Zeeland and Cornwall have not met this target, with Zeeland in fourth and last place out of all of the sites and Bavaria in first place. This is a direct reversal of the first guest satisfaction KPI. It could indicate that Bavaria has been focussing on this financial measure and cutting costs to ensure a high contribution margin at the expense of their guest satisfaction and service levels. This is also borne out by the low staffing ratio of Bavaria.

Overall, Bavaria's contribution per tent is significantly lower than each of the other sites. Looking into the monthly data for Bavaria it appears that their contribution margin is significantly higher in summer months, when Bavaria is close to or at full capacity for three months. Occupancy is clearly an important factor for BC and their performance, and although this feeds into contribution margin it might be better for BC if occupancy was one of their KPIs, in order to encourage sites to increase year-round occupancy.

The picture is mixed across the other sites. Cornwall, like Bavaria, is a large site but it does not manage to meet the target margin despite a higher occupancy. It does however have the highest contribution per tent in absolute terms at £11,102. This indicates that Cornwall's costs are higher which may indicate regional variations on costs. Despite this in absolute terms Cornwall's absolute contribution is £77,711 which is the second largest behind Dyfed.
Therefore, despite not meeting the target it gives a strong contribution to the overall profit of the company.

(3) **Environmental impact**

Overall, BC does not meet the target of <15.55kg CO_2 emissions as it has an overall figure of 16.23kg (See Appendix). This however is still well below the industry benchmark of 31.1kg and is only just above the target. It would be useful to have information about emissions for similar campsites to compare rather than a whole hospitality industry standard.

For individual sites the picture is mixed, with only two out of the four sites, Bavaria and Dyfed, meeting the target on emissions. These two sites are the two biggest sites with eight tents so there may be a certain level of emissions for a site overall and additional tents do not add to this in the same degree. Indeed, it is the smallest site, Zeeland, which performs the worst in this KPI, ranked fourth.

Although not the worst performing, it is worth pointing out that Cornwall again does not meet this target. Despite being a larger site, with seven tents, it does not appear to have any of the benefits of the larger sites like Bavaria and Dyfed with regard to emissions. This may indicate that their environmental impact needs to be reviewed and lessons learnt from other sites which are managing to meet this target, like Bavaria.

Conclusion

The picture is mixed, for KPI 1 (guest satisfaction) all sites with the exception of Bavaria meet the target. However, Bavaria then goes on to be the best performer in both KPI 2 (contribution margin) and 3 (CO_2 emissions per tent per night). Bavaria has the lowest overall occupancy which should perhaps be a KPI itself. If it could increase occupancy, possibly through more focus on activities and guest satisfaction, it would be a high performing site.

However, for other sites such as Zeeland which have a very high guest satisfaction score this seems to have been focussed on at the expense of margins and KPI 3.

Only Dyfed has met all three of the KPIs set by the board, with all other sites failing to meet at least one of them. This indicates there is room for improvement across the board. However, it should be noted that all of the sites are making a positive contribution to the business.

	A	B	C	D	E	F	G	H
8			**Bavaria**	**Cornwall**	**Dyfed**	**Zeeland**	**Total**	**KPI reference**
9		No of tents	**8**	**7**	**8**	4	**27**	
10		**AVERAGE** no of employees	3.6	3.6	4.0	2.8	**13.9**	**1**
11								
12		**AVERAGE** % of tents occupied	55%	70%	62%	82%	**67%**	**2**
13		**RANK tenant occupancy**	4	2	3	1		**2**
14		**AVERAGE % satisfaction score**	**79%** (2)	**82%**	**82%**	**91%**	**83%**	**1**
15		**RANK guest satisfaction**	4	2	2	1		**1**
16		**IF meets target KPI?**	NO	YES	YES	YES	**Yes**	**1**
17		SUM accommoda tio n revenue	£60,745 (1)	£102,950	£103,230	£50,170	**£317,095**	**2**

	A	B	C	D	E	F	G	H
18		SUM activity revenue	£8370	£11,278	£11,310	£12,593	**£43,551**	2
19		% of revenue from activities	12%	10%	10%	20%	**12%**	1
20		**SUM** total revenue	£69,115	£114,228	£114,540	£62,763	**£360,646**	2
21		**SUM** contribution	£50,262	£77,711	£81,038	£39,504	**£248,515**	2
22		Contribution per tent	£6283	£11,102	£10,130	£9876	**£9204**	2
23		**RANK contribution per tent**	4	1	2	3		2
24		**Contribution margin**	**73%**	**68%**	**71%**	**63%**	**69%**	2
25		**RANK contribution margin**	**1** (3)	**3**	**2**	**4**		2
26		**IF meets target KPI?**	**YES**	**NO**	**YES**	**NO**	**NO**	2
27		**SUM** CO2 emission estimate (kg)	40,201	44,919	44,919	29,882	**159,921**	3
28		**CO2 per tent per night**	**13.77** (3)	**17.54**	**15.38**	**20.47**	**16.23**	3
29		**RANK**	**1**	**3**	**2**	**4**		3
30		**IF meets target KPI?**	**YES** (4) (6)	**NO** (6)	**YES**	**NO**	**NO**	2
31								

Calculations students could do, Key calculations to see if data meets KPIs (could use Rank or IF but not both), Spreadsheet notes

(1) **SUM**

The Sum function adds values together. Here we have monthly data which can be added to get the total for the year. This can be done for ranges of cells eg, total accommodation revenue for Bavaria =SUM(C66:N66)

(2) **AVERAGE**

Instead of calculating an average manually, we can use the AVERAGE function. For example, when calculating the average Guest satisfaction score, we could add the monthly satisfaction scores and then divide by the number of months. However, it is quicker to select the AVERAGE function then select either the individual cells or a range of cells.

For example, Bavaria's guest satisfaction score is =AVERAGE(C65:N65)

(3) **RANK**

To rank the sites against each other you can use the RANK function, to quickly see the best and worst performers in that particular category. For example, for the contribution margin for Bavaria using the RANK function quickly compares their contribution margin of 73% to the other three sites.

To do this, select the cell to be compared, (C24) and then cells to compare it against C24:F24) The formula is =RANK(C24,$C$24:$F$24).

The four numbers which are being ranked are set as absolute cell references using the $ sign (see note 6). The default for RANK is to go from highest to lowest, however if a lower score is better (as with CO2 emissions) then '1' needs to be inserted at the end of the formula to rank from lowest to highest. For example, Bavaria's ranking for CO2 emissions is

=RANK(C28,$C$28:$F$28,1)

(4) **IF**

To quickly see whether the sites have met their KPI, the IF function can be used. To do this we reference the cell with the KPI calculated, then the target KPI, then identify what we want to show when the target has been met and when it has not been met. For example, with the CO2 emissions target, for Bavaria we want to see if CO2 per tent per night (cell C28) is less than the KPI target of 15.55 (cell C5). Instruct the spreadsheet to display YES if the target is met and NO if it is not.

The formula is: =IF(C28<$C$5,"YES","NO"). So if C28 is less than 15.55 the cell will show YES, if it exceeds the target the cell will show NO.

(5) **Relative cell references**

When doing calculations it is possible to copy and paste across the calculations into other cells so we don't have to do the same calculation multiple times. This can be done using CtrlC on the relevant cell then CtrlV, or by dragging the cell.

When we calculate the contribution margin for Bavaria the method is the same for the other sites so we can copy the formula across all columns so that the formula changes from =C21/C20 to

=D21/D20

(6) **Absolute cell references**

When doing calculations, it is possible to copy and paste across the calculations into other cells so we don't have to do the same calculation multiple times. This can be done using CtrlC on the relevant cell then CtrlV, or by dragging the cell.

Each site's KPI for CO2 emissions is judged against the same target of 15.55, so a relative cell reference (see note 5) in the IF formula will not work if we copy and paste it across columns. Instead use an absolute cell reference to fix the cell referring to the target (cell C5)

For example, Bavaria's CO2 emissions IF function is =IF(C28<$C$5,"YES","NO"). When we copy and paste this across to Cornwall it will change to IF(D28<$C$5,"YES","NO").

The CO2 per tent per night figure changes because C28 is a relative cell reference, but the target $C$5 does not change as it is an absolute cell reference.



Worked example: Chrono Ltd

Chrono Ltd (Chrono) manufactures three types of upmarket watches. It operates throughout England and Wales, selling to jewellery retailers.

Chrono's selling prices and variable costs per watch for each product type were as follows for June 2021:

Product code	Product type	Selling price per watch	Variable cost per watch
Product 1	Elite Gold	£935	£561
Product 2	Luxury Silver	£750	£525
Product 3	Fashion Bronze	£520	£416

Chrono has a geographical organisational structure, operating as three largely autonomous regions. Each region has a regional manager who is responsible for performance.

Region code	Region
Region A	North
Region B	Midlands and Wales
Region C	South

In recent years, the relative monthly performance of the South region has been significantly worse than the other two divisions. However, in May 2021 a new manager was appointed in the South region.

Data analysis for a board meeting

You are an assistant to the finance director. The board requires an analysis of performance for June 2021 for products and for regions.

The finance director has provided you with a **pre-populated spreadsheet** for June 2021 showing daily volumes of sales transactions (each row shows a day's sales of one product type in one region).

Requirements

1 Prepare a report for the board which analyses the data in the pre-populated spreadsheet, provided by the finance director, as follows:

(1) Assimilate and analyse the data according to:

(a) product type

(b) region

(Your answer should include a structured analysis of the data relating to sales volumes, revenue, and contribution.)

2 Referring to your analysis of the data in part (a) above, compare and explain the relative performance in June 2021 of:

(1) the three product types

(2) the three regions

Exhibit: Pre-populated spreadsheet data

	A	B	C	D
1	SALES FOR JUNE 2021			
2			**Quantity**	
3	**Delivery date**	**Region code**	**Delivered**	**Product type**
4				
5	01 June 2022	B	22	2
6	02 June 2022	B	54	3
7	03 June 2022	A	9	1
8	04 June 2022	C	42	3
9	04 June 2022	A	58	3
10	05 June 2022	B	33	2
11	06 June 2022	C	12	1
12	06 June 2022	B	39	3
13	07 June 2022	A	31	2
14	08 June 2022	B	27	2
15	08 June 2022	A	60	3
16	09 June 2022	B	5	1
17	10 June 2022	C	29	2
18	10 June 2022	A	41	3
19	11 June 2022	C	14	1
20	12 June 2022	A	48	3
21	12 June 2022	C	47	3
22	13 June 2022	B	37	2
23	14 June 2022	C	11	1
24	14 June 2022	A	34	2
25	15 June 2022	B	45	3
26	16 June 2022	A	7	1
27	16 June 2022	C	33	2
28	17 June 2022	A	47	3
29	17 June 2022	B	15	1
30	18 June 2022	A	43	3
31	18 June 2022	C	49	3

32	19 June 2022	A	8	1
33	19 June 2022	C	14	1
34	20 June 2022	C	31	2
35	21 June 2022	A	6	1
36	22 June 2022	B	18	2
37	23 June 2022	C	33	3
38	23 June 2022	A	9	1
39	24 June 2022	C	16	2
40	25 June 2022	A	7	1
41	26 June 2022	C	43	3
42	27 June 2022	B	66	3
43	27 June 2022	C	23	2
44	28 June 2022	C	7	1
45	29 June 2022	B	48	3
46	30 June 2022	A	17	2

Solution

1 Summary data table

	A	B	C	D	E
1	**Summary data**				
2		**Price**	**VC**	**Sales margin**	**Margin**
3		**£**	**£**	**£**	
4	Product 1	935	561	374	40%
5	Product 2	750	525	225	30%
6	Product 3	520	416	104	20%

Note: Columns A–C have been copied from the question. Columns D–E are calculations that candidates are expected to complete.

June 2021

Analysis by product

	A	B	C	D	E
65	**PRODUCTS**				
66		**Product 1**	**Product 2**	**Product 3**	**Total**
67	Volume	124	351	763	1,238

68	Price	£935	£750	£520	
69	**Revenue**	**£115,940**	**£263,250**	**£396,760**	**£775,950**
70					
71	**Contribution**	**£46,376**	**£78,975**	**£79,352**	**£204,703**
72					
73	**Mix %**				
74	Volume	10%	28%	62%	
75	Revenue	15%	34%	51%	
76	Contribution	23%	39%	39%	

June 2021 Analysis by region

	A	B	C	D	E
79	**REGIONS**				
80					
81		**Region A**	**Region B**	**Region C**	**Total**
82	**Volume**	425	409	404	1238
83					
84	**Revenue**	**£258,950**	**£252,490**	**£264,510**	**£775,950**
85					
86	**Contribution**	**£66,542**	**£64,513**	**£73,648**	**£204,703**
87					
88	**Margin**	26%	26%	28%	26%
89					
90	**Mix %**				
91	Volume	34%	33%	33%	
92	Revenue	33%	33%	34%	
93	Contribution	33%	32%	36%	

Notes

1. Mix figures are subject to rounding differences.
2. The analysis shown above should be copied from the spreadsheet software into the word processing software in the exam, in order to be awarded marks.
3. Spreadsheet functionality has been used to produce the analysis (see tutorial notes below).

Tutorial Note Summary

data table

The initial summary data table is important in determining some base data of price and unit contribution for each product type. The candidate should copy it into the spreadsheet area from the exam paper. The data in the summary table is a key reference for later analysis and should be referred to with absolute cell references (rather than relative cell references).

Thus, for example, for the sales price of Product 1 this should be referred to as $B$7.

Analysis by product

In carrying out the analysis by product, there are a number of methods that can be used. The method used here is the =SUMIF function which is perhaps the easiest.

For example, in determining the sales volume for Product 1, the function is =SUMIF(D17:D58,"1",C17:C58). This is saying that the items in the quantity delivered column should be added but only selecting those amounts to be added where there is a '1' in the product type column (ie, for Product 1 – Elite Gold watches).

WORKING

	A	B	C	D	E	F
14	SALES FOR JUNE 2021					
15			**Quantity**		**Regional**	**Regional**
16	**Delivery date**	**Region code**	**Delivered**	**Product type**	**Revenue (£)**	**Contributio n (£)**
17	01 June 2022	B	22	2	16,500	4,950
18	02 June 2022	B	54	3	28,080	5,616
19	03 June 2022	A	9	1	8,415	3,366
20	04 June 2022	C	42	3	21,840	4,368
21	04 June 2022	A	58	3	30,160	6,032
22	05 June 2022	B	33	2	24,750	7,425
23	06 June 2022	C	12	1	11,220	4,488
24	06 June 2022	B	39	3	20,280	4,056
25	07 June 2022	A	31	2	23,250	6,975
26	08 June 2022	B	27	2	20,250	6,075

	A	B	C	D	E	F
27	08 June 2022	A	60	3	31,200	6,240
28	09 June 2022	B	5	1	4,675	1,870
29	10 June 2022	C	29	2	21,750	6,525
30	10 June 2022	A	41	3	21,320	4,264
31	11 June 2022	C	14	1	13,090	5,236
32	12 June 2022	A	48	3	24,960	4,992
33	12 June 2022	C	47	3	24,440	4,888
34	13 June 2022	B	37	2	27,750	8,325
35	14 June 2022	C	11	1	10,285	4,114
36	14 June 2022	A	34	2	25,500	7,650
37	15 June 2022	B	45	3	23,400	4,680
38	16 June 2022	A	7	1	6,545	2,618
39	16 June 2022	C	33	2	24,750	7,425
40	17 June 2022	A	47	3	24,440	4,888
41	17 June 2022	B	15	1	14,025	5,610
42	18 June 2022	A	43	3	22,360	4,472
43	18 June 2022	C	49	3	25,480	5,096
44	19 June 2022	A	8	1	7,480	2,992
45	19 June 2022	C	14	1	13,090	5,236
46	20 June 2022	C	31	2	23,250	6,975

	A	B	C	D	E	F
47	21 June 2022	A	6	1	5,610	2,244
48	22 June 2022	B	18	2	13,500	4,050
49	23 June 2022	C	33	3	17,160	3,432
50	23 June 2022	A	9	1	8,415	3,366
51	24 June 2022	C	16	2	12,000	3,600
52	25 June 2022	A	7	1	6,545	2,618
53	26 June 2022	C	43	3	22,360	4,472
54	27 June 2022	B	66	3	34,320	6,864
55	27 June 2022	C	23	2	17,250	5,175
56	28 June 2022	C	7	1	6,545	2,618
57	29 June 2022	B	48	3	24,960	4,992
58	30 June 2022	A	17	2	12,750	3,825
59	**Total**		**1,238**		**775,950**	**204,703**

Note: Columns A–D are copied from the pre-populated data. Columns E–F are calculated using spreadsheet functionality (see tutorial notes below).

Tutorial Note

To generate the amounts in the regional revenue column (Column E) it is necessary to use the =IF function. This is a conditional statement which, in its simplest form, would look like =IF(D19=1,C19*$B$7,""). This would say that if the product type is 1 (ie, Product 1) then multiply the price for Product 1 (given as an absolute reference to cell B7 as $B$7) by the sales quantity in the row for this day. If it is not Product 1 in the sales transaction, then the cell is left blank (as indicated by the "" instruction).

However, this simple type of =IF statement does not generate revenues for products 2 and 3 – this would require separate columns for Products 2 and 3 and create a more complicated assimilation task. By nesting the three =IF statements together we can generate revenues for all the rows for each product and price. This is just extending the use of the IF statement, rather than requiring any new functions. Alternatively we can use IFS to test multiple conditions, which delivers the same results as nested IF statements.

So, for example in the first row of the regional revenue column (Row 17); Using a nested IF statement the function is:
=IF(D17=1,$B$7*C17,IF(D17=2,$B$8*C17,IF(D17=3,$B$9*C17,IF))).

Using IFS the function is:

=IFS(D17=1,$B$7*C17,D17=2,$B$8*C17,D17=3,$B$9*C17)

This is saying that the figure to be inserted into the regional revenue column (Column E) is based on the quantity in that row, multiplied by the price according to the product type (1, 2 or 3). The price is determined by the IF statement. So, given the first row of the regional revenue column is product 2, the =IF statement refers back to the price of product 2 in the initial summary data table using an absolute cell reference $B$8. Products 1 and 3 are not relevant to this row and so are ignored by the =IF statement.

Nesting the =IF functions (ie, grouping three simple =IF statements together) therefore inserts the relevant regional revenue figure for each row according to the relevant product price. The relevant amounts can then be selected from this column using the =SUMIF function as previously.

The same procedures can then be applied to the regional contribution column (Column F).

2

Product types Revenue and volumes

Despite having the lowest price, the dominant product in terms of sales revenues and sales volumes is the Fashion Bronze watch. In June 2021, it generated 62% of the total sales volume of Chrono Ltd and over half (51%) its sales revenue.

In contrast, the Elite Gold generated only 10% of the total sales volume of Chrono Ltd and only 15% of its sales revenue.

The Luxury Silver product is positioned between the other two products, both in terms of revenue generated (34%) and volume sold (28%).

Analysing this data, starting with the Elite Gold watch perspective, indicates a surprising degree of price sensitivity. As the price decreases from £925 for the Elite Gold watch to £520 for the Fashion Bronze watch (a 44% decrease), the volume sold increases from only 124 Elite Gold watches sold to 763 Fashion Bronze watches sold to (a 515% increase).

The extent of the difference in the quantity demanded relative to the price difference between the two watches is slightly surprising, as the higher price watch is better quality, so customers are obtaining more for their money.

This calculation is similar to price elasticity of demand, but it is not the same, as these are different products at one point in time, not changes in the price of the same product over time.

Some investigation is needed for the differences in quantity demanded between the three products and whether price alone is an adequate explanation. Surveys of customers (jewellery retailers) and consumers are needed to obtain further information.

Other explanations and types of analysis of the sales volume and price differences are possible:

- The Elite Gold may not have an attractive design or other attributes with consumer appeal, compared with the other two watches, despite its quality.
- Sales channels may discriminate against high value watches (eg, some jewellery retailers may not believe high value items like the Elite Gold are appropriate for the price range of their target customer market).
- The one month of June 2021 is a small data set and may be atypical. Similar analysis is needed over a longer period.

Also, a key issue with data analysis is that the same data can be analysed in different ways and care is needed not to allow different data perspectives to bias the analysis or the decision making arising from that analysis. An alternative analysis of the above data, starting with the Fashion Bronze watch perspective, gives different figures as follows:

As the price increases from £520 for the Fashion Bronze watch to £925 for the Elite Gold watch (a 78% increase) the demand falls from 763 Fashion Bronze watches sold to 124 Elite Gold watches sold (an 84% decrease).

Starting from this Fashion Bronze watch perspective, the same data on the differences in price and volume demanded for the two watches now appear much more in line.

Contribution – volumes and margins

Revenue is a key driver of contribution and profit. However, total contribution is a more appropriate measure of performance than revenue alone, as it considers both revenue and costs.

The total contribution is a function of sales margin per item (in £s) and sales volume.

The sales margin is greatest on the highest price product Elite Gold at £374 (40%) and the lowest on the lowest price product Fashion Bronze at £104 (20%).

Performance in terms of sales volumes is the inverse of margin, where Elite Gold only sold 124 watches in June 2021, compared with the Fashion Bronze sales volume of 763 items.

The Luxury Silver watch is positioned between these extremes, where it sold 351 watches in June 2021, at a sales margin per watch of £225 (30%).

Conclusion

In terms of total contribution, the mix data in the table above shows that the Luxury Silver watch and the Fashion Bronze watch have performed equally, each having a 39% share of total contribution. This arises as the lower sales volume for the Luxury Silver watch is compensated by the higher sales margin per item, compared with the Fashion Bronze watch.

Whilst the Elite Gold watch has the highest percentage margin and absolute margin per item, this is not sufficient to compensate for the much lower sales volume. As a result, in terms of total contribution it is the worst performing product, with only 23% of the total contribution in the mix table above.

Regions

Revenue and volumes

All three regions have a similar share of sales revenue. In the revenue mix data table, all three regions fall in the narrow range of 33%–34%, with the South having, very slightly, the highest revenue.

Similarly, for sales volumes, all three regions fall in the narrow range of 33–34%, with the North having, very slightly, the highest sales volume at 34%.

Given that the key drivers of sales revenues are product sales price and product sales volumes, this suggests that each region has a similar mix of the three products, in relative and absolute terms.

As such, the three regions have performed approximately equally, with no clear signs of excess, or inferior, performance by any one division.

Even though the relative performance of the three regions is approximately equal, the new manager for the South has performed well in comparison with historical performance, even though they have only just been appointed.

Contribution

All three regions have a similar share of total contribution although there is a little more dispersion (a range of 32% to 36% in the total contribution mix) than for revenue and volumes.

The South has just the highest contribution margin at 28%, compared with 26% each for the other two regions. This may indicate a slightly higher proportion of Elite Gold watch sales for the South region. Combined with the slightly higher revenue for the South, compared with the other two regions, this means the South contributes 36% of the total contribution in the

contribution mix, compared with 33% for the North and 32% for the Midlands.

Conclusion

There is little to choose between the performance of the three regions, with the South having only a slightly higher total contribution.

The new manager for the South has therefore performed well relative to the other two regions, but more particularly in comparison with historical performance, even though they have only just been appointed.

However, this is only one month's data, so care needs to be exercised in drawing strong conclusions. This data analysis needs to be combined with more past data and with future data to establish stronger evidence of relative performance.



Worked example: Home and Garden Furniture Ltd

Home and Garden Furniture Ltd (HGF) is a wholesaler of furniture, selling to retailer customers (customers) in the UK. HGF's operations are located near the port of Hull, where it has offices plus its single, large warehouse to hold inventory.

Customers order online using HGF's website. The 'HGF delivery promise' is that a customer's goods will be delivered to them two days after the order date, at the latest. Customers can place an order for goods only when they are held in inventory.

HGF outsources customer deliveries to three small, low-cost independent distribution companies (distributors), which all have their main operations near Hull. These three distributors are:

(1) Nexprezz

(2) Roadee

(3) Hullier

The HGF distribution manager, Matt Moore, is responsible for allocating customer orders to the three distributors. The selected distributor for a delivery is required to collect the goods ordered by a customer from the HGF warehouse in Hull on the day following the order date. The distributor is then required to make the delivery to the customer anywhere in the UK on the next day.

It has become apparent that, for longer distance journeys, the distributors are sometimes delaying delivery to HGF customers, so they can add the goods collected from other companies to their load. This is to avoid making two separate journeys and enables travelling with a fully loaded van.

Using an online questionnaire, HGF's customers are asked to score the overall quality of the service provided by the distributor (10 = excellent; 1 = very poor). Issues that customers are asked to consider in giving their overall score include timing of delivery; condition of the goods and packaging; and politeness and efficiency of delivery staff.

HGF has the following KPIs and targets with respect to its delivery operations:

KPI	Targets for each distributor and for HGF overall
Delivery to customer is on, or before, the promised date.	85% of deliveries are on or before the promised date
High customer satisfaction.	Average customer survey score of over 7

A board meeting is to be held next month. One of the agenda items is to consider the level of service being provided to HGF and its customers by the three distributors.

You work in the operations management division of HGF and you have been asked to analyse data for the board which has been provided by Matt Moore. Matt was asked to select a typical three-week period and to provide data on delivery dates and customer survey results. Matt has provided you with data for the three weeks beginning 1 June 2021 in the **pre-populated spreadsheet**.

Requirements

1 Prepare a report for the HGF board which analyses the data in the pre-populated spreadsheet as follows:

(1) Assimilate and analyse the data in relation to:

(a) delivery dates achieved

(b) customer satisfaction, as denoted by survey scores

Evaluate and explain the relative performance of the three distribution companies relating to delivery dates and customer satisfaction.

State any further information that would be required to make a more complete assessment of the performance of the three distributors.

2 Calculate the standard deviation of the customer survey scores for HGF as a whole.

Briefly comment on the extent to which this standard deviation may give useful information for management.

Note: You are **not** required to calculate the standard deviation of the survey scores of each individual distributor.

3 Assess the extent to which the data collected in the customer survey is sufficiently reliable and unbiased to inform the HGF board's assessment of the distributors' performance.

Exhibit: Pre-populated spreadsheet data

	A	B	C	D
1				
2	Distributors are:			
3	A.	Nexprezz		
4	B.	Roadee		
5	C.	Hullier		
6				
7	**DELIVERIES FOR THE THREE WEEKS BEGINNING 1 JUNE 2021**			
8				
9	**Order date**	**Actual Delivery date**	**Distributor**	**Survey score**
10				
11	01 June 2022	03 June 2022	A	7
12	01 June 2022	03 June 2022	B	8
13	01 June 2022	03 June 2022	C	8
14	02 June 2022	05 June 2022	A	3
15	02 June 2022	04 June 2022	C	8
16	03 June 2022	05 June 2022	A	4
17	03 June 2022	05 June 2022	B	8

	A	B	C	D
18	03 June 2022	05 June 2022	C	9
19	04 June 2022	06 June 2022	B	4
20	05 June 2022	07 June 2022	A	7
21	05 June 2022	08 June 2022	B	3
22	05 June 2022	07 June 2022	C	8
23	06 June 2022	08 June 2022	A	6
24	07 June 2022	09 June 2022	A	4
25	07 June 2022	09 June 2022	B	7
26	08 June 2022	10 June 2022	A	3
27	08 June 2022	09 June 2022	B	9
28	08 June 2022	10 June 2022	C	9
29	09 June 2022	11 June 2022	B	8
30	09 June 2022	11 June 2022	C	9
31	10 June 2022	15 June 2022	A	1
32	10 June 2022	11 June 2022	B	9
33	10 June 2022	12 June 2022	C	6
34	11 June 2022	13 June 2022	A	8
35	11 June 2022	13 June 2022	C	8
36	12 June 2022	14 June 2022	C	9
37	13 June 2022	16 June 2022	B	3
38	14 June 2022	18 June 2022	A	2
39	14 June 2022	16 June 2022	B	8
40	14 June 2022	17 June 2022	C	4
41	15 June 2022	17 June 2022	A	5
42	15 June 2022	17 June 2022	B	3
43	15 June 2022	17 June 2022	C	4
44	16 June 2022	18 June 2022	A	5
45	17 June 2022	18 June 2022	B	9
46	17 June 2022	18 June 2022	C	9
47	18 June 2022	21 June 2022	A	2
48	18 June 2022	20 June 2022	B	8
49	18 June 2022	20 June 2022	C	8
50	19 June 2022	21 June 2022	B	7
51	19 June 2022	21 June 2022	C	8
52	20 June 2022	22 June 2022	C	4
53	21 June 2022	24 June 2022	A	3
54	21 June 2022	23 June 2022	B	7
55	21 June 2022	23 June 2022	C	9

Solution

WORKING

	A	B	C	D	E	F	G	H	I	J
1										
3										
4	DELIVERIES FOR THE THREE WEEKS BEGINNING 1 JUNE 2021									
5							SURVEY	SURVEY	SURVEY	
6	Order date	Actual Delivery date	Distributor	Survey score	Days early (-) late(+)	Delivery by promised date	A	B	C	
7					(with 2 days lag promised)					
8	01 June 2022	03 June 2022	A	7	0	YES	7			
9	01 June 2022	03 June 2022	B	8	0	YES		8		
10	01 June 2022	03 June 2022	C	8	0	YES			8	
11	02 June 2022	05 June 2022	A	3	1	NO	3			
12	02 June 2022	04 June 2022	C	8	0	YES			8	
13	03 June 2022	05 June 2022	A	4	0	YES	4			
14	03 June 2022	05 June 2022	B	8	0	YES		8		
15	03 June 2022	05 June 2022	C	9	0	YES			9	
16	04 June 2022	06 June 2022	B	4	0	YES		4		
17	05 June 2022	07 June 2022	A	7	0	YES	7			
18	05 June 2022	08 June 2022	B	3	1	NO		3		
19	05 June 2022	07 June 2022	C	8	0	YES			8	
20	06 June 2022	08 June 2022	A	6	0	YES	6			
21	07 June 2022	09 June 2022	A	4	0	YES	4			
22	07 June 2022	09 June 2022	B	7	0	YES		7		
23	08 June 2022	10 June 2022	A	3	0	YES	3			
24	08 June 2022	09 June 2022	B	9	-1	YES		9		
25	08 June 2022	10 June 2022	C	9	0	YES			9	
26	09 June 2022	11 June 2022	B	8	0	YES		8		
27	09 June 2022	11 June 2022	C	9	0	YES			9	
28	10 June 2022	17 June 2022	A	1	5	NO	1			
29	10 June 2022	11 June 2022	B	9	-1	YES		9		
30	10 June 2022	12 June 2022	C	6	0	YES			6	
31	11 June 2022	13 June 2022	A	8	0	YES	8			
32	11 June 2022	13 June 2022	C	8	0	YES			8	
33	12 June 2022	14 June 2022	C	9	0	YES			9	
34	13 June 2022	16 June 2022	B	3	1	NO		3		
35	14 June 2022	19 June 2022	A	2	3	NO	2			
36	14 June 2022	16 June 2022	B	8	0	YES		8		
37	14 June 2022	17 June 2022	C	4	1	NO			4	
38	15 June 2022	17 June 2022	A	5	0	YES	5			
39	15 June 2022	17 June 2022	B	3	0	YES		3		
40	15 June 2022	17 June 2022	C	4	0	YES			4	
41	16 June 2022	18 June 2022	A	5	0	YES	5			
42	17 June 2022	18 June 2022	B	9	-1	YES		9		
43	17 June 2022	18 June 2022	C	9	-1	YES			9	
44	18 June 2022	23 June 2022	A	2	3	NO	2			
45	18 June 2022	20 June 2022	B	8	0	YES		8		
46	18 June 2022	20 June 2022	C	8	0	YES			8	
47	19 June 2022	21 June 2022	B	7	0	YES		7		
48	19 June 2022	21 June 2022	C	8	0	YES			8	
49	20 June 2022	22 June 2022	C	4	0	YES			4	
50	21 June 2022	24 June 2022	A	3	1	NO	3			
51	21 June 2022	23 June 2022	B	7	0	YES		7		
52	21 June 2022	23 June 2022	C	9	0	YES			9	
53							Nexprezz	Roadee	Hullier	Total
54			Average	6.2		AVERAGE	4.3	6.7	7.5	6.2
55										
56			Standard deviation	2.5		KPI target	7	7	7	7.0
57	Delivery times					Achieved	no	no	yes	no
58		A	B	C	Total					
59										
60	Late	5	2	1	8					
61	Ontime	9	13	15	37					
62										
63	Total	14	15	16	45					
64										
65	KPI ontime %	64.3%	86.7%	93.8%	82.2%					
66										

Tutorial Note

Late deliveries (column E)

To determine whether the delivery is late, or not, the dates given the in the pre-populated spreadsheet are compared. Dates can be treated as any other number, and should be deducted from each other to show the gap between the two dates in terms of the number of days.

The question says that the due date is two days after the order date. One approach would be to add 2 to the order date to produce a new column of due dates for delivery. However, this is not necessary as it can be done in a single step by taking the order dates in column A, and adding 2, then deducting the actual delivery date from this total.

Thus, for cell E8, to calculate days early/late this is =B8-(A8+2).

Column E shows not just whether the delivery is late or not, but also the number of days by which it is late (or early), which is useful management information in this context.
As the KPI is whether or not the delivery is late, column F shows this (as "YES" or "NO") according to whether column E is greater than zero using an =IF statement. So for the cell F8 it is =IF(E3>0,"NO","YES")

To count the total number of on-time and late deliveries the =COUNTIF function is needed. In the workings above, the total column (column E) uses the function as: =COUNTIF(F8:F52,"no") for late deliveries in E60 and =COUNTIF(F8:F52,"yes") for on-time or early deliveries in E61.

Counting the number of on-time and late deliveries for each distributor is slightly more difficult as there are now two conditions: (a) whether it is late (b) whether it is distributor A, B or C. It is therefore necessary to use the =COUNTIFS function (note the extra S in the function title).

So, for example, in determining the number of late deliveries for Distributor A the function used in B60 is: =COUNTIFS(E8:E52,">0",C8:C52,"A") This is simply saying that the item should be counted if both: (a) the figure in Column E is greater than zero (ie, it is late); AND (b) the letter in column C is "A" (ie, it is Distributor A that has made the late delivery).

This function then counts the number of late deliveries by Distributor A, and the next columns do the same for Distributors B and C.

Customer survey

To determine the simple average (the mean) of all survey scores (ie, for all three distributors), the function used in D54 is =AVERAGE(D8:D52) To determine the average survey score for each distributor requires separate identification of each distributor's scores. There are a number of ways to do this. In the table above this has been done using the =IF function. This is not necessarily the quickest method of determining the separate averages, but it is a flexible approach that can be used for extracting and analysing data in a variety of circumstances.

The **key technique** is to insert the correct formulae in the first row of the three columns relating to the customer survey. These can then be easily copied into all the remaining rows.

So, in the first row of the customer survey column for Distributor A the formula in G8 is =IF(C8="A", D8,"") This is saying that if column C states "A" (ie, it is Distributor A) then insert the survey score from column D into column G. The double inverted commas "" means that if it is not "A" in column D, then the cell is left blank.

So, in the first row of the customer survey column for Distributor B the formula in H8 is =IF(C8="B", D8,"") As the first row relates to Distributor A (not B), H8 is left blank.

In this way the three-column analysis selects the appropriate customer survey scores for each of the three distributors. The average scores for each of the three columns are determined using the
=AVERAGE function.

Standard deviation

The standard deviation can be calculated using the =STDEV function in cell D56.

This function is applied to the column of customer survey scores using =STDEV(D8:D52).

1 Delivery times (see workings)

	A	B	C	D	E
57	**Delivery times**				
58		**A**	**B**	**C**	**Total**
59					
60	Late	5	2	1	8
61	Ontime	9	13	15	37
62					
63	**Total**	**14**	**15**	**16**	**45**
64					
65	**KPI ontime %**	64.3%	86.7%	93.8%	82.2%

The KPI target of 85% of deliveries being on time (ie, within the promised two-day period from the order date) has been met by two distributors: Roadee and Hullier.

Roadee has marginally satisfied the KPI target, with two late deliveries out of 15. Hullier has comfortably satisfied the KPI target, with only one late delivery out of 16.
In contrast, Nexprezz has missed the KPI target significantly with 5 out of 14 deliveries being late (ie, only 64.3% of deliveries were on time – less than 2 in 3).

Moreover, in the three-week sample period, visual inspection of the data shows three deliveries were late by more than one day. These were all by Nexprezz. One of these was five days late.

The deliveries data can be linked to the customer survey data by visual inspection as the late deliveries also all received low customer survey scores (no late delivery was scored more than 4 out of 10). This indicates that late delivery matters to HGF's customers.

Customer survey scores (see workings)

	F	G	H	I	J
53		**Nexprezz**	**Roadee**	**Hullier**	**Total**
54	**AVERAGE**	4.3	6.7	7.5	6.2
55					
56	KPI Target	7	7	7	7.0
57	Achieved	no	no	yes	no

The average satisfaction score actually achieved for the company as a whole is 6.2, which is below the target of 7.

The overall score hides considerable variance between the customer survey scores achieved by individual distributors, although the total number of deliveries made by each distributor in the three-week sample period is approximately equal.

Roadee has marginally fallen below the KPI target. Low customer survey scores for the two late deliveries have been a key factor in not achieving the KPI target.

Hullier has comfortably satisfied the KPI target, with good customer survey scores for most deliveries and some very good scores.

In contrast, Nexprezz has missed the KPI target significantly, with an average customer survey score of only 4.3 out of 10. The five late deliveries have been a factor in not achieving the KPI target but also, only three of the other customer survey scores for Nexprezz were at 7 or above.

Nexprezz is clearly the weakest of the three distributors in terms of customer survey scores. Indeed, taken together, Roadee and Hullier would jointly have achieved the target KPI of 7, if Nexprezz had been excluded.

Further information

Before conclusions can be drawn, further information and explanations are required from each distributor to identify the underlying causes of the differences in relative performance in the sample period.

For example:

- Policies – do any of the distributors have a policy of delaying deliveries to obtain a full load or shared load?
- Practices – why have delays arisen in the three-week period? Is it due to inefficient practices or inappropriate procedures (eg, for collection, storage or delivery)?
- Capacity – do the distributors lack staff or a sufficient number of vehicles to meet peak demand? Do they accept assignments from HGF when they are already at peak demand?
- Capability – is there appropriate resource for staff (training and recruitment with necessary skills); IT (meets industry standards and appropriate for purpose); vehicles (right size and type; reliable)?
- Allocations of long-distance journeys – are all three distributors allocated an equal number of the longest journeys?
- Events – have there been random unavoidable or unforeseeable events which have caused delays (eg, traffic delays, staff illness, access to retailer customers' premises, vehicle break down)?

2 The standard deviation (SD) of the customer survey scores for HGF overall is 2.5 (see workings).

The broad indication for management is therefore that there is significant variability in the customer survey scores around the mean of 6.2.

However, the validity of the SD is limited in these circumstances. The distribution only has 10 points and it is not identified as a normal distribution. Conclusions about proportions of the distribution within one SD of the mean are not therefore valid.

Moreover, the distribution is not random or normally distributed but, as noted above, is related to each particular distribution company's performance. Analysing this data for individual distributors is therefore more useful to management than exploring the overall SD for the company. The overall SD shows little beyond the basic notion that there is variability in the overall customer survey scores, without linking this to underlying causes related to variability between individual distributors.

3 The sample of customer survey scores for HGF as whole is small at 45 observations. It may therefore be atypical of a larger sample over a longer period. Moreover, once the overall sample of 45 is segmented into three sub samples for the three distributors, the sample sizes become even smaller (14 to 16 observations) and perhaps even more unrepresentative.

For example, Nexprezz could have been having particular difficulties in the three-week sample period (eg. IT, staffing, vehicles break-downs) and the customer survey scores may therefore have been unusually poor.

Aside from sample size, a degree of professional scepticism needs to be applied to the data as Matt Moore extracted the sample, and the performance of the distributors is their responsibility. They may have selected a particularly favourable three-week period, resulting in bias. This may mean that more KPIs were not actually achieved in typical weeks than indicated by the data.

They may even have distorted the data itself or omitted data, introducing favourable biasing. It would have been more appropriate to have greater objectivity in data collection by using somebody to capture data who would not be affected by the consequences of the data analysis.

The parameters in the data may also lead to bias in the assessment of distributors' performance. In particular, there may be omitted variables relating to aspects of performance that are not seen, or do not affect, customers, for example the price charged to HGF for delivery and efficiency of collection from HGF.

Overall customer survey scores are one piece of data that forms part of the assessment of distributors' performance, but more data and more reliable data is needed.

Glossary of terms

Acquisition:

Is the purchase of a controlling interest in another company.

Alliance: An agreement between firms to share a commercial opportunity characterised by each member of the alliance retaining autonomy and pursuing its own commercial interests.

Artificial intelligence (AI): Is the field of study and application involving the creation and use of advanced computer systems to perform tasks traditionally requiring human intelligence. This includes learning from data; reasoning and problem-solving; sensory understanding; language processing; and recognising patterns. AI systems are characterised by their ability to adapt to new information or environments, operate with a degree of autonomy, and make informed decisions. AI often, but not always, involves the use of machine learning.

Asset specificity: Where investments are made to support the relationship which have the effect of locking parties into a relationship to some degree.

Automation: 'The creation and application of technology to monitor and control the production and delivery of products and services' (The International Society of Automation, 2018).

BCG matrix: A tool which examines market growth rates and an organisation's relative market strength in its markets to assess the attractiveness and balance of a business portfolio.

Balanced scorecard: An integrated set of performance measures linked to the achievement of strategic objectives. The balanced scorecard consists of four perspectives: customer, internal business, innovation and learning, financial.

Benchmarking: The establishment, through data gathering, of targets and comparators, through whose use relative levels of performance (and particularly areas of underperformance) can be identified. By the adoption of identified best practices it is hoped that performance will improve.

Big data: The term that describes those 'datasets whose size is beyond the ability of typical database software to capture, store, manage and analyse' (McKinsey Global Institute, 2011).

An alternative definition is provided by Gartner: big data concerns 'high-volume, high velocity and high-variety information assets that demand cost-effective, innovative forms of information processes for enhanced insight and decision making.'

Blockchain: Is a specific form of **distributed ledger technology** that records data in a decentralised and immutable manner across multiple nodes. This structure makes records tamper-proof and enhances security by removing single points of failure. Blockchains are often public, allowing anyone to verify the data, though private configurations exist for tailored organisational needs. This technology supports **cryptocurrencies** and has many other applications, including supply chain oversight, digital identity, and secure voting.

Bots: Software programs, also known as software robots, that can automatically execute tasks and actions. Bots are useful for performing repetitive, logic-based tasks, following a set of pre-programmed rules. Some bots also make use of artificial intelligence (AI) to mimic human behaviour. For example, chatbots use AI to enable conversation with humans.

Boundary-less organisations: Are entities which are structured in such a way as to make it easier to collaborate with external parties. Such organisations remove the internal barriers that often exist between different functions and departments.

Brand equity: An intangible asset that adds value to a business through positive associations made by the consumer between the brand and benefits to themselves.

Business continuity planning: A process through which a business details how and when it will recover and restore any operations interrupted by the occurrence of a rare, but massive, risk event.

Business plan: Is a document which is usually prepared as part of the process of applying for funding. It is a critical document for the potential investor. The contents of a business plan may include: a statement of purpose, a description of the business, financial data and supporting documents.

Business risk: Is the variability of returns due to how a business trades or operates, its exposure to markets, competitors, etc.

Business trust: Exists when stakeholders act as willing and positive participants in their dealings with an organisation because their experience of the organisation, and its reputation, support open and mutually beneficial relations. Loss by an organisation of business trust can have multiple and serious consequences.

Centralisation: Decision-making authority is concentrated in one place in an organisation; that is, in the centre or strategic apex.

Change agent: Is an individual (sometimes called a champion of change), a group or external consultancy with the responsibility for driving and 'selling' change.

Chatbot: Is a software application designed to simulate conversation with human users. It uses **natural language processing** to understand and respond to user queries.

Climate change adaptation: Refers to adjustments of strategies and actions in response to the effects and future risks of climate change. These risks may be linked to the physical impacts of climate change; but also, the economic impact of a transition towards net zero. Identifying, managing and adapting to these risks is key for the development of a resilient business that can survive, and even thrive in the unprecedented physical and transitional changes ahead.

Climate change mitigation: Means avoiding and reducing emissions of greenhouse gases into the atmosphere to prevent the planet from warming to more extreme temperatures. This involves a transition away from fossil fuels to clean, renewable energy; halting deforestation and restoring natural habitats to reach net zero emissions.

Climate risk: The risk posed to organisations and society at large from changes in global temperatures, resulting in changing and potentially damaging weather patterns and systems. Organisations can both contribute to and be negatively affected by climate risk. Scenario planning, which is useful where there are several key factors influencing the long-term strategy, can be used to better understand and plan for climate risk.

Cloud accounting: 'Refers to performing basic accounting tasks, like managing and balancing the books, using software that resides in the cloud and is often delivered in an as-a-service model.' (**Source:** http://www.netsuite.com/portal/resource/articles/accounting/cloud-accounting.shtml#.~:text=Cloud%20accounting%20refers%20to%20performing,service%20model)

Cloud computing: 'Is on-demand access, via the internet, to computing resources – applications, servers (physical servers and virtual servers), data storage, development tools, networking capabilities, and more-hosted at a remote data centre managed by a cloud services provider (or CSP).' (**Source**: https://www.ibm.com/topics/cloud-computing#:~:text=Cloud%20computing%20is%20on,or%20CSP)

Co-efficient of variation: Is the ratio of the standard deviation to the mean, calculated as (standard deviation ÷ mean)*100

Code of ethics: A codification of principles or rules to help prevent wrong-doing and to avoid conflicts of interest and to promote honest and ethical conduct among staff in an organisation.

Cognitive bias: Relates to human perception and describes the way individuals interpret and understand information differently based on their own background, experiences and beliefs.

Competitive strategy: The way that the firm will seek to win customers and secure profitability against rivals.

Compliance risk: The risk arising from non-compliance with laws or regulations. This includes breach of laws/regulations by the company, or breaches by a stakeholder (eg, customer or supplier) which may have consequences for the company. It may relate to financial laws/regulations (eg, contracts, tax, financial reporting, pensions and social security, company law etc) or to non-financial laws/regulations (eg, health and safety, employment law etc).

Conditional probability: Measures the likelihood of an event occurring, given that another event has already happened. It answers the question: 'what is the probability of event A happening given that event B has occurred?'

The probability of two dependent events occurring is calculated by multiplying the individual probabilities together. Contingency tables can be useful for dealing with conditional probability.

For example, the probability of rolling a six, followed by another six is:

$1/6 \times 1/6 = 1/36$

Probability is used to help to calculate risk in decision making. The higher the probability of an event occurring, the lower the associated risk will be.

Risk involves situations or events which may or may not occur, but whose probability of occurrence can be calculated statistically and the frequency predicted.

Uncertainty involves situations or events whose outcome cannot be predicted with statistical confidence.

A popular quote that illustrates an element of the distinction between risk and uncertainty was given by the United States ex Secretary of Defence, Donald Rumsfeld:

'There are known knowns; there are things we know that we know. There are known unknowns; that is to say there are things that we now know we don't know. But there are also unknown unknowns – there are things we do not know, we don't know.'

Confirmation bias: This occurs when people see data that confirms their beliefs and they ignore (consciously or sub-consciously) data that disagrees with their beliefs.

Consortia: Organisations that co-operate on specific business prospects.

Controllable risk: Those risk factors which are within the ability of management to control to some extent. Often controllable risks relate to factors internal to the organisation and, as a result, may be easier to manage directly, however, this may not always be the case.

Core competences: The critical activities and processes which enable an organisation to achieve its critical success factors (CSFs) and therefore achieve a sustainable competitive advantage. The core competences must be **better** than those of competitors and **difficult to replicate**.

Corporate Sustainability Reporting Directive (CSRD): In tandem with the development of standards by the International Sustainability Standards Board (ISSB), the European Union (EU) has adopted the Corporate Sustainability Reporting Directive (CSRD). The CSRD introduces detailed sustainability reporting requirements for large and listed companies and paves the way for mandatory sustainability assurance.

Corporate appraisal: A critical assessment of an entity's internal strengths and weaknesses, and the opportunities and threats it faces in the external environment, to establish its condition prior to preparation of its long-term plan.

Corporate governance: The set of rules which governs the structure and determines the objectives of an organisation and regulates the relationship between the organisation's management, its board of directors and its shareholders.

Corporate strategy: 'Is concerned with an organisation's basic direction for the future, its purpose, its ambitions, its resources and how it interacts with the world in which it operates' (Lynch).

Correlation: Is a measure of the extent to which changes in the dependent variable are explained by changes in the independent variable.

Correlation coefficient: A statistical measure of the strength of the relationship between the relative movements of two variables. It shows what portion of the change in the dependent variable can be explained by changes in the independent variable.

Cost leadership strategy: A strategy that seeks to achieve the position of lowest-cost producer in the industry as a whole.

Critical success factors (CSFs): Are a small number of key goals vital to the success of an organisation ie, 'things that must go right.'

An alternative definition is provided by Johnson, Scholes and Whittington.

'Those product features that are particularly valued by a group of customers, and, therefore, where the organisation must excel to outperform the competition.'

This second definition is more restricted because it deals merely with product features rather than considering additional factors which are vital to commercial success such as product availability, competitive knowledge or cost and performance control.

Crowdfunding: Is a way of raising finance by asking a large number of people for a small amount of money (The UK Crowdfunding Association).

Crowdsourcing: The process of getting work or funding, usually online, from a network of people (Oxford Dictionary).

Cryptocurrency: A digital asset that is designed to function as a medium of exchange and store of value. It operates through a decentralised computer network independent of any central authority, such as a central bank, which does not issue or regulate it. Cryptocurrencies use blockchain technology to achieve decentralisation, transparency, and immutability, distinguishing them from traditional monetary systems and financial assets. While they perform roles similar to currency, under IFRS, they are not recognised as cash or financial instruments, but may be accounted for as intangible assets or inventory depending on the circumstances.

(Source: https://www.merriam-webster.com/dictionary/cryptocurrency)

Cyber risk: The risk of financial loss, disruption or damage to the reputation of an organisation from some sort of failure of its information technology systems.

Cyber-attack: Is an attempt by individuals or groups to breach the **cybersecurity** of a specific organisation or individual. This includes any unauthorised access, manipulation, or damage to computers, networks, digital systems, or data. Cyber-attacks are carried out to steal, alter, or destroy information, or to disrupt the operation of systems and services. Common forms of cyber-attacks include phishing, malware, ransomware, denial of service (DoS), and advanced persistent threats (APTs).

Cybersecurity: Is the collection of tools, policies, security concepts, security safeguards, guidelines, risk management approaches, actions, training, best practices, assurance and technologies that can be used to protect an organisation's and user's assets against **cyber-attack**.

Data analytics: The process of collecting, organising and analysing large sets of data to discover patterns and other information which an organisation can use for its future business decisions.

Data bias: Data is biased when it is not representative of the population that is being analysed. Bias can be inherent in the data collected or introduced by those analysing the data.

Data mining: Is the process of analysing large datasets to discover patterns, relationships, and insights. It employs AI and statistical algorithms to identify correlations and trends that transform raw data into actionable information.

Database: A structured collection of records or data that is stored in a computer system along with rules as to the information that will be sought from it, so that queries may be answered by interrogating the database.

Decentralisation: The ability to make decisions (ie, commit people, money and resources) is passed down to lower levels of the hierarchy.

Deliberate strategies: This is the term used to describe those intended strategies which are implemented.

Dependencies: Consider how the current and future environmental, social and governance issues can affect the organisation's ability to create and maintain value. Examples which may affect an organisation are worker health, workplace diversity, climatic conditions, resource availability, regulation, consumer expectations, other stakeholder expectations and risks to organisational reputation.

Descriptive analytics: A technique using historic quantitative or qualitative data to provide an account of what has happened, via reports, dashboard, bar charts and other visualisations. For example, descriptive analytics can be used to analyse revenue in more detail to understand and compare sales trends over time. Descriptive analytics is the simplest of the data analytics techniques and answers the question, 'what happened?'

Desk research: Is the gathering and analysis of existing or secondary data. This may use existing company reports and other information from both internal and external sources.

Development strategy: The decision on how to gain access to the chosen products and markets.

Diagnostic analytics: This takes a deeper look at data to understand the causes of events and behaviours. By using techniques such as drill-down, data mining and correlations, it is possible to examine and better understand why something happened. For example, diagnostic analytics could be used to understand the reasons behind changing demand for a product. Diagnostic analytics answers the question, 'why did this happen?'

Differentiation strategy: A differentiation strategy assumes that competitive advantage can be gained through **particular characteristics** of a firm's products.

Digital assets: Assets which are held in digital form, that is to say assets which are not available in physical form. Common examples of digital assets include: PDF files, images, audio and video files.

Digital marketing: This refers to any marketing methods conducted through electronic devices, including online marketing via the internet. A business may use websites, search engines, blogs, social media, video, email and similar channels to reach customers.

Distributed ledger: Distributed ledger technology (DLT) encompasses a range of decentralised database systems where transactions are recorded and validated across multiple sites. Unlike traditional ledgers, DLT does not rely on a central authority, enhancing transparency and reducing the need for audits. While **blockchain** is a well-known type of DLT noted for its chained data structure, other forms of DLT may organise data differently and offer varying levels of access and privacy.

Divisionalisation: The division of a business into autonomous regions (geographic divisionalisation) or product businesses (product/brand divisionalisation), each with its own revenues, expenditures, capital asset purchase programmes and profit responsibility.

Double materiality: Means considering the sustainability issues that might create financial risks for the company (financial materiality) and also sustainability issues where an organisation's activities materially impact on people and the environment (impact materiality). This concept is fundamental to the Corporate Sustainability Reporting Directive (CSRD).

Downside risk (pure risk): The possibility that the outcome will be worse than expected ie, something will go wrong (the worst-case scenario).

ESG: Environmental, social and governance (ESG) examines how organisations can approach sustainability-related issues. Therefore, ESG considers the impact of these risks on business and enterprise value (rather than on society more broadly). ESG is frequently used terminology in corporate and investment communities (see double materiality).

ESG is another characterisation of sustainability, providing the three lenses of environment, social and corporate governance to use when exploring relevant sustainability factors.

Economies of scale: Economies arising within the business from the organisation of production (internal economies), or attainable by the business because of the growth of the industry as a whole (external economies).

Economy: A measure of the actual resources used (inputs) eg, cost of labour per employee or the rent per square metre.

Ecosystem: A business ecosystem is the network of organisations that enable the delivery of a product or service. Participants in the ecosystem, such as suppliers, distributors, customers, competitors and government agencies, interact through competition or collaboration.

Effectiveness: A measure of the impact achieved and considers outputs in relation to objectives. It looks at the extent to which the processes used by the business deliver the right results, eg, percentage of products delivered without faults, market share.

Efficiency: A measure of productivity. It considers the relationship between the goods or services produced (outputs) and the resources used to produce them (inputs) eg, output per person or per £ of labour, sales per square metre.

Emergent change: Views change as a series of continuous open-ended adjustments to the environment.

Emergent strategies: Are those behaviours which are adopted and which have a strategic impact. Emergent strategies adapt to human needs and evolve continuously.

Entity risks: Are all of the risks that affect an entity, how it trades, the markets and countries it operates in, the decisions that management make, the operational, strategic, hazard, financial and compliance risks the entity faces. It is important to recognise that 'entity risk' is wider than the term 'business risk.' All of the risks listed below make-up an organisation's entity risk:

Ethics: A system of behaviour which is deemed acceptable in the society or context under consideration.

Explicit knowledge: Is knowledge that the company knows that it has. This includes facts, transactions and events that can be clearly stated and stored in management information systems.

Field research: Involves the collection of new (primary) information direct from respondents. As such it is usually more expensive than desk research and so is only performed if desk research fails to answer all questions asked.

Financial risk: Is due to the financial structure and policies of a business. It comprises controllable and uncontrollable considerations. The controllable element of financial risk is concerned with how the business chooses to finance itself. Issues cover the entity's choice of finance structure including the level of gearing or leverage, its exposure to credit risk and liquidity risks. The uncontrollable element of financial risk is likely to relate to factors beyond the entity's control such as interest and exchange rates, taxes and the state of economy. These factors may contribute to the risk of business failure; though they can be managed to some extent they cannot be controlled as such by the entity.

Focus (niche) strategy: A firm concentrates its attention on one or more particular segments or niches of the market, and does not try to serve the entire market with a single product. Within the niche, the firm can either aim to be a cost leader or pursue differentiation.

Franchising: A method of expanding the business on less capital than would otherwise be possible. Franchisers include IKEA, McDonald's, Starbucks and Pizza Hut.

Functional (operational) strategies: This refers to the main functions within each SBU, such as production, purchasing, finance, human resources and marketing, and how they deliver effectively the strategies determined at the corporate and business levels.

Gap analysis: The comparison between an entity's ultimate objective and the expected performance from projects both planned and under way, identifying means by which any identified difference, or gap, might be filled.

Generative AI: Refers to a type of **artificial intelligence (AI)** that is designed to generate new content, such as images, text, audio, or video, based on patterns learned from a training dataset. Unlike predictive models that focus on making predictions or classifications based on existing data, generative AI models create entirely new data instances that mimic the characteristics of the training data.

Generative AI is advancing rapidly. ChatGPT was released in November 2022 and, since that point, new iterations of generative AI technology have been released several times a month. McKinsey research has found that generative AI features could add up to US$4.4 trillion annually to the global economy and it is now estimated that the technology will achieve a median level of human performance for many capabilities, such as creativity, logical reasoning and problem solving, by 2030 (McKinsey 2023). Generative AI will likely have the biggest impact on knowledge-based work with the fields of education, law, technology and the arts seeing many roles and tasks being automated.

(Source: McKinsey & Company (2023) *What's the future of generative AI? An early view in 15 charts*. [Online] Available from: https://www.mckinsey.com/featured-insights/mckinsey-explainers/whats- the-future-of-generative-ai-an-early-view-in-15-charts [Accessed 19 June 2024])

Giffen goods: Are consumed more as the price increases, effectively reversing the law of demand. Giffen goods are generally inferior goods without any clear substitute.

Gig economy: Employment is temporary or part-time involving freelance and independent contractors. Businesses and contractors can benefit from the flexibility it provides but contractors also experience fluctuating working hours and pay.

Globalisation: The production and distribution of products and services of a homogenous type and quality on a worldwide basis.

Green finance: Can refer to financial investments into environmental goods and services, for example the protection of biodiversity. It can also refer to schemes to encourage sustainability, such as policies to encourage investment in renewable energies. Green finance can also specifically relate to the financial systems that deal with green investments, which are known as green bonds and green funds.

Harmon's process-strategy matrix: Is used in the analysis of business processes. It assesses the strategic importance of a process as well as its complexity and dynamics to determine how that process should be managed.

Hazard risk: The exposure a business may have to natural events and their impacts. Such risks may arise from factors outside of the control of the entity. They may also arise from factors that would ordinarily be within the control of the organisation, such as the actions of employees, the consequences of accidents etc, be it on the business, its trading partners or customers.

Human capital: The knowledge, skills, competencies and attributes embodied in individuals that contribute to improved performance and wellbeing. Human capital also refers to the economic value of a worker's experience and skills. Investing in people, through health care, quality education, jobs and skills can develop human capital.

Human capital: Is concerned with the collective attributes of an organisation's human resources. It includes the capabilities, creativity, skills and knowledge of an organisation's staff which combine to enable the organisation to create economic value.

Human resource management (HRM): 'A strategic and coherent approach to the management of an organisation's most valued assets: the people working there who individually and collectively contribute to the achievement of its objectives for sustainable competitive advantage' (Armstrong).

Hybrid working: A flexible working arrangement whereby employees work in the office occasionally but work remotely for the remainder of their contracted hours.

Impacts: Consider how the decisions an organisation makes or its actions either positively or negatively affect environmental, societal and governance issues. Examples include worker rights, human rights, health-and-safety policy, waste, greenhouse gas (GHG) emissions, water usage, land usage and biodiversity.

Incremental change: Is characterised by a series of small steps. It is a gradual process.

Industry: A group of organisations supplying a market offering similar products using similar technologies to provide customer benefits.

Industry life cycle: Describes the phases of development that an industry goes through. The key stages of the industry life cycle are: introduction; growth; shakeout; maturity; and decline.

Information management strategy: Strategy specifying who controls and uses the technology provided.

Information systems (IS): All systems and procedures involved in the collection, storage, production and distribution of information.

Information systems strategy: Strategy specifying how hardware, software and telecommunications can achieve delivery of the information systems strategy.

Information technology strategy: Strategy specifying the systems that will best enable the use of information to support the business strategy.

Infrastructure-as-a-service (IaaS): Provides virtual computing resources over the internet. It allows businesses to rent servers, storage space, and networking capabilities, without needing to own physical infrastructure. Users can manage operating systems, storage, and deployed applications, while the service provider handles the physical hardware.

Innovation: Is concerned with the generation of new ideas of how to do business. It is primarily a creative activity.

Intended strategies: Are those which are conscious plans imposed by management. If they are implemented, they are referred to as deliberate strategies.

Internet of things: A system of interrelated computing devices, mechanical and digital machines, or objects that are provided with unique identifiers and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction.

Joint venture: Is a contractual arrangement whereby two or more parties undertake an economic activity which is subject to joint control.

Just-in-time manufacturing: An approach to planning and control based on the idea that goods or services should be produced only when they are ordered or needed. Also called lean manufacturing.

Key performance indicator (KPI): A measure of the level of performance in an area where a target level must be achieved for the business to outperform rivals and achieve competitive advantage.

Knowledge: Is the potential for action based on data, information, intuition and experience.

Knowledge management: Describes a range of strategies and tools that capture all the knowledge that is valuable to an organisation, and deliver it to the people in such a way that it can be acted on quickly, to the competitive advantage of the business.

Licensing agreement: Permission from one party to another to manufacture or sell a product, or to use a brand name.

Limiting factor: A factor which at any time, or over a period, may limit the activity of an entity, often occurring where there is shortage or difficulty of availability.

Machine learning: Is the capability of computer systems to learn from and make predictions or decisions based on large volumes of training data, without being explicitly programmed to do so. It involves two primary methods: supervised and unsupervised learning. In supervised learning, the model is trained on labelled data, meaning it learns to predict outcomes from input data that has an associated correct answer. Unsupervised learning involves training on data without labelled responses, allowing the model to identify patterns and relationships in the data independently.

Market segmentation: A tool of marketing strategy that helps management to identify new markets, and to use a marketing mix to arrive at a desirable market position.

Market segmentation: The division of the market into homogeneous groups of potential customers who may be treated similarly for marketing purposes.

Marketing: Can be defined as the set of human activities directed at facilitating and consummating exchanges. From this it can clearly be seen that it covers the whole range of a firm's activities.

Alternatively, marketing can be defined as 'the management process which identifies, anticipates and supplies customer requirements efficiently and profitably' *(Chartered Institute of Marketing)*.

Marketing mix: The set of controllable marketing variables that a firm blends to produce the response it wants in the target market.

Marketing research: Is the systematic gathering, recording and analysing of information about problems relating to the marketing of goods and services.

Mean: The mean (or average) of a set of data is calculated by taking the sum of all the values and dividing by the number of values in the distribution. The mean can be taken as the expected value of a data set.

Merger: Concerns the joining of two separate companies to form a single company.

Mission: The values and expectations of those who most strongly influence strategy about the scope and posture of the organisation (Johnson, Scholes and Whittington).

Modern slavery: Is the severe exploitation of people for commercial gain. Modern slavery covers a wide range of abuse and exploitation including sexual exploitation, domestic servitude, forced labour, criminal exploitation and organ harvesting.

Natural capital: The stock of renewable and non-renewable natural resources that combine to yield a flow of benefits or 'services' to people (eg, biodiversity such as plants and animals, air, water, soils, minerals).

The flows can be ecosystem services or abiotic services; which provide value to business and to society.

Natural language processing: Is a branch of **AI** that enables computers to understand, interpret, and generate human language in a way that is both meaningful and useful. By employing techniques from linguistics and computer science, NLP helps machines process and analyse large amounts of natural language data, facilitating tasks such as translation, sentiment analysis, voice-activated services and **chatbot** responses.

Network architecture: The network of relational **contracts**, within or around, the firm.

Normal distribution: A frequency distribution which is important because it arises frequently in 'real- life.' It is any distribution that is symmetrical around the mean.

Not-for-profit organisations: The **primary goals** of such organisations are **non-financial** and can vary enormously. Goals often include meeting members' needs, contributing to social well-being and pressing for political and social change. Secondary goals will include the economic goal of not going bankrupt and, in some cases, generating a financial surplus to invest in research or give to the needy.

Often, the goals of NFP organisations will reflect the need to maximise the benefit derived from limited resources, such as funds. Their objectives may be more heavily influenced by external stakeholders such as the government.

Observer bias: This relates to interpretation and occurs when observing and recording results. The researcher allows their assumptions (which may be unconscious) to influence their observations.

Omitted variable bias: This occurs when a variable is excluded from the data model and therefore the cause of a change in one variable is incorrectly attributed to another variable in the model. For example, analysis of sales incorrectly attributes an increase to a new advertising campaign when it is in fact caused by a fall in interest rates that has encouraged consumer expenditure.

Operational risk: Variability arising from the effectiveness of how the business is managed and controlled on a day-to-day basis, the accuracy and effectiveness of its information/accounting systems, its reporting systems and its management and control structures. Operational risk also encompasses compliance with issues such as health and safety, consumer protection etc.

Operations management: Is concerned with the **design**, **implementation** and **control** of the **processes** in an organisation that transform inputs (materials, labour, other resources, information and customers) into output products and services.

Opportunity cost: The cash flow foregone if a unit of the resource is used on the contract/project instead of in the best alternative way.

Organic growth: Involves the expansion of a firm's size, profits, and activities through the use of its own resources and capabilities without taking over other firms.

Organisational structure: How the various functions of an organisation (operations, marketing, HR, finance etc,) are formally arranged into departments or sections, and how responsibility and authority are allocated.

Outsourcing: The strategic use of outside resources to perform activities traditionally handled by internal staff and resources (thebalance.com).

Partnership: Joint participation between two or more organisations in the serving of a market or project, characterised by the close interrelationship of operations, exchange of staff and mutual trust and commitment to working with the other(s).

Physical risks: Physical risks arise from the physical effects of climate change such as storms, extreme temperatures, wildfires, flooding.

Planned change: Involves following a series of pre-planned steps.

Platform-as-a-service (PaaS): Offers a development and deployment environment delivered over the internet, allowing users to build, test, and deploy applications without the complexity of managing the underlying infrastructure. Providers supply a framework that developers can build upon, including support for software development kits, databases, and development tools. PaaS is designed for developers who want to create applications without worrying about hardware, software updates, or storage.

Population: The entire set of data from which a sample is selected for analysis.

Porter's Diamond: Helps organisations to understand the determinants of national competitive advantage. It identifies these as key determinants: firm strategy, structure, rivalry; demand conditions; related and supporting industries; factor conditions.

Porter's Five Forces: According to Michael Porter, five competitive forces influence the state of competition in an industry: threat of new entrants; bargaining power of customers; threat of substitute products or services; bargaining power of suppliers; rivalry among existing firms.

Positioning: The overall location of a product or service in the buyer's mind in relation to other competing products, services /brands.

Predictive analytics: This technique uses data to predict future trends and events. Historical data is used to make assumptions about the future, using techniques such as regression analysis. For example, predictive analytics could be used to predict a company's future annual cash flows.
Predictive analytics answers the question, 'what might happen in the future?' Closely linked to the term data analytics is data mining.

Prescriptive analytics: Data is used to determine the best course of action. Machine-learning algorithms are an important part of prescriptive analytics because they can sort through large quantities of data to make recommendations. For example, prescriptive analytics is used to curate customers' social media feeds and recommendations based on content individuals have previously engaged with. Prescriptive analytics answers the question, 'what should we do next?'

Price discrimination: Sometimes also referred to as differential pricing, involves setting different prices for a similar product in different parts of the market.

Price elasticity of demand: Measures how far demand for a good will change in response to a change in its price.

$$\text{PED} = \frac{\text{Change in quantity demanded, as a \% of original demand}}{\text{Change in price, as a \% of original price}}$$

Probability: Is a measure of **likelihood** and can be stated as a percentage, a ratio, or more usually as a number from 0 to 1. It is a measure of the likelihood of an event happening in the long run, or over a large number of times.

$$\text{Probability of achieving the desired results} = \frac{\text{Number of ways of achieving desired result}}{\text{Total number of possible outcomes}}$$

Number of ways of achieving desired result Total number of possible outcomes

Product/market strategy: Concerns the decision on what products to offer and the markets to be served. This is closely linked to Ansoff's growth vector matrix.

Profit seeking organisations: The **primary goal** for such organisations is the delivery of economic value to the owners, ie, to increase shareholder wealth. Goals such as satisfying customers, building market share, cutting costs, and demonstrating commitment to ESG are **secondary goals** which enable economic value to be delivered.

Purchasing: Is the acquisition of material resources and business services for use by the organisation.

Qualitative data: Is fundamentally concerned with trends in opinions, thoughts and motivations, as a result it tends to be subjective. Qualitative data cannot be calculated or computed.

Quality assurance: Focuses on the way a product or service is produced. Procedures and standards are devised with the aim of ensuring defects are eliminated (or at least minimised) during the development and production process).

Quality control: Is concerned with checking and reviewing work that has been done. Quality control is focused on detecting defects in the output produced, as a result it has a narrower focus than quality assurance.

Quantitative data: Is focused on numerical data which can be calculated and computed.

Rational planning model: A prescriptive approach to strategy formulation. There are three main stages to the model: strategic analysis, strategic choice and strategy implementation.

Re-positioning: Changing the identity of a product or service, relative to the identity of competing products or services, in the collective minds of the target market.

Regression analysis: Aims to specify the relationship between two or more variables. One of these variables is the dependent variable, whose value depends on the independent variable(s).

Relationship marketing: Management process that seeks to attract, maintain and enhance customer relationships by focusing on the whole satisfaction experienced by the customer when dealing with the firm.

Relevant cash flows: Future, incremental, cash flows arising from the decision being made.

Remote working: Also known as working from home, refers to the practice of employees working outside of a traditional office setting. Remote workers have the flexibility to work from their homes, coworking spaces or any other location that is productive for the employee.

Residual income (RI): Is a divisional performance measure. It is calculated as: Divisional profit – Net assets of division (required rate)

Resource audit: Cataloguing the assets of an organisation by considering physical resources, intangibles, human resources, technological resources and financial resources.

Return on capital employed (ROCE): Is a divisional performance measure whose principle is that the return (profit) should be in excess of the cost of capital, in order to increase shareholder wealth.

Risk: The possible variation in outcome from what is expected to happen.

Risk appetite: Is the extent to which a company is prepared to take on risks to achieve its objectives.

Risk assessment: Involves establishing the financial consequences of each risk event (severity) and its likelihood of occurrence (frequency).

Risk evaluation: The process by which an organisation determines the significance of any risk and whether those risks need to be addressed.

Risk management: The process of identifying and assessing (analysing and evaluating) risks and the development, implementation and monitoring of a strategy to respond to those risks.

Risk register: Lists and prioritises the main risks the organisation faces. It is used as the basis for decision-making on how to deal with risks.

Robotic process automation (RPA): Is a solution to replicate a business process, designed to do a task that would otherwise be performed manually. It is programmed to follow 'if this, then that' instructions. It runs with other programs rather than replacing them.

(Source: https://www.icaew.com/technical/technology/technology-and-the-profession/risks-and- assurance-of-emerging-technologies/what-is-cognitive-technology)

Robots: Are programmable machines capable of carrying out complex actions autonomously or semi-autonomously. Typically equipped with sensors and actuators, robots are used in various industries for tasks that require precision, endurance, or involve hazardous conditions.

Sample: A subset of a larger population of data, facilitating more timely analysis of the data.

Scenario planning: Concerns the development of pictures of potential futures for the purposes of managerial learning and the development of strategic responses.

Selection bias: This describes bias caused by a lack of random data selection so that the sample of data is not representative of the population.

Self-selection bias: This occurs when individuals have the chance to select themselves to appear in a population of data. For example, a customer satisfaction survey is most likely to be answered by individuals who are either extremely happy or extremely unhappy with the service received. The majority of customers will simply not reply so the results are not representative of all customers.

Shared service centre: A number of internal transaction processing activities which had previously been conducted in a number of different departments, or business units, are brought together into one site within an organisation.

Skewness: A statistical measure that describes the asymmetry of a probability distribution. A normal distribution is symmetrical with its left and right sides being mirror images of each other. However, when a distribution is asymmetrical, its left and right sides are not identical.

Social capital: The networks together with shared norms, values and understanding that facilitate cooperation within and among groups. Social capital enables people to work together effectively to achieve a common purpose or goal.

Software-as-a-service (SaaS): Is a delivery model that allows users to access software applications over the internet. These applications are hosted and maintained by external providers, offering convenience and reduced IT overhead for users. SaaS solutions are typically subscription-based, with common applications including email, customer relationship management (CRM) and collaboration tools.

Span of control: The number of subordinates reporting to one person.

Stakeholders: Are those groups or persons with an interest in what an organisation does.

Standard deviation: This shows the amount of variability in a data set, showing how far, on average, each result lies from the mean (or expected value).

Step change: Describes an unexpected jump (upwards) or drop (downwards) in the pace of change. The step is caused by an unexpected event (eg, environmental disaster, unexpected change in government etc).

Strategic business unit: This is a section, within a larger business, which is responsible for planning, developing, producing and marketing its own products or services for a distinct external market.

Strategic risk: Risks associated with the long-term strategic objectives of the business, potential variability of business returns arising as a result of the company's strategy and its strategic position with respect to competitors, customers, reputation, legal or regulatory change, political change.
Strategic risk also encompasses knowledge management, ie, the effective management and control of the knowledge resources including key personnel, intellectual property and production technology.

Strategy: 'Is the direction and scope of an organisation over the long term, which achieves advantage for the organisation through its configuration of resources within a changing environment, to meet the needs of markets and to fulfil stakeholder expectations' (Johnson, Scholes, Whittington).

Supply chain management (SCM): The management of all supply activities from the suppliers to a business through to delivery to customers.

Survivorship bias: This is the tendency towards studying successful outcomes while excluding unsuccessful outcomes.

Sustainability: The ability to 'meet the needs of the present without compromising the ability of future generations to meet their own needs' (Brundtland Report 1987).

Sustainable development: Is the process by which we achieve sustainability.

Sustainable enterprise: A company, institution or entity that generates continuously increasing stakeholder value through the application of sustainable practices through the entire base activity – products and services, workforce, workplace, functions/processes, and management/governance (Deloitte Creating the Wholly Sustainable Enterprise).

Synergy: The benefits gained from two or more businesses combining that would not have been available to each independently. Sometimes expressed as the 2 + 2 = 5 effect. Synergy arises because of complementary resources which are compatible with the products or markets being developed and is often realised by transferring skills or sharing activities.

Tacit knowledge: Is personal knowledge and expertise held by people within the organisation that has not been formally documented.

Targeting: Involves selecting the most attractive market segments.

The Taskforce on Climate-related Financial Disclosures (TCFD): Was formed in 2015 by the Financial Stability Board (FSB) with a goal of developing a set of voluntary climate-related financial risk disclosures. The disclosure recommendations are structured around four core elements: governance; strategy; risk management; and metrics and targets.

The business partner model: Has resulted in finance professionals adopting a more commercially- focused, action-orientated approach. Increasingly, finance professionals are being embedded within operational departments and business units so that they can provide financial insights in a more integrated manner than if they were based in the finance department alone.

Threshold competences: The activities and processes involved in using and linking the firm's resources **necessary to stay in business**.

Threshold resources: The **basic** resources needed by **all** firms in the market.

Transactions marketing: A management approach that focuses on the product, and develops marketing mixes for a product according to the needs customers satisfy when they buy it.

Transfer price: The price at which one division in a group sells its products or services to another division in the same group.

Transformational change: Is characterised by major, significant change being introduced relatively quickly.

Transition risks: Relate to social and economic shifts to a low-carbon economy such as changes to policy, regulation, technology and markets.

In the face of increasing climate-related impacts and dependencies, organisations need to focus on both climate change mitigation and climate adaptation activities.

Uncertainty: The inability to predict the outcome from an activity due to a lack of information.

Uncontrollable risk: Risks outside of the organisation's direct control and tend to be driven by external changes; factors include changes in economic conditions and advances in technology.

Unique resources: Those resources which give the firm a sustainable competitive advantage over its competitors, enabling it to meet the CSFs. They are resources which are **better** than those of the competition and **difficult to replicate**.

Upside risk (speculative risk or opportunity): The possibility that something could go better than expected (best case scenario).

Value chain: The sequence of business activities by which value is added to the products or services produced by an entity.

Veblen goods: Demand for the good rises as price increases due to consumer perception that a higher price reflects better quality. It is also argued that a higher price may make a product desirable as a status symbol. Veblen goods are generally luxury or designer products.

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